Electronic Principles and Devices · Unit_01 · Unit-1 Semiconductor Diode

DRAFT — UE26EC141A · Unit_01 — Unit-1 Semiconductor Diode · Complete Unit Master

Source-complete searchable master · transcript, slide text, page visuals and archive references

Hugging Face PESU_Materials source tree, official units.json catalogue, transcript sidecars, slide text/PDF/PPT sources and the selected First_Year_8enterprise archive; see the provenance datasets in the release ZIP. Generated 18 October 2026.

· How to use this pack

Search this single unit master by lesson, timestamp, slide text or source page. Read the transcript in timestamp order, then compare extracted slide text with its page-faithful source figure. Formula candidates and archive question mappings are explicitly flagged for review. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

Source text	Verbatim or OCR-derived text, kept separate from editorial notes.
Candidate	Automated topic/source association requiring manual confirmation.
Low ASR confidence	Transcript segment requires audio verification.
DRAFT	Human review remains incomplete.

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M1

Unit_01 · Unit-1 Semiconductor Diode

1 UE26EC141A · Unit_01: Unit-1 Semiconductor Diode

This unit master organizes source material for Electronic Principles and Devices, Unit_01, officially titled Unit-1 Semiconductor Diode. It includes 11 video transcripts and 13 source slide decks, with page-level text and rendered images where available. This draft preserves source wording and page order; it does not claim a complete human subject review or full audiovisual review. EV0000001

SHORT-NOTE RECAP

Official catalogue entry: UE26EC141A, Unit_01 (Unit-1 Semiconductor Diode). Bucket folder present: yes. Videos: 11; slide decks: 13.

EV0000001

2 Introduction to semiconductors, intrinsic and extrinsic semiconductors

This lesson is indexed from the source video titled Introduction to semiconductors, intrinsic and extrinsic semiconductors. Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or rewritten. A candidate slide deck is grouped here according to the current matching score; the exact path and page locators are listed in the source evidence.

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Graphs and diagrams

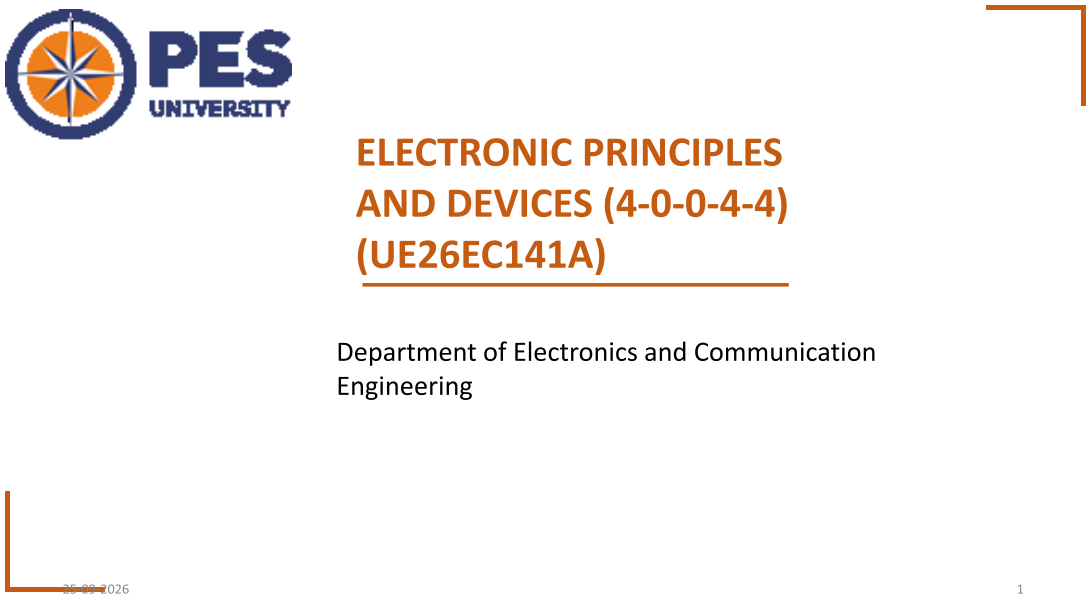

Figure G000001. Introduction to semiconductors intrinsic and extrinsic semiconductors · source slide page 1.

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How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

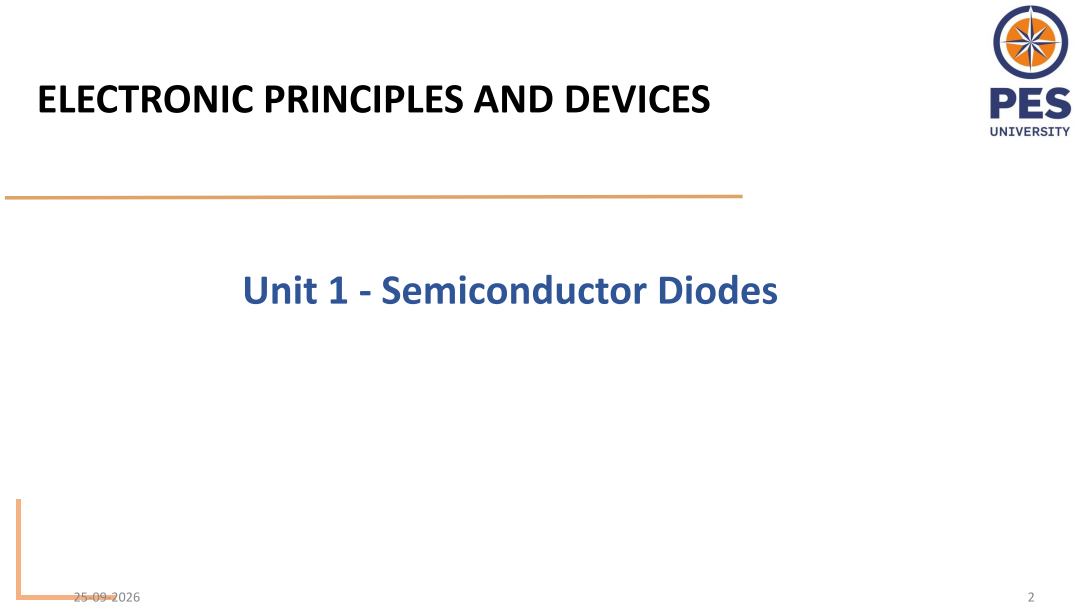

Figure G000002. Introduction to semiconductors intrinsic and extrinsic semiconductors · source slide page 2.

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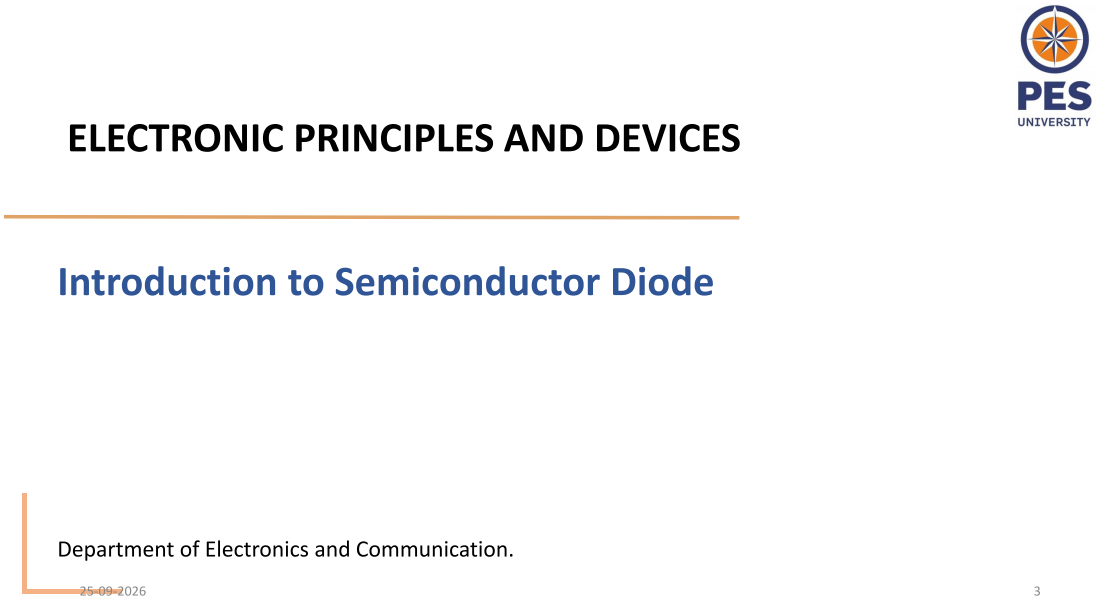

Figure G000003. Introduction to semiconductors intrinsic and extrinsic semiconductors · source slide page 3.

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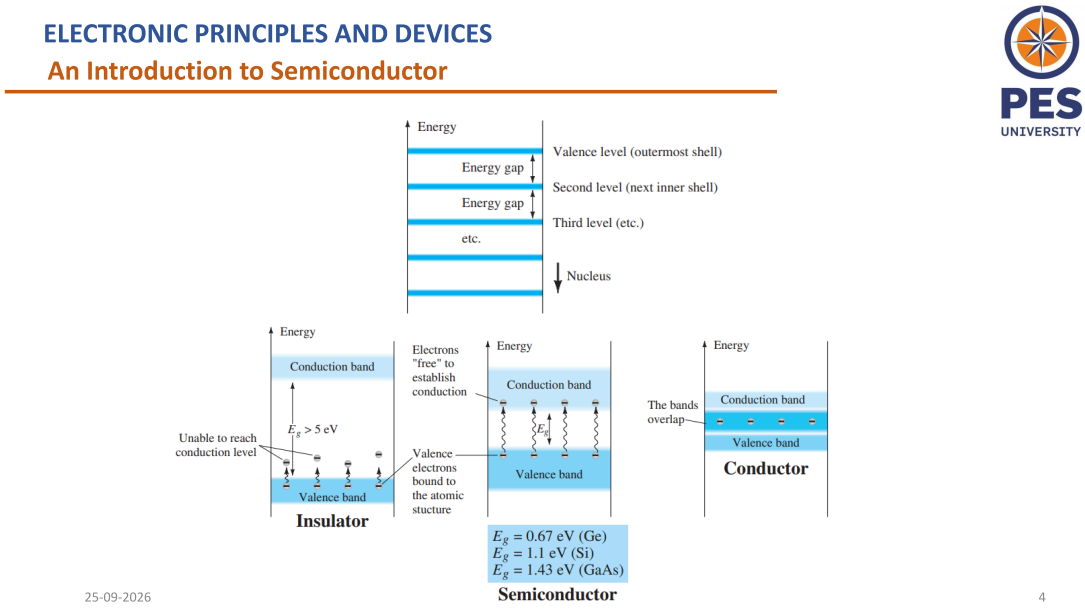

Figure G000004. Introduction to semiconductors intrinsic and extrinsic semiconductors · source slide page 4.

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> ***How to read it.*** Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

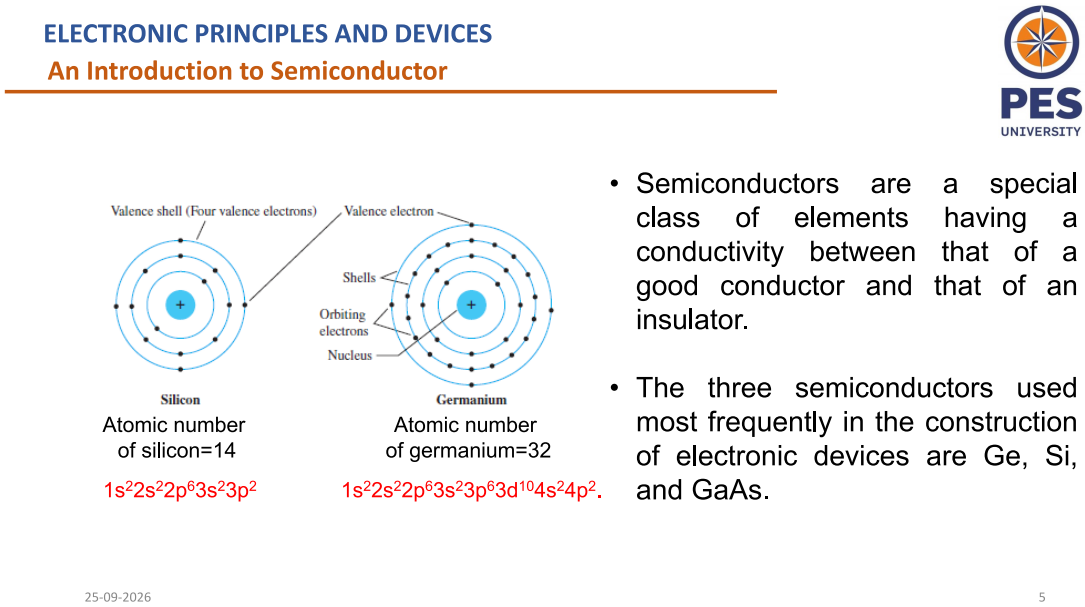

Figure G000005. Introduction to semiconductors intrinsic and extrinsic semiconductors · source slide page 5.

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ELECTRONIC PRINCIPLES AND DEVICES

An Introduction to Semiconductor

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- The term intrinsic is applied to any semiconductor material that has been carefully refined to reduce the number of impurities to a very low level—essentially as pure as can be made available through modern technology.
- A semiconductor material that has been subjected to the doping process is called an extrinsic material.

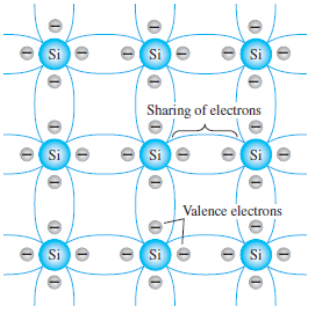

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Figure G000006. Introduction to semiconductors intrinsic and extrinsic semiconductors · source slide page 6.

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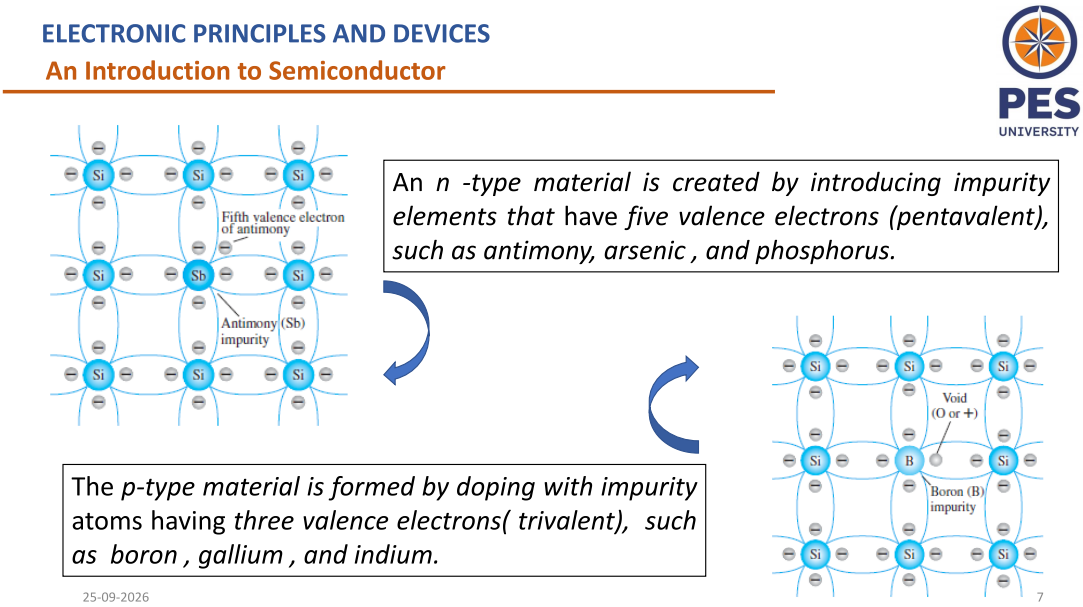

Figure G000007. Introduction to semiconductors intrinsic and extrinsic semiconductors · source slide page 7.

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How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Figure G000008. Introduction to semiconductors intrinsic and extrinsic semiconductors · source slide page 8.

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Important points and why

1. Source slide page 1 (extracted text): ELECTRONIC PRINCIPLES AND DEVICES (4-0-0-4-4) (UE26EC141A) Department of Electronics and Communication Engineering 25-09-2026 1 EV0000102 · EV0000103 · EV0000104
2. Source slide page 2 (extracted text): ELECTRONIC PRINCIPLES AND DEVICES Unit 1 - Semiconductor Diodes 25-09-2026 2 EV0000105 · EV0000106 · EV0000107
3. Source slide page 3 (extracted text): ELECTRONIC PRINCIPLES AND DEVICES Introduction to Semiconductor Diode Department of Electronics and Communication. 25-09-2026 3 EV0000108 · EV0000109 · EV0000110
4. Source slide page 4 (extracted text): ELECTRONIC PRINCIPLES AND DEVICES An Introduction to Semiconductor 25-09-2026 4 EV0000111 · EV0000112 · EV0000113
5. Source slide page 5 (extracted text): ELECTRONIC PRINCIPLES AND DEVICES An Introduction to Semiconductor • Semiconductors are a special class of elements having a conductivity between that of a good conductor and that of an insulator. • The three semiconductors used Atomic number Atomic number most frequently in the construction of silicon=14 of germanium=32 of electronic devices are Ge, Si, 1s22s22p63s23p2 1s22s22p63s23p63d104s24p2. and GaAs. 25-09-2026 5 EV0000114 · EV0000115 · EV0000116

6. Source slide page 6 (extracted text): ELECTRONIC PRINCIPLES AND DEVICES An Introduction to Semiconductor • The term intrinsic is applied to any semiconductor material that has been carefully refined to reduce the number of impurities to a very low level— essentially as pure as can be made available through modern technology. • A semiconductor material that has been subjected to the doping process is called an extrinsic material. 25-09-2026 6 EV0000117 · EV0000118 · EV0000119
7. Source slide page 7 (extracted text): ELECTRONIC PRINCIPLES AND DEVICES An Introduction to Semiconductor An n -type material is created by introducing impurity elements that have five valence electrons (pentavalent), such as antimony, arsenic , and phosphorus. The p-type material is formed by doping with impurity atoms having three valence electrons(trivalent), such as boron , gallium , and indium. 25-09-2026 7 EV0000120 · EV0000121 · EV0000122
8. Source slide page 8 (extracted text): THANK YOU Department of Electronics and Communication 25-09-2026 8 EV0000123 · EV0000124 · EV0000125

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:05.040] Good morning, and [00:00:07.890] welcome to Electronic Principles and Devices. [00:00:11.791] Today we are going to learn about [00:00:16.380] unit one. In that, topic one, [00:00:20.100] an introduction to semiconductor. [00:00:24.180] So semiconductors are special class of elements, having a conductivity between that [00:00:29.460] of a good conductor and that of an insulator. [00:00:34.590] The three semiconductors which is most commonly or most frequently used in the [00:00:39.600] construction of electronic devices are germanium, silicon, and gallium arsenide. [00:00:46.590] And we also see that the atomic structure of silicon and germanium. [00:00:52.140] First, if you look at the atomic structure of silicon, the atomic number is 14, and [00:00:57.990] it's represented in terms of SPDF model one S two, two S two, two P six, three S [00:01:02.700] two. [00:01:04.560] And if you start arranging with respect to the shell around, [00:01:10.440] we call it as orbits. So the first orbit can have two, second can have eight, [00:01:15.961] and the remaining four, that is three S two, three P two are at the outside. [00:01:21.510] We call it as a valence shell, which has four valence electron. [00:01:25.650] Similarly, any semiconductor material, if you look at example again, germanium [00:01:31.530] 32, the outermost valence electron will be four again. [00:01:36.029] That means to say, [00:01:40.711] they are not stable. So they themselves will try to be stable [00:01:46.650] with the help of a bond called covalent bond. [00:01:51.810] But before that, they [00:01:54.330] try to balance by themselves by moving from lower energy level to the higher energy [00:01:59.250] level. So semiconductor in comparison with an insulator, if you look at, there is a [00:02:05.130] large energy gap between the conduction band and the valence band in the case of an [00:02:09.961] insulator. There is a no gap at all. [00:02:13.380] There's an overlapping in the case of a conductor between the conduction band and [00:02:17.055] the valence band. Whereas in the case of a semiconductor, as we see, there is a [00:02:22.500] little gap compared to the insulator and conductor. [00:02:27.240] So they could, with a change in the temperature or with the applied voltage, the [00:02:34.380] charges will start move from lower energy level to higher energy level. [00:02:39.960] So keeping this in mind now, [00:02:44.130] if you go back, the outermost valence electron, they try to move from lower band to [00:02:49.590] higher band. [00:02:52.351] So the [00:02:54.421] semiconductor has been classified into intrinsic and extrinsic semiconductor. [00:02:58.530] The term intrinsic is applied to any semiconductor material that carefully refine [00:03:04.020] to reduce the number of impurities to a very low level. [00:03:07.530] That is, we can

call it as a pure form, made available through modern technology. [00:03:12.870] Whereas extrinsic semiconductor are those that can be subjected to doping process [00:03:17.550] with the impurities being added to that. [00:03:20.880] So as we see the structure of silicon with, [00:03:28.140] as we see here, it's having an outermost four valence bond, forming a covalent bond [00:03:33.421] by sharing of an electron with a neighboring. [00:03:36.360] Even in the upper part, even in the lower, even in the lower and also the side we [00:03:40.201] see that there is a covalent bond where valence electrons are shared. [00:03:45.645] So this is the concept on which the semiconductor working principle depends [00:03:51.541] on. With the increase in the voltage, the semiconductor act like a conductor where [00:03:56.010] covalent bond is ruptured, with the temperature reduced [00:04:01.921] act like an insulator where the covalent bonds are not ruptured. [00:04:09.090] So keeping this in mind now, [00:04:12.840] these extrinsic semiconductor has been classified based on the impurity level [00:04:18.810] added. For example, if you look at the first diagram here, as marked here, where [00:04:24.570] the impurity added is the antimony impurity has been added, where outermost five [00:04:30.150] valence electrons are there. We know the four valence electrons are shared with the [00:04:34.140] neighboring silicon atom, and one is free, [00:04:38.370] the fifth valence electron of the antimony is free. [00:04:41.550] And if you apply the voltage to this [00:04:45.960] bonding arrangement, and covalent bond is ruptured, one more electrons will get [00:04:49.920] free up. So there are two [00:04:53.760] free electrons with one hole left over. [00:04:57.630] That means the number of electrons are more compared to the holes here.

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Transcript block 2 (source interval 000300-000599 seconds): [00:05:02.970] So electrons are N-type. So the type of the [00:05:06.900] structure obtained is N-type material, which is created by introducing the impurity [00:05:10.980] element that have five valence electron pentavalent, and this could be possible [00:05:14.880] with the addition of an impurity such as antimony, arsenic, and phosphorus. [00:05:20.220] Similarly, if I look at [00:05:24.240] here, there is a structure which has a boron as an impurity introduced here [00:05:30.240] with three valence electron, if the trivalent impurity has been added and one is [00:05:34.320] void. No covalent bond is formed, so already there exists one bond. [00:05:39.600] With the addition of an impurity over there, [00:05:43.440] results in one hole. With the further [00:05:49.290] change, means you say with applied voltage, one more covalent bond is ruptured, and [00:05:54.510] a free electron is released and a hole is introduced over there. [00:05:58.904] So there are two holes and one electron where the number of hole charges are more. [00:06:02.850] The positive charge material is formed with the introduction of a trivalent [00:06:06.210] impurity with the three valence electron such as boron, gallium, and indium [00:06:11.341] materials. [00:06:16.800] Next, [00:06:19.680] let us see what is a diode, where diode is one which has a two airports. [00:06:25.170] We can construct our first solid state electronic device, that is a semiconductor [00:06:29.610] diode, by sandwiching P-type and N-type material together to form a depletion [00:06:34.800] region between the two. It is [00:06:37.980] a fabrication of one material with majority charge carrier of [00:06:43.260] electron to one with majority carrier of holes. [00:06:47.730] How do we do a biasing of a diode? [00:06:51.210] Where we go for forward bias and reverse bias. [00:06:55.380] In a forward bias, a material type connected to the same polarity [00:07:02.701] terminal of a voltage source. Whereas the reverse bias, a material type connected [00:07:06.765] to a opposite polarity terminal of the voltage source. [00:07:11.430] As shown in the diagram now, a P-type and N-type material sandwich with the [00:07:16.680] depletion region form. Now a P-type is connected to the same material, that is

[00:07:22.440] P terminal and a N [00:07:26.760] region is connected to the negative terminal, then it's said to be the forward [00:07:30.990] bias. And if you reverse this, it's called as a reverse bias. [00:07:35.730] We see the different type of diodes in the diagram here. [00:07:38.760] Various companies have come up with the various [00:07:41.370] structures of diodes here as we see. [00:07:47.130] It could be a two terminal or a three terminal also sometime. [00:07:50.910] So that could depend on the company and even the representation here, we see [00:07:56.401] the normal resistor like or a [00:08:01.770] IC-based [00:08:03.720] diodes, which could be designed by the different companies. [00:08:09.570] Thank you.

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Full transcript: 1092 normalized words, 105 timestamped segments, 495.872 seconds duration. Low-confidence segments flagged: 0.

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3 Semiconductor diode under no bias, forward bias and reverse bias conditions

This lesson is indexed from the source video titled Semiconductor diode under no bias, forward bias and reverse bias conditions. Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or rewritten. A candidate slide deck is grouped here according to the current matching score; the exact path and page locators are listed in the source evidence.

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Graphs and diagrams

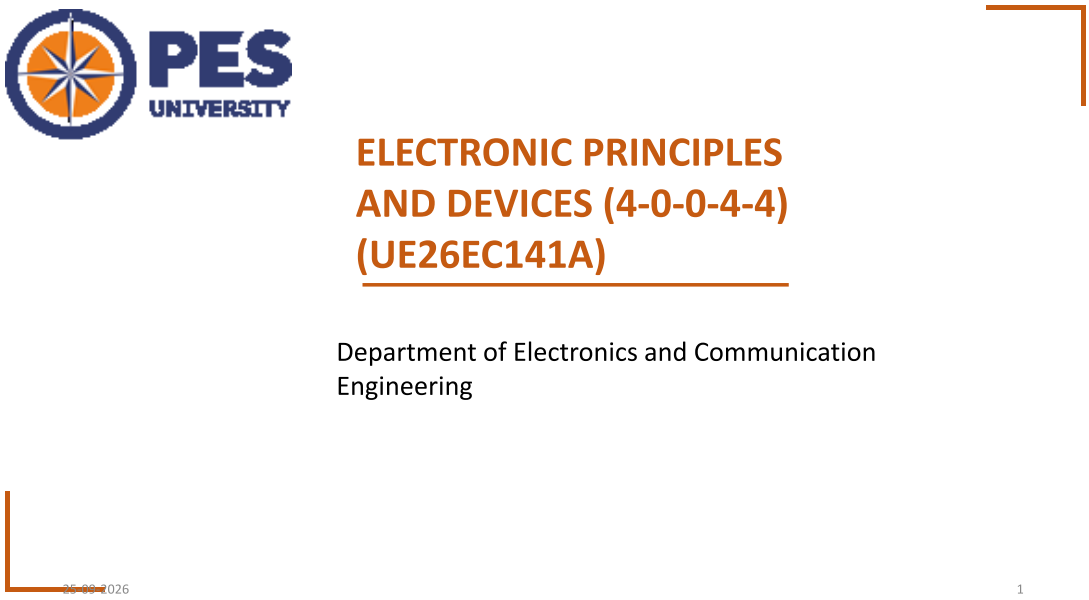

Figure G000009. Semiconductor diode under no bias forward bias and reverse bias conditions · source slide page 1.

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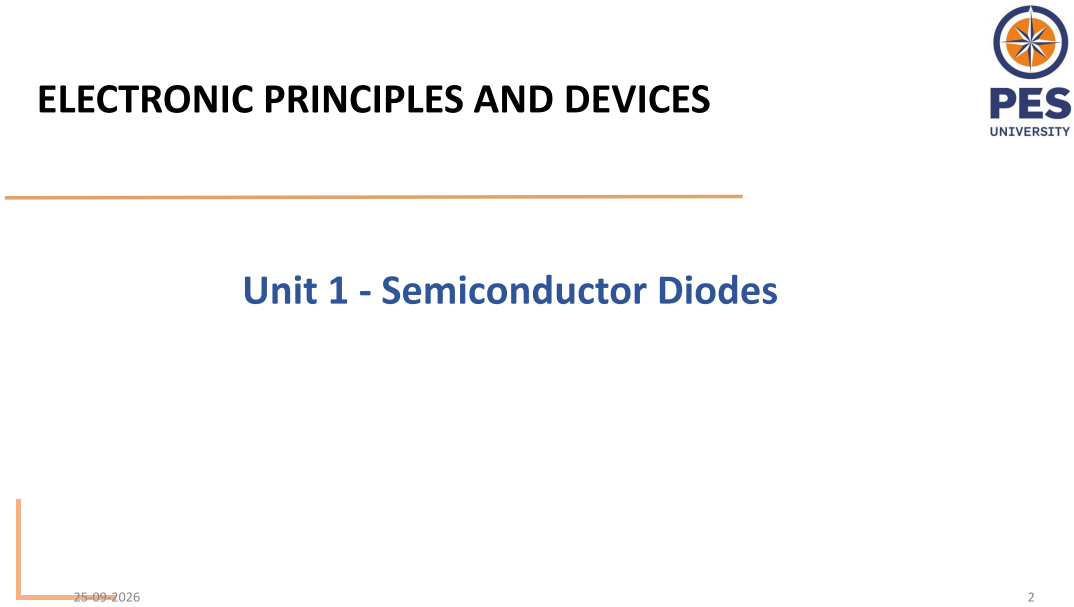

Figure G000010. Semiconductor diode under no bias forward bias and reverse bias conditions · source slide page 2.

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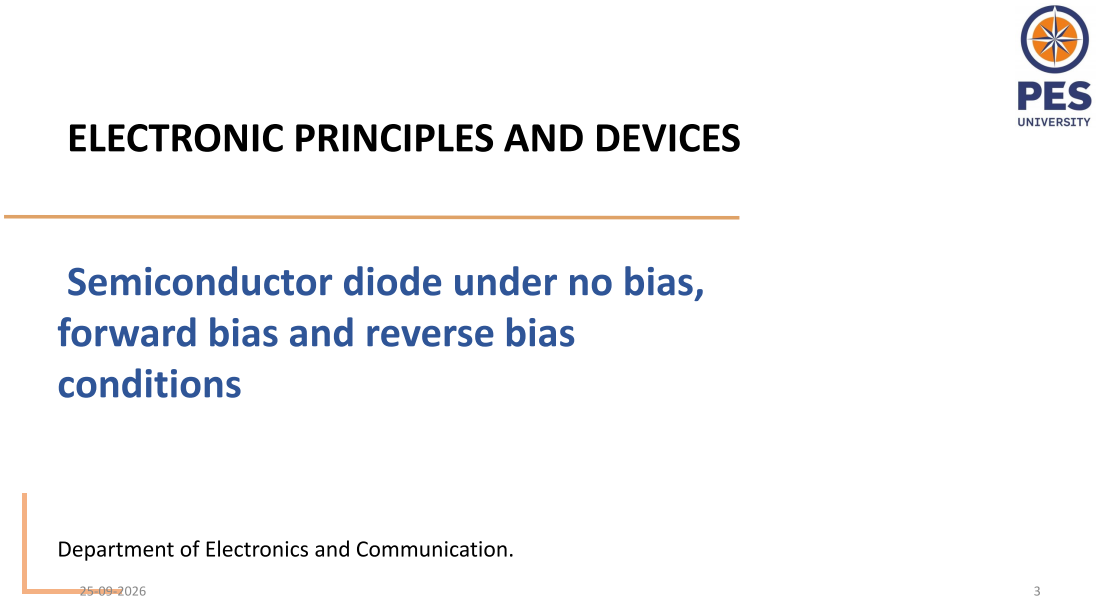

Figure G000011. Semiconductor diode under no bias forward bias and reverse bias conditions · source slide page 3.

EV0000133

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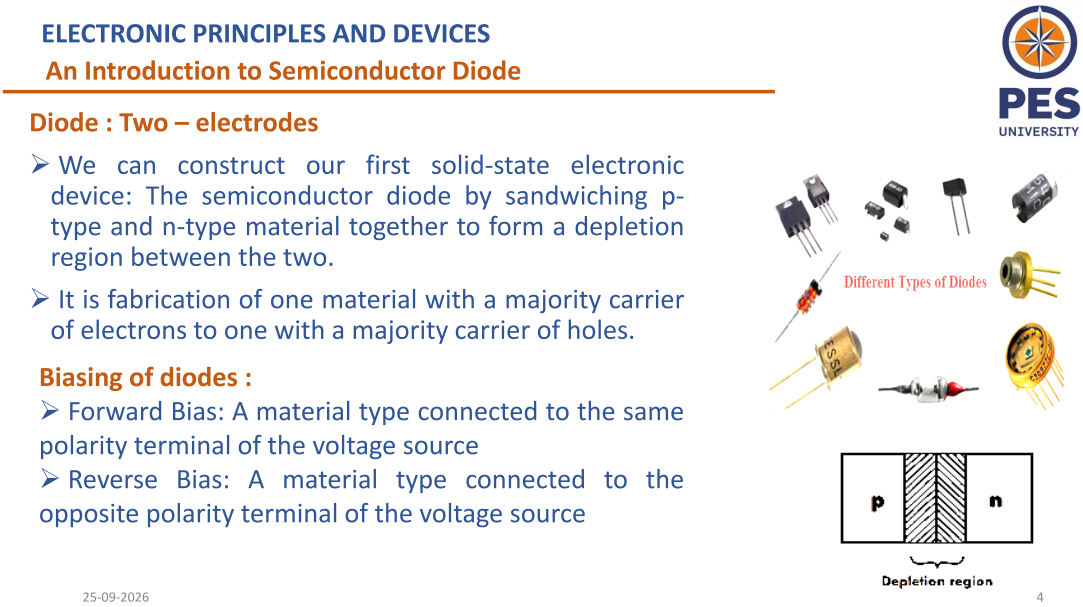

Figure G000012. Semiconductor diode under no bias forward bias and reverse bias conditions · source slide page 4.

EV0000136

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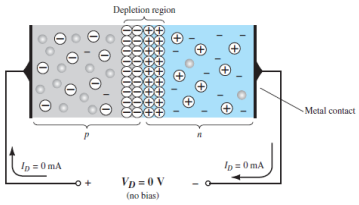
ELECTRONIC PRINCIPLES AND DEVICES

Diode operation under different biasing conditions

PES UNIVERSITY



No bias

In the absence of an applied bias across a semiconductor diode, the net flow of charge in one direction is zero.

No – bias condition:

- At the instant the two materials are "joined" the electrons and the holes in the region of the junction will combine, resulting in a lack of free carriers in the region near the junction, as shown in Fig.
- Note in Fig. that the only particles displayed in this region are the positive and the negative ions remaining once the free carriers have been absorbed.
- ***This region of uncovered positive and negative ions is called the depletion region due to the "depletion" of free carriers in the region.***

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Figure G000013. Semiconductor diode under no bias forward bias and reverse bias conditions · source slide page 5.

EV0000139

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

ELECTRONIC PRINCIPLES AND DEVICES

Diode operation under different biasing conditions

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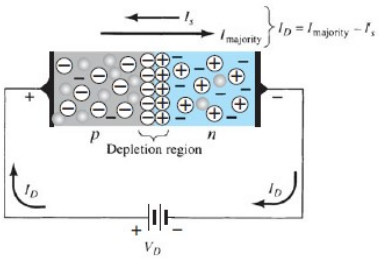

Forward Bias

The current that exists under forward-bias conditions is called the forward-bias current and is represented by I_D.

Forward – bias condition

- A *forward-bias* or "on" condition is established by applying the positive potential to the *p*-type material and the negative potential to the *n*-type material as shown in Fig.
- The application of a forward-bias potential V_D will "pressure" electrons in the *n*-type material and holes in the *p*-type material to recombine with the ions near the boundary and reduce the width of the depletion region as shown in Fig.

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Figure G000014. Semiconductor diode under no bias forward bias and reverse bias conditions · source slide page 6.

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ELECTRONIC PRINCIPLES AND DEVICES

Diode operation under different biasing conditions

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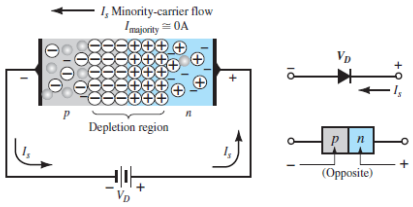

Reverse Bias

The current that exists under reverse-bias conditions is called the reverse saturation current and is represented by I_s.

Reverse – bias condition:

- When a P-N junction is reverse-biased as shown in Fig., the number of uncovered positive ions in the depletion region of the *n*-type material will increase due to the large number of free electrons drawn to the positive potential of the applied voltage.
- For similar reasons, the number of uncovered negative ions will increase in the *p*-type material.
- The net effect, therefore, is a widening of the depletion region. This widening of the depletion region will establish too great a barrier for the majority carriers to overcome, effectively reducing the majority carrier flow to zero.

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Figure G000015. Semiconductor diode under no bias forward bias and reverse bias conditions · source slide page 7.

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How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

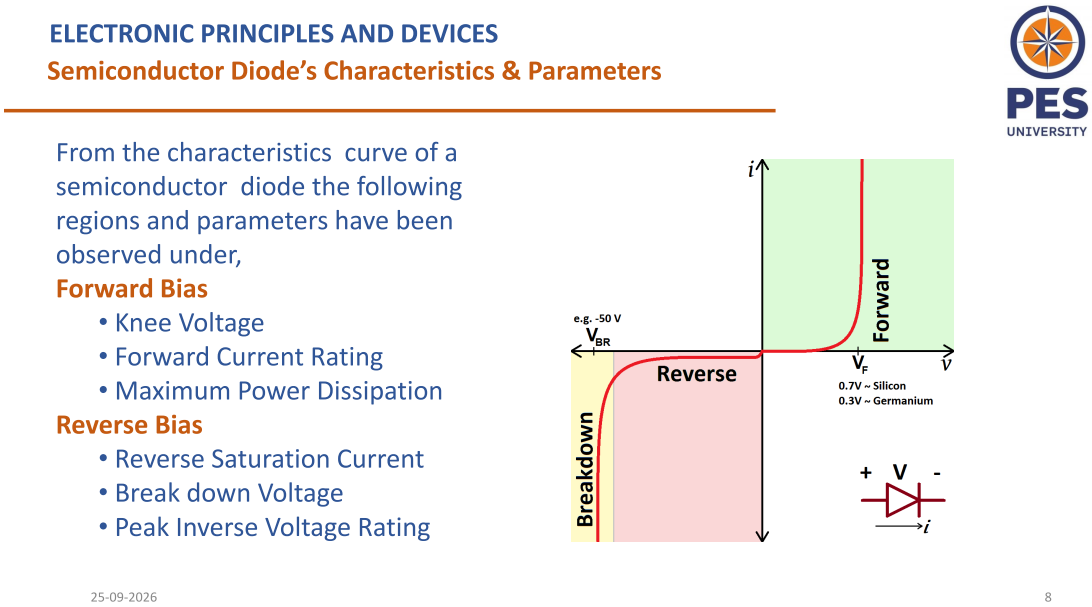

Figure G000016. Semiconductor diode under no bias forward bias and reverse bias conditions · source slide page 8.

EV0000148

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Figure G000017. Semiconductor diode under no bias forward bias and reverse bias conditions · source slide page 9.

EV0000151

> ***How to read it.*** Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Important points and why

1. Source slide page 1 (extracted text): ELECTRONIC PRINCIPLES AND DEVICES (4-0-0-4-4) (UE26EC141A) Department of Electronics and Communication Engineering 25-09-2026 1 EV0000126 · EV0000127 · EV0000128
2. Source slide page 2 (extracted text): ELECTRONIC PRINCIPLES AND DEVICES Unit 1 - Semiconductor Diodes 25-09-2026 2 EV0000129 · EV0000130 · EV0000131
3. Source slide page 3 (extracted text): ELECTRONIC PRINCIPLES AND DEVICES Semiconductor diode under no bias, forward bias and reverse bias conditions Department of Electronics and Communication. 25-09-2026 3 EV0000132 · EV0000133 · EV0000134
4. Source slide page 4 (extracted text): ELECTRONIC PRINCIPLES AND DEVICES An Introduction to Semiconductor Diode Diode : Two – electrodes ☒We can construct our first solid-state electronic device: The semiconductor diode by sandwiching p- type and n-type material together to form a depletion region between the two. ☒It is fabrication of one material with a majority carrier of electrons to one with a majority carrier of holes. Biasing of diodes : ☒Forward Bias: A material type connected to the same polarity terminal of the voltage source ☒Reverse Bias: A material type connected to the opposite polarity terminal of the voltage source 25-09-2026 4 EV0000135 · EV0000136 · EV0000137

5. Source slide page 5 (extracted text): ELECTRONIC PRINCIPLES AND DEVICES Diode operation under different biasing conditions No – bias condition: ☒At the instant the two materials are “joined” the electrons and the holes in the region of the junction will combine, resulting in a lack of free carriers in the region near the junction, as shown in Fig. No bias ☒Note in Fig. that the only particles displayed in this In the absence of an applied bias region are the positive and the negative ions across a semiconductor diode, the net flow of charge in one direction remaining once the free carriers have been absorbed. is zero. ☒This region of uncovered positive and negative ions is called the depletion region due to the “depletion” 25-09-2026 of free carriers in the region. 5 EV0000138 · EV0000139 · EV0000140
6. Source slide page 6 (extracted text): ELECTRONIC PRINCIPLES AND DEVICES Diode operation under different biasing conditions Forward – bias condition ☒A forward-bias or “on” condition is established by applying the positive potential to the p-type material and the negative potential to the n-type material as shown in Fig. ☒The application of a forward-bias potential VD will “pressure” electrons in the n-type material and holes in Forward Bias The current that exists under the p-type material to recombine with the ions near forward-bias conditions is called the boundary and reduce the width of the depletion the forward-bias current and is represented by ID. region as shown in Fig. 25-09-2026 6 EV0000141 · EV0000142 · EV0000143
7. Source slide page 7 (extracted text): ELECTRONIC PRINCIPLES AND DEVICES Diode operation under different biasing conditions Reverse – bias condition: ☒When a P-N junction is reverse-biased as shown in Fig., the number of uncovered positive ions in the depletion region of the n-type material will increase due to the large number of free electrons drawn to the positive potential of the applied voltage. ☒For similar reasons, the number of uncovered negative ions Reverse Bias will increase in the p-type material. The current that exists under reverse-bias conditions is called ☒The net effect, therefore, is a widening of the depletion the reverse saturation current region. This widening of the depletion region will establish tooand is represented by Is. great a barrier for the majority carriers to overcome, 25-09-2026 effectively reducing the majority carrier flow to zero. 7 EV0000144 · EV0000145 · EV0000146
8. Source slide page 8 (extracted text): ELECTRONIC PRINCIPLES AND DEVICES Semiconductor Diode’s Characteristics & Parameters From the characteristics curve of a semiconductor diode the following regions and parameters have been observed under, Forward Bias • Knee Voltage • Forward Current Rating • Maximum Power Dissipation Reverse Bias • Reverse Saturation Current • Break down Voltage • Peak Inverse Voltage Rating 25-09-2026 8 EV0000147 · EV0000148 · EV0000149
9. Source slide page 9 (extracted text): THANK YOU Department of Electronics and Communication 25-09-2026 9 EV0000150 · EV0000151 · EV0000152

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:03.600] Good morning, and welcome back [00:00:06.090] to Electronic Principles and Devices. [00:00:09.541] Today we will go with the topic three, Introduction to Semiconductor [00:00:15.360] Diode. [00:00:17.760] In the last class we have looked at the diode and biasing of the diode, [00:00:23.670] where we know that diode has two regions, P and N type, where P side is connected [00:00:29.460] with the biasing. P side is connected to the positive of the terminal and N side is [00:00:34.470] connected to negative of the battery terminal. [00:00:37.440] Now before we go understanding about the biasing part, let

us understand the basic [00:00:42.930] principle and concept of bias and no bias. [00:00:46.755] So as we see here in the diagram, we have a no bias setup with a [00:00:53.011] P [00:00:55.440] and N region not connected to the voltage supply. We call it as a no bias. [00:00:59.850] So in the absence of an applied bias across the semiconductor diode, the net flow [00:01:04.621] of charges in one direction is zero. [00:01:07.801] That means to say, with no voltage applied to the diode, we see that the boundary, [00:01:13.711] there is a depletion region with the charges being accumulated. [00:01:17.430] This happens [00:01:19.410] because at the instant the two metals are joined, the electrons and the holes in [00:01:24.330] the region of the junction will combine, resulting in lack of free charges in the [00:01:29.520] region near the junction, as shown, which we call it as a depletion region. [00:01:34.590] And note in the figure that only particle displayed in this region are positive and [00:01:39.090] negative ion remaining once the free charges have been observed. [00:01:43.260] That means to say, because of the concentration gradient, the charge carriers will [00:01:46.980] start move from [00:01:50.250] P side to N side and N side to P side. As the moment they [00:01:57.570] start moving, they forms a charges [00:02:02.851] which is been settled at the boundaries of the P and N, [00:02:08.671] where negative charge is settled at the P region boundary and [00:02:13.891] positive charge is settled at the N region boundary. [00:02:17.070] This region of uncovered positive and negative ion is called depletion region. [00:02:22.020] This is due to the depletion of free charge [00:02:28.050] carrier in the region. So which we call it as a no bias condition. [00:02:34.680] Now let us go with forward bias. As the name indicates, a forward bias or [00:02:40.110] 'ON' condition. The current that exists under the forward bias is called the [00:02:45.960] forward bias current and is represented by ID in the diagram. [00:02:50.430] As the name indicates, forward bias, we connect P region to positive terminal of [00:02:54.510] the battery and N region to negative terminal of the battery. [00:02:58.830] As the moment we connect it, we call it as an 'ON' condition, that is [00:03:02.790] established by applying a positive potential to the P terminal and a negative [00:03:08.370] potential to the N terminal, as shown in the diagram. [00:03:11.520] The application of the forward bias potential that is VD we call here, will [00:03:16.620] pressurize or will pressure the electron in the N-type material and hole in the [00:03:21.840] P-type material to recombine with ion near the boundary and reduces the width [00:03:27.824] of the depletion region, as shown here. [00:03:30.601] If compared to the previous diagram, we have a very big depletion region which is [00:03:36.270] shown here. And because of the application of the voltage, the depletion region is [00:03:40.980] reduced and we see only a few charge carrier. [00:03:43.291] This happens due to the application of the voltage on both P and N region, and [00:03:48.690] which results in the ID current which is a sum of the eye majority and the leakage [00:03:54.300] current. [00:03:56.400] This can be represented by a V-A characteristic, as shown, the forward bias current [00:04:01.380] is shown here. There is a very less resistance is there in the forward bias and we [00:04:05.490] see that there is a current flow with application of the voltage. [00:04:12.510] This is about the forward bias. Then what is reverse bias? [00:04:16.291] The reverse bias condition, the P side is connected to the negative terminal of the [00:04:20.130] battery, N side is connected to the positive terminal of the battery with applied [00:04:23.880] voltage VD. [00:04:26.430] So when a PN junction is reverse biased, the number of uncovered positive ions in [00:04:30.720] the depletion region of the N-type material [00:04:33.600] will increase, as shown here. It's increasing due to the large number of free [00:04:39.210] electron drawn from the positive terminal of the applied voltage. [00:04:42.390] Similarly, the negative terminal of the applied voltage here. [00:04:46.201] For a similar reason, the number of uncovered

ion will increase in the P-type [00:04:50.460] material. The net effect therefore is widening of a depletion region, as shown. [00:04:56.070] This widening of the depletion region will establish too a great

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Transcript block 2 (source interval 000300-000599 seconds): [00:05:04.500] a barrier for majority carrier to overcome, effectively reducing the majority [00:05:09.330] carrier flow to zero. [00:05:13.920] So the current that exists under a reverse bias condition is called the reverse [00:05:18.750] saturation current and is represented as IS. [00:05:24.510] And this also represented in the V-A characteristic as shown, with the initial [00:05:30.330] applied voltage because of the depletion region will not be able to see the current [00:05:34.380] flow which has been [00:05:38.010] shown here. And once after the reverse breakdown region, once after the [00:05:43.680] voltage called VBR, there is going to be a breakdown with the current increase [00:05:49.560] as shown. [00:05:51.480] And this could be [00:05:54.210] realized from a reverse bias condition. [00:05:56.430] So today's class we have seen with a no bias condition, forward bias condition and [00:06:01.350] a reverse bias condition. Thank you.

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Full transcript: 891 normalized words, 80 timestamped segments, 368.64 seconds duration. Low-confidence segments flagged: 0.

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4 Diode characteristics, Shockley’s equation

This lesson is indexed from the source video titled Diode characteristics, Shockley’s equation. Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or rewritten. A candidate slide deck is grouped here according to the current matching score; the exact path and page locators are listed in the source evidence. EV0000018 · EV0000019

Graphs and diagrams

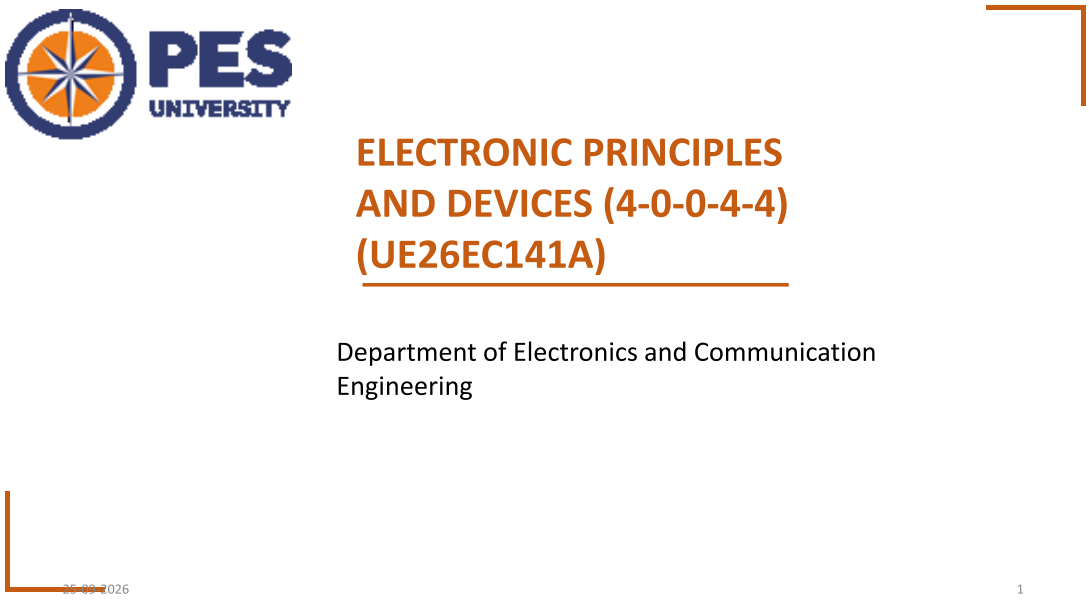

Figure G000018. Diode characteristics Shockley s equation · source slide page 1.

EV0000154

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

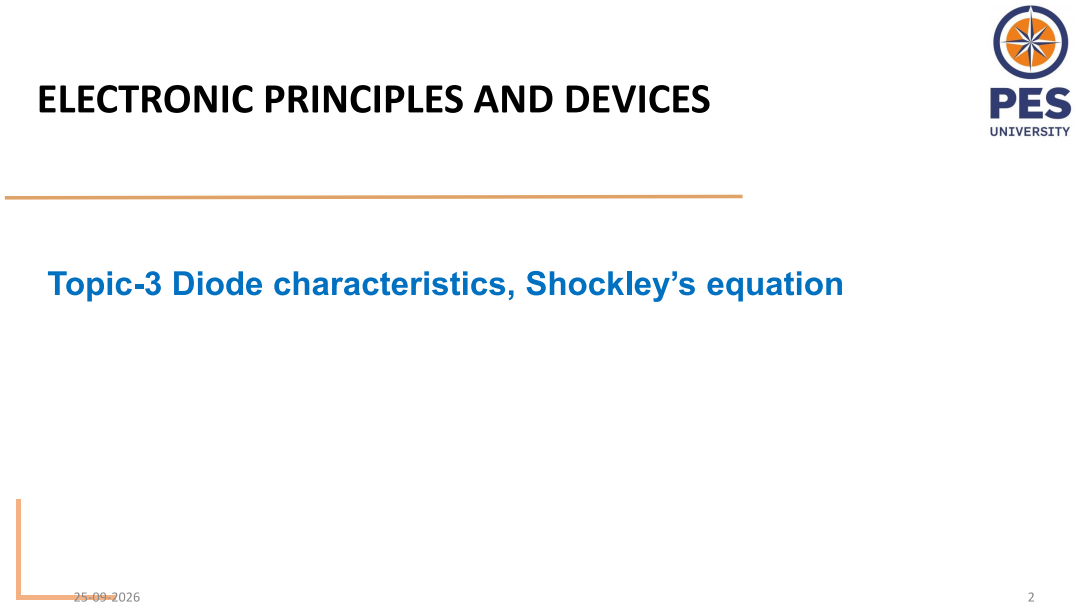

Figure G000019. Diode characteristics Shockley s equation · source slide page 2.

EV0000157

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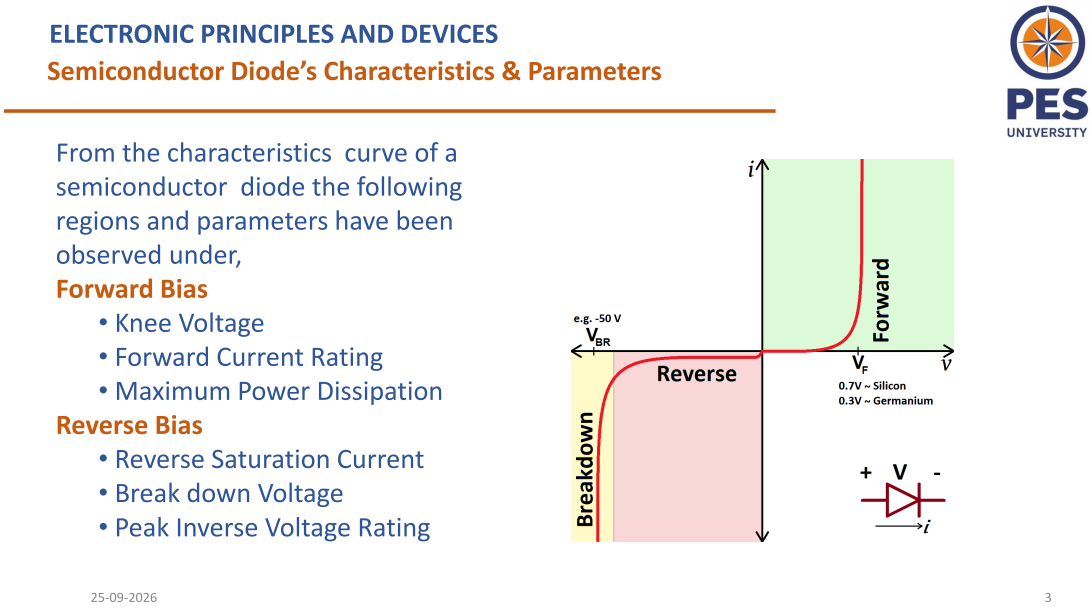

Figure G000020. Diode characteristics Shockley s equation · source slide page 3.

EV0000160

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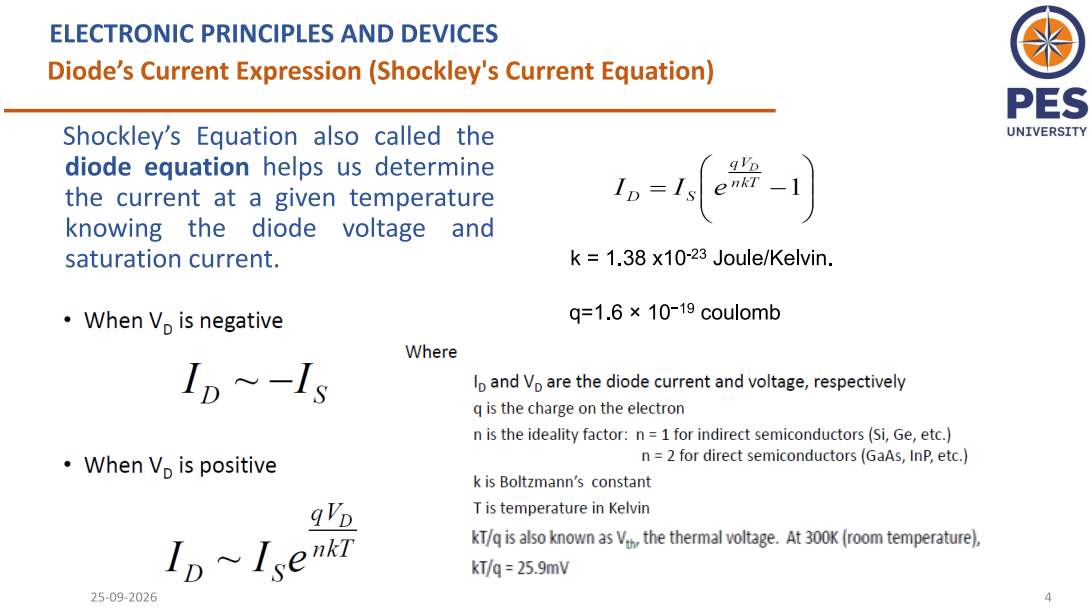

Figure G000021. Diode characteristics Shockley s equation · source slide page 4.

EV0000163

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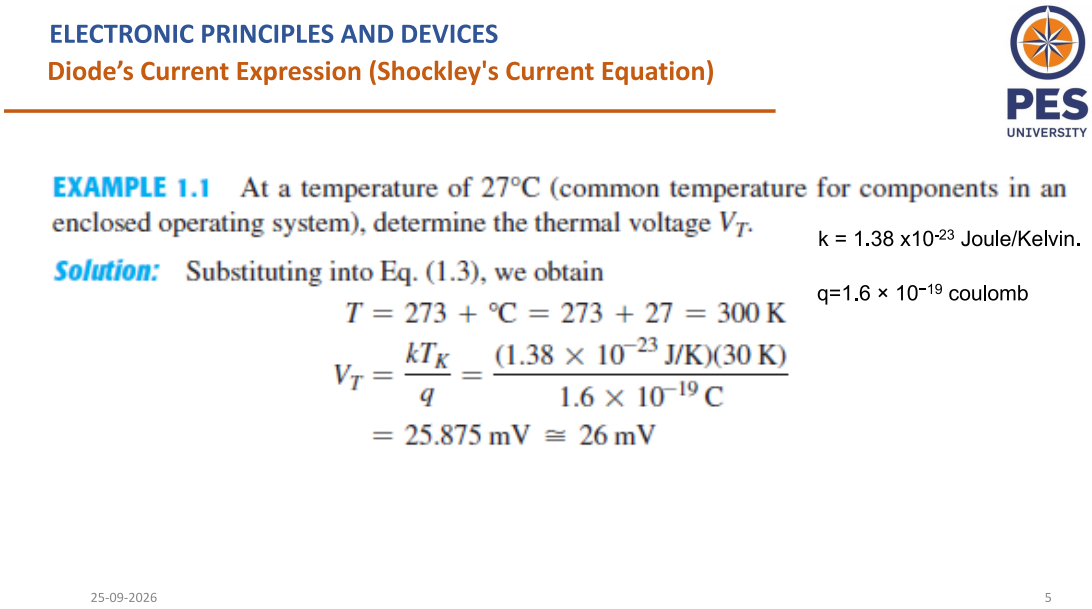

Figure G000022. Diode characteristics Shockley s equation · source slide page 5.

EV0000166

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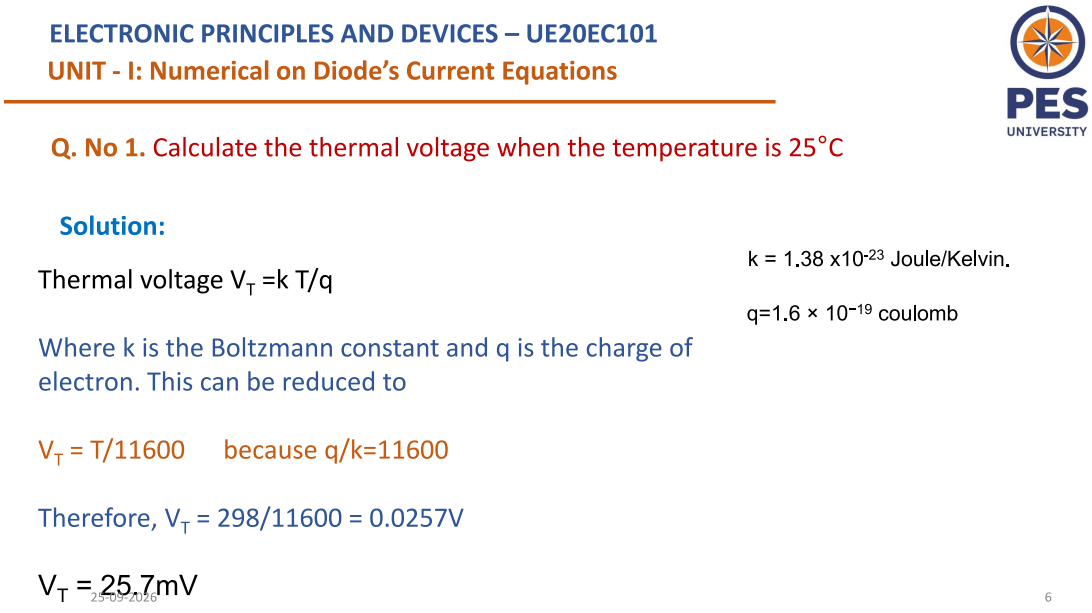

Figure G000023. Diode characteristics Shockley s equation · source slide page 6.

EV0000169

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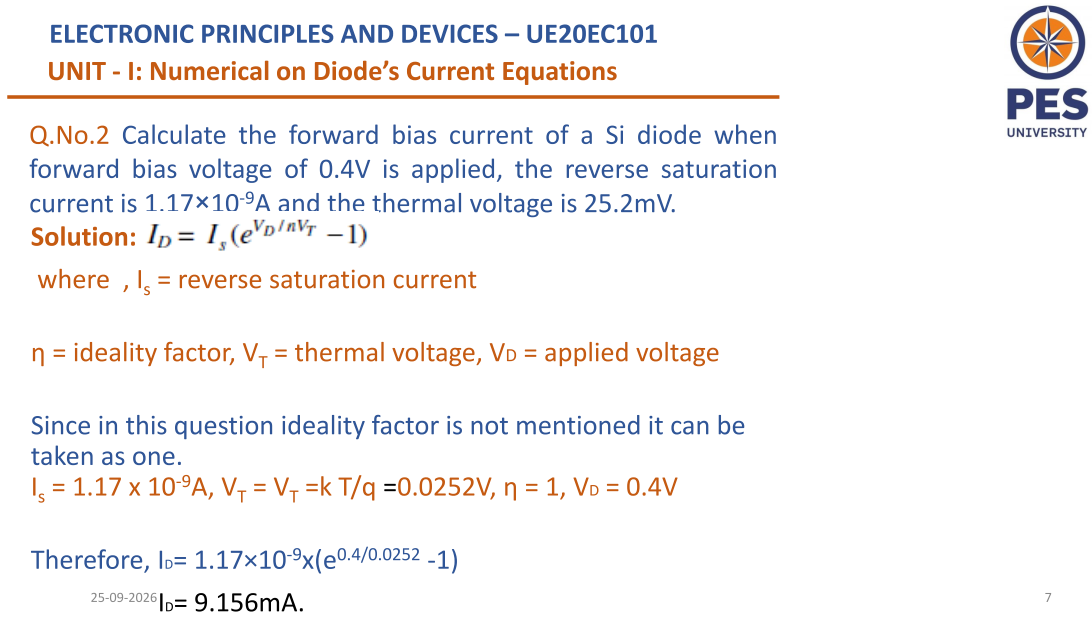

Figure G000024. Diode characteristics Shockley s equation · source slide page 7.

EV0000172

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

ELECTRONIC PRINCIPLES AND DEVICES

Numerical on Diode's Current Equations

PES UNIVERSITY

Q. No 3 Given a diode current of 8 mA and η =1, find the reverse saturation current (I_s) if the applied voltage is 0.5 V and the room temperature is 25°C.

Solution: Equation for diode current (Shockley's Equation) $I_D = I_s(e^{V_D/\eta V_T} - 1)$

Given: I_D = **diode current = 8 mA**, η = ideality factor = 1, V_D = Applied voltage = 0.5V

We have to find V_T = Thermal voltage and Reverse saturation current I_S

$$V_T = \frac{kT_K}{q} = \frac{(1.38\times10^{-23}\ \text{J/K})(25°\text{C} + 273°\text{C})}{1.6\times10^{-19}\ \text{C}}$$

$$= 25.70\ \text{mV}$$

$$8\text{mA} = I_s(e^{(0.5\text{V})/(1)(25.70\ \text{mV})} - 1) = I_s(28\times10^8)$$

$$I_s = \frac{8\ \text{mA}}{2.8\times10^8} = \mathbf{28.57\ pA}$$

25-09-2026 8

Figure G000025. Diode characteristics Shockley s equation · source slide page 8.

EV0000175

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

ELECTRONIC PRINCIPLES AND DEVICES

Numerical on Diode's Current Equations

PES UNIVERSITY

Q. No 4: Calculate the applied voltage V_D, if diode current is 6 mA, Thermal Voltage is 26 mV, Ideality factor is 1, and Reverse saturation current is 1 nA.

Solution: **Given:** I_D = 6 mA, V_T = 26 mV, η = 1, I_S = 1 nA, V_D = Applied voltage = ...V

Equation for diode current (Shockley's Equation)

$$I_D = I_s(e^{V_D/\eta V_T} - 1)$$

$$6\ \text{mA} = 1\ \text{nA}(e^{V_D/(1)(26\ \text{mV})} - 1)$$

$$6\times10^6 = e^{V_D/26\ \text{mV}} - 1$$

$$e^{V_D/26\ \text{mV}} = 6\times10^6 - 1 \cong 6\times10^6$$

$$\log_e e^{V_D/26\,\text{mV}} = \log_e 6\times10^6$$

$$\frac{V_D}{26\ \text{mV}} = 15.61$$

$$V_D = 15.61(26\ \text{mV}) \cong \mathbf{0.41\ V}$$

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Figure G000026. Diode characteristics Shockley s equation · source slide page 9.

EV0000178

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

ELECTRONIC PRINCIPLES AND DEVICES – UE20EC101

UNIT - I: Numerical on Diode's Current Equations

PES UNIVERSITY

Q. No 5 Consider a silicon diode with η=1.2. Find change in voltage if the current changes from 0.1mA to 10mA.

Solution: Equation for diode current $I=I_0\times(e^{(V/\eta V_T)}-1)$

Where I_0 = reverse saturation current, η = ideality factor
V_T = thermal voltage, V = applied voltage

η = 1.2, I_2 = 10mA, I_1 = 0.1mA and take V_T = 0.026V

Change in voltage $\Delta V = \eta V_T \ln(\frac{I_2}{I_1}) = 1.2\text{x}0.026\text{x}\ln(10\times\frac{10^{-3}}{0.1\times10^{-3}}) = 0.143V$

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Figure G000027. Diode characteristics Shockley s equation · source slide page 10.

EV0000181

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

ELECTRONIC PRINCIPLES AND DEVICES

Comparison of diode characteristics of Ge, Si, and GaAs

PES UNIVERSITY

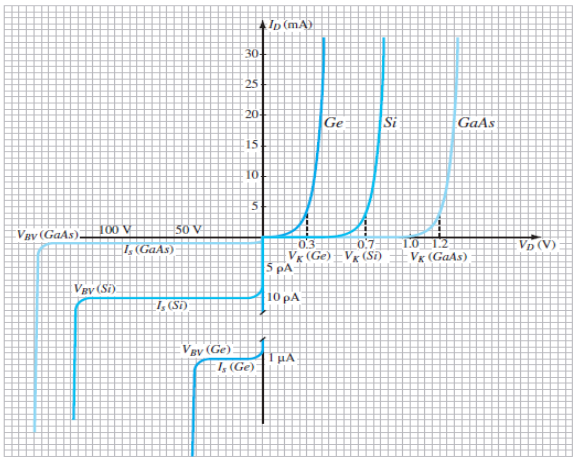

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Figure G000028. Diode characteristics Shockley s equation · source slide page 11.

EV0000184

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Figure G000029. Diode characteristics Shockley s equation · source slide page 12.

EV0000187

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Important points and why

1. Source slide page 1 (extracted text): ELECTRONIC PRINCIPLES AND DEVICES (4-0-0-4-4) (UE26EC141A) Department of Electronics and Communication Engineering 25-09-2026 1 EV0000153 · EV0000154 · EV0000155
2. Source slide page 2 (extracted text): ELECTRONIC PRINCIPLES AND DEVICES Topic-3 Diode characteristics, Shockley's equation 25-09-2026 2 EV0000156 · EV0000157 · EV0000158
3. Source slide page 3 (extracted text): ELECTRONIC PRINCIPLES AND DEVICES Semiconductor Diode's Characteristics & Parameters From the characteristics curve of a semiconductor diode the following regions and parameters have been observed under, Forward Bias • Knee Voltage • Forward Current Rating • Maximum Power Dissipation Reverse Bias • Reverse Saturation Current • Break down Voltage • Peak Inverse Voltage Rating 25-09-2026 3 EV0000159 · EV0000160 · EV0000161
4. Source slide page 4 (extracted text): ELECTRONIC PRINCIPLES AND DEVICES Diode's Current Expression (Shockley's Current Equation) Shockley's Equation also called the diode equation helps us determine the current at a given temperature knowing the diode voltage and saturation current. k = 1.38 x10-23 Joule/Kelvin. q=1.6 × 10−19 coulomb 25-09-2026 4 EV0000162 · EV0000163 · EV0000164
5. Source slide page 5 (extracted text): ELECTRONIC PRINCIPLES AND DEVICES Diode's Current Expression (Shockley's Current Equation) k = 1.38 x10-23 Joule/Kelvin. q=1.6 × 10−19 coulomb 25-09-2026 5 EV0000165 · EV0000166 · EV0000167

6. Source slide page 6 (extracted text): ELECTRONIC PRINCIPLES AND DEVICES – UE20EC101 UNIT - I: Numerical on Diode's Current Equations Q. No 1. Calculate the thermal voltage when the temperature is 25°C Solution: k = 1.38 x10-23 Joule/Kelvin. Thermal voltage VT =k T/q q=1.6 × 10−19 coulomb Where k is the Boltzmann constant and q is the charge of electron. This can be reduced to VT = T/11600 because q/k=11600 Therefore, VT = 298/11600 = 0.0257V VT =25-09-202625.7mV 6 EV0000168 · EV0000169 · EV0000170
7. Source slide page 7 (extracted text): ELECTRONIC PRINCIPLES AND DEVICES – UE20EC101 UNIT - I: Numerical on Diode's Current Equations Q.No.2 Calculate the forward bias current of a Si diode when forward bias voltage of 0.4V is applied, the reverse saturation current is 1.17×10-9A and the thermal voltage is 25.2mV. Solution: where , Is = reverse saturation current η = ideality factor, VT = thermal voltage, VD = applied voltage Since in this question ideality factor is not mentioned it can be taken as one. Is = 1.17 x 10-9A, VT = VT =k T/q =0.0252V, η = 1, VD = 0.4V Therefore, ID= 1.17×10-9x(e0.4/0.0252 -1) 25-09-2026 ID= 9.156mA. 7 EV0000171 · EV0000172 · EV0000173
8. Source slide page 8 (extracted text): ELECTRONIC PRINCIPLES AND DEVICES Numerical on Diode's Current Equations Q. No 3 Given a diode current of 8 mA and η =1, find the reverse saturation current if the applied voltage is 0.5 V and the room temperature is 25°C. (Is) Solution: Equation for diode current (Shockley's Equation) Given: ID = diode current = 8 mA, η = ideality factor = 1, VD = Applied voltage = 0.5V We have to find VT = Thermal voltage and Reverse saturation current IS 25-09-2026 8 EV0000174 · EV0000175 · EV0000176
9. Source slide page 9 (extracted text): ELECTRONIC PRINCIPLES AND DEVICES Numerical on Diode's Current Equations Q. No 4: Calculate the applied voltage VD, if diode current is 6 mA, Thermal Voltage is 26 mV, Ideality factor is 1, and Reverse saturation current is 1 nA. Solution: Given: ID = 6 mA, VT = 26 mV, η = 1, IS = 1 nA, VD = Applied voltage = ...V Equation for diode current (Shockley's Equation) 25-09-2026 9 EV0000177 · EV0000178 · EV0000179
10. Source slide page 10 (extracted text): ELECTRONIC PRINCIPLES AND DEVICES – UE20EC101 UNIT - I: Numerical on Diode's Current Equations Q. No 5 Consider a silicon diode with η=1.2. Find change in voltage if the current changes from 0.1mA to 10mA. Solution: Equation for diode current I=I0×(e(V/ηVT)-1) Where I0 = reverse saturation current, η = ideality factor VT = thermal voltage, V = applied voltage η = 1.2, I2 = 10mA, I1 = 0.1mA and take VT = 0.026V 25-09-2026 10 EV0000180 · EV0000181 · EV0000182
11. Source slide page 11 (extracted text): ELECTRONIC PRINCIPLES AND DEVICES Comparison of diode characteristics of Ge, Si, and GaAs 25-09-2026 11 EV0000183 · EV0000184 · EV0000185
12. Source slide page 12 (extracted text): THANK YOU Department of Electronics and Communication 25-09-2026 12 EV0000186 · EV0000187 · EV0000188

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:04.170] Good morning, and welcome to Electronic Principle and Devices. [00:00:08.580] I'm Professor Ananda, Department of Electronics and Communication Engineering. [00:00:12.780] Today, we are going to see the semiconductor diode characteristics and parameters [00:00:17.070] and with some numericals. [00:00:19.980] So, as we see in the previous class, the diode characteristics, we have seen the [00:00:25.710] forward and the reverse bias characteristics. [00:00:28.680] As we look at the forward

bias here, [00:00:32.820] initially for the applied voltage, the diode was off, [00:00:38.880] and at particular voltage [00:00:42.330] of Vf, the current start increasing. [00:00:45.151] Similarly, in the reverse bias, as applied voltage is increased, at one particular [00:00:49.740] voltage, which is a breakdown voltage, the current start increasing. [00:00:54.060] So to understand this, let us [00:00:57.301] consider the characteristics once again with the I and V, where we call this region [00:01:02.670] as a forward bias, we call this region as a reverse bias. [00:01:07.321] So from the characteristics curve of the semiconductor diode, the following region [00:01:10.620] and parameter have been observed here. [00:01:12.810] So in a forward bias region, [00:01:15.540] the voltage at which the diode start conducting is called as a knee voltage. [00:01:20.700] So for a silicon, it is 0.7, for germanium it is 0.3. [00:01:24.900] So the voltage at which the diode start conducting is called a forward, as a knee [00:01:28.560] voltage. Then, we also come to another parameter called forward current rating. [00:01:34.620] The forward current rating, which is I. [00:01:37.635] It's the maximum current [00:01:40.200] at which the diode starts conducting. [00:01:42.330] It depends on the voltage at which the diode is on. [00:01:45.601] So for silicon, it's 0.7 volts, for germanium, it's 0.3. [00:01:49.650] So the current can vary based on the device, because it [00:01:54.510] depends on the fabrication of the device. [00:01:57.360] Then the maximum power expression P is equal to V into I. [00:02:02.610] So once we know the voltage and the current, we get the power value. [00:02:08.400] Similarly, coming to the reverse bias here, [00:02:11.550] the breakdown voltage plays important role. The first is breakdown voltage. [00:02:15.240] Until there is a breakdown, there is a leakage current flowing, which is a reverse [00:02:19.380] saturation current, which is [00:02:22.230] less than the reverse saturation current, which is called as [00:02:28.200] a leakage current. We need to remember this. [00:02:31.455] Then, the reverse saturation current is this region, where the diode start [00:02:36.150] conducting even with the reverse bias. That happens at a reverse bias voltage. [00:02:41.520] It could vary based on the device. It could be five volts reverse saturation [00:02:45.060] current. That depends on the fabrication again. [00:02:47.700] Then we also need to calculate what the peak inverse voltage, the voltage at which [00:02:51.510] the device starts conducting, which is a peak inverse voltage rating. [00:02:56.040] And the diode representation symbol is this, where it looks like an arrow lying [00:03:01.770] here. The arrow indicate the conventional current flow from P region to N region. [00:03:07.050] Next, we also learn here the diode current expression or the Shockley's current [00:03:12.900] equation. The Shockley's equation is also called as a [00:03:17.010] diode equation, help us to determine the current at the given temperature, knowing [00:03:21.090] the diode voltage and saturation current. [00:03:23.460] That is given by Id is equal to Is into qVd by NqT minus one. [00:03:28.740] When Vd is positive, we represent Id is equal to minus Is. [00:03:32.490] When Vd is [00:03:34.710] negative, we represent Id is equal to minus Is. [00:03:37.080] When Vd is positive, we represent Id is approximately equal to Is into E to the [00:03:40.830] power of qVd by NmkBT, where Id and Vd are the diode current and voltage [00:03:46.320] respectively, where q is the charge of the electron, where eta is the ideality [00:03:51.480] factor, where eta N is equal to one for indirect semiconductor like silicon, [00:03:56.460] germanium, et cetera. For a direct semiconductor, N is equal to two, where gallium [00:04:00.210] arsenide, if you use gallium arsenide and indium phosphate. [00:04:03.421] Where k is the Boltzmann constant, T is the temperature in Kelvin. [00:04:08.160] Then, k is the Boltzmann constant given by value. [00:04:11.010] So entire this course now, we will be using k value is 1.38 into 10 to the power of [00:04:15.390] -3, and q is equal to 1.6 into 10 to the power of -19 coulombs. [00:04:20.400] So in all our numerical, we are going to use the k and q. [00:04:23.100] We are going

to use these k and the q values. [00:04:27.300] Where the kT by q is also called as a thermal voltage, where kT by q is equal to [00:04:31.890] 25.9 millivolts. [00:04:36.330] Let us solve few numericals now with an example from the textbook. [00:04:41.341] At a temperature 27 degrees centigrade, a common temperature for a component in an [00:04:45.630] enclosed operating system, determine the thermal voltage. [00:04:48.570] As we know the thermal voltage, how to calculate? [00:04:50.490] VT is equal to kT by q, where we need to calculate the T [00:04:56.250] here. Given temperature is 27, so add this 27 to 273, we get 300

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Transcript block 2 (source interval 000300-000599 seconds): [00:05:01.680] Kelvin. Then substitute that here, where we know the k value, 1.38 into 10 to the [00:05:06.360] power of -23, multiply with [00:05:09.000] 300 Kelvin [00:05:14.850] here, divide by 1.6 into 10 to the power of -19 coulombs. [00:05:20.010] So we get around 26 millivolts. [00:05:23.220] So this is a standard value [00:05:27.330] what we obtain at a room temperature for a thermal voltage. [00:05:32.640] Then coming to the numerical two, calculate the thermal voltage when the [00:05:36.540] temperature is 25 degree. The previous numerical was at 27 degree, now calculate at [00:05:40.920] 25 degree. So [00:05:43.410] if you substitute that value kT by q, we need to calculate, so we get 0.25, that is [00:05:48.240] 25.7 millivolts. Similarly, coming to the next numerical. [00:05:54.720] Calculate the forward bias current of silicon diode [00:05:58.591] when forward voltage is 0.4 applied and the reverse saturation current is 1.17 and [00:06:03.510] the thermal voltage is 25.2 millivolt. [00:06:06.540] So this is related to the Shockley's equation, which is given by Id is equal to Is [00:06:10.980] into E to the power of qVd. So we need to calculate here forward current, that is [00:06:16.770] Id. We need to calculate Id here. So make a note of each values [00:06:23.610] given here. Since in this equation, ideality factor is not mentioned, it can be [00:06:27.599] taken as one. So please note, whenever an ideality factor is not given, we'll [00:06:32.790] always take as one. So given here, Is is equal to 1.17 into 10 to the power of -19. [00:06:38.760] Vd is equal to, which is kT by q, with a given value, they have already given 25.2, [00:06:44.040] we know that n is equal to one, Vd is equal to 0.4. Then calculate. [00:06:47.880] Substitute all the values here and calculate Id, which is given by 9.156 [00:06:53.460] milliamps, and which is going to be solved in a class by the faculty. [00:06:58.845] Coming to numerical three. Given a diode current of eight milli and N is equal to [00:07:03.870] one, find the reverse saturation current if the diode voltage is 0.5 [00:07:09.360] and the room temperature is 25 degree centigrade. [00:07:12.720] So the equation for the diode Shockley is Id is equal to Is into E to the power of [00:07:17.520] Vd NK T minus one. Given Id is equal to eight milli, ideality factor one, [00:07:23.250] applied voltage is 0.5. We have to find VT thermal voltage and reverse saturation [00:07:28.140] current. That is given by the equation, we get 25.70 millivolts. [00:07:32.701] Then substitute all the values for Id, for Is. [00:07:37.170] For Id, you have to calculate. For Is, we need to calculate. [00:07:40.710] So Is, we get as 28.57 picoamps, and Is is the reverse saturation [00:07:46.651] current we will get in terms of picoamps. [00:07:48.960] The summary is the current should be less compared to the forward current. [00:07:52.710] So the forward current is eight milli here, the given. [00:07:55.290] The reverse current is pico in 10 to the power of minus 12. [00:07:58.260] So this answer is correct. So you have to verify every time the given answer is [00:08:02.401] correct or not based on the theoretical concepts. [00:08:05.190] Coming to the numerical four, calculate the applied voltage Vd if the diode current [00:08:08.970] is six milliamps, thermal voltage is 26 millivolts, ideality is one, reverse [00:08:13.201] saturation current is one. So the given values have been noted here.

[00:08:16.920] So we need to calculate here the applied voltage Vd. [00:08:20.880] This numerical is not similar to the previous what we solved. [00:08:24.690] So we need to change over the places of the numerator, denominator, and we need to [00:08:29.430] use a log E on both the sides, so finally, we get 0.41 volts. [00:08:35.160] Next, coming to the numerical five. Consider silicon with n is equal to 1.2. [00:08:39.151] Find the change in the voltage if current changes from 0.1 [00:08:43.980] milliamps to 10 milliamps. If such a type of question is given, we need to use a [00:08:48.420] formula, the change in the voltage delta V is equal to nVT ln I2 by I1, along [00:08:54.345] with the diode current equation. So we already know the value of eta here. [00:08:58.800] Eta is one [00:09:01.470] or 1.2 given. So since they given 1.2, if they don't give, we substitute as one. [00:09:06.480] Then the VT, they have already given, 26 milli. We need to calculate 26 milli. [00:09:12.990] So [00:09:14.850] yeah, VT is 26. We can directly use that value 26 ln 10, and [00:09:20.850] the change, the I2 by I1. So the given I2 is 10 milli, so 10 milli by [00:09:26.790] 0.1 milli we have taken here. 10 milli by 0.1, so we get 0.143 volts. [00:09:32.970] So this is how we can solve the numerical and diode current equation. [00:09:37.170] And finally, we can see the comparison of the diode characteristics of germanium, [00:09:40.770] silicon, and gallium arsenide with respect to the forward cutting voltages. [00:09:45.480] So the cutting voltage of gallium arsenide is 1.2, cutting voltage for silicon is [00:09:50.790] 0.7. Similarly, the cutting voltage for germanium is 0.3. [00:09:55.561] And also the reverse saturation [00:09:58.230] current also has been shown. The breakdown voltage will vary for type of the device

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Transcript block 3 (source interval 000600-000899 seconds): [00:10:02.550] being used. We see that the breakdown voltage for germanium is very high, 100 [00:10:06.901] volts. [00:10:09.210] Similarly, the silicon is around, it could be 50, and gallium [00:10:15.090] arsenide is maybe less than the 50 volts. [00:10:21.030] So thank you.

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Full transcript: 1680 normalized words, 145 timestamped segments, 627.776 seconds duration. Low-confidence segments flagged: 0.

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5 Zener and Avalanche breakdown, Temperature effects

This lesson is indexed from the source video titled Zener and Avalanche breakdown, Temperature effects. Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or rewritten. A candidate slide deck is grouped here according to the current matching score; the exact path and page locators are listed in the source evidence. EV0000030 · EV0000031

Graphs and diagrams

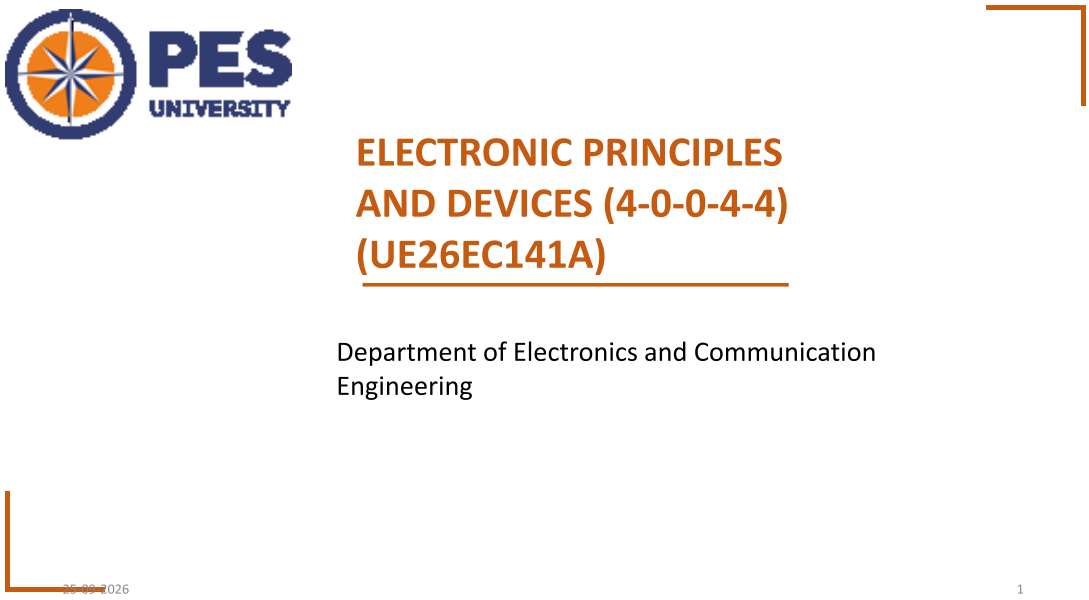

Figure G000030. Zener and Avalanche breakdown Temperature effects · source slide page 1.

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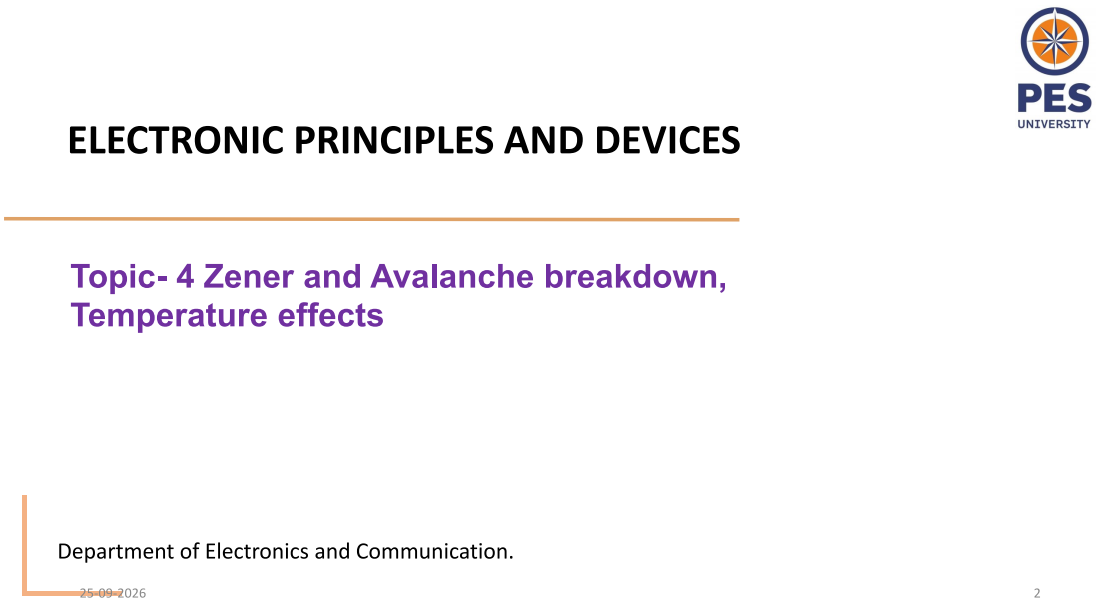

Figure G000031. Zener and Avalanche breakdown Temperature effects · source slide page 2.

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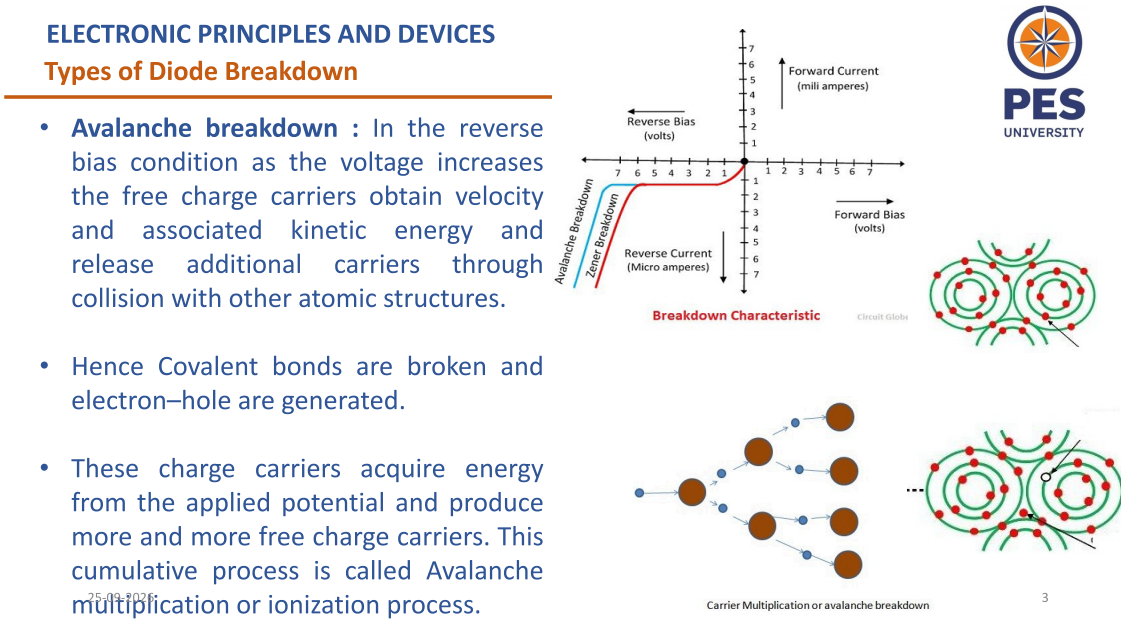

Figure G000032. Zener and Avalanche breakdown Temperature effects · source slide page 3.

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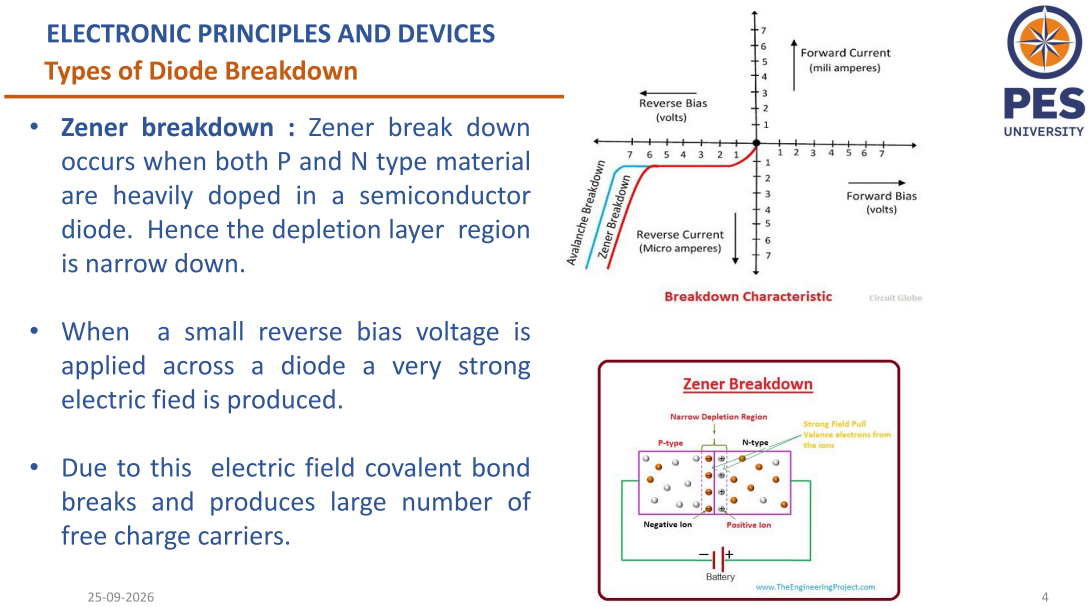

Figure G000033. Zener and Avalanche breakdown Temperature effects · source slide page 4.

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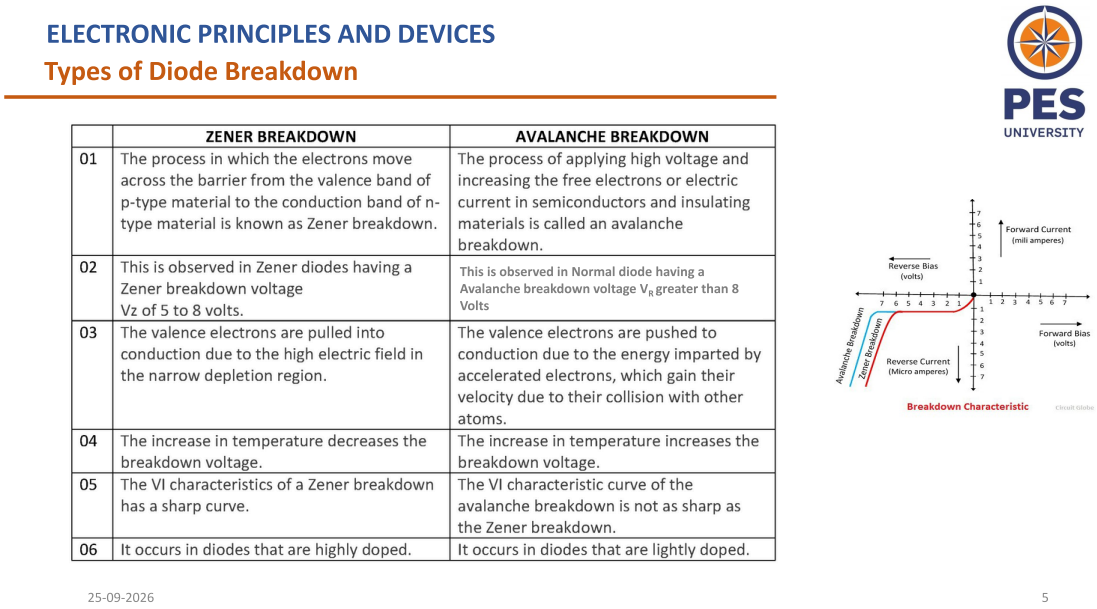

	ZENER BREAKDOWN	AVALANCHE BREAKDOWN
01	The process in which the electrons move across the barrier from the valence band of p-type material to the conduction band of n-type material is known as Zener breakdown.	The process of applying high voltage and increasing the free electrons or electric current in semiconductors and insulating materials is called an avalanche breakdown.
02	This is observed in Zener diodes having a Zener breakdown voltage Vz of 5 to 8 volts.	This is observed in Normal diode having a Avalanche breakdown voltage V_R greater than 8 Volts
03	The valence electrons are pulled into conduction due to the high electric field in the narrow depletion region.	The valence electrons are pushed to conduction due to the energy imparted by accelerated electrons, which gain their velocity due to their collision with other atoms.
04	The increase in temperature decreases the breakdown voltage.	The increase in temperature increases the breakdown voltage.
05	The VI characteristics of a Zener breakdown has a sharp curve.	The VI characteristic curve of the avalanche breakdown is not as sharp as the Zener breakdown.
06	It occurs in diodes that are highly doped.	It occurs in diodes that are lightly doped.

Figure G000034. Zener and Avalanche breakdown Temperature effects · source slide page 5.

EV0000202

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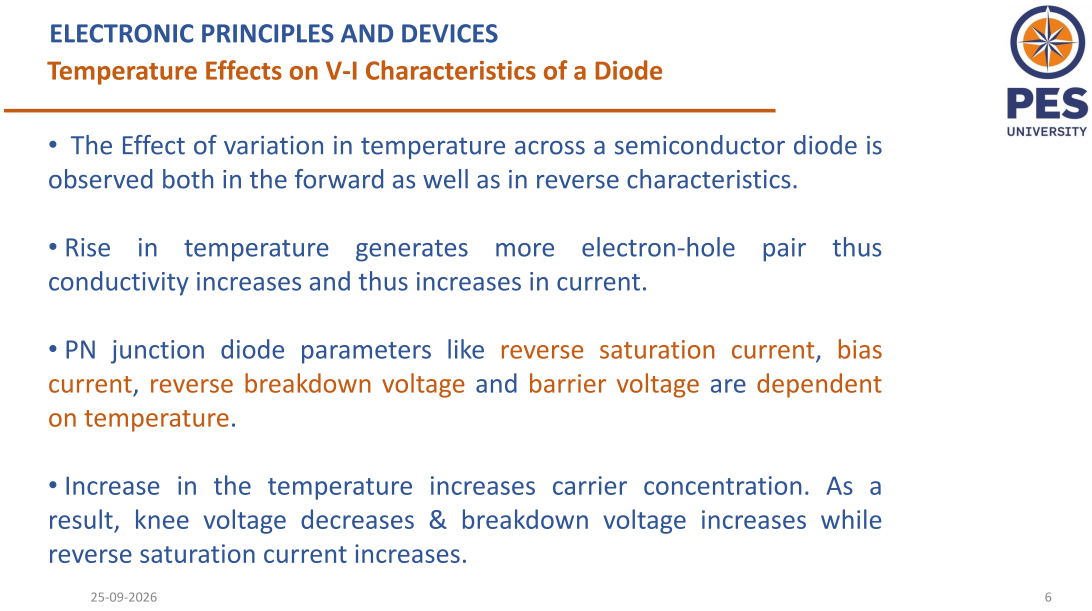

Figure G000035. Zener and Avalanche breakdown Temperature effects · source slide page 6.

EV0000205

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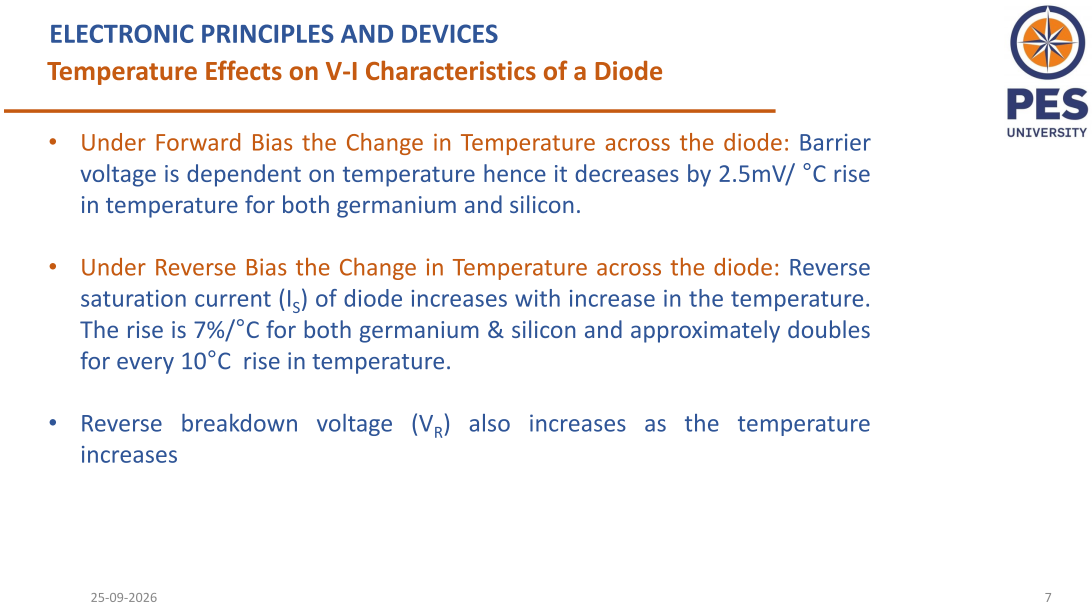

Figure G000036. Zener and Avalanche breakdown Temperature effects · source slide page 7.

EV0000208

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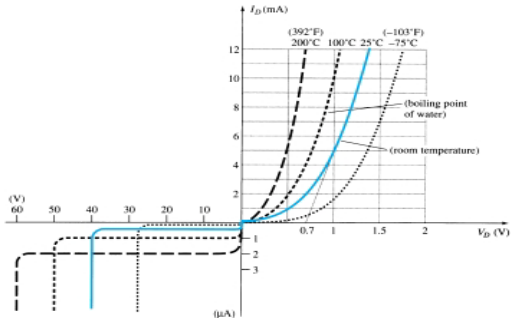
ELECTRONIC PRINCIPLES AND DEVICES

Temperature Effects on A Semiconductor diode

PES UNIVERSITY

1. The Knee voltage shifts to the left at a rate of 2.5mV/^{0}C increase in temperature



2. The reverse saturation current I_s will double in magnitude for every 10°C increase in temperature.

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Figure G000037. Zener and Avalanche breakdown Temperature effects · source slide page 8.

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ELECTRONIC PRINCIPLES AND DEVICES

An Example - Temperature Effects on A Semiconductor diode

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Example: The Reverse saturation current of silicon diode at 20°C is 0.1μA. Determine its value if the temperature is increased by 40 °C .

ANS:

$$Given\ Data\ at\ T = 20^o C : I_s = 0.1\mu A$$
$$T = 30^o C : I_s = 0.2\mu A$$
$$T = 40^o C : I_s = 0.4\mu A$$
$$T = 50^o C : I_s = 0.8\mu A$$
$$\mathrm{T} = 60^{\mathrm{o}}\mathrm{C} : \mathrm{I_s} = 1.6\mu\mathrm{A}$$

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Figure G000038. Zener and Avalanche breakdown Temperature effects · source slide page 9.

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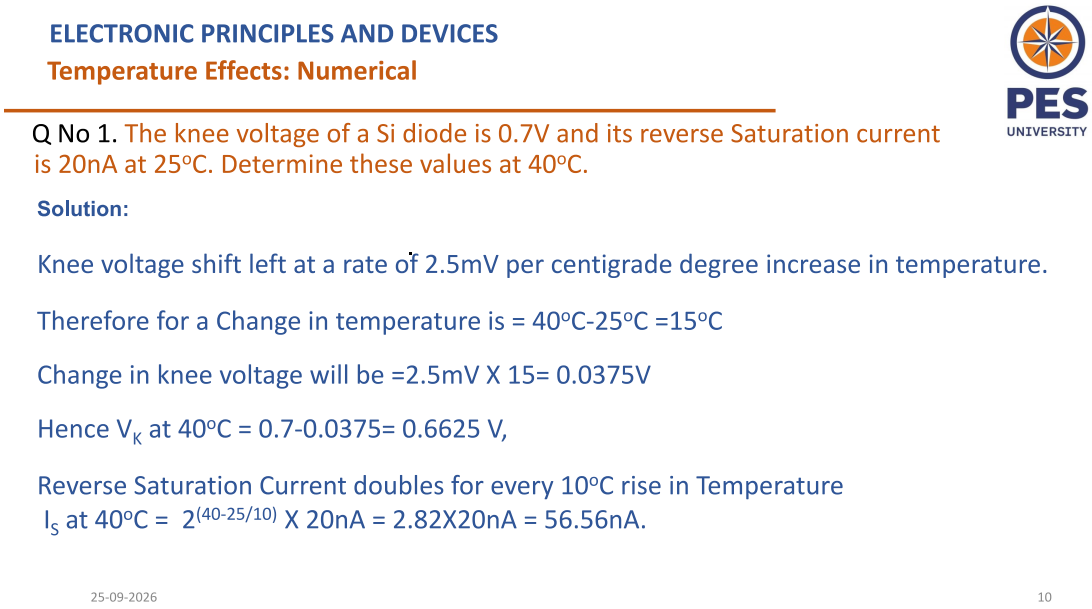

Figure G000039. Zener and Avalanche breakdown Temperature effects · source slide page 10.

EV0000217

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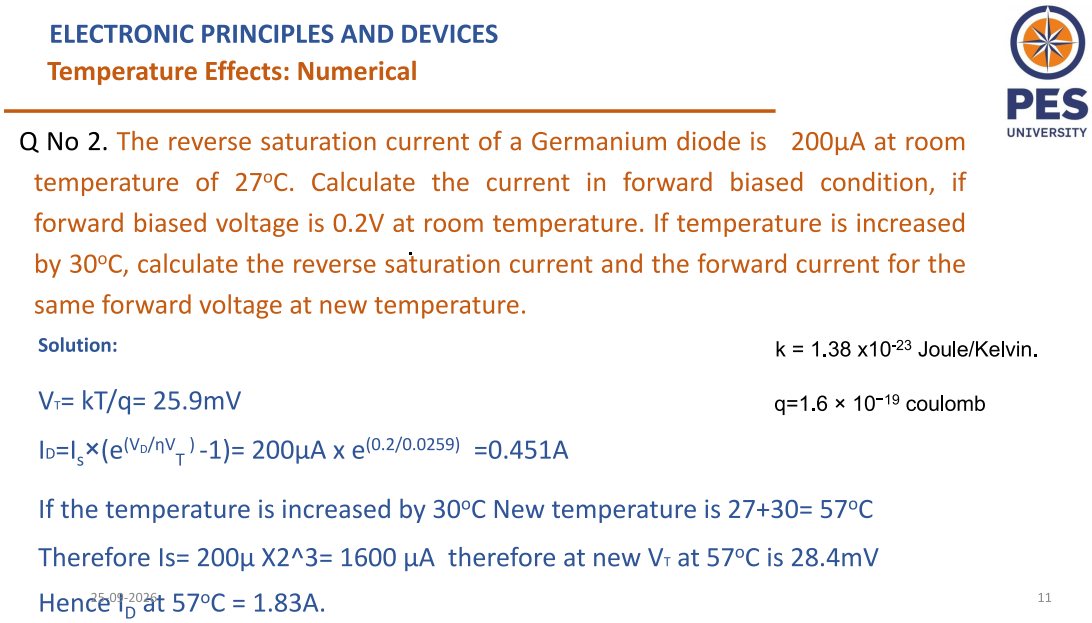

Figure G000040. Zener and Avalanche breakdown Temperature effects · source slide page 11.

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ELECTRONIC PRINCIPLES AND DEVICES

Temperature Effects: Numerical

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Q No 3. A Ge diode has a reverse saturation current of 5µA at temperature 300K find diode current at 40°C . When forward bias voltage is 0.27.

Solution:

Given Is at 27°C is 5µA

k = 1.38 x10^{-23} Joule/Kelvin.

q=1.6 × 10^{-19} coulomb

Therefore, I_s at 40°C is = 5µ x $2^{(40-27/10)}$
= 5µ X 2.462
=12.31 µ A

Hence V_T at 40°C = kT/q= 26.9mV

Therefore $I_D=I_s\times(e^{(VD/\eta V_T)}-1)$= 12.31µA x $e^{(V_D/\eta V_T)}$ = 281.49mA

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Figure G000041. Zener and Avalanche breakdown Temperature effects · source slide page 12.

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ELECTRONIC PRINCIPLES AND DEVICES

Temperature Effects: Numerical

PES UNIVERSITY

Q No 4. The reverse saturation current of a Si diode is 2pA at 27°C. Determine the forward biased voltage across the diode at 57°C, if the forward current through the diode at 57°C is 50mA.

Solution:

Reverse saturation current doubles for every 10°C rise in temperature

Hence at if reverse saturation current at 27°C is 2pA

Then reverse saturation current at 37°C is 4pA

Then reverse saturation current at 47°C is 8pA

Then reverse saturation current at 57°C is 16pA

k = 1.38 x10^{-23} Joule/Kelvin.

q=1.6 × 10^{-19} coulomb

Therefore V_T 57°C is =k x 330/q= 28.42mV

Hence by using $I_D=I_s\times(e^{(V_D/\eta V_T)}-1)$ = 50mA

$50/16= e^{V_D/\eta V_T}$ → $\ln(50/16)=V_D/28.42m$ → V_D = 0.686V at 57°C.

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Figure G000042. Zener and Avalanche breakdown Temperature effects · source slide page 13.

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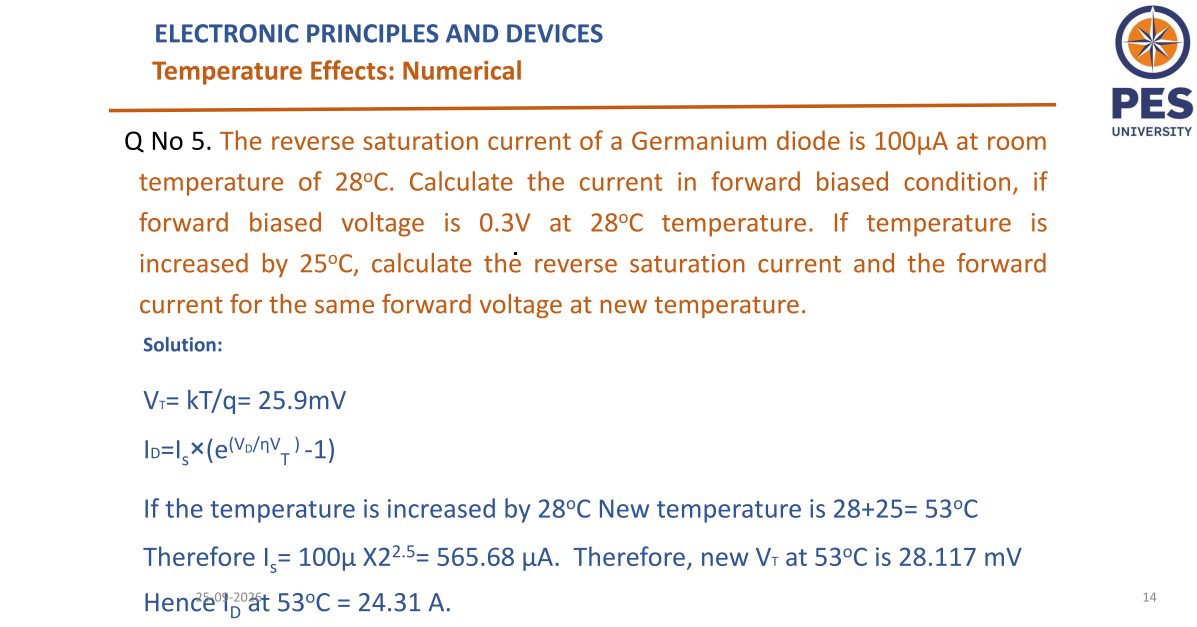

Figure G000043. Zener and Avalanche breakdown Temperature effects · source slide page 14.

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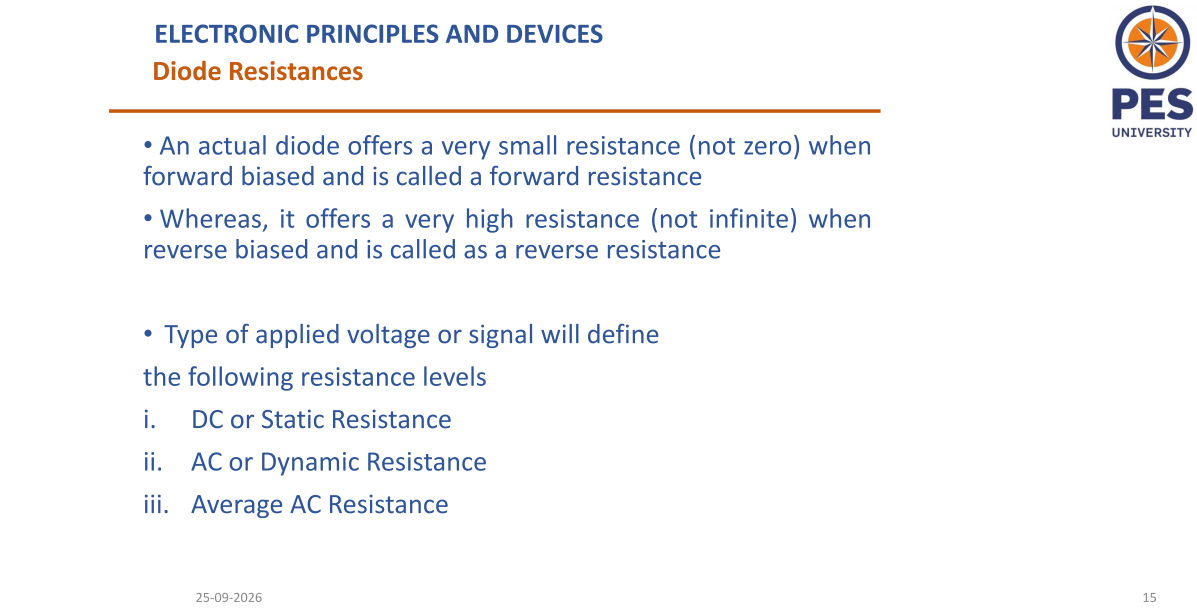

Figure G000044. Zener and Avalanche breakdown Temperature effects · source slide page 15.

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Figure G000045. Zener and Avalanche breakdown Temperature effects · source slide page 16.

EV0000235

> ***How to read it.*** Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Important points and why

1. Source slide page 1 (extracted text): ELECTRONIC PRINCIPLES AND DEVICES (4-0-0-4-4) (UE26EC141A) Department of Electronics and Communication Engineering 25-09-2026 1 EV0000189 · EV0000190 · EV0000191
2. Source slide page 2 (extracted text): ELECTRONIC PRINCIPLES AND DEVICES Topic- 4 Zener and Avalanche breakdown, Temperature effects Department of Electronics and Communication. 25-09-2026 2 EV0000192 · EV0000193 · EV0000194
3. Source slide page 3 (extracted text): ELECTRONIC PRINCIPLES AND DEVICES Types of Diode Breakdown • Avalanche breakdown : In the reverse bias condition as the voltage increases the free charge carriers obtain velocity and associated kinetic energy and release additional carriers through collision with other atomic structures. • Hence Covalent bonds are broken and electron–hole are generated. • These charge carriers acquire energy from the applied potential and produce more and more free charge carriers. This cumulative process is called Avalanche multiplication25-09-2026 or ionization process. 3 EV0000195 · EV0000196 · EV0000197
4. Source slide page 4 (extracted text): ELECTRONIC PRINCIPLES AND DEVICES Types of Diode Breakdown • Zener breakdown : Zener break down occurs when both P and N type material are heavily doped in a semiconductor diode. Hence the depletion layer region is narrow down. • When a small reverse bias voltage is applied across a diode a very strong electric fied is produced. • Due to this electric field covalent bond breaks and produces large number of free charge carriers. 25-09-2026 4 EV0000198 · EV0000199 · EV0000200

5. Source slide page 5 (extracted text): ELECTRONIC PRINCIPLES AND DEVICES Types of Diode Breakdown This is observed in Normal diode having a Avalanche breakdown voltage VR greater than 8 Volts 25-09-2026 5 EV0000201 · EV0000202 · EV0000203
6. Source slide page 6 (extracted text): ELECTRONIC PRINCIPLES AND DEVICES Temperature Effects on V-I Characteristics of a Diode • The Effect of variation in temperature across a semiconductor diode is observed both in the forward as well as in reverse characteristics. • Rise in temperature generates more electron-hole pair thus conductivity increases and thus increases in current. • PN junction diode parameters like reverse saturation current, bias current, reverse breakdown voltage and barrier voltage are dependent on temperature. • Increase in the temperature increases carrier concentration. As a result, knee voltage decreases & breakdown voltage increases while reverse saturation current increases. 25-09-2026 6 EV0000204 · EV0000205 · EV0000206
7. Source slide page 7 (extracted text): ELECTRONIC PRINCIPLES AND DEVICES Temperature Effects on V-I Characteristics of a Diode • Under Forward Bias the Change in Temperature across the diode: Barrier voltage is dependent on temperature hence it decreases by 2.5mV/ °C rise in temperature for both germanium and silicon. • Under Reverse Bias the Change in Temperature across the diode: Reverse saturation current (IS) of diode increases with increase in the temperature. The rise is 7%/°C for both germanium & silicon and approximately doubles for every 10°C rise in temperature. • Reverse breakdown voltage also increases as the temperature (VR) increases 25-09-2026 7 EV0000207 · EV0000208 · EV0000209
8. Source slide page 8 (extracted text): ELECTRONIC PRINCIPLES AND DEVICES Temperature Effects on A Semiconductor diode 1. The Knee voltage shifts to the left at a rate of 2.5mV/0C increase in temperature . 2. The reverse saturation current Is will double in magnitude for every 10°C increase in temperature. 25-09-2026 8 EV0000210 · EV0000211 · EV0000212
9. Source slide page 9 (extracted text): ELECTRONIC PRINCIPLES AND DEVICES An Example - Temperature Effects on A Semiconductor diode Example: The Reverse saturation current of silicon diode at 20°C is 0.1µA. Determine its value if the temperature is increased by 40 °C . ANS: Given Data at T ☒ 20o C : I ☒ 0. 1☒A s T ☒ 30o C : I ☒ 0. 2☒A s T ☒ 40o C : I ☒ 0. 4☒A s T ☒ 50o C : I ☒ 0. 8☒A s T ☒ 60o C : I ☒ 1.6☒A s 25-09-2026 9 EV0000213 · EV0000214 · EV0000215
10. Source slide page 10 (extracted text): ELECTRONIC PRINCIPLES AND DEVICES Temperature Effects: Numerical Q No 1. The knee voltage of a Si diode is 0.7V and its reverse Saturation current is 20nA at 25oC. Determine these values at 40oC. Solution: . Knee voltage shift left at a rate of 2.5mV per centigrade degree increase in temperature. Therefore for a Change in temperature is = 40oC-25oC =15oC Change in knee voltage will be =2.5mV X 15= 0.0375V Hence VK at 40oC = 0.7-0.0375= 0.6625 V, Reverse Saturation Current doubles for every 10oC rise in Temperature IS at 40oC = 2(40-25/10) X 20nA = 2.82X20nA = 56.56nA. 25-09-2026 10 EV0000216 · EV0000217 · EV0000218

11. Source slide page 11 (extracted text): ELECTRONIC PRINCIPLES AND DEVICES Temperature Effects: Numerical Q No 2. The reverse saturation current of a Germanium diode is 200µA at room temperature of 27oC. Calculate the current in forward biased condition, if forward biased voltage is 0.2V at room temperature. If temperature is increased . by 30oC, calculate the reverse saturation current and the forward current for the same forward voltage at new temperature. Solution: k = 1.38 x10-23 Joule/Kelvin. VT= kT/q= 25.9mV q=1.6 × 10−19 coulomb 200µA x e(0.2/0.0259) =0.451A ID=Is×(e(VD/ηVT)-1)= If the temperature is increased by 30oC New temperature is 27+30= 57oC Therefore Is= 200µ X2^3= 1600 µA therefore at new VT at 57oC is 28.4mV Hence25-09-2026ID at 57oC = 1.83A. 11 EV0000219 · EV0000220 · EV0000221
12. Source slide page 12 (extracted text): ELECTRONIC PRINCIPLES AND DEVICES Temperature Effects: Numerical Q No 3. A Ge diode has a reverse saturation current of 5µA at temperature 300K find diode current at 40oC . When forward bias voltage is 0.27. Solution: . k = 1.38 x10-23 Joule/Kelvin.Given Is at 27oC is 5µA q=1.6 × 10−19 coulomb Therefore, Is at 40oC is = 5µ x 2(40-27/10) = 5µ X 2.462 =12.31 µ A Hence VT at 40oC = kT/q= 26.9mV Therefore ID=Is×(e(VD/ηVT) -1)= 12.31µA x e(VD/ηV T) = 281.49mA 25-09-2026 12 EV0000222 · EV0000223 · EV0000224
13. Source slide page 13 (extracted text): ELECTRONIC PRINCIPLES AND DEVICES Temperature Effects: Numerical Q No 4. The reverse saturation current of a Si diode is 2pA at 27oC. Determine the forward biased voltage across the diode at 57oC, if the forward current through the diode at 57oC is 50mA. Solution: . Reverse saturation current doubles for every 10oC rise in temperature Hence at if reverse saturation current at 27oC is 2pA k = 1.38 x10-23 Joule/Kelvin.Then reverse saturation current at 37oC is 4pA Then reverse saturation current at 47oC is 8pA q=1.6 × 10−19 coulomb Then reverse saturation current at 57oC is 16pA Therefore VT 57oC is =k x 330/q= 28.42mV Hence by using ID=Is×(e(VD/ηVT) -1) = 50mA 50/16= eVD/ηV ☒ln(50/16)=VD/28.42m = 0.686V at 57oC. T ☒VD 25-09-2026 13 EV0000225 · EV0000226 · EV0000227
14. Source slide page 14 (extracted text): ELECTRONIC PRINCIPLES AND DEVICES Temperature Effects: Numerical Q No 5. The reverse saturation current of a Germanium diode is 100µA at room temperature of 28oC. Calculate the current in forward biased condition, if forward biased voltage is 0.3V at 28oC temperature. If temperature is . increased by 25oC, calculate the reverse saturation current and the forward current for the same forward voltage at new temperature. Solution: VT= kT/q= 25.9mV ID=Is×(e(VD/ηVT)-1) If the temperature is increased by 28oC New temperature is 28+25= 53oC Therefore Is= 100µ X22.5= 565.68 µA. Therefore, new VT at 53oC is 28.117 mV Hence25-09-2026ID at 53oC = 24.31 A. 14 EV0000228 · EV0000229 · EV0000230
15. Source slide page 15 (extracted text): ELECTRONIC PRINCIPLES AND DEVICES Diode Resistances • An actual diode offers a very small resistance (not zero) when forward biased and is called a forward resistance • Whereas, it offers a very high resistance (not infinite) when reverse biased and is called as a reverse resistance • Type of applied voltage or signal will define the following resistance levels i. DC or Static Resistance ii. AC or Dynamic Resistance iii. Average AC Resistance 25-09-2026 15 EV0000231 · EV0000232 · EV0000233
16. Source slide page 16 (extracted text): THANK YOU Department of Electronics and Communication 25-09-2026 16 EV0000234 · EV0000235 · EV0000236

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:02.960] Good morning, and welcome to Electronic Principle and Devices. [00:00:07.520] Today we're going to learn about avalanche breakdown and Zener breakdown. [00:00:10.870] Before that, we'll go for refreshing of the last topic, comparison of the diode [00:00:14.780] characteristics of germanium, silicon, and gallium arsenide. [00:00:17.980] We see that [00:00:20.280] there are different voltages at which the [00:00:23.740] different semiconductor device has different conducting [00:00:28.900] voltages and with respect to forward bias and reverse bias. [00:00:33.040] In the forward bias, we see the germanium, silicon, and [00:00:37.100] gallium arsenide, which conduct at different voltages. [00:00:40.000] The reverse bias also, the [00:00:43.300] breakdown also occurs at different voltages. [00:00:45.880] So now let us try to understand about what are the different types of [00:00:49.860] breakdowns are there. The first is avalanche breakdown for a normal [00:00:53.240] diode we look at. [00:00:58.140] So as we know already, the VI characteristics is as shown. [00:01:01.740] So we're keeping the reference as VI characteristics here. [00:01:05.360] The forward bias region is this, the reverse bias region is this. [00:01:08.720] So in the reverse bias condition, as the voltage increases, the free charge [00:01:13.320] carriers obtain velocity and associated kinetic energy [00:01:17.340] and release additional carriers through collision with the other atomic [00:01:20.980] structure. Hence, the covalent bonds are broken and electron-hole pair are [00:01:24.940] generated. So as we see here from the diagram, the electron [00:01:28.820] hole before application of the voltage, and we see that the electron hole [00:01:32.800] is getting disappeared with the application of the voltage. [00:01:36.020] This electron [00:01:38.040] will hit to the walls of an atom, as we see here, and acquire a [00:01:41.780] energy from applied voltage. [00:01:44.980] This will hit based on the energy acquired from applied voltage and produce [00:01:49.000] more and more free electrons. This free electron again will hit [00:01:52.940] the another atoms, and this cumulative process is called [00:01:56.900] avalanche multiplication or ionization process. [00:02:01.620] Similarly, we have another type of breakdown, which is Zener [00:02:05.240] breakdown. The Zener breakdown occurs when both P and N-type materials [00:02:09.301] are heavily doped in a semiconductor diode, hence depletion region is [00:02:13.220] narrowed down, [00:02:15.280] as we see from the diagram here. When a small reverse voltage is [00:02:19.140] applied across the diode, a very strong electric field is produced. [00:02:23.280] Due to this electric field, covalent bonds breaks and produce a large number of [00:02:27.280] free charge carriers. [00:02:30.300] So coming to the comparison between the Zener breakdown and avalanche breakdown, [00:02:34.980] first we [00:02:36.520] start with Zener, and on the right-hand side we have avalanche. [00:02:39.920] The first point is, the process in which the [00:02:44.060] electron move across a barrier from valence band of the [00:02:48.960] p-type material to the conduction band of n-type material is known as breakdown [00:02:52.520] voltage, whereas the process of applying a high voltage [00:02:56.580] and increasing the free electron. [00:02:58.400] So we apply a high voltage, that will result in increase in the free electron or [00:03:02.020] electric current in the semiconductor and insulating material is called as a [00:03:05.260] avalanche breakdown. Coming to the second point, this is observed in a [00:03:09.180] Zener diode having a Zener breakdown, which varies from five to [00:03:13.060] eight volts, whereas in the case of an avalanche breakdown, this is observed in [00:03:16.880] the normal diode [00:03:20.000] having a [00:03:22.620] voltage greater than the eight volts. [00:03:31.660] Number three, the valence electrons are pulled out into a conduction due to the [00:03:35.480] high electric field in the narrow depletion region. [00:03:38.120] Here, the valence electrons are pushed to the conduction due to the energy imparted [00:03:41.260] with the accelerated electron which gain

their velocity due to the collision with [00:03:44.580] other atom. Coming to the fourth point, the increase in temperature [00:03:48.660] decreases with the breakdown voltage. [00:03:50.700] The increase in temperature will increase the breakdown voltage here. [00:03:54.400] The VI characteristics of Zener diode breakdown has sharp value. [00:03:57.680] The VI characteristics of avalanche breakdown is not sharp as [00:04:01.280] Zener breakdown, as we see from the diagram. [00:04:03.920] Then in the Zener, it occurs in diode that are lightly doped. [00:04:12.600] Let us also consider the effect of [00:04:15.140] temperature on VI characteristics of the diode. [00:04:18.680] The effect of variation in the temperature across the semiconductor diode is [00:04:21.899] observed in both forward as well as the reverse characteristics. [00:04:27.800] The rise in temperature generates more [00:04:30.360] electron-hole pair, thus conductivity increases and [00:04:34.280] thus increase in [00:04:36.200] the current. The PN junction diode parameter like reverse [00:04:40.100] saturation current, bias current, reverse breakdown voltage, and barrier [00:04:43.940] voltage are dependent on the temperature. [00:04:46.600] Increase in temperature increases the carrier concentration. [00:04:49.820] As a result, the knee voltage decreases, breakdown voltage increases, while the [00:04:53.760] reverse saturation current also increases. [00:04:56.720] This we can understand with the help of a forward bias and a reverse

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Transcript block 2 (source interval 000300-000599 seconds): [00:05:00.660] bias. So in a forward bias, [00:05:04.320] whenever there is a change in the temperature across the diode, the barrier voltage [00:05:07.440] depended on the temperature increase, hence it decreases by 2.5 [00:05:11.300] volts for every degree [00:05:13.540] rise, the temperature for both germanium and silicon. [00:05:17.080] Under reverse bias, the change in the temperature across the diode, the reverse [00:05:20.080] saturation current of the diode increases with increase in temperature. [00:05:23.260] That is, there is a rise of 7% for every degree [00:05:27.920] rise in temperature for both germanium and silicon, and approximately doubles [00:05:31.820] for every 10 degree rise in the temperature. [00:05:34.480] The reverse breakdown voltage, VR, also increases with the temperature [00:05:38.140] increase. [00:05:42.540] So [00:05:43.640] the temperature effect on VI characteristics of the diode is as shown, [00:05:47.840] with the different rise in the temperature here. [00:05:50.820] Starting from right to the left, initially we have -75 degree centigrade, [00:05:55.040] where the cut-in voltage is at this point. [00:05:58.280] Now as the moment we increase from -75 to 125 degree, [00:06:02.500] the cut-in voltage is [00:06:04.700] ... is decreased. It may be around 0.1 or [00:06:08.600] even less than 0.1. Similarly, coming to the reverse bias here, [00:06:13.340] at -75 degree centigrade, the breakdown is [00:06:17.600] very less. [00:06:20.180] Whereas at 125 degree, the breakdown voltage is increasing, and [00:06:24.000] reverse saturation current also doubles for every 10 degree rise in the [00:06:27.220] temperature. So, to understand this, let us try to solve a few examples [00:06:31.180] on this and try to understand the concept of temperature [00:06:35.000] effect on semiconductor diode. [00:06:37.000] In the example, the reverse saturation current of silicon diode [00:06:40.860] is at 20 degree [00:06:44.600] is 0.1 microamp. Determine the value if the temperature is increased by 40. [00:06:48.700] We know that for every 10 degree rise in the temperature, the reverse saturation [00:06:52.420] current double. So we use the same principle here, and hence we get at 40 [00:06:56.400] degree, the required is 40 degree, we see that it's a [00:06:59.360] 0.4 microamps. Similarly, if there is an increase in the temperature, [00:07:03.360] we see that the different variations. [00:07:06.040] Now let us solve a few numericals on temperature effects on a [00:07:09.940] semiconductor diode [00:07:12.760] using the concept of

[00:07:14.720] what we studied. [00:07:16.300] So number one, the knee voltage of silicon diode is 0.7 and the [00:07:20.240] reverse saturation current is 20 nanoamps at 25 [00:07:24.020] degree. Determine the values at 40 degree centigrade. [00:07:28.860] So as we see that there's a rise in the temperature from 25 to 40. [00:07:32.620] The difference is 15 degree. So the change in the knee voltage will be [00:07:36.560] 25 millivolts into 15 is [00:07:38.760] 0.037 volts. So if you subtract at [00:07:42.600] 40 degree, we get 0.33 is the breakdown voltage, new breakdown voltage. [00:07:46.900] So now calculate the-- [00:07:51.260] At 40 degree, what is the current? So we get [00:07:54.680] 56.6 nanoamps, the reverse saturation current. [00:07:59.400] Similarly, [00:08:00.860] the numerical two, the reverse saturation current of germanium diode is [00:08:04.520] 200 micro at room temperature of 27. [00:08:07.160] We need to calculate the current in the forward bias condition if the forward bias [00:08:10.640] is 0.2 at room temperature. If the temperature is increased to 30 degree [00:08:14.360] centigrade. So keeping in mind here the temperature is increased to [00:08:18.280] 30 degree, so it's 57 degree centigrade, and at this [00:08:22.120] point, what is the IS? So 1,600 microamp. [00:08:25.940] Therefore, VT is 0.8 and actual current [00:08:29.840] calculation is 1.83 amps. [00:08:34.400] So similarly, question number three, the germanium diode has reverse [00:08:38.400] saturation current of five microamp at temperature 300 degree [00:08:42.500] Kelvin. Find the current at 40 [00:08:45.920] degree centigrade. So [00:08:49.800] given at 27 degree centigrade is five micro. [00:08:52.630] So we need to calculate for change in the temperature at 40 degree. [00:08:56.320] So five microamp, two to the power of T2 minus T1 by 10. [00:08:59.480] We use the whole formula last time what we used. [00:09:02.900] So we get at 40 degree, VT is 26.9. Substitute that [00:09:06.810] and get the ID which is 281.49 milliamps. [00:09:12.440] Next, [00:09:13.720] fourth, the reverse saturation current of germanium is two pico at 27 degree. [00:09:17.560] Similarly, calculate at 27 degree. [00:09:19.780] So we use the same principle concept where IS is calculated [00:09:23.580] first. Substitute the value and calculate the [00:09:27.540] reverse saturation current. Determine the forward bias also. [00:09:31.020] The forward bias voltage VA. So it's [00:09:33.060] 0.686 volts at 57 degree [00:09:36.980] centigrade. [00:09:39.220] Similarly, we can solve the numerical five, and this problem also will [00:09:43.140] be solved physically in the class. [00:09:46.720] Thank you.

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Full transcript: 1472 normalized words, 164 timestamped segments, 592.981 seconds duration. Low-confidence segments flagged: 0.

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6 Ideal versus Practical diode, Diode resistances

This lesson is indexed from the source video titled Ideal versus Practical diode, Diode resistances. Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or rewritten. A candidate slide deck is grouped here according to the current matching score; the exact path and page locators are listed in the source evidence. EV0000038 · EV0000039

Graphs and diagrams

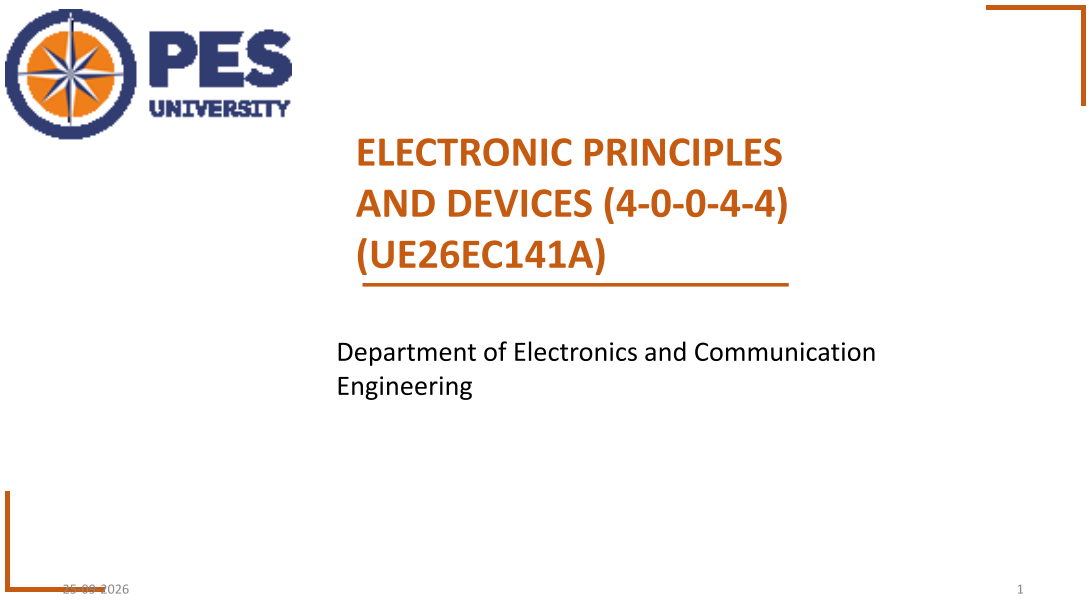

Figure G000046. Ideal versus Practical diode Diode resistances · source slide page 1.

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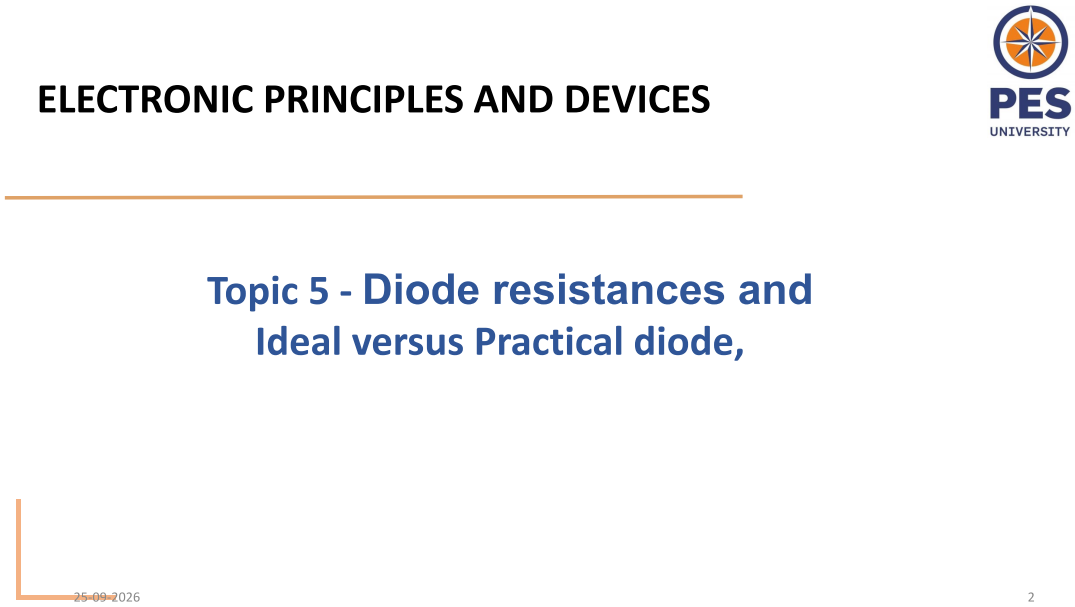

Figure G000047. Ideal versus Practical diode Diode resistances · source slide page 2.

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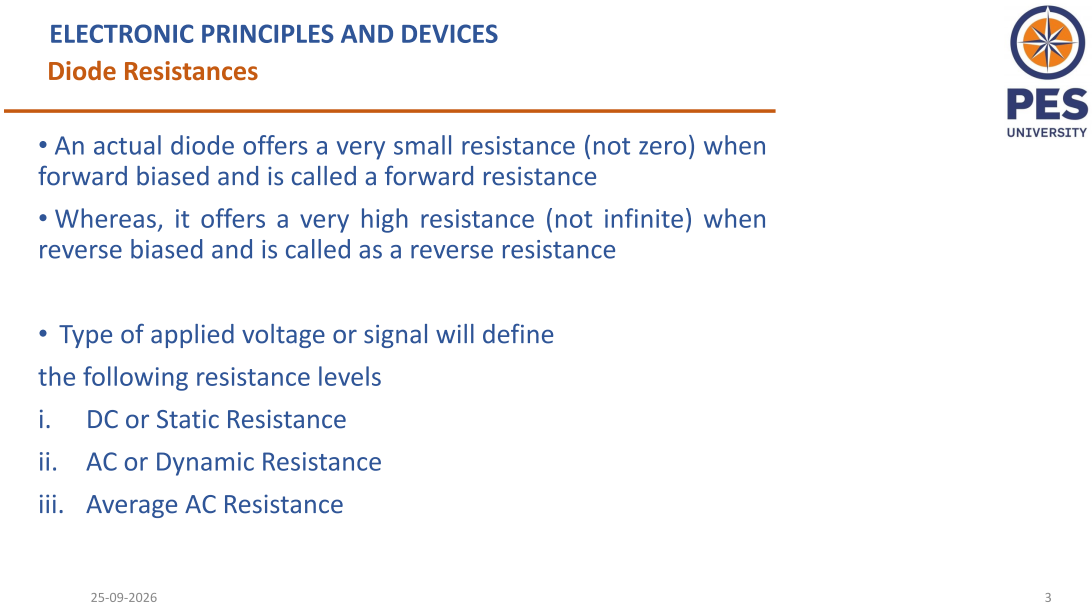

Figure G000048. Ideal versus Practical diode Diode resistances · source slide page 3.

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ELECTRONIC PRINCIPLES AND DEVICES

DC or Static Resistance

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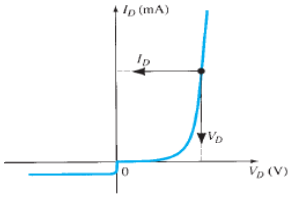
- The application of a dc voltage to a circuit containing a semiconductor diode will result in an operating point on the semiconductor diode characteristic curve.
- Finding the corresponding levels of V_D and I_D at particular operating point gives the DC resistance.



$$R_D = \frac{V_D}{I_D}$$

- Typically, the dc resistance of a diode in:

Forward bias : 10 Ω to 80 Ω

Reverse Bias: 10 MΩ

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Figure G000049. Ideal versus Practical diode Diode resistances · source slide page 4.

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ELECTRONIC PRINCIPLES AND DEVICES

AC or Dynamic Resistance

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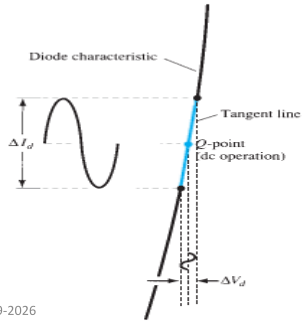
- The application of a small AC voltage to a circuit containing a semiconductor diode offers AC resistance.
- A straight line drawn tangent to the curve through the Q –point will define a particular change in voltage and current determines the ac or dynamic resistance for this region of the diode characteristics.



$$r_d = \frac{\Delta V_d}{\Delta I_d}$$

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Figure G000050. Ideal versus Practical diode Diode resistances · source slide page 5.

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ELECTRONIC PRINCIPLES AND DEVICES

AC or Dynamic Resistance

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- Dynamic resistance can be found simply by substituting the quiescent value of the diode current into the equation also

$$r_d = \frac{26\text{ mV}}{I_D} \text{ ohms}$$

$$r'_d = \frac{26\text{ mV}}{I_D} + r_B \text{ ohms}$$

- Where, r_B is bulk or body resistance

- Typically, the ac resistance of a diode in the active region will range from 1 Ω to 100 Ω.

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Figure G000051. Ideal versus Practical diode Diode resistances · source slide page 6.

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ELECTRONIC PRINCIPLES AND DEVICES

Average AC Resistance

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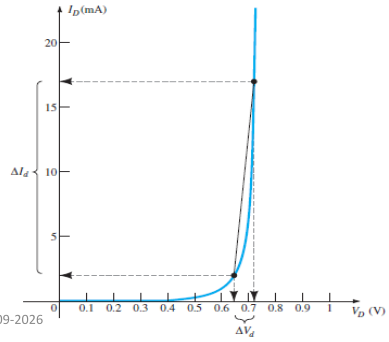
- If applied input signal is sufficiently large to produce a broad swing and the resistance determined by a straight line drawn between the two intersections established by the maximum and minimum values of input voltage is called Average AC Resistance



$$r_{av} = \left.\frac{\Delta V_d}{\Delta I_d}\right|_{\text{pt. to pt.}}$$

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Figure G000052. Ideal versus Practical diode Diode resistances · source slide page 7.

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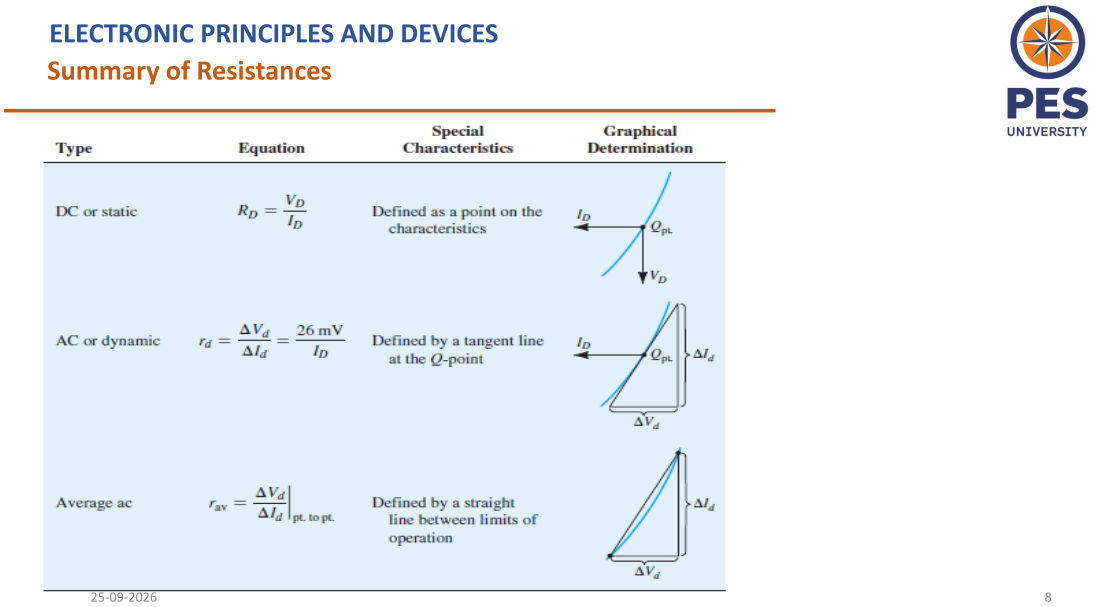

Figure G000053. Ideal versus Practical diode Diode resistances · source slide page 8.

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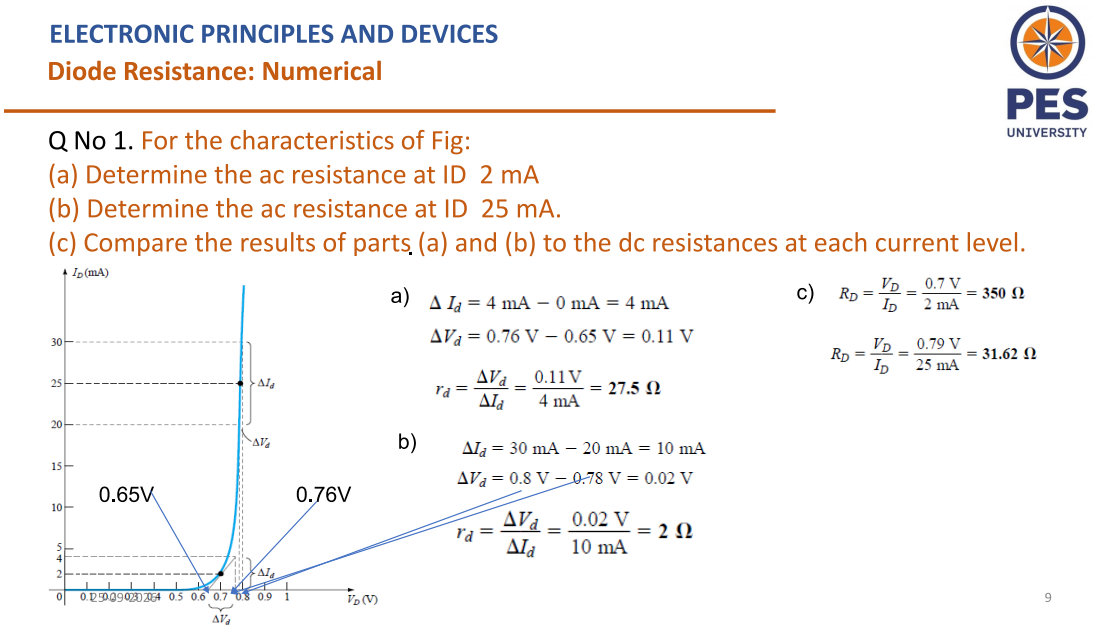

Figure G000054. Ideal versus Practical diode Diode resistances · source slide page 9.

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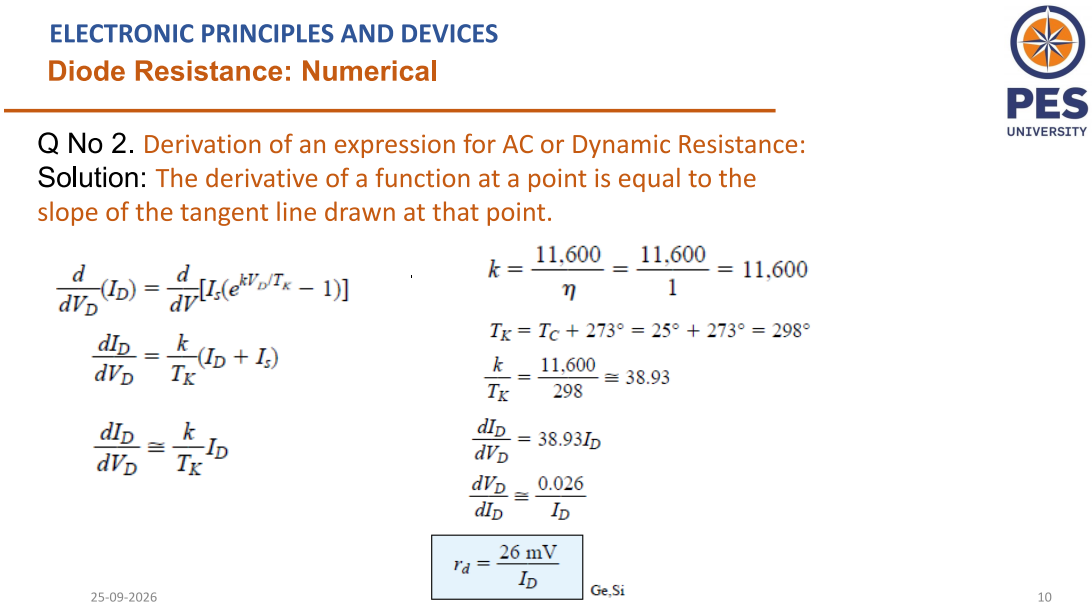

Figure G000055. Ideal versus Practical diode Diode resistances · source slide page 10.

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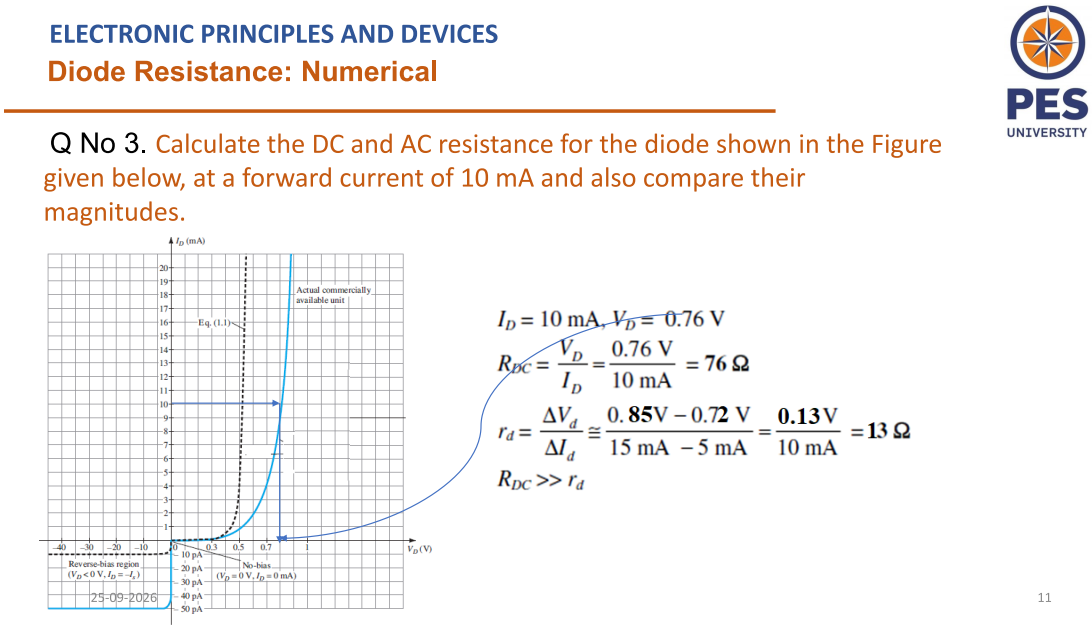

Figure G000056. Ideal versus Practical diode Diode resistances · source slide page 11.

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ELECTRONIC PRINCIPLES AND DEVICES

Diode Resistance: Numerical

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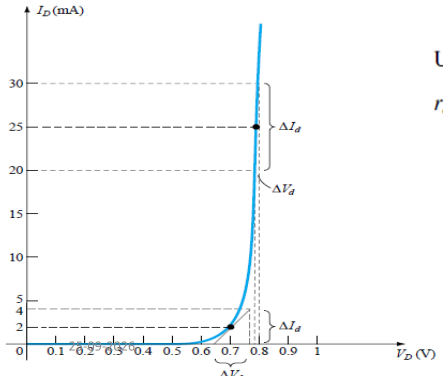
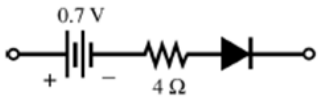
Q No 4. Find the piecewise-linear equivalent circuit for the diode shown in the Fig. given below. Use a straight line segment that intersects the horizontal axis at 0.7 V and best approximates the curve for the region greater than 0.7 V.



Using the best approximation to the curve beyond $V_D = 0.7$ V:

$$r_{av} = \frac{\Delta V_d}{\Delta I_d} \cong \frac{0.8\text{ V} - 0.7\text{ V}}{25\text{ mA} - 0\text{ mA}} = \frac{0.1\text{ V}}{25\text{ mA}} = 4\ \Omega$$



12

Figure G000057. Ideal versus Practical diode Diode resistances · source slide page 12.

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ELECTRONIC PRINCIPLES AND DEVICES

Diode Resistance: Numerical

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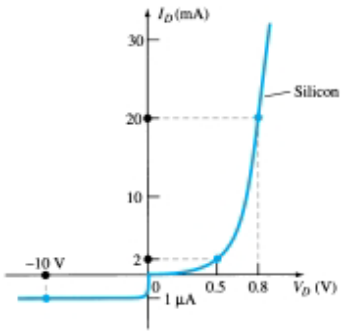
Q No 5. Determine the DC resistance levels for the diode of Fig. given below at (a) I_D =2 mA (b) I_D =20 mA (c) V_D = -10 V



(a) At $I_D = 2$ mA, $V_D = 0.5$ V (from the curve) and

$$R_D = \frac{V_D}{I_D} = \frac{0.5\text{ V}}{2\text{ mA}} = 250\ \Omega$$

(b) At $I_D = 20$ mA, $V_D = 0.8$ V (from the curve) and

$$R_D = \frac{V_D}{I_D} = \frac{0.8\text{ V}}{20\text{ mA}} = 40\ \Omega$$

(c) At $V_D = -10$ V, $I_D = -I_s = -1\ \mu$A (from the curve) and

$$R_D = \frac{V_D}{I_D} = \frac{10\text{ V}}{1\ \mu\text{A}} = 10\ \text{M}\Omega$$

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Figure G000058. Ideal versus Practical diode Diode resistances · source slide page 13.

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ELECTRONIC PRINCIPLES AND DEVICES

Diode Equivalent circuits /Diode Approximations

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- Diode approximation is a mathematical method used to approximate the nonlinear behavior of real diodes to enable calculations and circuit analysis.
- There are three different approximations used to analyze the diode circuits, namely

I. Ideal Diode Characteristics
II. Simplified Diode Characteristics
III. Piecewise-linear equivalent Circuit

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Figure G000059. Ideal versus Practical diode Diode resistances · source slide page 14.

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ELECTRONIC PRINCIPLES AND DEVICES

Ideal diode Characteristics

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- The diode is considered as a closed switch with zero voltage drop when forward-biased (V_d = 0) and as an open switch when reverse biased

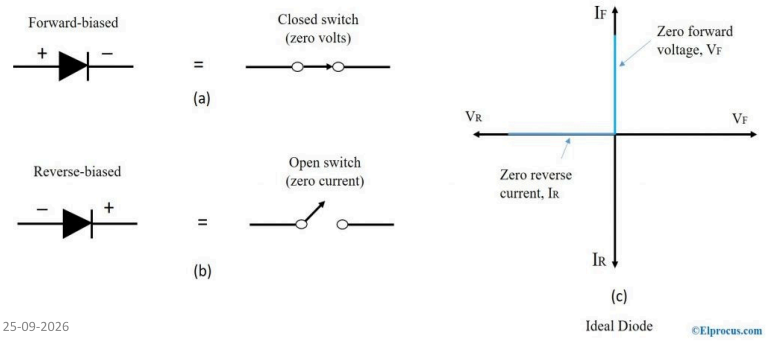

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Figure G000060. Ideal versus Practical diode Diode resistances · source slide page 15.

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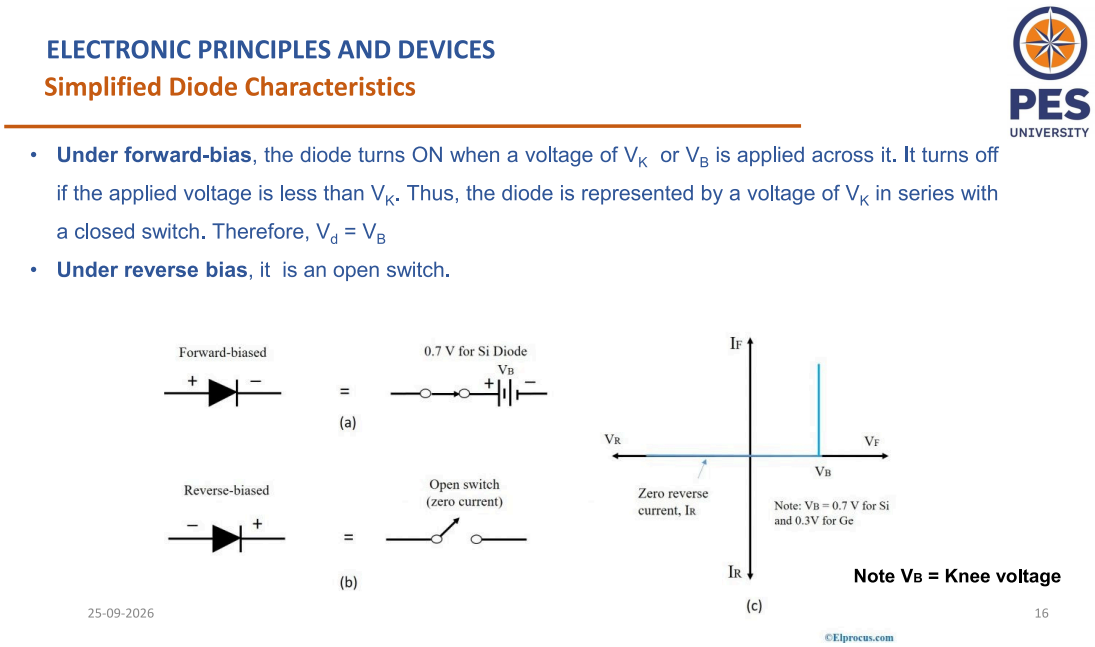

Figure G000061. Ideal versus Practical diode Diode resistances · source slide page 16.

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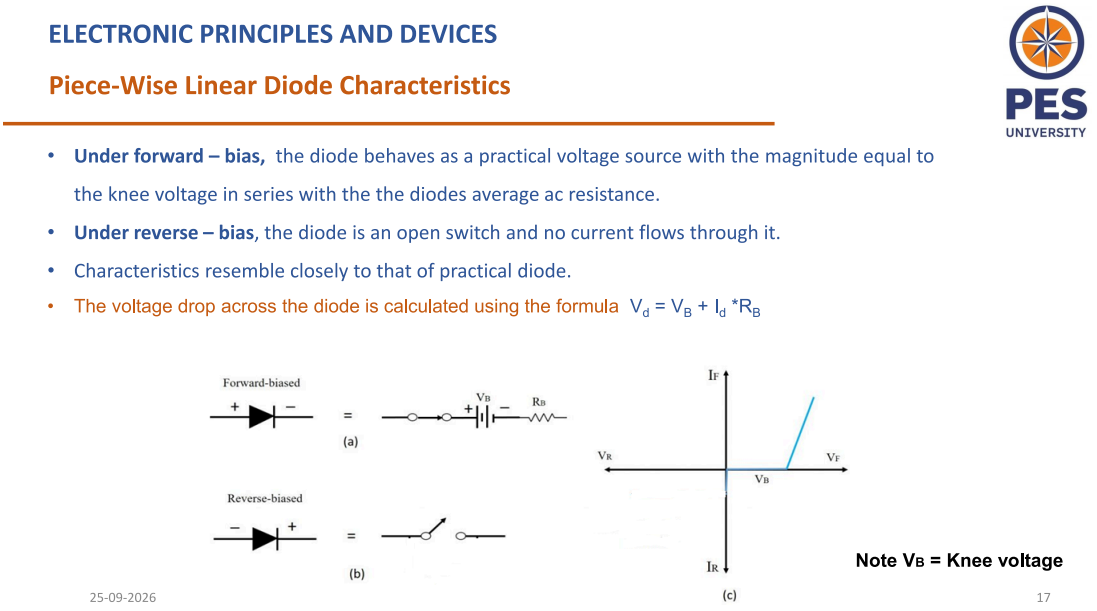

Figure G000062. Ideal versus Practical diode Diode resistances · source slide page 17.

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Figure G000063. Ideal versus Practical diode Diode resistances · source slide page 18.

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> ***How to read it.*** Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Important points and why

1. Source slide page 1 (extracted text): ELECTRONIC PRINCIPLES AND DEVICES (4-0-0-4-4) (UE26EC141A) Department of Electronics and Communication Engineering 25-09-2026 1 EV0000237 · EV0000238 · EV0000239
2. Source slide page 2 (extracted text): ELECTRONIC PRINCIPLES AND DEVICES Topic 5 - Diode resistances and Ideal versus Practical diode, 25-09-2026 2 EV0000240 · EV0000241 · EV0000242
3. Source slide page 3 (extracted text): ELECTRONIC PRINCIPLES AND DEVICES Diode Resistances • An actual diode offers a very small resistance (not zero) when forward biased and is called a forward resistance • Whereas, it offers a very high resistance (not infinite) when reverse biased and is called as a reverse resistance • Type of applied voltage or signal will define the following resistance levels i. DC or Static Resistance ii. AC or Dynamic Resistance iii. Average AC Resistance 25-09-2026 3 EV0000243 · EV0000244 · EV0000245
4. Source slide page 4 (extracted text): ELECTRONIC PRINCIPLES AND DEVICES DC or Static Resistance • The application of a dc voltage to a circuit containing a semiconductor diode will result in an operating point on the semiconductor diode characteristic curve. • Finding the corresponding levels of VD and ID at particular operating point gives the DC resistance. • Typically, the dc resistance of a diode in: Forward bias : 10 Ω to 80 Ω Reverse Bias: 10 MΩ 25-09-2026 4 EV0000246 · EV0000247 · EV0000248

5. Source slide page 5 (extracted text): ELECTRONIC PRINCIPLES AND DEVICES AC or Dynamic Resistance • The application of a small AC voltage to a circuit containing a semiconductor diode offers AC resistance. • A straight line drawn tangent to the curve through the Q –point will define a particular change in voltage and current determines the ac or dynamic resistance for this region of the diode characteristics. 25-09-2026 5 EV0000249 · EV0000250 · EV0000251
6. Source slide page 6 (extracted text): ELECTRONIC PRINCIPLES AND DEVICES AC or Dynamic Resistance • Dynamic resistance can be found simply by substituting the quiescent value of the diode current into the equation also • Where, rB is bulk or body resistance •Typically, the ac resistance of a diode in the active region will range from 1 Ω to 100 Ω. 25-09-2026 6 EV0000252 · EV0000253 · EV0000254
7. Source slide page 7 (extracted text): ELECTRONIC PRINCIPLES AND DEVICES Average AC Resistance • If applied input signal is sufficiently large to produce a broad swing and the resistance determined by a straight line drawn between the two intersections established by the maximum and minimum values of input voltage is called Average AC Resistance 25-09-2026 7 EV0000255 · EV0000256 · EV0000257
8. Source slide page 8 (extracted text): ELECTRONIC PRINCIPLES AND DEVICES Summary of Resistances 25-09-2026 8 EV0000258 · EV0000259 · EV0000260
9. Source slide page 9 (extracted text): ELECTRONIC PRINCIPLES AND DEVICES Diode Resistance: Numerical Q No 1. For the characteristics of Fig: (a) Determine the ac resistance at ID 2 mA (b) Determine the ac resistance at ID 25 mA. (c) Compare the results of parts. (a) and (b) to the dc resistances at each current level. a) c) b) 0.65V 0.76V 25-09-2026 9 EV0000261 · EV0000262 · EV0000263
10. Source slide page 10 (extracted text): ELECTRONIC PRINCIPLES AND DEVICES Diode Resistance: Numerical Q No 2. Derivation of an expression for AC or Dynamic Resistance: Solution: The derivative of a function at a point is equal to the slope of the tangent line drawn at that point. . 25-09-2026 10 EV0000264 · EV0000265 · EV0000266
11. Source slide page 11 (extracted text): ELECTRONIC PRINCIPLES AND DEVICES Diode Resistance: Numerical Q No 3. Calculate the DC and AC resistance for the diode shown in the Figure given below, at a forward current of 10 mA and also compare their magnitudes. . 25-09-2026 11 EV0000267 · EV0000268 · EV0000269
12. Source slide page 12 (extracted text): ELECTRONIC PRINCIPLES AND DEVICES Diode Resistance: Numerical Q No 4. Find the piecewise-linear equivalent circuit for the diode shown in the Fig. given below. Use a straight line segment that intersects the horizontal axis at 0.7 V and best approximates the curve for the region greater than 0.7 V. . 25-09-2026 12 EV0000270 · EV0000271 · EV0000272
13. Source slide page 13 (extracted text): ELECTRONIC PRINCIPLES AND DEVICES Diode Resistance: Numerical Q No 5. Determine the DC resistance levels for the diode of Fig. given below at (a) ID =2 mA (b) ID =20 mA (c) VD = -10 V . 25-09-2026 13 EV0000273 · EV0000274 · EV0000275

14. Source slide page 14 (extracted text): ELECTRONIC PRINCIPLES AND DEVICES Diode Equivalent circuits /Diode Approximations • Diode approximation is a mathematical method used to approximate the nonlinear behavior of real diodes to enable calculations and circuit analysis. • There are three different approximations used to analyze the diode circuits, namely I. Ideal Diode Characteristics II. Simplified Diode Characteristics III. Piecewise-linear equivalent Circuit 25-09-2026 14 EV0000276 · EV0000277 · EV0000278
15. Source slide page 15 (extracted text): ELECTRONIC PRINCIPLES AND DEVICES Ideal diode Characteristics • The diode is considered as a closed switch with zero voltage drop when forward-biased and as an open switch when reverse biased (Vd = 0) 25-09-2026 15 EV0000279 · EV0000280 · EV0000281
16. Source slide page 16 (extracted text): ELECTRONIC PRINCIPLES AND DEVICES Simplified Diode Characteristics • Under forward-bias, the diode turns ON when a voltage of VK or VB is applied across it. It turns off if the applied voltage is less than VK. Thus, the diode is represented by a voltage of VK in series with a closed switch. Therefore, Vd = VB • Under reverse bias, it is an open switch. Note VB = Knee voltage 25-09-2026 16 EV0000282 · EV0000283 · EV0000284
17. Source slide page 17 (extracted text): ELECTRONIC PRINCIPLES AND DEVICES Piece-Wise Linear Diode Characteristics • Under forward – bias, the diode behaves as a practical voltage source with the magnitude equal to the knee voltage in series with the the diodes average ac resistance. • Under reverse – bias, the diode is an open switch and no current flows through it. • Characteristics resemble closely to that of practical diode. • The voltage drop across the diode is calculated using the formula Vd = VB + Id *RB Note VB = Knee voltage 25-09-2026 17 EV0000285 · EV0000286 · EV0000287
18. Source slide page 18 (extracted text): THANK YOU Department of Electronics and Communication 25-09-2026 18 EV0000288 · EV0000289 · EV0000290

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:02.520] Good morning and welcome to Electronic Devices, Electronic Principles and Devices. [00:00:07.380] I'm Professor, I'm in the Department of Electronics and Communication Engineering. [00:00:11.400] Today, we're going to learn the ideal and practical diode characteristics along [00:00:15.510] with the diode resistances. [00:00:18.030] As we see from the diagram, we're going to learn about the equivalent [00:00:22.470] representation of the diode, this has been [00:00:28.110] written based on the characteristics curve. [00:00:33.570] There are two types of practical diodes out there. [00:00:36.030] There are two types of diodes representations are there. One is ideal diode. [00:00:40.440] When an ideal diode is forward biased, as shown in the diagram here, it [00:00:46.140] offers no resistance and act like a closed switch. [00:00:51.270] Likewise, the ideal diode under reverse bias [00:00:57.151] offers infinite resistance, hence it is a open switch. [00:01:01.440] Coming to the practical diode [00:01:04.140] as shown, the diode which is said to be forward biased, it starts conducting at a [00:01:08.850] new voltage, which is 0.7 for silicon [00:01:14.010] and reverse bias, no current, and this practical diode is considered [00:01:19.920] as, again, the open switch, similar to that of an ideal diode. [00:01:24.000] So this 0.7 is for silicon. For germanium, it would be 0.3. [00:01:27.900] So in general, we represent it as a [00:01:32.940] VK. [00:01:36.211] So the next point, what we are going to understand is diode resistances. [00:01:41.970] An actual diode offers a

very small resistance, non-zero, when forward biased, and [00:01:46.680] is called the forward resistance. [00:01:48.750] Whereas it offers very high resistance, not infinite, when reverse biased, and is [00:01:54.660] also called as a reverse resistance. [00:01:58.530] Type of applied voltage or a signal will define the following resistance levels. [00:02:03.810] So we can have DC or a static resistances, [00:02:07.980] then AC or a dynamic resistance, and average AC resistance. [00:02:12.300] Coming to the first DC or a static resistance, the application of an DC voltage to [00:02:17.220] a circuit containing a semiconductor diode will result in operating point on a [00:02:21.720] semiconductor diode characteristics curve, which is x-axis is VD, y-axis is [00:02:27.540] ID. So finding the corresponding levels of VD and ID at a particular operating [00:02:33.180] point gives the DC resistance, RD is equal to VD by ID. [00:02:37.891] Typically, the DC resistance of a diode is for forward bias, 10 ohm to 80 ohm. [00:02:43.320] Reverse bias, 10 megaohm. [00:02:46.800] Coming to the AC or a dynamic resistances, the application of a small AC voltage [00:02:53.160] to a circuit containing a semiconductor diode offers AC resistance. [00:02:57.480] A straight line drawn tangent to the curve through Q point will define a [00:03:01.530] particular [00:03:10.590] change in voltage and a current. [00:03:13.860] That change in the voltage and change in the current determine the AC or a dynamic [00:03:18.601] resistance, which is shown that is RD is equal to delta VD by ID for this region of [00:03:23.851] the diode characteristics. [00:03:26.610] Coming to AC or a dynamic resistance, dynamic resistance can be found simply by [00:03:31.470] substituting the quotient value of the diode current into the equation as RD is [00:03:36.031] equal to 26 millivolts by ID, or RD dash is equal to 26 millivolts by ID plus [00:03:41.820] RB. This RB is the bulk current body resistances. [00:03:45.090] Typically, the AC resistance of diode in active region will range from 1 ohm to 100 [00:03:50.010] ohm. [00:03:53.400] Further, the average AC resistance can also be calculated with a [00:03:58.951] diagrammatic from the diode characteristics as shown. [00:04:03.000] If the applied input voltage is sufficiently large to produce a broad swing and a [00:04:07.800] resistance determined by the straight line drawn between the two intersection [00:04:13.770] established by a maximum and a minimum value of the input voltage is called the [00:04:18.240] average AC resistance. That is, RAV is equal to delta VD by delta V at point to [00:04:22.830] point. So these points are very, very important for us to calculate the average AC [00:04:28.410] resistance. So coming to the summary of the resistance, DC or a static, RD is equal [00:04:33.150] to VD by ID, defined as a point on the characteristics. [00:04:38.850] Similarly, AC or a dynamic, RD is equal to delta VD by ID, 26 millivolts by ID, [00:04:44.550] defined by the tangent line at a Q point. [00:04:50.010] Then average AC, [00:04:52.440] defined by the [00:04:54.841] straight line between the limits of the operation. [00:04:58.140] So with this now, we'll try to solve numerical for the given quotient.

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Transcript block 2 (source interval 000300-000599 seconds): [00:05:03.630] So quotient number one, for a characteristics of the figure, determine the AC [00:05:07.440] resistance at ID is equal to 2 milli. [00:05:10.080] Determine the AC resistance ID is equal to 25 milli. [00:05:13.290] Then compare the results of and to the DC resistance at the each level. [00:05:18.300] So keeping this in mind, from the characteristics, we find out the different values [00:05:21.630] here. So delta ID value [00:05:25.710] for 2 milli, it is this. [00:05:28.440] 2 milli is the center point. So the maximum is four, minimum is zero here, so we [00:05:33.300] get [4-0] . Similarly, delta VD, maximum is 0.76 on the characteristics curve, then [00:05:39.210] minimum is 0.65, we get 0.11. Then substitute the value, we get 27.5. [00:05:43.860] Similarly, calculate the value for 25 milli, which is the center is

25. [00:05:48.240] So maximum is 30, minimum is 20. Similarly, at 30, we have 0.8, at 20, we have 0.7. [00:05:54.510] And substitute here, we get 2 milli. [00:05:56.460] So [00:05:58.080] 0.1 and 0.2 is calculated. Coming to the point 3, compare the results of and to the [00:06:02.580] DC resistance. For a DC resistance at 2 milli, we have 0.7, we get 30, [00:06:08.640] 50, RD is equal to 31. So we see that the resistance value for DC [00:06:14.341] value is more compared to the AC here. [00:06:17.730] Similarly, [00:06:20.130] these are at 2 milli and 25 milli. This is because in AC, we use a small value [00:06:25.890] change, whereas in DC, we don't take a small value change, we take a large value [00:06:29.730] change. [00:06:32.341] Similarly, we have a second numerical. [00:06:37.081] Before we go for second numerical, there could be one derivation of expression for [00:06:41.040] AC or a dynamic resistance. We know that RD is equal to 26 millivolts by ID. [00:06:46.800] How to derive this? So we start from the current equation, ID is equal to IS. [00:06:52.230] So differentiate with respect to the applied voltage ID equation, so we get K by [00:06:57.990] KT ID plus IS. Then [00:07:01.561] arrange, whereas we know that this is where IS is reverse saturation, very less [00:07:06.540] current. So K by KT into ID. We know that K is equal to 11,600, TK is [00:07:12.570] 298. After substitution, we get RD is equal to 26 millivolts by ID. [00:07:16.591] This could be asked in the exam for some formulas. [00:07:21.000] Then coming to question number three, calculate the DC and AC resistance of the [00:07:24.690] diode shown in the figure given below at a forward current of 10 milli, and also [00:07:29.430] compare their magnitudes. So at ID 10 milli, we have VD is equal to 0.76, as shown [00:07:35.160] in the diagram here. So RDZ, we calculate VD, we get 76 milli. [00:07:39.660] Similarly, calculate RD, we get 13 ohms. So RDC is more than the small RD. [00:07:45.930] Similarly, the fourth numerical also can be solved with the values given over [00:07:49.140] there. And the fifth numerical, [00:07:53.401] here first they're asking, determine the DC resistance levels. [00:07:56.160] In the previous numerical, we solved for AC, now for the DC at 2 milli, at 20 [00:08:01.140] milli, and -10 milli, these are the reverse point. [00:08:05.880] After substitution, we get the different values over there. [00:08:08.341] In this comparison, for forward bias, the resistance is very less. [00:08:11.370] Reverse bias, the resistance is very high, around 10 megaohms. [00:08:18.811] Thank you.

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Full transcript: 1244 normalized words, 111 timestamped segments, 504.384 seconds duration. Low-confidence segments flagged: 0.

EV0000038 · EV0000039 · EV0000040 · EV0000041

7 Diode equivalent circuits Series diode configurations

This lesson is indexed from the source video titled Diode equivalent circuits Series diode configurations. Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or rewritten. A candidate slide deck is grouped here according to the current matching score; the exact path and page locators are listed in the source evidence. EV0000046 · EV0000047

Graphs and diagrams

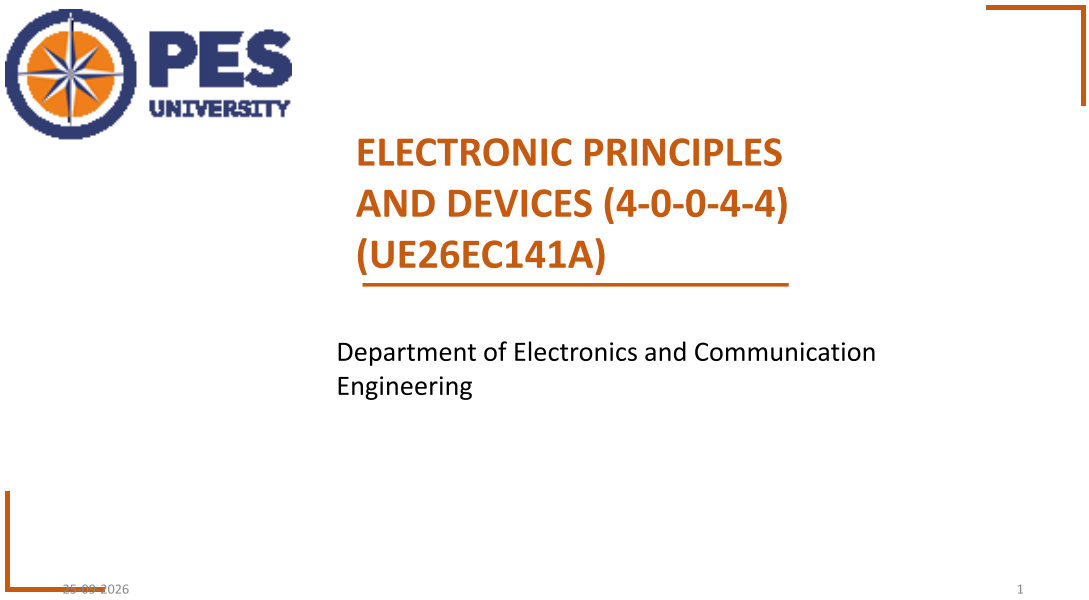

Figure G000064. Diode equivalent circuits Series diode configurations · source slide page 1.

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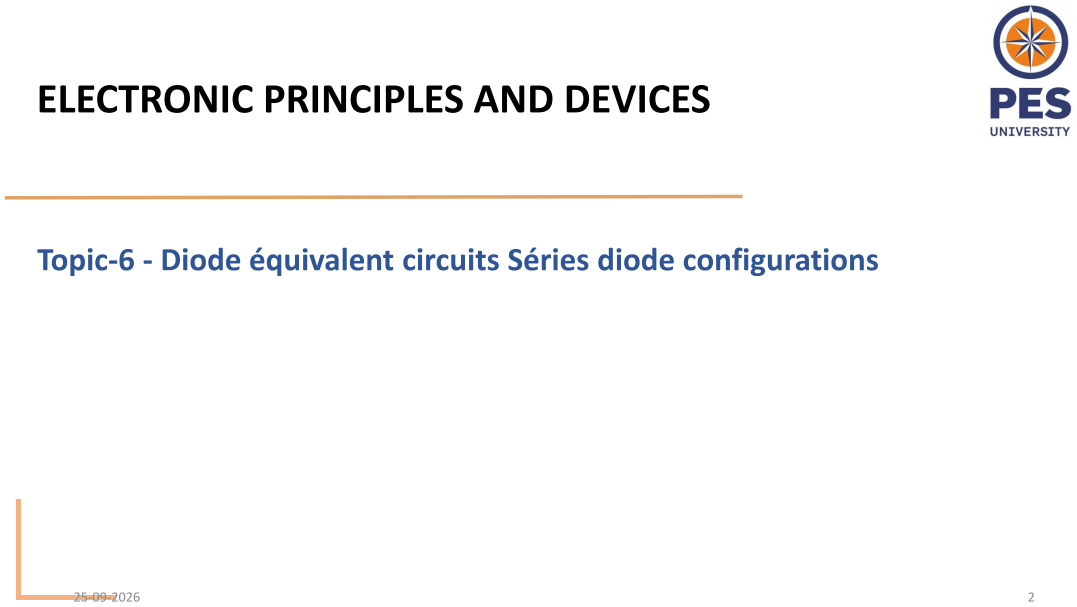

Figure G000065. Diode equivalent circuits Series diode configurations · source slide page 2.

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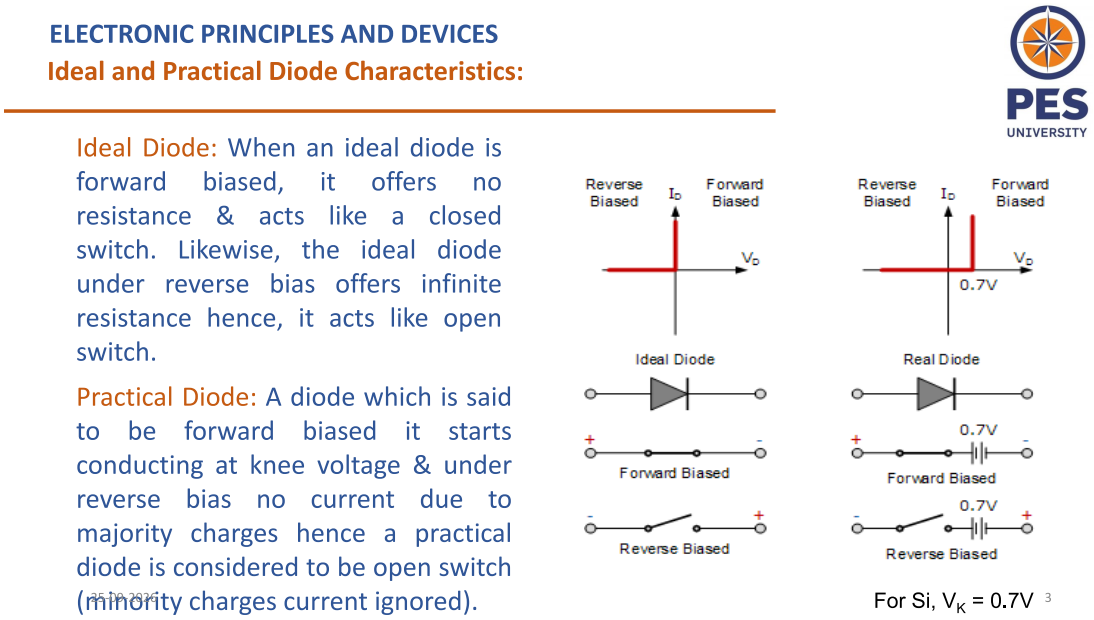

Figure G000066. Diode equivalent circuits Series diode configurations · source slide page 3.

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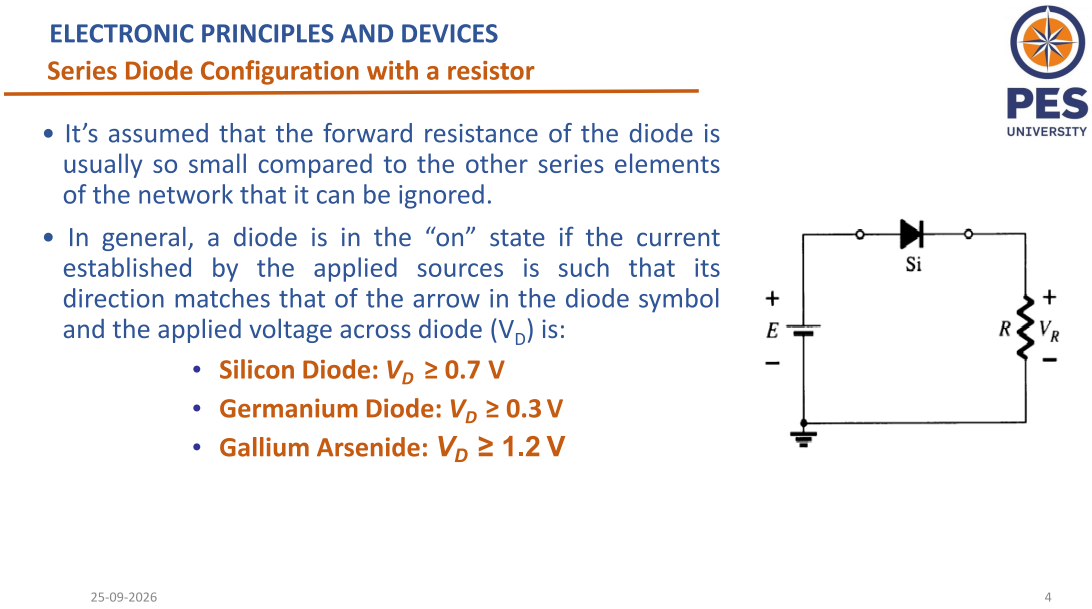

Figure G000067. Diode equivalent circuits Series diode configurations · source slide page 4.

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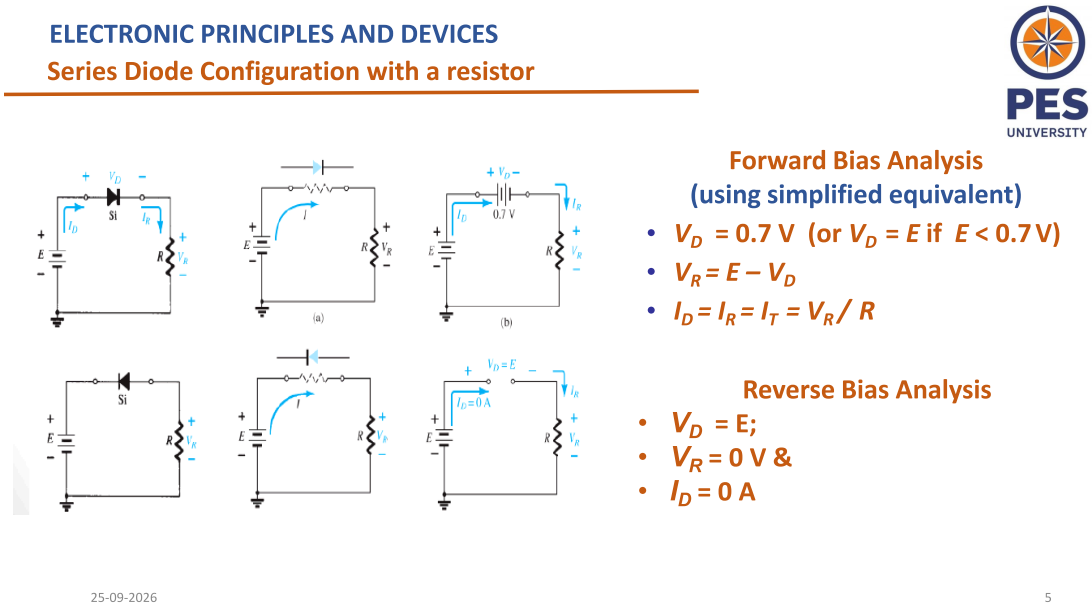

Figure G000068. Diode equivalent circuits Series diode configurations · source slide page 5.

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ELECTRONIC PRINCIPLES AND DEVICES – UE20EC101

Series Diode Configuration – A Solved Problem

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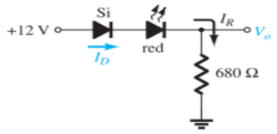
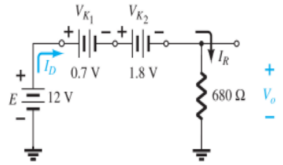
1. Determine Vout and Id for the given series diode Circuit. Assume the LED Voltage as 1.8 V.



Solution: Re-draw the given circuit based on simplified diode's configuration, we get



$$V_o = E - V_{K_1} - V_{K_2} = 12\text{ V} - 2.5\text{ V} = \mathbf{9.5\text{ V}}$$

$$I_D = I_R = \frac{V_R}{R} = \frac{V_o}{R} = \frac{9.5\text{ V}}{680\ \Omega} = \mathbf{13.97\text{ mA}}$$

25-09-2026 6

Figure G000069. Diode equivalent circuits Series diode configurations · source slide page 6.

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ELECTRONIC PRINCIPLES AND DEVICES

Series Diode Configuration

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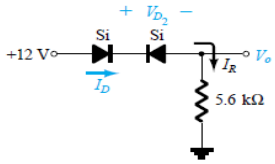
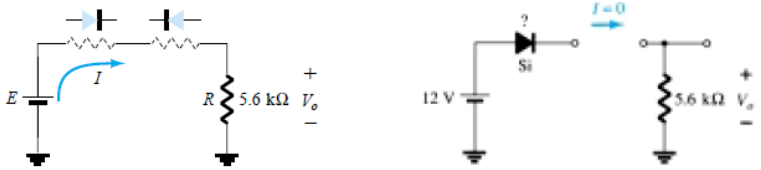
2. Determine I_D, V_{D2} and V_o



Solution: Re-draw the given circuit based on simplified diode's configuration, we get



25-09-2026 7

Figure G000070. Diode equivalent circuits Series diode configurations · source slide page 7.

EV0000310

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ELECTRONIC PRINCIPLES AND DEVICES

Series Diode Configuration

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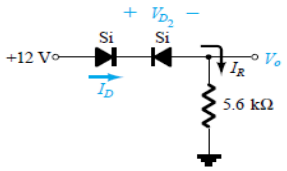
$$V_o = I_R R = I_D R = (0\ \text{A})R = 0\ \text{V}$$



$$E - V_{D_1} - V_{D_2} - V_o = 0$$

$$V_{D_2} = E - V_{D_1} - V_o = 12\text{V} - 0.7 - 0$$

$$= 11.3\ \text{V}$$

$$V_o = \mathbf{0\ V}$$

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Figure G000071. Diode equivalent circuits Series diode configurations · source slide page 8.

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ELECTRONIC PRINCIPLES AND DEVICES

Series Diode Configuration

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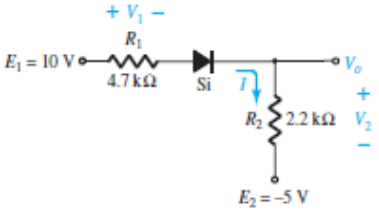
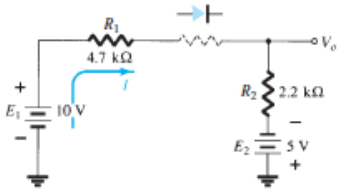
3. Determine I, V_1, V_2 and V_0





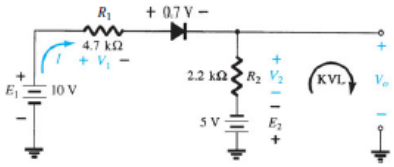


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Figure G000072. Diode equivalent circuits Series diode configurations · source slide page 9.

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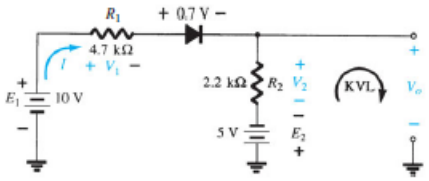
ELECTRONIC PRINCIPLES AND DEVICES

Series Diode Configuration



$$I = \frac{E_1 + E_2 - V_D}{R_1 + R_2} = \frac{10\text{ V} + 5\text{ V} - 0.7\text{ V}}{4.7\text{ k}\Omega + 2.2\text{ k}\Omega} = \frac{14.3\text{ V}}{6.9\text{ k}\Omega}$$
$$\cong \mathbf{2.07\text{ mA}}$$

$$V_1 = IR_1 = (2.07\text{ mA})(4.7\text{ k}\Omega) = \mathbf{9.73\text{ V}}$$
$$V_2 = IR_2 = (2.07\text{ mA})(2.2\text{ k}\Omega) = \mathbf{4.55\text{ V}}$$

$$-E_2 + V_2 - V_o = 0$$
$$V_o = V_2 - E_2 = 4.55\text{ V} - 5\text{ V} = \mathbf{-0.45\text{ V}}$$

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Figure G000073. Diode equivalent circuits Series diode configurations · source slide page 10.

EV0000319

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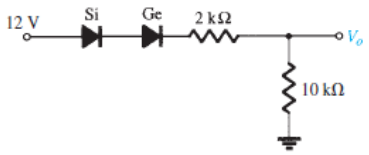
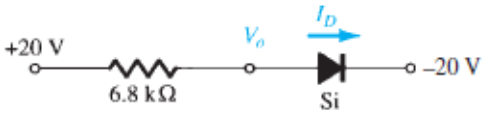
ELECTRONIC PRINCIPLES AND DEVICES

Series Diode Configuration

4. Determine the level of V_0 for each network



(a) $$V_o = \frac{10\text{ k}\Omega(12\text{ V} - 0.7\text{ V} - 0.3\text{ V})}{2\text{ k}\Omega + 10\text{ k}\Omega} = \mathbf{9.17\text{ V}}$$



(b) Diode forward-biased

$$I_D = \frac{20\text{ V} + 20\text{ V} - 0.7\text{ V}}{6.8\text{ k}\Omega} = \mathbf{5.78\text{ mA}}$$

Kirchhoff's voltage law (CW):

$$+V_o - 0.7\text{ V} + 20\text{ V} = 0$$
$$V_o = \mathbf{-19.3\text{ V}}$$

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Figure G000074. Diode equivalent circuits Series diode configurations · source slide page 11.

EV0000322

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ELECTRONIC PRINCIPLES AND DEVICES

Series Diode Configuration

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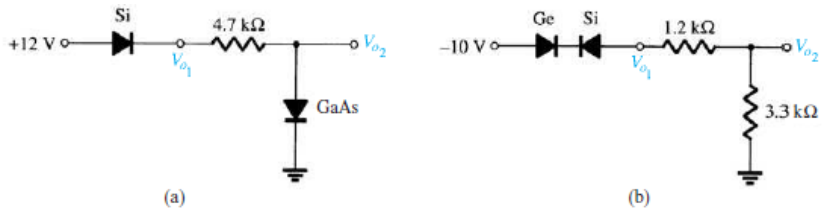
5. Determine V_{01} and V_{02} for the circuits



(a) $V_{o_1} = 12\text{ V} - 0.7\text{ V} = \mathbf{11.3\text{ V}}$

$V_{o_2} = \mathbf{1.2\text{ V}}$

(b) $V_{o_1} = \mathbf{0\text{ V}}$

$V_{o_2} = \mathbf{0\text{ V}}$

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Figure G000075. Diode equivalent circuits Series diode configurations · source slide page 12.

EV0000325

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CELEBRATING 50 YEARS

THANK YOU

Department of Electronics and Communication

25-09-2026 13

Figure G000076. Diode equivalent circuits Series diode configurations · source slide page 13.

EV0000328

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Important points and why

1. Source slide page 1 (extracted text): ELECTRONIC PRINCIPLES AND DEVICES (4-0-0-4-4) (UE26EC141A) Department of Electronics and Communication Engineering 25-09-2026 1 EV0000291 · EV0000292 · EV0000293
2. Source slide page 2 (extracted text): ELECTRONIC PRINCIPLES AND DEVICES Topic-6 - Diode équivalent circuits Séries diode configurations 25-09-2026 2 EV0000294 · EV0000295 · EV0000296
3. Source slide page 3 (extracted text): ELECTRONIC PRINCIPLES AND DEVICES Ideal and Practical Diode Characteristics: Ideal Diode: When an ideal diode is forward biased, it offers no resistance & acts like a closed switch. Likewise, the ideal diode under reverse bias offers infinite resistance hence, it acts like open switch. Practical Diode: A diode which is said to be forward biased it starts conducting at knee voltage & under reverse bias no current due to majority charges hence a practical diode is considered to be open switch (minority25-09-2026 charges current ignored). For Si, VK = 0.7V 3 EV0000297 · EV0000298 · EV0000299
4. Source slide page 4 (extracted text): ELECTRONIC PRINCIPLES AND DEVICES Series Diode Configuration with a resistor • It's assumed that the forward resistance of the diode is usually so small compared to the other series elements of the network that it can be ignored. • In general, a diode is in the "on" state if the current established by the applied sources is such that its direction matches that of the arrow in the diode symbol and the applied voltage across diode (VD) is: • Silicon Diode: VD ≥ 0.7 V • Germanium Diode: VD ≥ 0.3 V • Gallium Arsenide: VD ≥ 1.2 V 25-09-2026 4 EV0000300 · EV0000301 · EV0000302
5. Source slide page 5 (extracted text): ELECTRONIC PRINCIPLES AND DEVICES Series Diode Configuration with a resistor Forward Bias Analysis (using simplified equivalent) • VD = 0.7 V (or VD = E if E < 0.7V) • VR = E – VD • ID = IR = IT = VR / R Reverse Bias Analysis • VD = E; • VR = 0 V & • ID = 0 A 25-09-2026 5 EV0000303 · EV0000304 · EV0000305
6. Source slide page 6 (extracted text): ELECTRONIC PRINCIPLES AND DEVICES – UE20EC101 Series Diode Configuration – A Solved Problem 1. Determine Vout and Id for the given series diode Circuit. Assume the LED Voltage as 1.8 V. Solution: Re-draw the given circuit based on simplified diode's configuration, we get 25-09-2026 6 EV0000306 · EV0000307 · EV0000308
7. Source slide page 7 (extracted text): ELECTRONIC PRINCIPLES AND DEVICES Series Diode Configuration 2. Determine ID, VD2 and Vo Solution: Re-draw the given circuit based on simplified diode's configuration, we get 25-09-2026 7 EV0000309 · EV0000310 · EV0000311
8. Source slide page 8 (extracted text): ELECTRONIC PRINCIPLES AND DEVICES Series Diode Configuration 12V - 0.7 -0 11.3 V 25-09-2026 8 EV0000312 · EV0000313 · EV0000314
9. Source slide page 9 (extracted text): ELECTRONIC PRINCIPLES AND DEVICES Series Diode Configuration 3. Determine I, V1, V2 and V0 25-09-2026 9 EV0000315 · EV0000316 · EV0000317
10. Source slide page 10 (extracted text): ELECTRONIC PRINCIPLES AND DEVICES Series Diode Configuration 25-09-2026 10 EV0000318 · EV0000319 · EV0000320
11. Source slide page 11 (extracted text): ELECTRONIC PRINCIPLES AND DEVICES Series Diode Configuration 4. Determine the level of V0 for each network 25-09-2026 11 EV0000321 · EV0000322 · EV0000323

12. Source slide page 12 (extracted text): ELECTRONIC PRINCIPLES AND DEVICES Series Diode Configuration 5. Determine V01 and V02 for the circuits 25-09-2026 12 EV0000324 · EV0000325 · EV0000326
13. Source slide page 13 (extracted text): THANK YOU Department of Electronics and Communication 25-09-2026 13 EV0000327 · EV0000328 · EV0000329

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:03.450] Good morning, and welcome to Electronic Principle and Devices. [00:00:07.830] Today's topic will be diode equivalent circuit or diode approximation and series [00:00:14.280] diode configuration. [00:00:18.210] So coming to the diode approximation, is a mathematical method used to approximate [00:00:23.190] the nonlinear behavior of the real diode to enable the calculations and circuit [00:00:28.290] analysis. There are three different approximations used to analyze the diode [00:00:33.540] circuits, namely the Piecewise linear equivalent circuits, [00:00:42.300] Simplified diode characteristics, and Ideal diode characteristics. [00:00:46.350] Let us see one by one today. [00:00:50.011] First is [00:00:52.290] Piecewise linear diode characteristics. [00:00:55.500] Under forward bias, with respect to the V-I characteristics, [00:01:00.660] the diode behaves as a practical voltage source with a magnitude equal to the knee [00:01:06.961] voltage in series with the diode average [00:01:11.280] series with the diode average AC resistance. This is shown the representation. [00:01:16.380] Under reverse bias, the diode is an open switch, [00:01:19.860] and no current flows here through it, [00:01:24.420] act like an ideal diode [00:01:26.850] reverse characteristics. So the characteristics resemble closely to that of a [00:01:30.301] practical diode. The voltage drop across the diode is calculated by the formula VD [00:01:34.950] is equal to Vb plus ID into RB. [00:01:39.000] So this is for the Piecewise linear diode characteristics. [00:01:41.940] Coming to the Simplified diode characteristics, under forward bias, the [00:01:47.760] diode turns on when a voltage VK or VB is applied across it. [00:01:51.900] It turns off if the voltage applied is less than the VK. [00:01:55.260] Thus, diode is represented by voltage of VK in a series with a closed path switch, [00:02:01.530] as shown. Therefore, VD is equal to VB. Okay. [00:02:04.590] Then coming to the reverse bias, it is an open switch similar to that of a [00:02:08.160] practical reverse bias voltage. [00:02:11.310] Coming to the Ideal diode characteristics here, we already know the diode is [00:02:14.940] considered as a closed switch with a zero voltage drop. [00:02:19.020] Then when in forward bias VD and open switch when it's in reverse bias. [00:02:23.910] So it's a closed switch in the forward bias, open switch in the reverse bias. [00:02:28.171] So with this note, we'll go for solving a numerical when a diode configure. [00:02:31.740] When the diode is connected in series, we call it as a diode configuration, a [00:02:36.570] series diode configuration with a resistor. [00:02:39.870] It is assumed that the forward characteristics or forward resistance of the diode [00:02:44.040] is usually so small compared to the other series elements of the network that can [00:02:49.171] be ignored. In general, the [00:02:54.360] diode is in on state if the current established by the applied sources is such that [00:02:59.490] its direction matches that of an arrow in the diode symbol and applied voltage [00:03:04.921] across the diode is silicon VD greater than 0.7, germanium VD greater than [00:03:10.950] 0.3, and gallium arsenide VD should be greater than 1.2 volts. [00:03:15.629] So now consider applied voltage is E across the silicon and the resistance VR. [00:03:21.990] So with this arrangement now, when it is forward bias, the analysis is when you [00:03:26.791] apply a voltage forward bias here, the current ID enter through the diode and diode [00:03:30.810] is on, we represent it as a resistance or a silicon if the 0.7 we need to [00:03:36.630] represent, and hence VD

is equal to 0.7 across that. [00:03:42.360] And when it's reverse bias, diode is reverse connected to the positive voltage here [00:03:47.101] and represented as an open circuit, and hence ID is current is [00:03:52.351] zero here, and VR also is equal to zero volts. [00:03:55.830] This is with respect to series diode configuration with a resistor. [00:03:59.640] Let us solve one numerical based on the series diode configuration. [00:04:03.870] The question is, determine the Vout and ID for a given series diode circuit. [00:04:09.330] Assume the LED voltage as 0.7. So the given circuit here, 12 [00:04:15.210] volts is applied voltage. We have a silicon and a LED here, and we have an output [00:04:20.580] resistance 680 ohm. So we need to redraw the circuit based on the simple [00:04:26.160] diode configuration. So 12 volts represented by [00:04:32.701] a positive and a negative here, followed by a diode represented as 0.7 volts, then [00:04:38.370] LED given as 0.8 volts, and the resistance. Then we need to apply the KVL here. [00:04:43.680] We need to remember the KVL equation, which we learned from 12th standard. [00:04:49.110] The sum of all the voltages across the closed loop network is always equal to zero. [00:04:53.460] So applying that. So Vout is equal to here the applied voltage, sum of all the [00:04:57.900] voltages. So entry here, the entry is negative, leaving is positive.

EV0000046 · EV0000047 · EV0000048 · EV0000049

Transcript block 2 (source interval 000300-000599 seconds): [00:05:02.550] So Vout is a leaving is negative here. So minus Vout here. [00:05:07.560] So we take on the other side. V is equal to E minus VK1 minus VK2. [00:05:12.690] We get 9.5 volts output. Then how to calculate ID? [00:05:16.890] Since we know the voltage across this, so VR by R. [00:05:21.360] That is Vout by R. Vout 9.5 divided by 680 ohm, which is 13.97 [00:05:26.610] milliamps. So any numerical given in this fashion, if the diode is connected in [00:05:30.841] series, we can calculate by applying a KVL first, calculate the output voltage, and [00:05:36.210] take that output voltage divided by the total resistance, we get a current flowing [00:05:40.201] through it. So we see that the current has been calculated here, and output voltage [00:05:46.110] also is calculated when the diode is connected in a series configuration. [00:05:51.540] Thank you.

EV0000050 · EV0000051 · EV0000052 · EV0000053

Full transcript: 878 normalized words, 75 timestamped segments, 357.483 seconds duration. Low-confidence segments flagged: 0.

EV0000046 · EV0000047 · EV0000048 · EV0000049

8 Parallel and Series- Parallel configurations

This lesson is indexed from the source video titled Parallel and Series- Parallel configurations. Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or rewritten. A candidate slide deck is grouped here according to the current matching score; the exact path and page locators are listed in the source evidence. EV0000054 · EV0000055

Graphs and diagrams

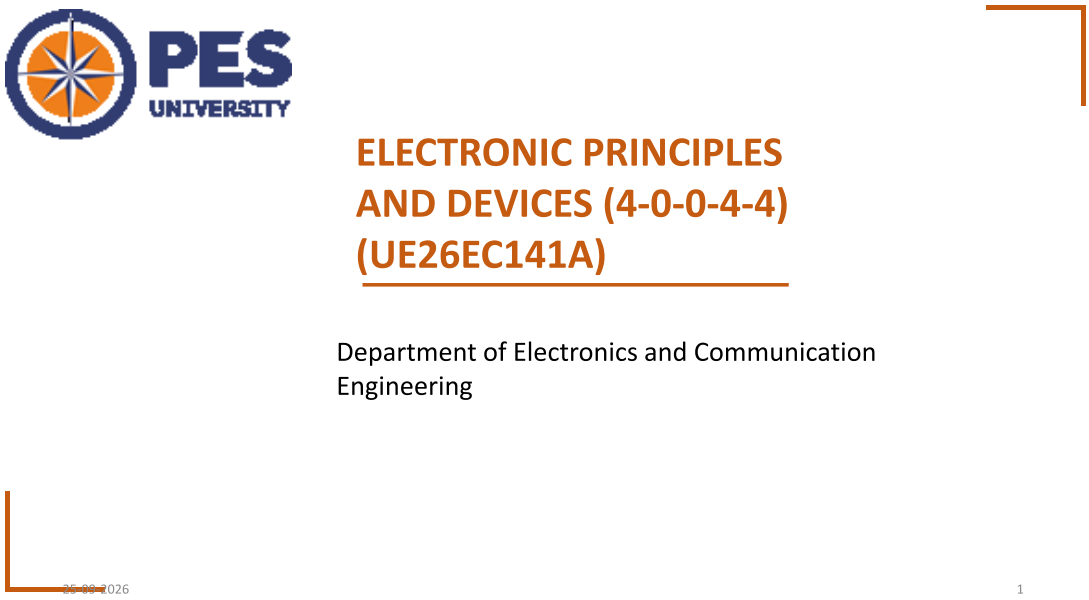

Figure G000077. Parallel and Series Parallel configurations · source slide page 1.

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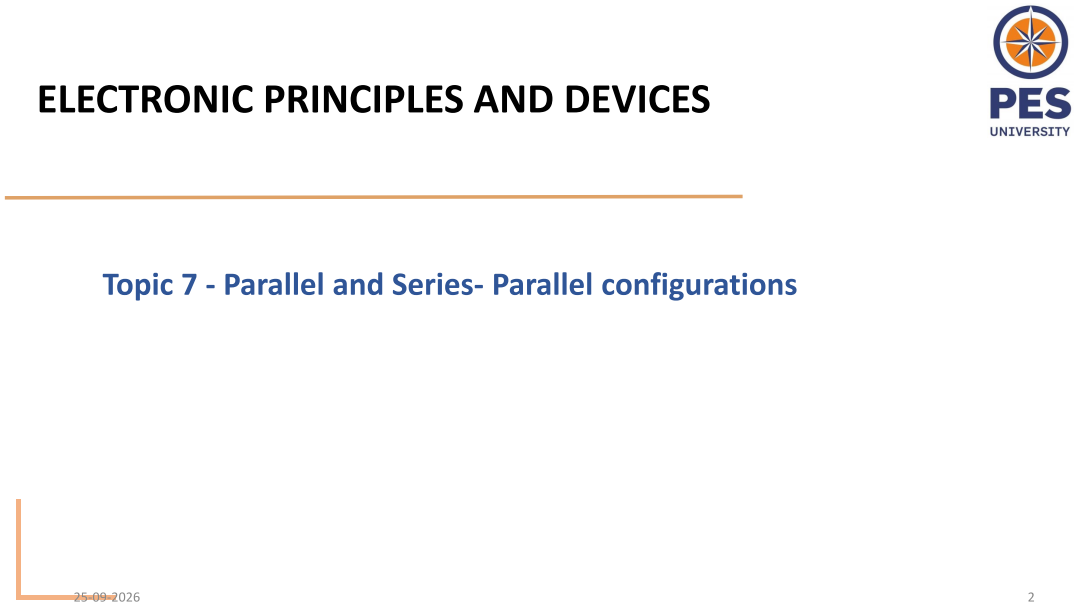

Figure G000078. Parallel and Series Parallel configurations · source slide page 2.

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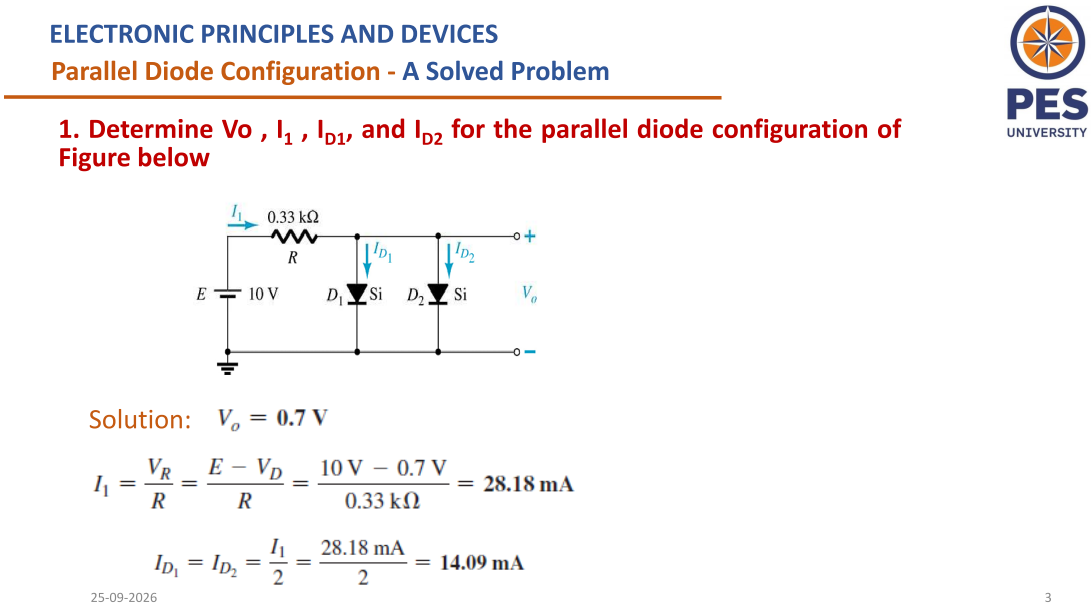

Figure G000079. Parallel and Series Parallel configurations · source slide page 3.

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ELECTRONIC PRINCIPLES AND DEVICES

Series-Parallel Diode Configuration

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2. Determine I_1, I_2, and I_{D2} for the series- parallel diode configuration of Figure below

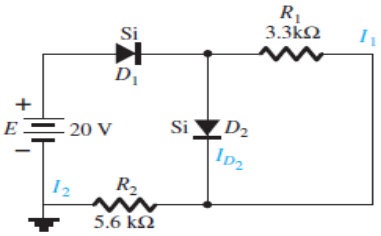

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Figure G000080. Parallel and Series Parallel configurations · source slide page 4.

EV0000340

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ELECTRONIC PRINCIPLES AND DEVICES

Series-Parallel Diode Configuration – An Exercise Problem

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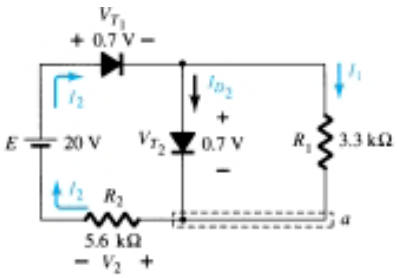

$$-V_2 + E - V_{T_1} - V_{T_2} = 0$$

$$V_2 = E - V_{T_1} - V_{T_2} = 20\text{ V} - 0.7\text{ V} - 0.7\text{ V} = 18.6\text{ V}$$

$$I_2 = \frac{V_2}{R_2} = \frac{18.6\text{ V}}{5.6\text{ k}\Omega} = \mathbf{3.32\text{ mA}}$$

$$I_1 = \frac{V_{T_2}}{R_1} = \frac{0.7\text{ V}}{3.3\text{ k}\Omega} = \mathbf{0.212\text{ mA}}$$

$$I_{D_2} + I_1 = I_2$$

$$I_{D_2} = I_2 - I_1 = 3.32\text{ mA} - 0.212\text{ mA} = \mathbf{3.108\text{ mA}}$$

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Figure G000081. Parallel and Series Parallel configurations · source slide page 5.

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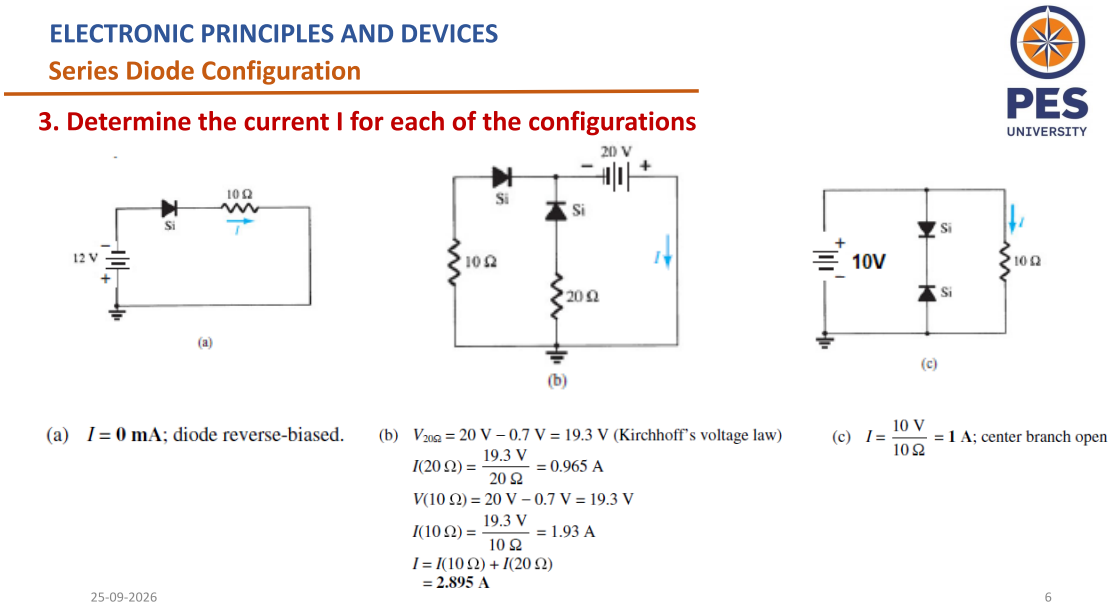

Figure G000082. Parallel and Series Parallel configurations · source slide page 6.

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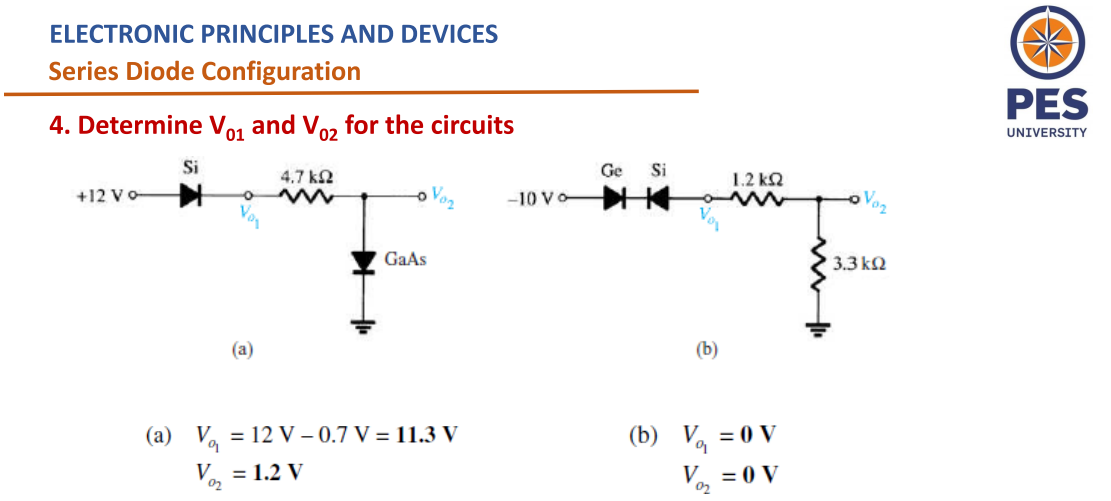

Figure G000083. Parallel and Series Parallel configurations · source slide page 7.

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ELECTRONIC PRINCIPLES AND DEVICES

Parallel Diode Configuration - A Solved Problem

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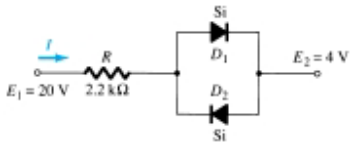
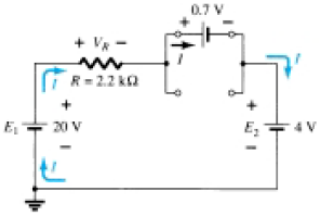
5. Determine I for the parallel diode configuration of Figure below



Solution:



$$I = \frac{E_1 - E_2 - V_D}{R} = \frac{20\text{ V} - 4\text{ V} - 0.7\text{ V}}{2.2\text{ k}\Omega} \cong \mathbf{6.95\text{ mA}}$$

25-09-2026 8

Figure G000084. Parallel and Series Parallel configurations · source slide page 8.

EV0000352

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ELECTRONIC PRINCIPLES AND DEVICES

Parallel Diode Configuration - A Solved Problem

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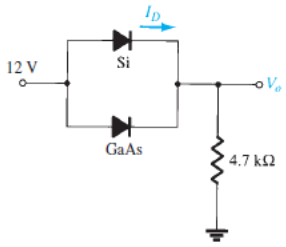
6. Determine I_D and V_0 for the circuits shown

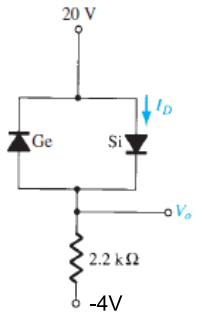


(a) Both diodes forward-biased
Si diode turns on first and locks in 0.7 V drop.

$$I_R = \frac{12\text{ V} - 0.7\text{ V}}{4.7\text{ k}\Omega} = 2.4\text{ mA}$$

$I_D = I_R = \mathbf{2.4\text{ mA}}$

$V_o = 12\text{ V} - 0.7\text{ V} = \mathbf{11.3\text{ V}}$



(b) Right diode forward-biased:

$$I_D = \frac{20\text{ V} + 4\text{ V} - 0.7\text{ V}}{2.2\text{ k}\Omega} = \mathbf{10.59\text{ mA}}$$

$V_o = 20\text{ V} - 0.7\text{ V} = \mathbf{19.3\text{ V}}$

25-09-2026 9

Figure G000085. Parallel and Series Parallel configurations · source slide page 9.

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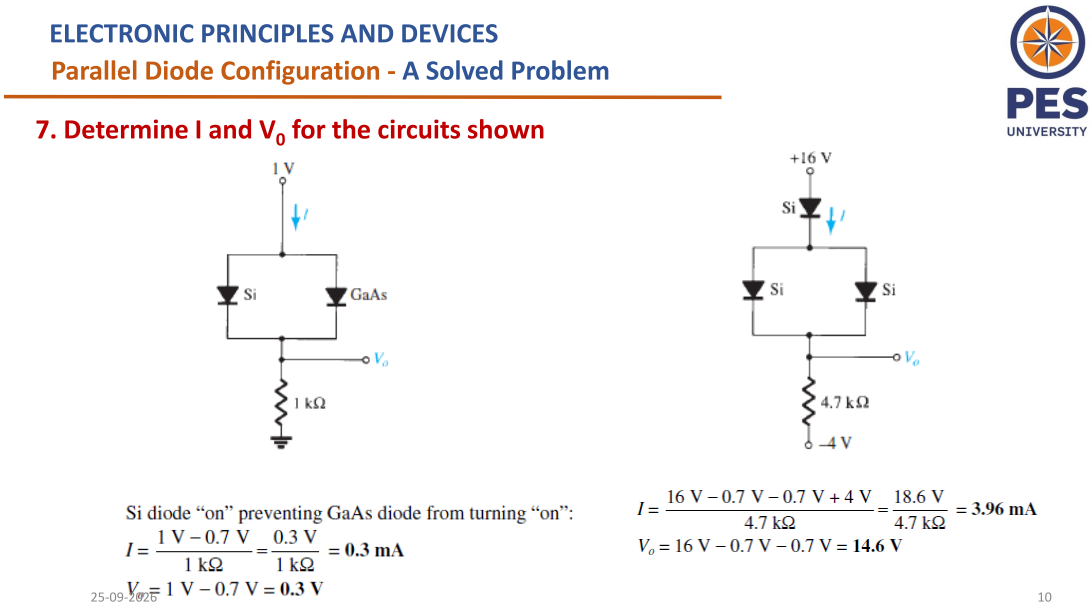

Figure G000086. Parallel and Series Parallel configurations · source slide page 10.

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> ***How to read it.*** Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

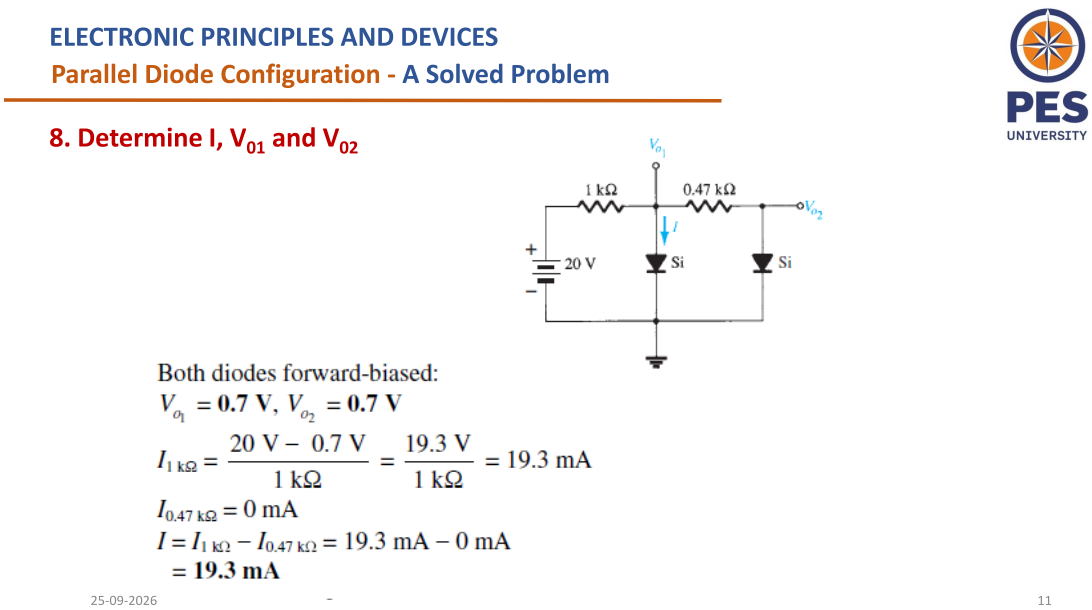

Figure G000087. Parallel and Series Parallel configurations · source slide page 11.

EV0000361

> ***How to read it.*** Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Figure G000088. Parallel and Series Parallel configurations · source slide page 12.

EV0000364

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Important points and why

1. Source slide page 1 (extracted text): ELECTRONIC PRINCIPLES AND DEVICES (4-0-0-4-4) (UE26EC141A) Department of Electronics and Communication Engineering 25-09-2026 1 EV0000330 · EV0000331 · EV0000332
2. Source slide page 2 (extracted text): ELECTRONIC PRINCIPLES AND DEVICES Topic 7 - Parallel and Series- Parallel configurations 25-09-2026 2 EV0000333 · EV0000334 · EV0000335
3. Source slide page 3 (extracted text): ELECTRONIC PRINCIPLES AND DEVICES Parallel Diode Configuration - A Solved Problem 1. Determine Vo , I1 , ID1, and ID2 for the parallel diode configuration of Figure below Solution: 25-09-2026 3 EV0000336 · EV0000337 · EV0000338
4. Source slide page 4 (extracted text): ELECTRONIC PRINCIPLES AND DEVICES Series-Parallel Diode Configuration 2. Determine I1, I2, and ID2 for the series- parallel diode configuration of Figure below 25-09-2026 4 EV0000339 · EV0000340 · EV0000341
5. Source slide page 5 (extracted text): ELECTRONIC PRINCIPLES AND DEVICES Series-Parallel Diode Configuration – An Exercise Problem 25-09-2026 5 EV0000342 · EV0000343 · EV0000344
6. Source slide page 6 (extracted text): ELECTRONIC PRINCIPLES AND DEVICES Series Diode Configuration 3. Determine the current I for each of the configurations 25-09-2026 6 EV0000345 · EV0000346 · EV0000347
7. Source slide page 7 (extracted text): ELECTRONIC PRINCIPLES AND DEVICES Series Diode Configuration 4. Determine V01 and V02 for the circuits 25-09-2026 7 EV0000348 · EV0000349 · EV0000350

8. Source slide page 8 (extracted text): ELECTRONIC PRINCIPLES AND DEVICES Parallel Diode Configuration - A Solved Problem 5. Determine I for the parallel diode configuration of Figure below Solution: 25-09-2026 8 EV0000351 · EV0000352 · EV0000353
9. Source slide page 9 (extracted text): ELECTRONIC PRINCIPLES AND DEVICES Parallel Diode Configuration - A Solved Problem 6. Determine ID and V0 for the circuits shown -4V 25-09-2026 9 EV0000354 · EV0000355 · EV0000356
10. Source slide page 10 (extracted text): ELECTRONIC PRINCIPLES AND DEVICES Parallel Diode Configuration - A Solved Problem 7. Determine I and V0 for the circuits shown 25-09-2026 10 EV0000357 · EV0000358 · EV0000359
11. Source slide page 11 (extracted text): ELECTRONIC PRINCIPLES AND DEVICES Parallel Diode Configuration - A Solved Problem 8. Determine I, V01 and V02 25-09-2026 11 EV0000360 · EV0000361 · EV0000362
12. Source slide page 12 (extracted text): THANK YOU Department of Electronics and Communication 25-09-2026 12 EV0000363 · EV0000364 · EV0000365

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:02.460] Good morning and welcome to Electronic Principles and Devices. [00:00:07.590] Today, we'll see parallel diode configuration. [00:00:09.690] In the last class, we have solved the numerical and serial diode configuration. [00:00:15.360] Today, completely, we look at the concept of parallel diode configuration with the [00:00:20.701] numerical examples. The first question here is: Determine the Vout, [00:00:26.430] ID, ID1, and ID2 for the parallel diode configuration of the figure. [00:00:33.000] So we need to calculate Vout here, I1. [00:00:37.231] I1 is this, ID2 is this, ID2 is this for the parallel diode. [00:00:43.050] Since we have two diodes here connected in parallel, we call it the parallel diode [00:00:47.100] configuration. Coming to the solution now. [00:00:50.520] Given, [00:00:52.051] if you measure the voltage from this point to this point, this is 0.7. [00:00:56.370] So Vout can be definitely calculated as 0.7. [00:00:59.731] Then, [00:01:01.500] how to calculate I1? So I1 is equal to the voltage applied here divided by R. [00:01:06.301] The voltage applied between these two points, it is E minus [00:01:13.080] VD. The applied voltage here and this voltage is 0.7. [00:01:19.230] So this is 0.7. We get 10 minus 0.7, which is 9.3 divided by 0.33 [00:01:25.200] Kelvin, which is 28.18. So we got the I1 now. [00:01:29.461] So now we need to calculate ID1 and ID2. [00:01:33.060] Coming to the ID1 and ID2, the potentials are same, and hence ID1 is equal to-- [00:01:38.220] Current also will be same, so I1 by I2, I1 by 2. [00:01:42.990] So it is the current entry here and leaving here. [00:01:46.560] So both current will be same now, because of the potential is same. [00:01:49.830] So what is the I1 current? 28.18 divided by 2, we get 14.09 milliamps. [00:01:56.490] So this type of numerical may be asked for four marks in the exam. [00:02:00.870] Coming to the question number three or the second [00:02:03.810] numerical example for [00:02:09.690] series parallel diode. In case if the numerical has one series and one parallel, [00:02:14.190] the previous one was both parallel. So both parallel. [00:02:17.610] Now one series and another parallel is there. How to go about? [00:02:20.851] So the given question, we need to calculate I1, I2. [00:02:24.270] So I1 is here, I2 is here, and ID2, we need to calculate [00:02:30.150] for a series parallel diode configuration of the figure. [00:02:33.000] So we will rewrite the diagram now as per our requirement and try to apply the [00:02:39.510] voltage divider rule or KVL

voltage. Then we apply the KVL here. [00:02:43.950] So the given voltage is 20 volts here. The diode is represented by 0.7 VD1, VD2. [00:02:50.340] So apply [00:02:55.830] the KVL within this loop for I2. So minus V2. [00:03:04.080] We are also able to apply the KVL this loop also. [00:03:08.851] So first go with this. So minus V2 voltage is across [00:03:15.541] this, across the resistance here. So leaving end is minus here, minus V2. [00:03:20.640] Then E applied voltage, leaving end is positive here. [00:03:23.730] Then VD1, minus VD1 here, leaving end is minus V2, VD1, and leaving [00:03:29.701] end is plus here, [00:03:32.820] plus minus or the minus here. So VD2. Then calculate the VD2 here. [00:03:38.250] After substituting, we get 18.6 volts. Then calculate I2 across this. [00:03:42.570] Since we know the voltage V2, calculate the value of the I2 with the resistance R2, [00:03:46.860] we get 3.32. Then how to calculate this I1? So I1 is here. [00:03:51.930] The applied voltage here is 0.7. The 0.7 by 3.2 kiloohm will give you 0.212 [00:03:57.570] milliamps. Then the total current given by ID2 plus I1 is equal to I2. [00:04:02.940] So calculate the ID2 from this, we get 3.108 milliamps. [00:04:07.650] One thing we need to notice, each numerical may not be same. [00:04:10.650] So based on the given question and the calculation, we need to apply the KVL and [00:04:16.110] Ohm's law, then we'll be able to get the answer. [00:04:20.940] Coming to one more numerical, where diodes are connected in process. [00:04:28.680] Till now what we have seen example for series parallel diode, series parallel. [00:04:33.360] So now let us solve few more numerical and series diode configuration. [00:04:38.701] We have a red LED connected 1.8. I think this numerical we solved. [00:04:43.080] Just to refresh to people, the diode can be connected in series. [00:04:46.110] We already solved this in topic four or five. [00:04:51.150] So now, with the same question, if the diode is connected in reverse bias, how do [00:04:55.500] you calculate the different values like ID, VD2, and Vout?

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Transcript block 2 (source interval 000300-000599 seconds): [00:05:00.600] So now in the previous numerical, the diode is connected in series with this, which [00:05:04.860] was in the same direction. Now they're in the different direction. [00:05:07.770] So when we apply the forward bias, what will happen? [00:05:10.320] When applied voltage is positive here, so the diode is reverse bias now. [00:05:15.060] So there's no connection between the input and the output stage. [00:05:17.880] So V output is equal to 0. Obviously, so we get V output is equal to 0 over there. [00:05:23.281] Then how to calculate the voltage across VD2? [00:05:26.250] In the given question, they are asking us to calculate the VD2. [00:05:29.610] So VD2 is across this. So coming to the VD2, how to calculate? [00:05:34.110] VD2 is 12 volts now. How it is 12 volts? Apply the KVL to the loop. [00:05:40.290] We get VD2 is equal to E minus VD1 minus VD2, which is the 12 volts. [00:05:45.360] Where Vout is equal to 0, yes, we'll be able to get, and VD1, we have 0.7 here. [00:05:51.810] For ideal diode, it is 0, but 0.7, we get here 11.3 volts. [00:05:55.830] Sorry, it is 11.3 volts we need to get. And Vout is equal to 0. [00:06:02.670] Similarly, for a series diode configuration, we can also solve few more numericals [00:06:07.020] as shown in the figure, which will be solved in the theory class. [00:06:12.330] So let us solve one more set of question here. [00:06:16.350] I have here, [00:06:18.990] which would be asked for one mark. See, it's not that the question could be asked [00:06:23.580] for only four marks. A question could be asked for one mark also. [00:06:26.820] Determine the current for each of the configuration. [00:06:29.160] For this configuration, calculate the I. [00:06:32.070] So what will be the value of the current here? [00:06:34.470] I is equal to, apply the [00:06:40.050] KVL, or first look at the negative voltage is connected here. [00:06:43.890] When you apply negative, diode is reverse bias. [00:06:45.990] When

it's reverse bias, the current I is equal to 0. [00:06:49.081] Similarly, determine the current for the given circuit here. [00:06:52.920] So we follow the same procedure how we solved in the previous. [00:06:56.970] So [00:06:58.290] by applying the KVL, calculate the current through each of those. [00:07:01.530] Similarly, I have one more numerical here. [00:07:03.390] C, calculate the [00:07:06.451] current in the given configuration. [00:07:08.610] Since the diodes are connected in reverse here, [00:07:12.270] the center branch will open here. So this diode is off and it open. [00:07:17.850] The current will pass only through the resistor. [00:07:20.341] So it is 10 by 10, which is 1 amps. [00:07:24.570] Similarly, we have various numericals which has been given here. [00:07:29.130] Try to follow the slides and solve the numerical as per the given [00:07:34.860] question, and we also will solve in the class. [00:07:38.820] Then coming to the few more numerical on parallel diode configuration. [00:07:42.720] If the diodes are connected, one in [00:07:46.770] same configuration, other in different configuration, one diode is ON, other diode [00:07:50.401] is OFF. So we follow like here, only one diode is conducting now. [00:07:54.390] So this is similar to that of a series diode configuration, even though diodes are [00:07:57.480] connected in parallel. So we apply a KVL here and calculate the current flowing and [00:08:02.130] the voltage over there. Similarly, if the two diodes are connected in parallel, [00:08:05.610] also we have seen. So few more numericals are also being given, and try to solve it [00:08:10.770] and [00:08:12.720] come back to us. Thank you.

EV0000058 · EV0000059 · EV0000060 · EV0000061

Full transcript: 1280 normalized words, 114 timestamped segments, 500.971 seconds duration. Low-confidence segments flagged: 0.

EV0000054 · EV0000055 · EV0000056 · EV0000057

9 Logical Operations using diode

This lesson is indexed from the source video titled Logical Operations using diode. Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or rewritten. A candidate slide deck is grouped here according to the current matching score; the exact path and page locators are listed in the source evidence. EV0000062 · EV0000063

Graphs and diagrams

Figure G000089. Logical Operations using diode · source slide page 1.

EV0000367

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

ELECTRONIC PRINCIPLES AND DEVICES

Unit-1 Introduction to Electronics and Semiconductor Diodes

Logical Operations

Dr. Ananda M
Department of Electronics and Communication.

Figure G000090. Logical Operations using diode · source slide page 2.

EV0000370

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ELECTRONIC PRINCIPLES AND DEVICES
Logical Operations

Logical operations:

It is an Operation that acts on binary numbers to produce a result according to the Laws of Boolean Logic.

A Logic Gate is an basic building block of Digital circuits.

Logic gate is considered as a device that has the ability to produce one output level with the combinations of input levels.

Example : AND,OR and NOT functions

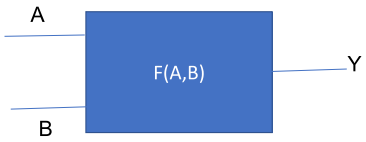

Figure G000091. Logical Operations using diode · source slide page 3.

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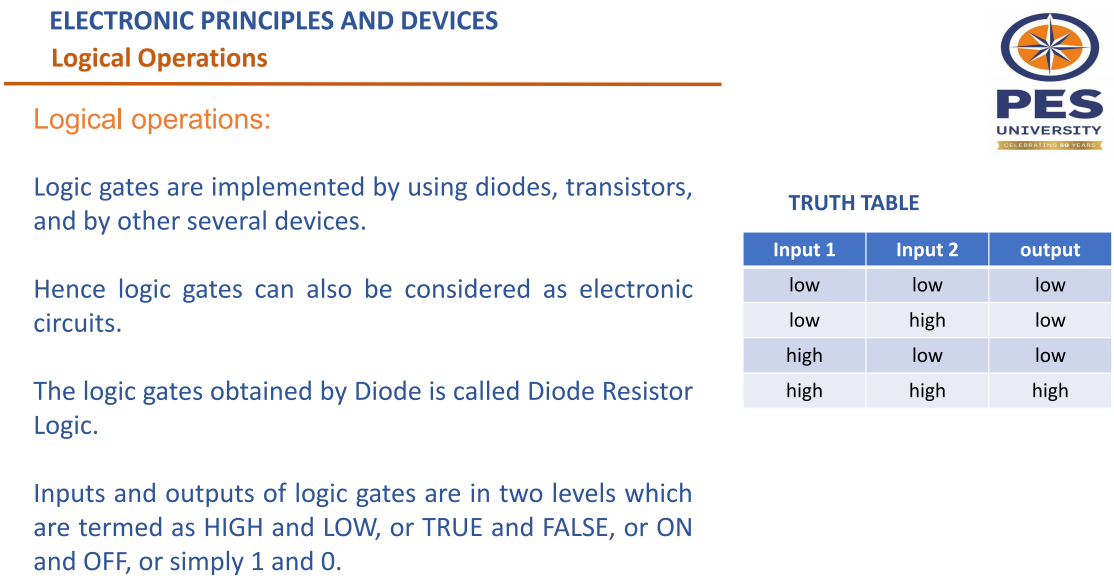

Figure G000092. Logical Operations using diode · source slide page 4.

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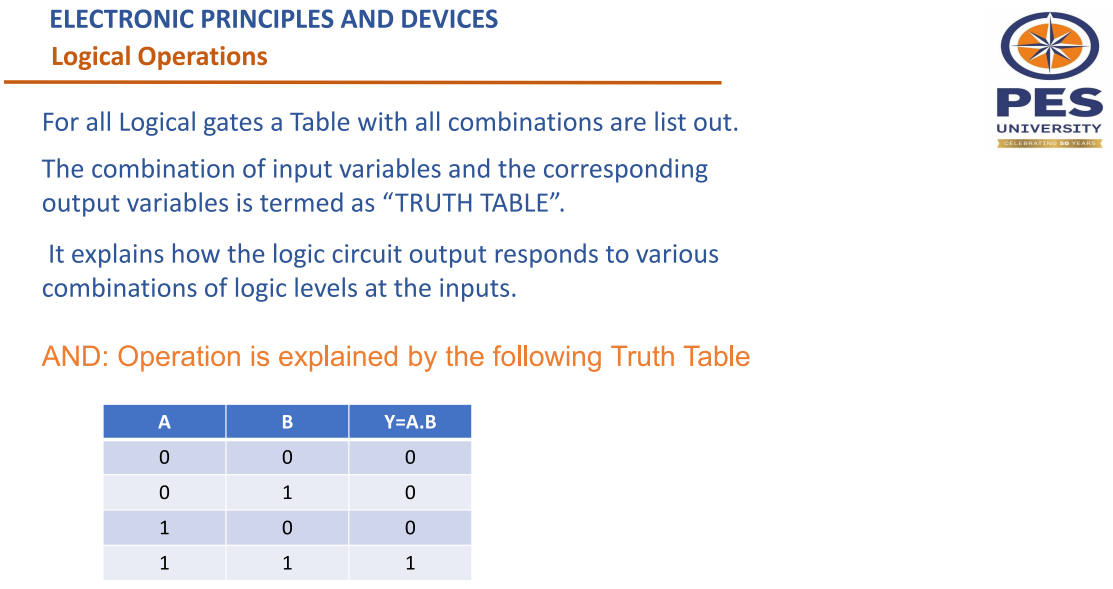

Figure G000093. Logical Operations using diode · source slide page 5.

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ELECTRONIC PRINCIPLES AND DEVICES

Logical Operations

An OR gate may also have two or more inputs but produce only one output.

The OR gate produces an output of logic 1 state even if any of its inputs is in logic 1 state and also produces an output of logic 0 state if any of its inputs is in logic 0 state.

OR Operation is explained by the following Truth Table

A	B	Y=A+B
0	0	0
0	1	1
1	0	1
1	1	1

Figure G000094. Logical Operations using diode · source slide page 6.

EV0000382

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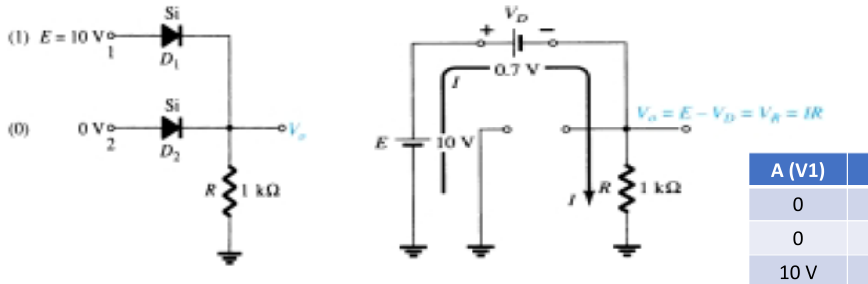
ELECTRONIC PRINCIPLES AND DEVICES

Logical Operations

(1)OR Operation Illustrated using Diode



A (V1)	B(V2)	Y (Vo)
0	0	0
0	10 V	9.3 V
10 V	0	9.3 V
10 V	10 V	9.3V

Figure G000095. Logical Operations using diode · source slide page 7.

EV0000385

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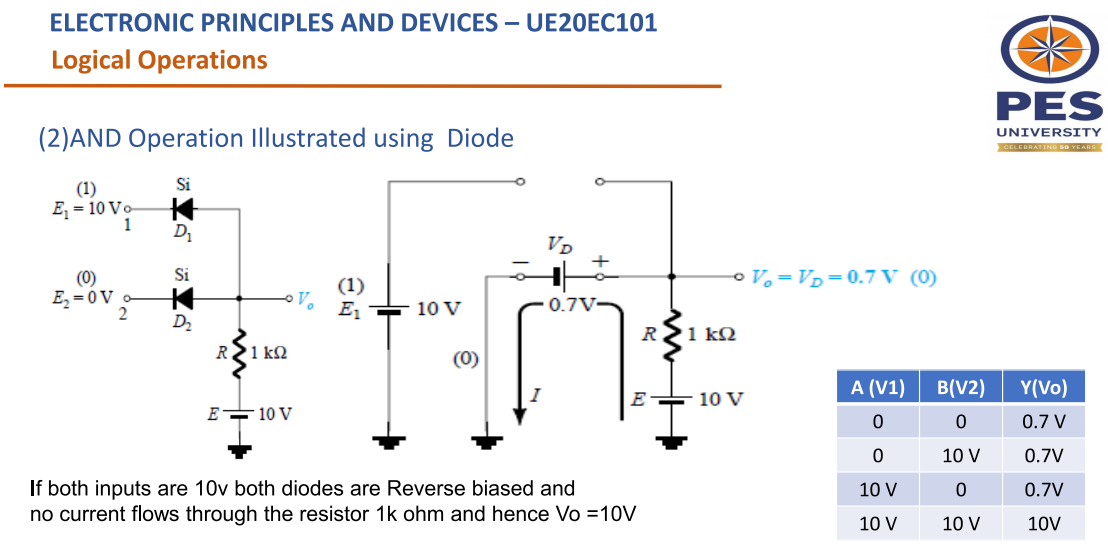

A (V1)	B(V2)	Y(Vo)
0	0	0.7 V
0	10 V	0.7V
10 V	0	0.7V
10 V	10 V	10V

Figure G000096. Logical Operations using diode · source slide page 8.

EV0000388

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

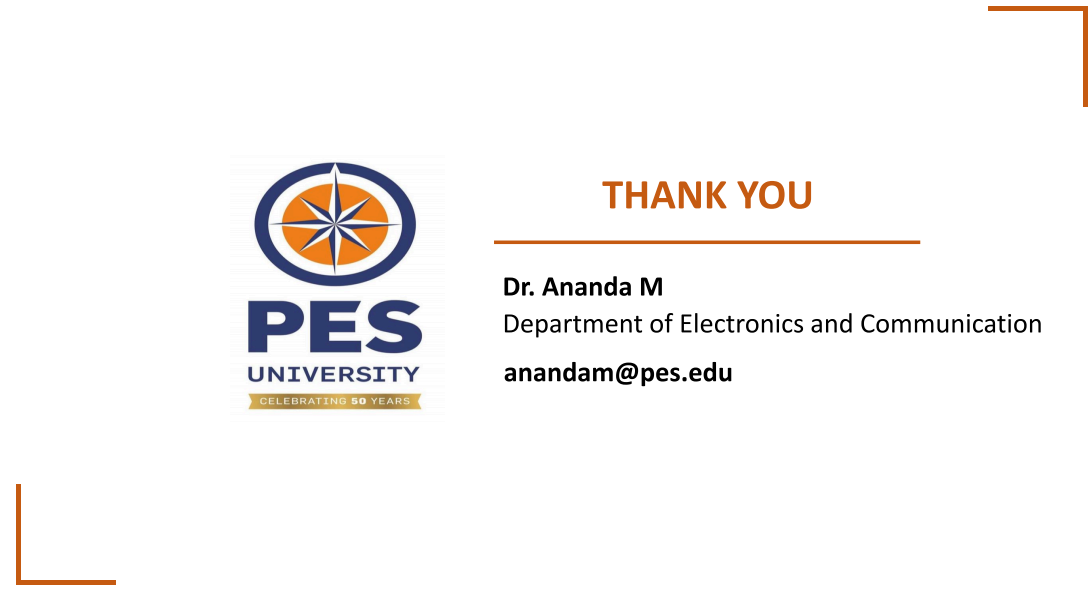

Figure G000097. Logical Operations using diode · source slide page 9.

EV0000391

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Important points and why

1. Source slide page 1 (extracted text): ELECTRONIC PRINCIPLES AND DEVICES Dr. Ananda M Department of Electronics and Communication. EV0000366 · EV0000367 · EV0000368
2. Source slide page 2 (extracted text): ELECTRONIC PRINCIPLES AND DEVICES Unit-1 Introduction to Electronics and Semiconductor Diodes Logical Operations Dr. Ananda M Department of Electronics and Communication. EV0000369 · EV0000370 · EV0000371
3. Source slide page 3 (extracted text): ELECTRONIC PRINCIPLES AND DEVICES Logical Operations Logical operations: It is an Operation that acts on binary numbers to produce a result according to the Laws of Boolean Logic. A A Logic Gate is an basic building block of Digital circuits. Y F(A,B) Logic gate is considered as a device that has the ability B to produce one output level with the combinations of input levels. Example : AND,OR and NOT functions EV0000372 · EV0000373 · EV0000374
4. Source slide page 4 (extracted text): ELECTRONIC PRINCIPLES AND DEVICES Logical Operations Logical operations: Logic gates are implemented by using diodes, transistors, TRUTH TABLE and by other several devices. Input 1 Input 2 output Hence logic gates can also be considered as electronic low low low circuits. low high low high low low The logic gates obtained by Diode is called Diode Resistor high high high Logic. Inputs and outputs of logic gates are in two levels which are termed as HIGH and LOW, or TRUE and FALSE, or ON and OFF, or simply 1 and 0. EV0000375 · EV0000376 · EV0000377
5. Source slide page 5 (extracted text): ELECTRONIC PRINCIPLES AND DEVICES Logical Operations For all Logical gates a Table with all combinations are list out. The combination of input variables and the corresponding output variables is termed as “TRUTH TABLE”. It explains how the logic circuit output responds to various combinations of logic levels at the inputs. AND: Operation is explained by the following Truth Table A B Y=A.B 0 0 0 0 1 0 1 0 0 1 1 1 EV0000378 · EV0000379 · EV0000380
6. Source slide page 6 (extracted text): ELECTRONIC PRINCIPLES AND DEVICES Logical Operations An OR gate may also have two or more inputs but produce only one output. The OR gate produces an output of logic 1 state even if any of its inputs is in logic 1 state and also produces an output of logic 0 state if any of its inputs is in logic 0 state. OR Operation is explained by the following Truth Table A B Y=A+B 0 0 0 0 1 1 1 0 1 1 1 1 EV0000381 · EV0000382 · EV0000383
7. Source slide page 7 (extracted text): ELECTRONIC PRINCIPLES AND DEVICES Logical Operations (1)OR Operation Illustrated using Diode A (V1) B(V2) Y (Vo) 0 0 0 0 10 V 9.3 V 10 V 0 9.3 V 10 V 10 V 9.3V EV0000384 · EV0000385 · EV0000386
8. Source slide page 8 (extracted text): ELECTRONIC PRINCIPLES AND DEVICES – UE20EC101 Logical Operations (2)AND Operation Illustrated using Diode A (V1) B(V2) Y(Vo) 0 0 0.7 V 0 10 V 0.7V If both inputs are 10v both diodes are Reverse biased and 10 V 0 0.7V no current flows through the resistor 1k ohm and hence Vo =10V 10 V 10 V 10V EV0000387 · EV0000388 · EV0000389
9. Source slide page 9 (extracted text): THANK YOU Dr. Ananda M Department of Electronics and Communication anandam@pes.edu EV0000390 · EV0000391 · EV0000392

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:01.381] Good morning and welcome to Electronic Principles and Devices. I'm Dr. [00:00:05.521] Anand, Department of Electronics and Communication. [00:00:08.010] Today, we're going to see the logical operations. [00:00:13.290] A logical operation is [00:00:15.720] an operation that acts on a binary number to produce the result according to the [00:00:21.030] laws of Boolean logic. A logic gate is a basic building block of [00:00:27.060] digital circuits. Logic gate is considered as a device that has the ability to [00:00:31.350] produce one output level with a combination of input levels. [00:00:36.090] So, as we see from the diagram, I have A and B are the inputs. [00:00:39.510] The function is represented f [00:00:42.840] and Y is the result. So we can build an example of AND, OR or NOT functions. [00:00:49.410] In continuation, logic gates are implemented by using diode transistors and by [00:00:54.660] other several devices. This logic gate can also be considered as electronic [00:00:58.890] circuits. The logic gate opened by diode is called diode resistive logic. [00:01:03.870] Inputs and outputs of logic gates are in two levels, which are termed as high and [00:01:09.030] low, [00:01:10.500] or true or false, or ON and OFF, simply one and zero. [00:01:15.690] Coming to the [00:01:17.850] next step. For all logic gates, a table with all the combination is listed below. [00:01:23.790] The combination of input variable and the corresponding output is termed as truth [00:01:27.930] table, which is shown here. [00:01:30.750] It explains how the logic circuit output responds to various combination of logic [00:01:36.689] level at the input. AND operation is explained by the following truth table, where [00:01:42.360] AND operation is A.B. When inputs A and B are zero, zero, B results in [00:01:48.360] zero. Input zero and one multiplied to get zero. One and zero, zero. [00:01:53.851] One and one is one. This is also explained by a Venn diagram here. [00:01:58.230] The common is being highlighted here with the red color. [00:02:04.980] Similarly, an AND/OR gate may also have two or more input, but produce only one [00:02:09.930] output. OR gate produces an output of logic one as shown in the diagram. [00:02:16.020] Even [00:02:19.171] if [00:02:22.200] one of the input is one state and produces an output of zero state if all the [00:02:27.030] inputs are in the logic zero state. [00:02:29.700] OR operation is explained with the following table. With zero, zero, we get zero. [00:02:34.290] With zero and one, and one and zero, and one and one, we get all one states. [00:02:39.690] Now let us consider the logical operation for OR operation illustrated using a [00:02:43.860] diode. The two diodes are connected in parallel here, and they're connected in [00:02:47.430] series with a resistance with 1K. With the input E is equal to 10, with input [00:02:53.250] E1 and E2 here. So OR voltage V1 and V2. [00:02:57.450] And that we can represent using a truth table with the input V1 in the one column, [00:03:03.450] V2 second column, and V output in the third column. [00:03:07.530] The input V1 is also called as an input A, and similarly input two respectively [00:03:13.351] with respect to V1 and V2. Now consider the first condition here, zero, zero. [00:03:18.300] When both the inputs are zero, zero, diodes are not forward biased and hence Vout [00:03:22.860] is equal to zero. If one of the input is 10 volts, so as shown in the diagram here, [00:03:28.110] the first V input is 10 volts here, the second is zero, the diode one is forward [00:03:33.360] biased hence we represent 0.7. [00:03:36.270] And what the difference between the applied voltage here and we calculate the Vout. [00:03:41.160] So Vout is equal to here E minus [00:03:45.390] VD. So which is 10 minus [00:03:49.770] 0.3, we get 9.3 is the voltage which is obtained. [00:03:55.680] So which is approximately equal to 10 volts and hence act like a OR gate. [00:03:59.281] Similarly, when both diodes are on, [00:04:02.880] so the corresponding voltage will be [00:04:07.291] 0.7. [00:04:11.910] It's not 0.3, it's 0.7. So we get 9.3 is the difference in the voltage, and [00:04:17.640] hence it

satisfies the table of OR operation. [00:04:22.591] Similarly, if you go for an AND operation as illustrated in the diagram. [00:04:29.370] Here diodes are reverse connected compared to the previous one, and we connect a [00:04:33.870] voltage E is equal to 10 volts here. And applied voltage E1 and E2. [00:04:38.730] So with the variation of E1 and E2, the zero volts applied here, both diodes are [00:04:43.620] forward biased and hence Vout is equal to [00:04:47.550] 0.7 here. Because the difference [00:04:53.281] between here is 0.7, so we get 0.7. If one of the voltage we consider 10 volts [00:04:59.160] here, so we consider this 10 volts and the second voltage is zero, diode is on.

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Transcript block 2 (source interval 000300-000599 seconds): [00:05:05.341] The previous case also diode was on, so now second diode is on, and hence the [00:05:09.060] difference in the voltage is 0.7 we get. [00:05:11.610] If both the diodes are applied with E1 and E2, 10 volts, then the both diodes are [00:05:16.710] off, and hence the voltage across this is 10 volts. [00:05:20.460] So we see the operation of an [00:05:23.160] AND with the help of a diode. So these logical operation helps us to understand [00:05:27.420] that diode is a basic fundamental block which is there in the [00:05:33.031] construction of a gate. So then this gate will [00:05:37.410] result in the logical operations of AND and OR. [00:05:41.160] Thank you.

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Full transcript: 864 normalized words, 75 timestamped segments, 343.915 seconds duration. Low-confidence segments flagged: 0.

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10 Diode Clippers : Series using Ideal diode

This lesson is indexed from the source video titled Diode Clippers : Series using Ideal diode. Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or rewritten. A candidate slide deck is grouped here according to the current matching score; the exact path and page locators are listed in the source evidence. EV0000070 · EV0000071

Graphs and diagrams

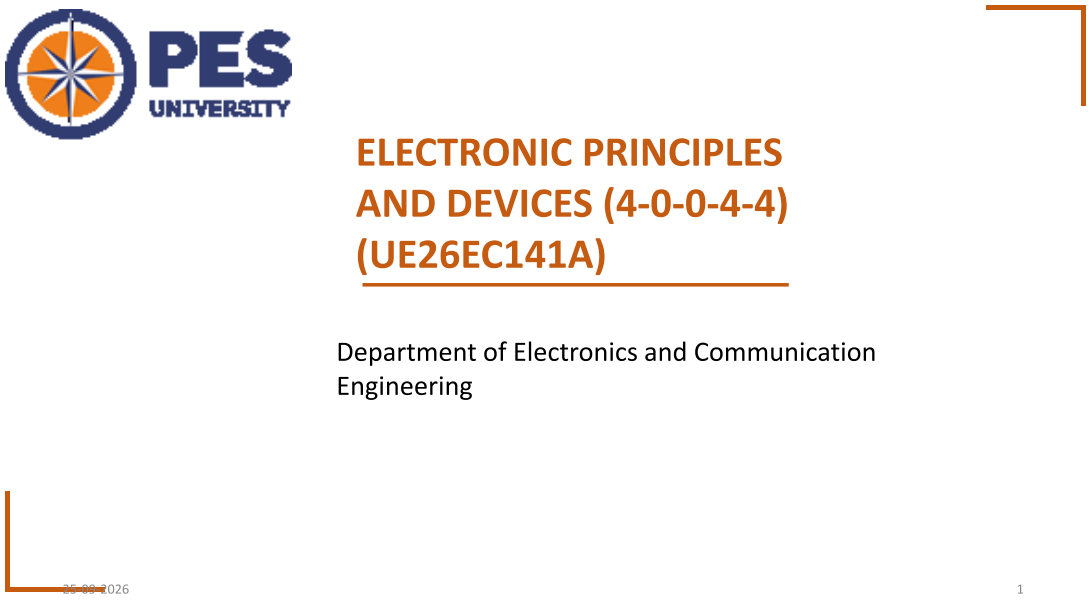

Figure G000098. Diode Clippers Series using Ideal diode · source slide page 1.

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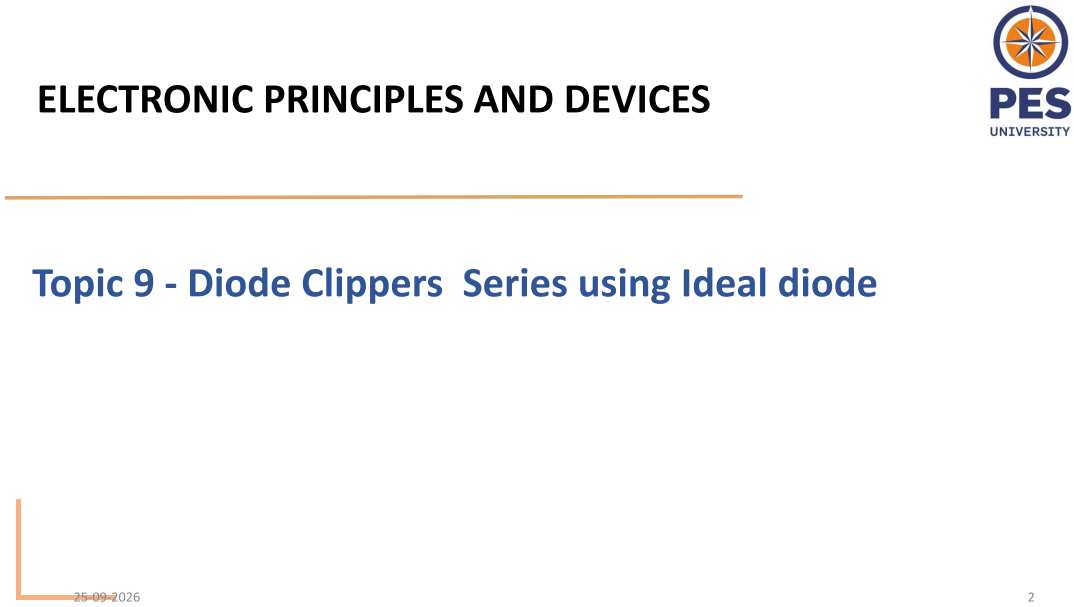

Figure G000099. Diode Clippers Series using Ideal diode · source slide page 2.

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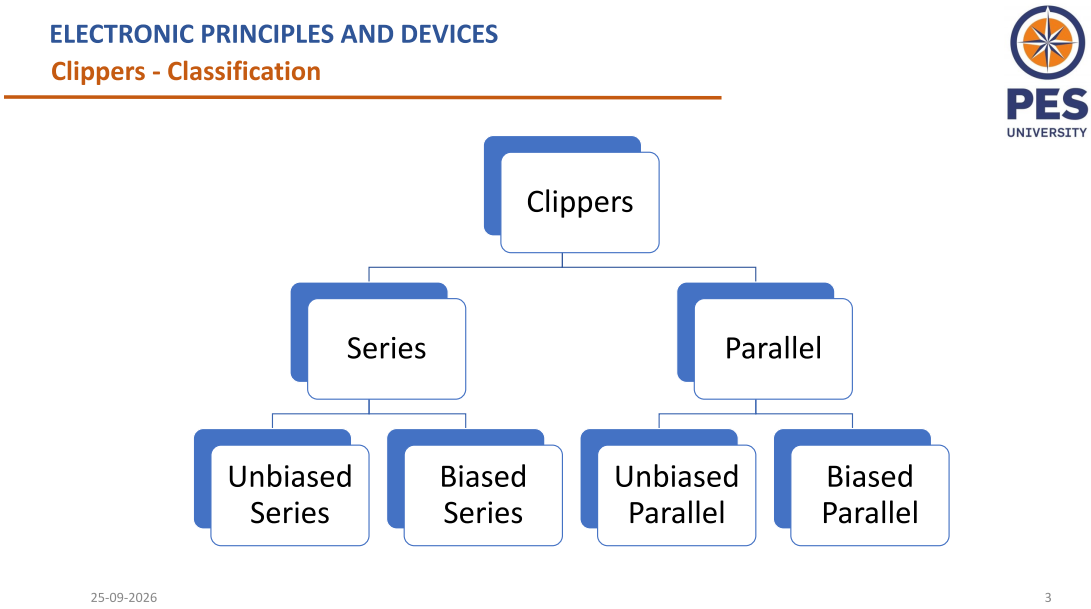

Figure G000100. Diode Clippers Series using Ideal diode · source slide page 3.

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ELECTRONIC PRINCIPLES AND DEVICES

Series Clippers

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- The **series configuration** is defined as one where the **diode is in series with the load.**
- Fig.1(a) shows the circuit of a series clipper.
- The response of the series configuration of Fig. 1(a) to a variety of alternating waveforms is shown in Fig. 1(b).
- There are **no boundaries on the type of signals** that can be applied to a clipper.

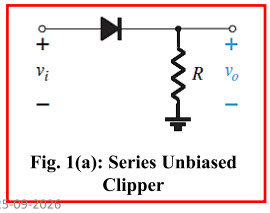

Fig. 1(a): Series Unbiased Clipper

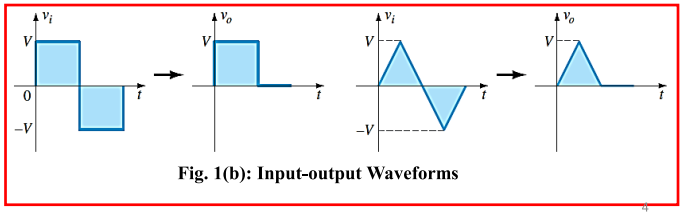

Fig. 1(b): Input-output Waveforms

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Figure G000101. Diode Clippers Series using Ideal diode · source slide page 4.

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- Fig.2 shows the **series clipper with DC supply**. This configuration is called as **biased series clipper**.
- The **addition of a dc supply** to the network as shown in Fig. 2 can have a pronounced effect on the analysis of the series clipper configuration.
- The dc supply can be in the **leg between the supply and output** or in the **branch parallel to the output**.
- The output response is not as obvious because the **dc supply can aid or work against the source voltage.**

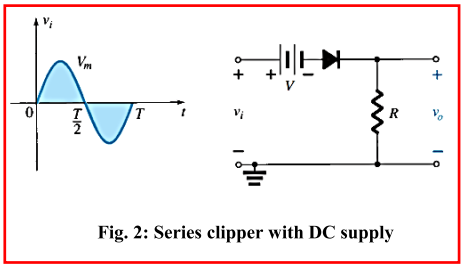

Fig. 2: Series clipper with DC supply

25-09-2026 5

Figure G000102. Diode Clippers Series using Ideal diode · source slide page 5.

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General instructions for analysis:

1. Take careful note of where the output voltage is defined.
2. Try to develop an overall sense of the response by simply noting the that how the input supply affects the conventional current direction through the diode.

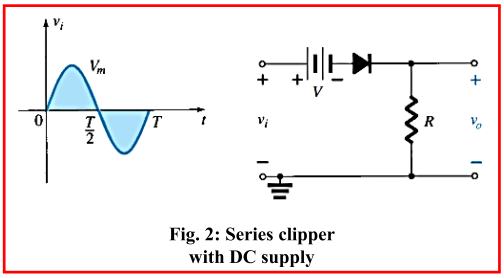

Fig. 2: Series clipper with DC supply

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Figure G000103. Diode Clippers Series using Ideal diode · source slide page 6.

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3. Determine the "transition voltage" that will result in a change of state for the diode from the "OFF" to the "ON" state.
   - Note the substitution of the short-circuit equivalent for the ideal diode in **Fig 3**, and the fact that the voltage across the resistor is 0 V because the diode current is 0 mA. The result is v_i **-V = 0 or** v_i **= V** is the applied (transition) voltage.
   - This permits drawing a line on the sinusoidal supply voltage as shown in **Fig. 4** to define the regions where the diode is on and off.

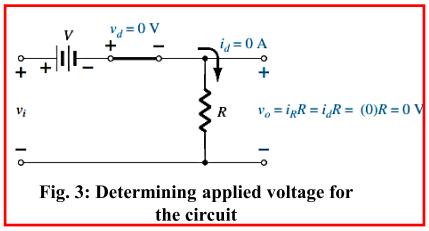

Fig. 3: Determining applied voltage for the circuit

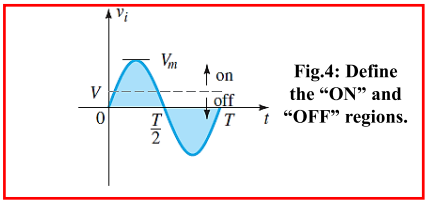

Fig.4: Define the "ON" and "OFF" regions.

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Figure G000104. Diode Clippers Series using Ideal diode · source slide page 7.

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- For the "ON" region, as shown in **Fig. 5**, the output voltage is defined by-
  - $v_o = v_i - V$ ..... (1)
- For the "OFF" region, the diode is an open circuit, I_D = 0 mA, and the output voltage is
  - $v_o = 0\ V$ .....(2)

4. **It is often helpful to draw the output waveform directly below the applied voltage using the same scales for the horizontal axis and the vertical axis.**

- For the "ON" condition, **Eq. (1)** can be used to find the output voltage when the applied voltage has its peak value:
  - $v_{o_peak} = V_m - V$ .....(3)
  - Output voltage, v_o is plotted in the **Fig. 6**.

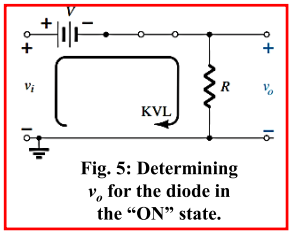

Fig. 5: Determining v_o for the diode in the "ON" state.

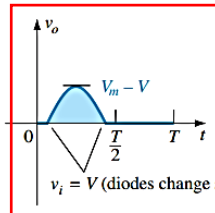

Fig. 6: Sketching the waveform of v_o above and below the transition level.

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Figure G000105. Diode Clippers Series using Ideal diode · source slide page 8.

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EXAMPLE 1: Determine the output waveform for the sinusoidal input of Fig. 7.

Step 1: The output is directly across the resistor R.

Step 2: The positive region of v_i and the dc supply are both working in same direction to turn the diode "ON". The result is that we can safely assume the diode is in the "on" state for the entire range of positive voltages for v_i .

Once the supply goes negative, it would have to exceed the dc supply voltage of 5 V before it could turn the diode "OFF".

Step 3: The transition model is substituted in **Fig. 8**, and we find that the transition from one state to the other will occur when,

$v_i + 5V = 0V$

Or $v_i = -5V$ .....(1)

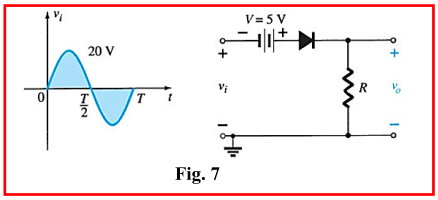

Fig. 7

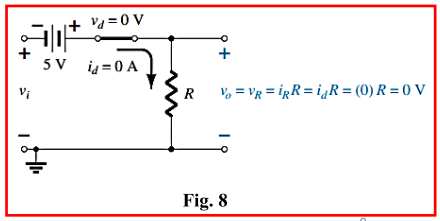

Fig. 8

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Figure G000106. Diode Clippers Series using Ideal diode · source slide page 9.

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EXAMPLE 1 (cont…..)

Step 4: In **Fig. 9** horizontal line is drawn through the applied voltage at the transition level.

For voltages less than 5V the diode is in the open-circuit state and the output is 0 V, as shown in the sketch of v_o .

Using **Fig. 9** , we find that for conditions when the diode is on and the diode current is established the output voltage will be the following, as determined using Kirchhoff's voltage law:

$$v_o = v_i + 5V \text{ .....(2)}$$

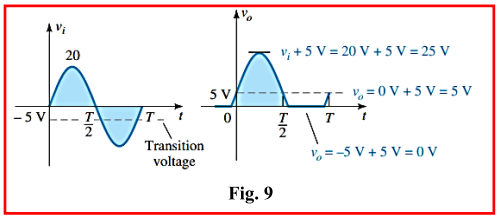

Fig. 9

Note: The analysis of clipper networks with square-wave inputs is actually easier than with sinusoidal inputs because only two levels have to be considered.

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Figure G000107. Diode Clippers Series using Ideal diode · source slide page 10.

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EXAMPLE 2: Find the output voltage for the network examined in previous Example, if the applied signal is the square wave of **Fig. 10**.

- For v_i = 20 V (0 → T/2) the network of **Fig. 11** results. The diode is in the short-circuit state i.e. "ON", and

  $v_o = 20\text{ V} + 5\text{ V} = 25\text{ V}$ .....(1)
- For v_i = -10 V (T/2 → T) the network of **Fig. 12**, results, moving the diode in the "OFF" state, and

  $v_o = i_R R = (0)R = 0\text{ V}$ .....(2)
- The resulting output voltage appears in **Fig. 13**.

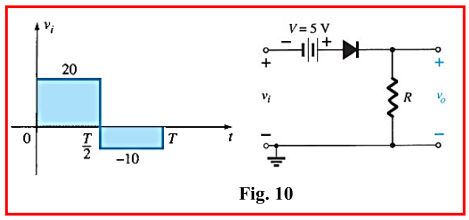

Fig. 10

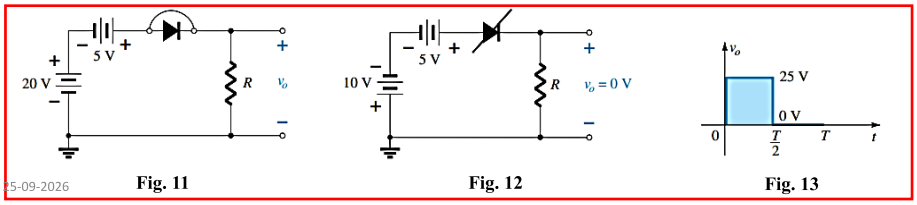

Fig. 11 Fig. 12 Fig. 13

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Figure G000108. Diode Clippers Series using Ideal diode · source slide page 11.

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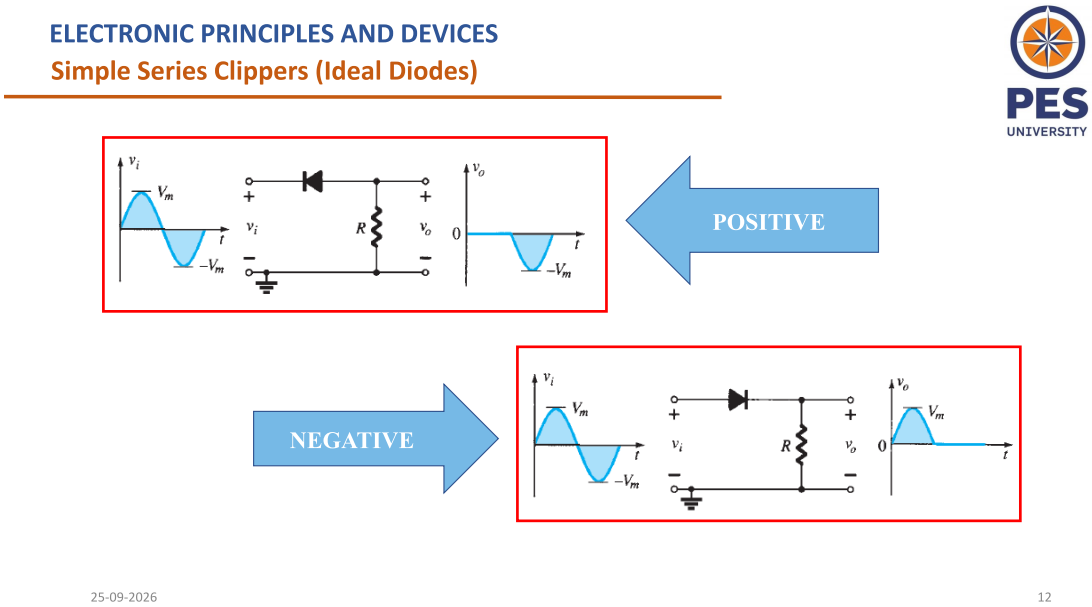

Figure G000109. Diode Clippers Series using Ideal diode · source slide page 12.

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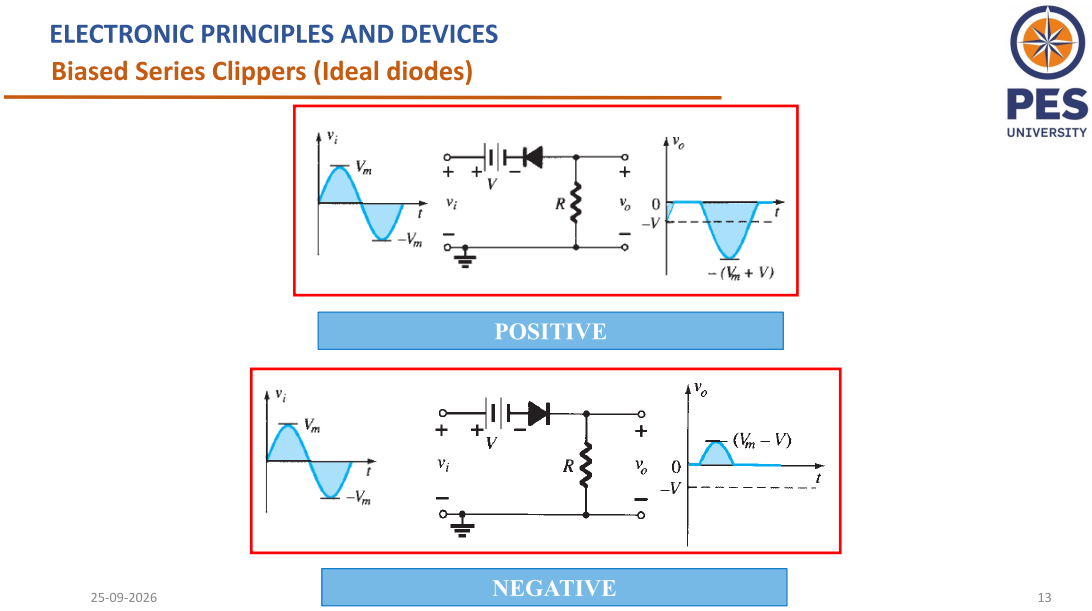

Figure G000110. Diode Clippers Series using Ideal diode · source slide page 13.

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Figure G000111. Diode Clippers Series using Ideal diode · source slide page 14.

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Important points and why

1. Source slide page 1 (extracted text): ELECTRONIC PRINCIPLES AND DEVICES (4-0-0-4-4) (UE26EC141A) Department of Electronics and Communication Engineering 25-09-2026 1 EV0000393 · EV0000394 · EV0000395
2. Source slide page 2 (extracted text): ELECTRONIC PRINCIPLES AND DEVICES Topic 9 - Diode Clippers Series using Ideal diode 25-09-2026 2 EV0000396 · EV0000397 · EV0000398
3. Source slide page 3 (extracted text): ELECTRONIC PRINCIPLES AND DEVICES Clippers - Classification Clippers Series Parallel Unbiased Biased Unbiased Biased Series Series Parallel Parallel 25-09-2026 3 EV0000399 · EV0000400 · EV0000401
4. Source slide page 4 (extracted text): ELECTRONIC PRINCIPLES AND DEVICES Series Clippers • The series configuration is defined as one where the diode is in series with the load. • Fig.1(a) shows the circuit of a series clipper. • The response of the series configuration of Fig. 1(a) to a variety of alternating waveforms is shown in Fig. 1(b). • There are no boundaries on the type of signals that can be applied to a clipper. Fig. 1(a): Series Unbiased Fig. 1(b): Input-output Waveforms Clipper 25-09-2026 4 EV0000402 · EV0000403 · EV0000404

5. Source slide page 5 (extracted text): ELECTRONIC PRINCIPLES AND DEVICES Series Clippers • Fig.2 shows the series clipper with DC supply. This configuration is called as biased series clipper. • The addition of a dc supply to the network as shown in Fig. 2 can have a pronounced effect on the analysis of the series clipper configuration. • The dc supply can be in the leg between Fig. 2: Series clipper with DC supply the supply and output or in the branch parallel to the output. • The output response is not as obvious because the dc supply can aid or work against the source voltage. 25-09-2026 5 EV0000405 · EV0000406 · EV0000407
6. Source slide page 6 (extracted text): ELECTRONIC PRINCIPLES AND DEVICES Series Clippers General instructions for analysis: 1. Take careful note of where the output voltage is defined. 2. Try to develop an overall sense of the response by simply noting the that how the input supply affects the conventional current direction through the diode. Fig. 2: Series clipper with DC supply 25-09-2026 6 EV0000408 · EV0000409 · EV0000410
7. Source slide page 7 (extracted text): ELECTRONIC PRINCIPLES AND DEVICES Series Clippers 3. Determine the “transition voltage” that will result in a change of state for the diode from the “OFF” to the “ON” state. • Note the substitution of the short-circuit equivalent for the ideal diode in Fig 3, and the fact that the voltage across the resistor is 0 V because the diode current is 0 mA. The result is vi -V = 0 or vi = V is the applied (transition) voltage. • This permits drawing a line on the sinusoidal supply voltage as shown in Fig. 4 to define the regions where the diode is on and off. Fig.4: Define the “ON” and “OFF” regions. Fig. 3: Determining applied voltage for the circuit 25-09-2026 7 EV0000411 · EV0000412 · EV0000413
8. Source slide page 8 (extracted text): ELECTRONIC PRINCIPLES AND DEVICES Series Clippers • For the “ON” region, as shown in Fig. 5, the output voltage is defined by- = vi – V ..... (1) • vo • For the “OFF” region, the diode is an open circuit, ID = 0 mA, and the output voltage is • vo = 0 V .....(2) Fig. 5: Determining 4. It is often helpful to draw the output vo for the diode in the “ON” state. waveform directly below the applied voltage using the same scales for the horizontal axis and the vertical axis. Fig. 6: Sketching the waveform of • For the “ON” condition, Eq. (1) can be used to find the output vo above and voltage when the applied voltage has its peak value: below the = Vm – V .....(3) transition level. • vo_peak • Output voltage, vo is plotted in the Fig. 6. 25-09-2026 8 EV0000414 · EV0000415 · EV0000416
9. Source slide page 9 (extracted text): ELECTRONIC PRINCIPLES AND DEVICES Series Clippers EXAMPLE 1: Determine the output waveform for the sinusoidal input of Fig. 7. Step 1: The output is directly across the resistor R. Step 2: The positive region of vi and the dc supply are both working in same direction to turn the diode “ON”. The result is that we can safely assume the diode is in the “on” state for the entire range of positive voltages for vi . Fig. 7 Once the supply goes negative, it would have to exceed the dc supply voltage of 5 V before it could turn the diode “OFF”. Step 3: The transition model is substituted in Fig. 8, and we find that the transition from one state to the other will occur when, vi + 5V = 0V Fig. 8 9 25-09-2026 Or vi = -5V .....(1) EV0000417 · EV0000418 · EV0000419

10. Source slide page 10 (extracted text): ELECTRONIC PRINCIPLES AND DEVICES Series Clippers EXAMPLE 1 (cont.....) Step 4: In Fig. 9 horizontal line is drawn through the applied voltage at the transition level. For voltages less than 5V the diode is in the open-circuit state and the output is 0 V, as shown in the sketch of . vo Using Fig. 9 , we find that for conditions when the diode is on and the diode current is established the output voltage will be the following, as determined using Kirchhoff's voltage Fig. 9 law: = + 5V .....(2) vo vi Note: The analysis of clipper networks with square-wave inputs is actually easier than with sinusoidal inputs because only two levels have to be considered. 25-09-2026 10 EV0000420 · EV0000421 · EV0000422
11. Source slide page 11 (extracted text): ELECTRONIC PRINCIPLES AND DEVICES Series Clippers EXAMPLE 2: Find the output voltage for the network examined in previous Example, if the applied signal is the square wave of Fig. 10. • For = 20 V (0 ☒T/2) the network of Fig. 11 vi results. The diode is in the short-circuit state i.e. "ON", and = 20 V + 5 V = 25 V .....(1) vo • For = -10 V (T/2 ☒T) the network of Fig. 12, vi results, moving the diode in the "OFF" state, and Fig. 10 = R = (0)R = 0 V .....(2) vo iR • The resulting output voltage appears in Fig. 13. 25-09-2026 Fig. 11 Fig. 12 Fig. 13 11 EV0000423 · EV0000424 · EV0000425
12. Source slide page 12 (extracted text): ELECTRONIC PRINCIPLES AND DEVICES Simple Series Clippers (Ideal Diodes) POSITIVE NEGATIVE 25-09-2026 12 EV0000426 · EV0000427 · EV0000428
13. Source slide page 13 (extracted text): ELECTRONIC PRINCIPLES AND DEVICES Biased Series Clippers (Ideal diodes) POSITIVE NEGATIVE 25-09-2026 13 EV0000429 · EV0000430 · EV0000431
14. Source slide page 14 (extracted text): THANK YOU Department of Electronics and Communication 25-09-2026 14 EV0000432 · EV0000433 · EV0000434

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:01.920] Good morning and welcome to Electronic Principles and Devices course. [00:00:06.690] Today, we're going to see the clippers. [00:00:10.650] Clippers are those circuits which are made up of electronic components such as [00:00:16.290] diodes, resistors, [00:00:19.260] and DC bias voltages, [00:00:22.470] which are used to remove a part of the waveform without disturbing the [00:00:29.131] other part of the waveform. That means to say, in case you [00:00:33.780] have a square waveform, as shown in the [00:00:39.630] figure here. [00:00:44.670] If I need only the positive part of the waveform [00:00:50.130] and I don't want the negative part, using the clipper circuit, we can get [00:00:56.011] only the positive part. Similarly, if I need a negative part and remove the [00:01:01.020] positive part, that is possible. So there are different types of clipper circuits [00:01:05.730] are there, which has been-- Where clippers are classified into mainly two types: [00:01:11.370] series clipper and parallel clipper. [00:01:14.160] Further, the series clipper can be grouped into unbiased series clipper, biased [00:01:19.711] series clipper, and in parallel, it's been classified into unbiased parallel and [00:01:25.020] biased parallel. [00:01:27.900] Now, let us go ahead with [00:01:32.250] series clipper. The series configuration is defined as one where diode is in series [00:01:38.160] with the load. Example in the [00:01:41.880] Figure 1A, [00:01:44.010] we are using a series unbiased clipper. First of all, what is this unbiased? [00:01:49.380] Unbiased means there is no DC supply used in the circuit. [00:01:52.650] Only we have in this block, we have only diode and the resistor. [00:01:57.360] In case if there is any bias voltage here along with the resistor or along with the [00:02:01.560] diode, other than the supply voltage. [00:02:04.380] Other than the

supply voltage, if we have a voltages here, a DC voltage, then it is [00:02:09.601] biased. Now we are going for unbiased circuit, which have a diode here and a [00:02:15.060] resistor. If the diode is connected in series with the resistor, it is called as a [00:02:19.440] series configuration. So the figure one shows the circuit of series clipper. [00:02:23.520] The response of the series configuration of Figure 1 to the variety of alternating [00:02:29.101] waveform is as shown in the figure B. [00:02:31.260] That means to say, variety of the input is first, in the diagram here, which I'm [00:02:36.150] pointing, it is a square waveform. [00:02:38.940] The output of the series clipper will be only the positive off cycle, negative has [00:02:44.101] been clipped off. Similarly, we have next, in the place [00:02:49.080] of square, we have a triangle waveform. [00:02:51.840] So the output of the clipper will be only the positive off cycle, the negative has [00:02:54.930] been clipped off. So there are no boundaries on type of the signal that can be [00:02:58.770] applied to a clipper circuit. That means we need to understand that there's no [00:03:04.200] boundary, where we can also use a sine waveform, [00:03:07.650] the sawtooth waveform, et cetera, as an input signal and based on the circuit, like [00:03:13.351] a clipper, the output [00:03:18.720] variations can be obtained. So now let us consider the figure two. [00:03:23.730] Figure shows here. [00:03:27.750] Figure shows a series clipper with a DC bias supply. [00:03:30.421] This is what I was referring previously. [00:03:33.090] When there is a DC bias here, it is called as a series clipper with a bias or a [00:03:38.340] series clipper with a DC supply. So figure shows the series clipper with a DC [00:03:42.060] supply. This configuration is called as a biased series clipper. [00:03:47.701] So we can easily differentiate between what is biased and what is unbiased series [00:03:51.510] clipper. The addition of a DC supply to the network as shown in the figure [00:03:56.610] can have a pronounced effect on an analysis of a series clipper configuration. [00:04:01.680] That means there could be some change in the output of which the DC supply can be [00:04:07.050] in the leg between the supply and the output or in a branch parallel to the output. [00:04:12.510] It can be at any place. One, it can be between the input and the diode, or diode [00:04:17.940] and the resistor, or resistor and the output. It can be anywhere. [00:04:22.320] The output response is not as obvious because the DC can aid or work as a [00:04:28.800] source voltage. That means the DC source, which is there here, [00:04:34.531] can aid in the sense it can have an addition with the voltage applied here or it [00:04:38.640] can subtract with the voltage applied here. [00:04:40.560] That we will understand with the analysis in the next slide. [00:04:45.210] So the general instruction for the analysis. [00:04:47.760] So before we go for an actual numerical, we'll take few steps in analysis. [00:04:52.980] The step one as shown here, take careful note of where the output voltage to be [00:04:58.680] defined. So the output voltage is here.

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Transcript block 2 (source interval 000300-000599 seconds): [00:05:00.990] So we need to first note that where we need to calculate the output voltage. [00:05:04.710] Try to develop an overall sense of the response by simply noting [00:05:08.701] that how the input supply affects the conventional current direction through [00:05:14.610] the diode. This means to say, check on the input voltage, how it has been applied [00:05:19.380] here, whether it's a positive or a negative. [00:05:22.050] So based on this, the current flow will start. [00:05:24.510] So the conventional current will flow from negative to the positive. [00:05:28.470] So we need to keep that in mind before we go for the analysis. [00:05:33.060] So now step three, determine the transition voltage that will result in change of [00:05:37.980] the state of the diode from off to on state. [00:05:40.800] Means to say, for example here, this DC voltage, what we apply, and along with the [00:05:46.020] input voltage,

whether it make the diode on or off. [00:05:49.201] And since we are working with an ideal diode here. [00:05:52.740] It is an ideal diode. Entire discussion will be with the ideal diode. [00:05:57.060] Means to say, the diode is on, even if the application of the voltage is zero. [00:06:01.170] If it's more than zero. Whereas for a non-ideal diodes, it will be [00:06:06.960] 0.7. So we need to keep in mind that if the input voltage is greater than [00:06:12.720] zero volts, even if there is a 0.1 volt, the diode will start conducting. [00:06:18.451] So now the applied voltage here, we're assuming that... [00:06:21.031] And here, one more thing we also we need to remember, Vi is greater than V. [00:06:26.099] Otherwise, the diode will not be on. [00:06:27.689] So we consider here Vi is greater than V and also Vi is greater than zero volts. [00:06:32.910] Obviously, this should be there for the diode to be on. [00:06:35.790] So during the positive half cycle of the input, so what we consider during the [00:06:39.540] positive half cycle of the input, my Vm is greater than V. [00:06:42.690] That means to say, diode is on. How do you represent when the diode is on? [00:06:45.990] By a close switch. That is, VD is equal to zero. [00:06:49.110] For ideal diode characteristics, if I look at the ideal diode characteristics, this [00:06:52.500] is what we looked at the previous class. This is V and forward bias. [00:06:58.140] And diode will start conducting at the zero. [00:07:01.591] So this VD is equal to zero. [00:07:06.811] So the voltage across the diode is zero. [00:07:09.210] That's what we have written here, VD is equal to zero. [00:07:13.410] Then the [00:07:15.091] current ID, [00:07:17.581] VD is equal to zero, [00:07:20.370] diode is on, but how about the current? [00:07:25.710] So we'll see that. So note that the substitution of a short circuit equivalent for [00:07:29.670] a diode is as shown in the figure three we discuss now, and the fact that the [00:07:33.630] voltage across the resistor is zero, [00:07:39.061] because the diode current is zero milliamps. [00:07:43.561] The result is Vi minus V is equal to zero or Vi is equal to V [00:07:48.870] and is [00:07:50.581] applied a transition voltage. This permits drawing a line on the sinusoidal supply [00:07:56.190] voltage as shown in the figure four. As shown in the figure four, on state. [00:08:01.350] So this is the on state, which we show it is on. [00:08:06.960] To define the region where the diode is on and off. [00:08:09.750] So the remaining region will be off. [00:08:11.730] So the red color which I mark now is on state, and which is not marked is off [00:08:16.020] state. So diode is on only at this point, greater than V and less than V. [00:08:22.020] So at this peak point, only we'll be able to get the results. [00:08:27.870] Now, for on-region, as shown in the figure five, so again, the repetition is there. [00:08:32.790] Figure five, the output voltage is defined by, apply the KVL here. [00:08:36.540] Apply the KVL in this direction. Vi, this is Vi plus Vi. [00:08:41.610] So we get plus Vi here, [00:08:44.490] minus V, [00:08:46.590] minus Vout is equal to zero. So what we get here? Vout is equal to... [00:08:51.840] What is Vout voltage? Vout is equal to Vi [00:08:55.770] minus V. That is given as equation number one here. Then for the off-region. [00:09:02.010] Now come back to the off-region here. [00:09:04.140] This is off-region and if you go back, here this is the off-region. [00:09:09.786] Off region, diode is in open circuit and ID current is equal to zero milliamps [00:09:16.027] and the output voltage is zero. Since there is no current there, output voltage is [00:09:21.066] zero. So coming to the fourth point, it is often helpful to draw the output voltage [00:09:25.686] waveform directly below the applied voltage using the same scales for horizontal [00:09:31.446] axis and vertical axis. This is shown in the figure. [00:09:34.116] It may not be the previous diagram. [00:09:36.216] It can also be written in this diagram with the equivalent scales. [00:09:39.396] So for on condition, equation one can be used to find the output voltage with the [00:09:43.536] applied voltage as its peak value. So VOP is equal to VM minus V. [00:09:49.476] So this equation one and three are same. [00:09:51.366] Output voltage VOUT is

plotted as shown in the figure six. [00:09:54.757] This is the figure six, how we plot it. [00:09:59.436] So this is about the analysis of the series clipper.

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Transcript block 3 (source interval 000600-000899 seconds): [00:10:01.716] Now let us solve a numerical example one. [00:10:05.766] The question is, determine the output waveform for a sinusoidal input of figure [00:10:10.027] seven. This is figure seven. I have a sinusoidal signal here. [00:10:15.007] VI 20 volts is a peak to peak of one cycle. [00:10:18.036] Then coming to the circuit here, we have a biased clipper here, series biased [00:10:23.046] clipper, where resistor is connected in series with the diode and voltage 5 volts [00:10:27.277] is given, and VI is equal to 20. So the applied voltage 20 is greater than the DC [00:10:33.186] bias 5 volts. So diode is on. So this is how we have to find out initially. [00:10:38.586] So the output is directly across the resistor. [00:10:43.056] Step two, the positive region of the VI and the DC are both working in the same [00:10:47.257] direction to turn on. So this voltage is now turned on. [00:10:51.036] So the applied total voltage is 10 [00:10:55.086] plus 5 because both are in the same direction. [00:10:57.576] We have plus is there, and we have plus is there, it get added. [00:11:00.096] So 25 is the voltage applied across the diode, and diode is on. [00:11:03.246] The result is that we can safely assume the diode is in on state for entire region [00:11:08.226] of positive cycle voltage. Once the supply goes negative, it will have exceeded the [00:11:14.196] DC supply voltage of 5, therefore it would turn off. [00:11:17.436] So during the negative cycle, it is off. [00:11:20.496] So the transition model is substituted in the figure eight. [00:11:23.346] So now come to figure eight, the transition model is substituted. [00:11:26.226] What is the transition model? If VI and the DC voltage is greater than the [00:11:31.986] diode cutting voltage, that is VD is equal to zero, then the diode is in on [00:11:36.906] condition. Diode is on. Therefore, we apply the KVL over there. [00:11:42.246] If you apply the KVL over there, VI plus 5 is equal to zero, so VI is equal to [00:11:47.286] minus 5 volts we get. [00:11:51.606] Minus 5 volts. [00:11:55.806] So in continuation, step four, in figure nine, horizontal line is drawn with the [00:12:00.186] applied voltage at the transition level. [00:12:04.626] For a voltage less than 5, the diode is in open circuit and the output is zero, as [00:12:09.966] shown in the sketch of the VOUT. This is what we want to prove here. [00:12:13.896] Using the figure number nine, we find that for a condition when a [00:12:19.566] diode is on and the diode current is established, the output voltage will be [00:12:23.436] following. So apply the KVL, VOUT is equal to VI plus 5 volts. [00:12:28.596] So the analysis of the clipper circuit with the square wave input is actually [00:12:31.656] easier than the sinusoidal because only two levels are to be considered. [00:12:36.696] That means to say in the place of sine, if you have a square, only two places if [00:12:40.656] you consider it is more than, whereas here we need to consider for all the places [00:12:43.956] here. The transition is important. [00:12:48.336] So this is the solution for example number one, where we see that during the [00:12:52.447] positive cycle, diode is on, during the negative cycle it is off. [00:12:56.166] So the negative part is clipped off here. It is clipped off. [00:13:01.116] Coming to the second example, find the output voltage for the network examined in [00:13:05.197] the previous example if the applied signal is square. [00:13:08.886] So if the applied signal is a square here for the same circuit being given, how you [00:13:13.416] calculate the output voltage. But here one thing we need to notice, for the [00:13:17.436] positive cycle 20 volts is given, negative cycle -10 volts is being shown. [00:13:21.456] So keeping that in mind now, VI is equal to 20. [00:13:24.426] For a [00:13:26.046] transition from 0 to T/2, [00:13:29.076] the figure 11. So we look at the figure 11 here. [00:13:33.216] Result, the diode is in short circuit state, that is on. So

apply the KVL here. [00:13:38.676] If we apply the KVL, we get VOUT is equal to 20 plus 20, 25 volts is the applied [00:13:44.496] voltage. [00:13:48.186] Then for VI is equal to -10 during the negative cycle, diode is off. [00:13:52.236] That is VOUT is equal to zero. So for the negative cycle, diode is off. [00:13:56.197] Unless we get a positive cycle only, and during the negative cycle, we won't see [00:14:00.336] the output waveform. It is clipped off. [00:14:04.986] So this is about the series clipper. Thank you.

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Full transcript: 2336 normalized words, 200 timestamped segments, 849.28 seconds duration. Low-confidence segments flagged: 0.

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11 Diode Clippers : Parallel using Ideal diode

This lesson is indexed from the source video titled Diode Clippers : Parallel using Ideal diode. Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or rewritten. A candidate slide deck is grouped here according to the current matching score; the exact path and page locators are listed in the source evidence. EV0000082 · EV0000083

Graphs and diagrams

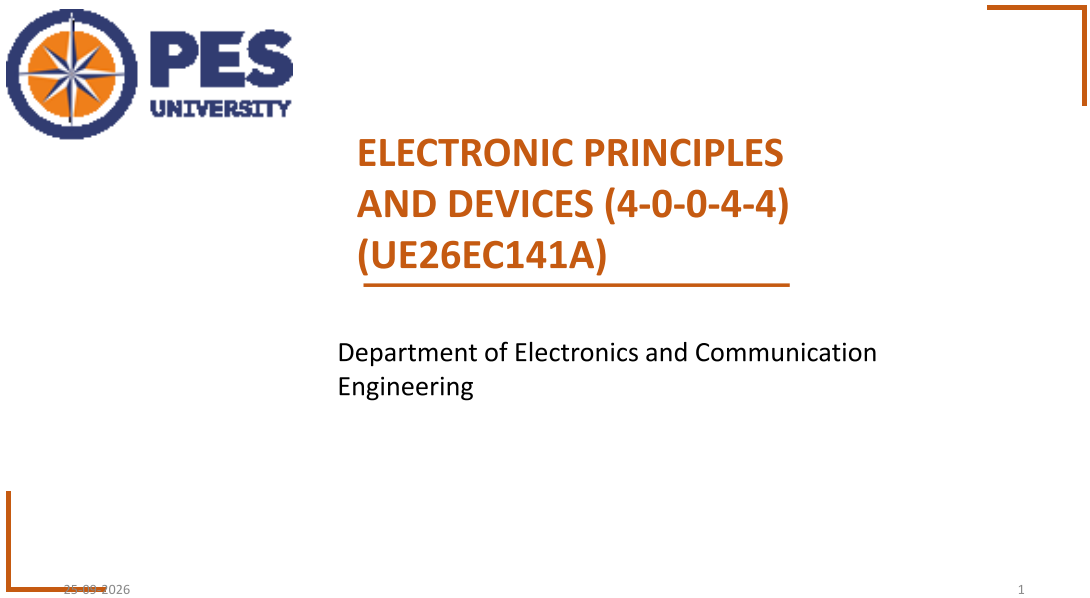

Figure G000112. Diode Clippers Parallel using Ideal diode · source slide page 1.

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ELECTRONIC PRINCIPLES AND DEVICES

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Topic 10 - Diode Clippers Parallel using Ideal diode

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Figure G000113. Diode Clippers Parallel using Ideal diode · source slide page 2.

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ELECTRONIC PRINCIPLES AND DEVICES

Parallel Clippers without bias

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- In parallel clippers, diode is connected in parallel with the load.
- The analysis of parallel configurations is very similar to that of series configurations.
- The simplest parallel clipper is shown in **Fig 14.**

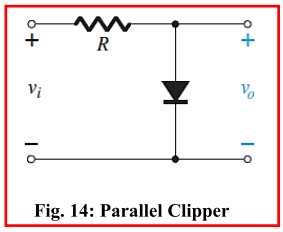

Fig. 14: Parallel Clipper

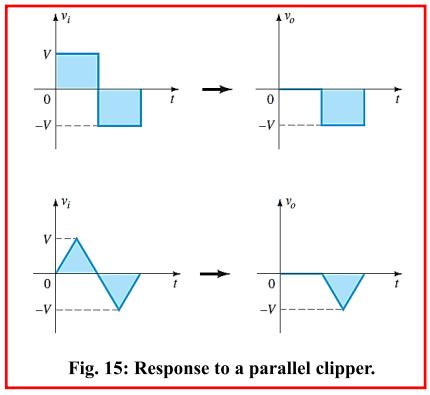

Fig. 15: Response to a parallel clipper.

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Figure G000114. Diode Clippers Parallel using Ideal diode · source slide page 3.

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ELECTRONIC PRINCIPLES AND DEVICES

Parallel Clippers with Bias

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EXAMPLE 3: Determine v_o for the network of Fig. 16.

- When the diode is in its short-circuit i.e. "ON" state the output voltage will be directly across the 4-V dc supply, requiring that the output be fixed at 4 V. In other words, when the diode is "ON" the output will be 4 V.
- When the diode is an open circuit i.e. "OFF" state, the current through the series network will be 0 mA and the voltage drop across the resistor will be 0 V. That will result in $v_o = v_i$.

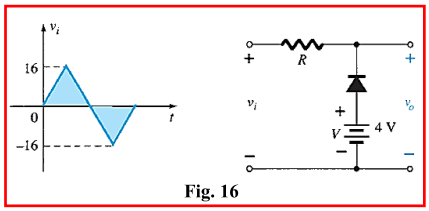

Fig. 16

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Figure G000115. Diode Clippers Parallel using Ideal diode · source slide page 4.

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ELECTRONIC PRINCIPLES AND DEVICES

Parallel Clippers

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Example 3 (cont….)

- The transition level of the input voltage can be found from **Fig. 17** by substituting the short-circuit equivalent and remembering the diode current is 0 mA at the instant of transition.
- The result is a change in state when v_i = 4 V
- In **Fig. 18** the transition level is drawn along with v_o = 4 V when the diode is "ON".

Therefore, for v_i >= 4 V, $v_o = v_i$ ……(1)

And, for v_i <= 4 V, v_o = 4 V, ……..……(2)

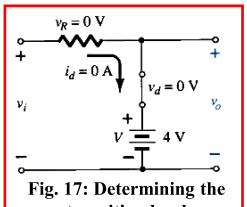

Fig. 17: Determining the transition level

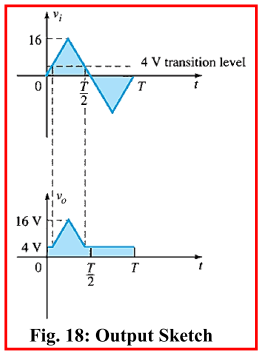

Fig. 18: Output Sketch

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Figure G000116. Diode Clippers Parallel using Ideal diode · source slide page 5.

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ELECTRONIC PRINCIPLES AND DEVICES

Parallel Clippers

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EXAMPLE 4: Repeat Example 3, using a **silicon diode with $V_K = 0.7$ V.**

- Determine Transition voltage
  - Using the condition $i_d = 0$ A at $v_d = V_D = 0.7$ V and obtaining the network of **Fig. 19** .
  - Applying KVL, $v_i + V_K - V = 0$
  - or $v_i = V - V_K = 4$ V - 0.7 V = 3.3 V .....(1)
- Diode will be "OFF", for $v_i > 3.3$ V (in Fig. 19) and
  - $v_o = v_i$ ..... (2)
- Diode will be "ON", for $v_i < 3.3$ V (in Fig. 20) and
  - $v_o = 4$ V - 0.7 V = 3.3 V .....(3)
- The resulting output waveform appears in **Fig. 21** .
- Note that the only effect of V_K was to drop the transition level to 3.3 from 4 V.

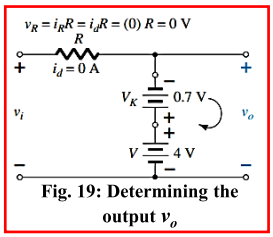

Fig. 19: Determining the output v_o

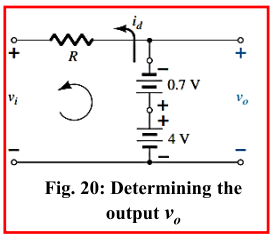

Fig. 20: Determining the output v_o

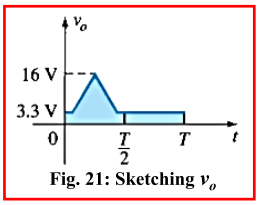

Fig. 21: Sketching v_o

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Figure G000117. Diode Clippers Parallel using Ideal diode · source slide page 6.

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ELECTRONIC PRINCIPLES AND DEVICES

Simple Parallel Clippers (Ideal diodes)

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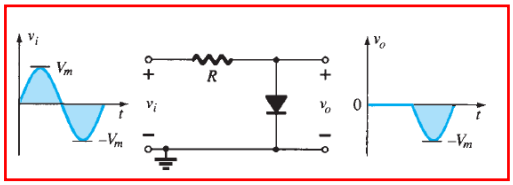
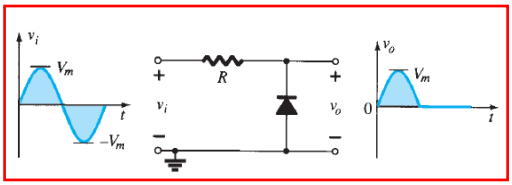


POSITIVE

NEGATIVE



25-09-2026 7

Figure G000118. Diode Clippers Parallel using Ideal diode · source slide page 7.

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Figure G000119. Diode Clippers Parallel using Ideal diode · source slide page 8.

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Important points and why

1. Source slide page 1 (extracted text): ELECTRONIC PRINCIPLES AND DEVICES (4-0-0-4-4) (UE26EC141A) Department of Electronics and Communication Engineering 25-09-2026 1 EV0000435 · EV0000436 · EV0000437
2. Source slide page 2 (extracted text): ELECTRONIC PRINCIPLES AND DEVICES Topic 10 - Diode Clippers Parallel using Ideal diode 25-09-2026 2 EV0000438 · EV0000439 · EV0000440
3. Source slide page 3 (extracted text): ELECTRONIC PRINCIPLES AND DEVICES Parallel Clippers without bias • In parallel clippers, diode is connected in parallel with the load. • The analysis of parallel configurations is very similar to that of series configurations. • The simplest parallel clipper is shown in Fig 14. Fig. 14: Parallel Clipper Fig. 15: Response to a parallel clipper. 25-09-2026 3 EV0000441 · EV0000442 · EV0000443
4. Source slide page 4 (extracted text): ELECTRONIC PRINCIPLES AND DEVICES Parallel Clippers with Bias EXAMPLE 3: Determine vo for the network of Fig. 16. • When the diode is in its short-circuit i.e. “ON” state the output voltage will be directly across the 4-V dc supply, requiring that the output be fixed at 4 V. In other words, when the diode is “ON” the output will be 4 V. • When the diode is an open circuit i.e. “OFF” state, the current through the series network will be 0 mA and the voltage drop across the resistor will be 0 V. That will result in vo = vi . Fig. 16 25-09-2026 4 EV0000444 · EV0000445 · EV0000446

5. Source slide page 5 (extracted text): ELECTRONIC PRINCIPLES AND DEVICES Parallel Clippers Example 3 (cont....) • The transition level of the input voltage can be found from Fig. 17 by substituting the short-circuit equivalent and remembering the diode current is 0 mA at the instant of transition. • The result is a change in state when vi = 4 V • In Fig. 18 the transition level is drawn along with vo = 4 V when the diode is “ON”. Fig. 17: Determining the transition level Therefore, for vi >= 4 V, vo = vi .....(1) Fig. 18: Output Sketch And, for vi <= 4 V, vo = 4 V, ...(2) 25-09-2026 5 EV0000447 · EV0000448 · EV0000449
6. Source slide page 6 (extracted text): ELECTRONIC PRINCIPLES AND DEVICES Parallel Clippers EXAMPLE 4: Repeat Example 3, using a silicon diode with VK = 0.7 V. • Determine Transition voltage • Using the condition id = 0 A at vd = VD = 0.7 V and obtaining the network of Fig. 19 . • Applying KVL, + VK - V = 0 vi • or = V - VK = 4 V - 0.7 V = 3.3 V .....(1) vi • Diode will be “OFF”, for > 3.3 V (in Fig. 19) and Fig. 19: Determining the Fig. 20: Determining the vi ..... (2) output vo output vo • vo = vi • Diode will be “ON”, for < 3.3 V (in Fig. 20) and vi = 4 V - 0.7 V = 3.3 V .....(3) • vo • The resulting output waveform appears in Fig. 21 . • Note that the only effect of VK was to drop the transition level to 3.3 from 4 V. Fig. 21: Sketching vo 25-09-2026 6 EV0000450 · EV0000451 · EV0000452
7. Source slide page 7 (extracted text): ELECTRONIC PRINCIPLES AND DEVICES Simple Parallel Clippers (Ideal diodes) POSITIVE NEGATIVE 25-09-2026 7 EV0000453 · EV0000454 · EV0000455
8. Source slide page 8 (extracted text): THANK YOU Department of Electronics and Communication 25-09-2026 8 EV0000456 · EV0000457 · EV0000458

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:01.470] Good morning and welcome to Electronic Principles and Devices course. [00:00:06.180] I'm Professor Ananda, Department of Electronics and Communication Engineering. [00:00:09.810] Today, we're going to see the parallel clipper, but the last class we have seen the [00:00:14.250] series clipper. Let us have a refresh of that. [00:00:16.860] The series clipper are those [00:00:19.741] which is used to clip off the waveform that is not required and without [00:00:25.320] disturbing the other part of the waveform. [00:00:27.870] So the last class we have seen the working principle of series clipper and [00:00:33.870] few examples. Now let us go ahead with the parallel clipper, first without bias, [00:00:38.670] then with bias. So in parallel clipper, diode is connected in parallel with the [00:00:43.260] load. So in comparison with the series clipper, [00:00:48.120] in series clipper diode was connected in series with the load, whereas now diode [00:00:51.481] will be connected in parallel with the load. [00:00:53.970] So now figure 14 shows the [00:00:57.240] simplest parallel clipper, where diode is connected in parallel with the load. [00:01:03.240] The analysis of the parallel configuration is very similar to that of the series [00:01:06.301] configuration. So the different responses have been shown in the figure 15. [00:01:12.360] Now we have an input connected, input VI given to the resistor connected to one end [00:01:16.980] of the resistor. The other end of the resistor is connected to the diode. [00:01:20.760] Well, connected to the positive terminal of the diode and negative terminal [00:01:23.640] connected to the ground point. Now between the positive [00:01:29.340] cycle of the input, diode is on. When the diode is on, we represent it by a short [00:01:33.900] circuit. So when you represent a short circuit, the voltage across the diode is VD [00:01:38.940] is equal to 0. [00:01:41.851] So hence we represent as the output voltage is equal to 0 here. [00:01:45.990] Similarly, during the negative cycle of the input,

diode is off. [00:01:50.700] When the diode is off, represented as an open circuit. [00:01:53.730] When it's represented as an open circuit, the voltage across the diode is same as [00:01:58.740] the input applied. So hence we will be able to see the negative cycle. [00:02:03.840] So we see that the positive part of the waveform is clipped off and negative part [00:02:07.891] is written as it is. The same thing applies to any input in the place of a square [00:02:13.171] if you use the triangle as shown. So the positive part is clipped off and the [00:02:17.490] negative part is retained as it is. [00:02:20.160] So this is about the basic principle of operation of parallel clipper without bias. [00:02:26.550] So let us go with the clipper with a bias, with an example. [00:02:30.990] Determine the Vout for the network of the figure 16. [00:02:34.110] Now in the waveform, we have the peak-to-peak voltage as 16 here. [00:02:37.830] A triangle waveform has been given to the [00:02:41.580] clipper with the bias. So now with the bias in the sense, the diode is connected in [00:02:46.260] series with a [00:02:48.240] DC voltage, V is equal to 4 volts, with the positive side connected to the positive [00:02:53.101] side of the diode and negative side connected to the ground. [00:02:56.490] And the diode positive side is connected to the resistor, where the voltage applied [00:03:01.080] VI is applied to the one end of the resistor, other end is connected to the supply [00:03:06.900] voltage. Now, the applied voltage is in series with the [00:03:13.860] DC voltage, and we see that the negative [00:03:17.610] is in series with the positive. That means the applied voltage here will get [00:03:23.820] minus with the V. So VI minus V is the voltage what we applied with the diode. [00:03:29.520] So when the diode is in its short circuit on state, the output [00:03:35.430] voltage will be directly across the 4 volts. That means to say... [00:03:41.281] So we'll try to understand that with the next step. [00:03:44.281] Then, [00:03:45.781] in other words, when the diode is on, the output will be 4 volts. [00:03:48.541] This we need to prove now. Then when the diode is in open circuit, that is the off [00:03:52.171] state, the current through the series network will be 0, and the voltage drop [00:03:55.980] across the resistor will be 0. That will be Vout is equal to VI. [00:03:59.791] So we'll try to prove this now with the help of a support from a diagram. [00:04:03.570] So now the transition level of the input voltage is shown from the figure 17. [00:04:08.070] So now determine the transition level when the input is positive half cycle. [00:04:12.270] During the positive half cycle, transition from 0 to T/2, diode is on. [00:04:16.680] How do we represent? VD=0. Now what is the voltage across this? 0+4. [00:04:21.660] So Vout=4 volts, isn't it? So we represent that 4 volts here. [00:04:31.110] 4 volts there. [00:04:33.990] And the diode is 0, and transition 0 instead of transition. [00:04:39.150] So in the figure 18, the transition level is drawn along with the Vout, Vout=4, [00:04:44.010] when the diode is on. That is why VI greater than 4 volts, Vout=VI. [00:04:48.510] That means to say, if VI is greater than 4 volts, then VA greater than 4 volts, [00:04:55.560] diode is off. How do you represent that? Open circuit. [00:04:59.580] So if the VI greater than 4, open circuit is there.

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Transcript block 2 (source interval 000300-000599 seconds): [00:05:05.701] Greater than 4, less than 4, 4 volts, greater than 4. [00:05:10.260] So we need to represent with a waveform there. So the clipped off part is this. [00:05:16.050] So we clipped off a negative part, but there is a 4 volts remaining as it is, given [00:05:19.890] from T/2 to T. So this is with respect to parallel clipper. [00:05:24.841] Now, we'll repeat the same numerical using a silicon diode. [00:05:30.180] VK is equal to 0.7. [00:05:33.360] During the transition voltage, using the ID is equal to 0 or VD is equal to 0.7 [00:05:38.940] volts and draw current in the network as shown in the figure 19. [00:05:42.660] So we represent along with applied voltage, the voltage drop across the diode [00:05:48.570] is 0.7.

In the previous case, it was 0, VD was 0. [00:05:52.230] Now this will add with the positive half cycle of the input. [00:05:56.281] The total voltage applied is, we'll add or subtract, so 3.3. [00:06:00.451] It's subtract here because I have minus plus or plus minus. It's 3.3. [00:06:03.990] In the previous case, it was 4 volts there. So if you look at here, it's 4 volts. [00:06:07.920] Whereas now, the total voltage, sum of the voltage here is 4 minus 0.7 is 3.3 we [00:06:12.930] get. Then during the negative half cycle what we notice is, same as an input. [00:06:18.031] The input voltage is 16 [00:06:21.930] here, and hence we follow up to 16 the same waveform, and T/2 to T, the same [00:06:27.630] 3.3 will be continued. [00:06:31.470] Now let us go with the transition diagrams for different type of input and [00:06:37.440] different configuration. The summary of the series clipper and the parallel [00:06:42.240] clipper. So in a summary, we see that consider a positive clipper [00:06:48.151] here. In a positive clipper, the positive part is missing here. [00:06:52.350] The circuit is this. Similarly, in the negative part, negative clipper is missing. [00:06:57.000] Hence we call it as a negative clipper. [00:07:00.660] Then, [00:07:04.350] coming to the bias. So this was for an unbiased setup. [00:07:09.000] Now summary for a biased with an ideal diode. [00:07:12.210] So we see that there is a peak in the voltage here instead of 0. [00:07:15.420] The positive was missing in the previous case. [00:07:17.490] Now positive is there, but it got clipped off because of the voltage available [00:07:22.020] here. Presence of the voltage over here. [00:07:24.300] Similarly, the negative part is missing, but it is at the upper cycle. [00:07:27.901] So positive part is also missing along with the negative. [00:07:30.030] This is because of the transition happened because of the clipper. [00:07:33.420] Because of the voltage applied there. [00:07:36.120] Then come to the simple parallel clipper with ideal diodes. [00:07:40.050] The positive part is missing, positive clipper. [00:07:41.820] Negative part is missing, negative clipper. [00:07:46.170] So this is about the series and the parallel clipper. Thank you.

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Full transcript: 1314 normalized words, 109 timestamped segments, 469.973 seconds duration. Low-confidence segments flagged: 0.

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12 Clampers using Ideal diode

This lesson is indexed from the source video titled Clampers using Ideal diode. Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or rewritten. A candidate slide deck is grouped here according to the current matching score; the exact path and page locators are listed in the source evidence. EV0000090 · EV0000091

Graphs and diagrams

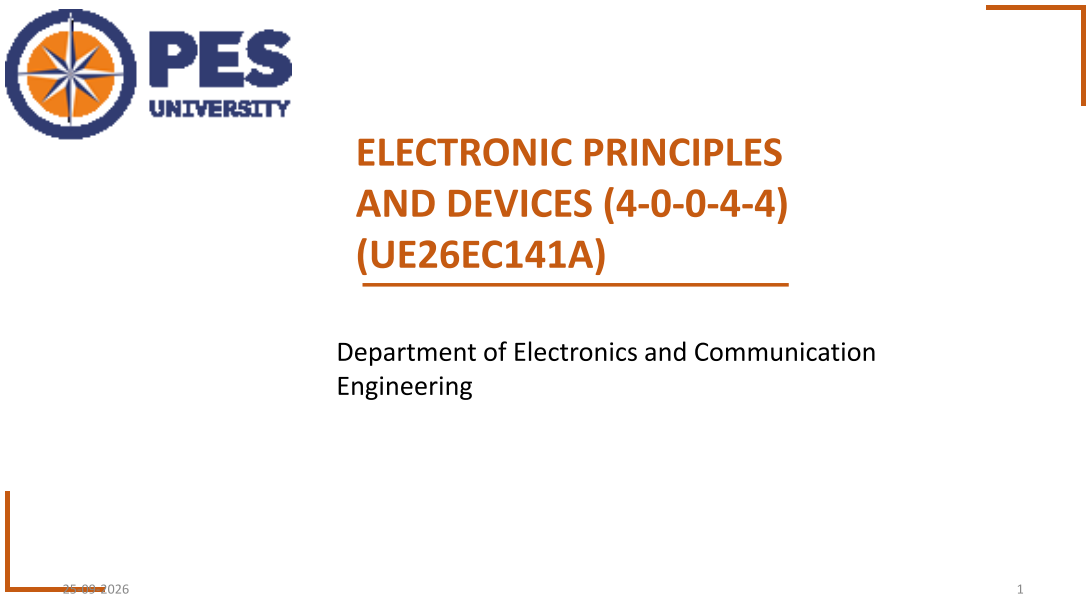

Figure G000120. Clampers using Ideal diode · source slide page 1.

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ELECTRONIC PRINCIPLES AND DEVICES

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Topic 11 - Clampers using Ideal diode

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Figure G000121. Clampers using Ideal diode · source slide page 2.

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ELECTRONIC PRINCIPLES AND DEVICES

Clampers

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- A clamper is a circuit with a diode, resistor, and capacitor that shifts a waveform to a new DC level without altering its shape.
- Also known as **DC restorer**.
- Clamping networks have a capacitor connected directly from input
- Resistive element connected in parallel with the output signal.
- The diode is also in parallel with the output signal but may or may not have a series dc supply as an added element.

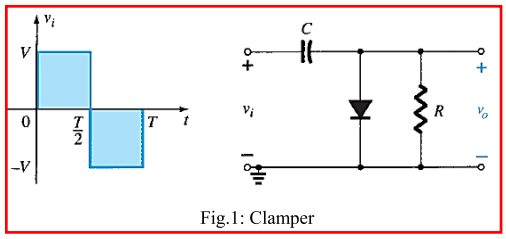

Fig.1: Clamper

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Figure G000122. Clampers using Ideal diode · source slide page 3.

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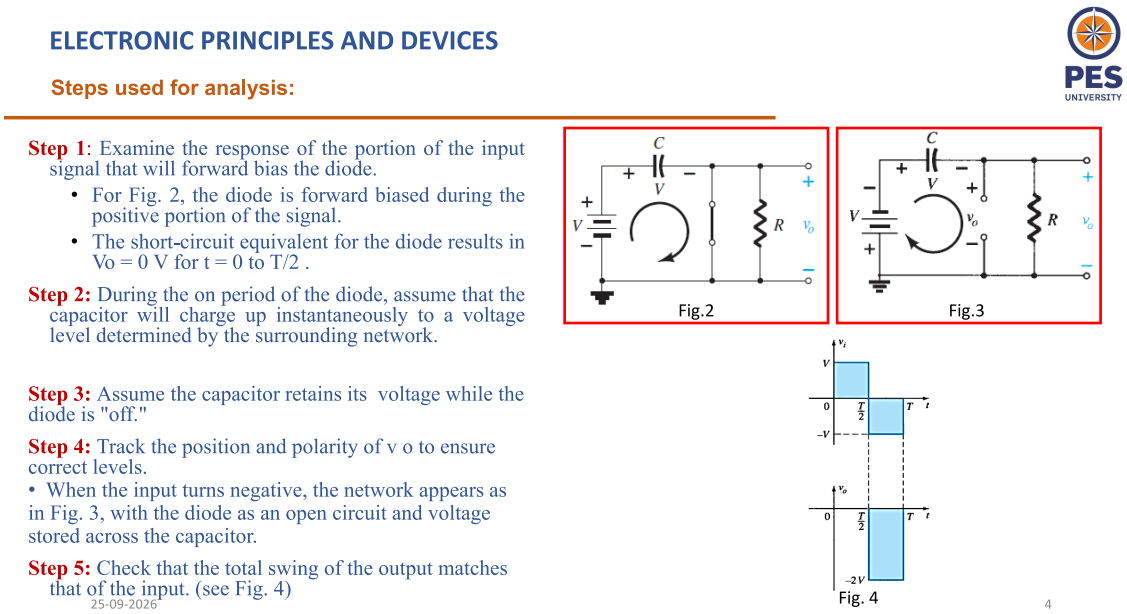

Figure G000123. Clampers using Ideal diode · source slide page 4.

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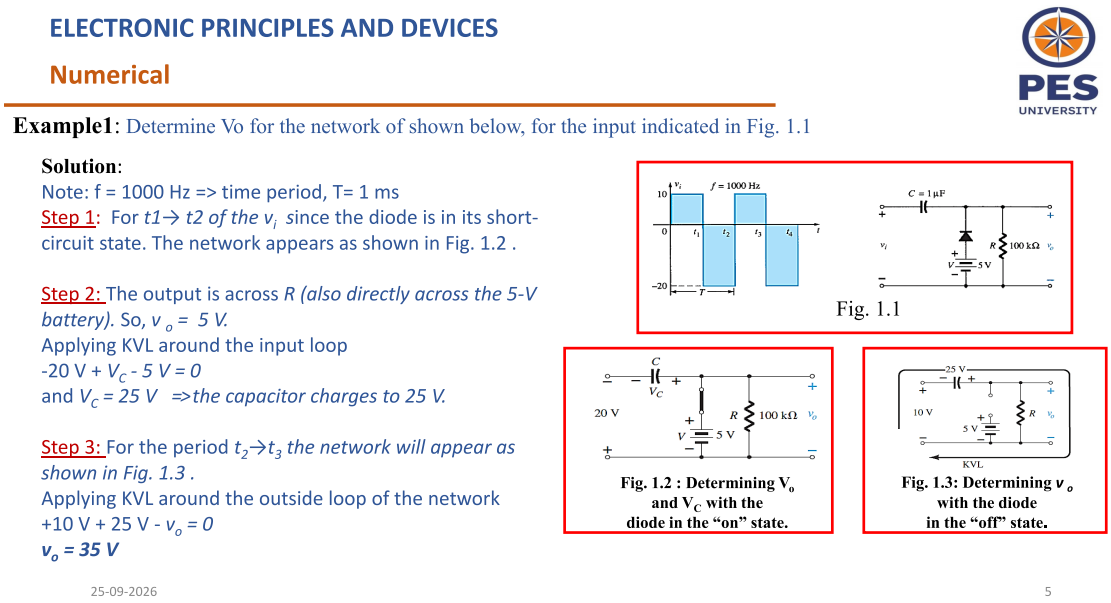

Figure G000124. Clampers using Ideal diode · source slide page 5.

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ELECTRONIC PRINCIPLES AND DEVICES

Numerical

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Step 4: The time constant of the network of Fig. 1.3 is determined by the product *RC and has the magnitude*
$\zeta = RC = (100\ k\Omega)(0.1\ \mu F) = 0.01\ s = 10\ ms$

The total discharge time is: $5\ \zeta = 5(10\ ms) = 50\ ms$

Step 5: The resulting output appears in Fig. 1.4 with the input signal.
Note: The output swing of 30 V matches the input swing.

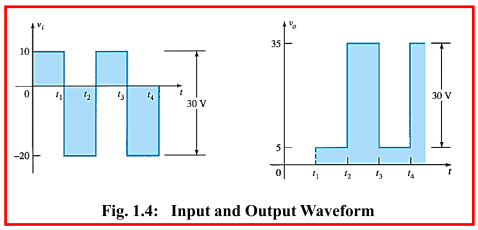

Fig. 1.4: Input and Output Waveform

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Figure G000125. Clampers using Ideal diode · source slide page 6.

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ELECTRONIC PRINCIPLES AND DEVICES

Numerical

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Example 2: Repeat example 1, using a silicon diode with $V_k = 0.7$ V.

Solution:

For the short-circuit state, the network appears as in Fig. 2.1 , and v_o *is determined by KVL in the output section as:*
$+5\ V - 0.7\ V - v_o = 0$
and $v_o = 4.3\ V$

For the input section, KVL results in
$-20\ V + V_C + 0.7\ V - 5\ V = 0$
and $V_C = 24.3\ V$

For the period $t_2 \rightarrow t_3$ *the network will now appear as in Fig. 2.2 , with the only change* being the voltage across the capacitor.
Applying KVL yields
$+10\ V + 24.3\ V - v_o = 0$
and $\boldsymbol{v_o = 34.3\ V}$

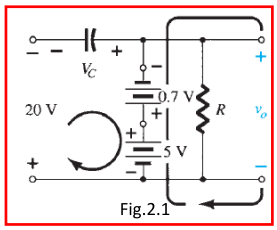

Fig.2.1

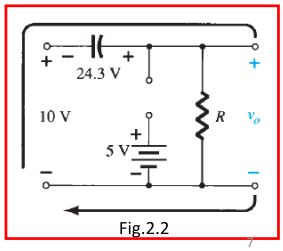

Fig.2.2

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Figure G000126. Clampers using Ideal diode · source slide page 7.

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ELECTRONIC PRINCIPLES AND DEVICES

Numerical

The resulting output appears in Fig. 2.3 , verifying the statement that the input and output swings are the same.

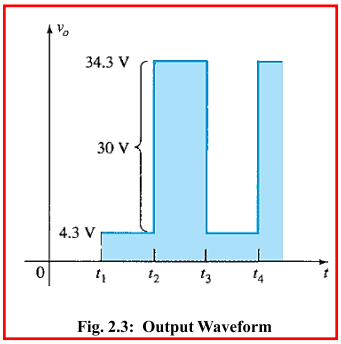

Fig. 2.3: Output Waveform

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Figure G000127. Clampers using Ideal diode · source slide page 8.

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ELECTRONIC PRINCIPLES AND DEVICES

Clamping Networks

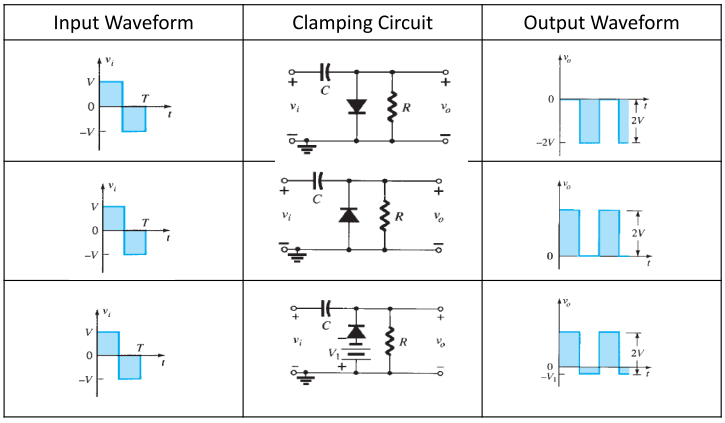

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Figure G000128. Clampers using Ideal diode · source slide page 9.

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Figure G000129. Clampers using Ideal diode · source slide page 10.

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Important points and why

1. Source slide page 1 (extracted text): ELECTRONIC PRINCIPLES AND DEVICES (4-0-0-4-4) (UE26EC141A) Department of Electronics and Communication Engineering 25-09-2026 1 EV0000459 · EV0000460 · EV0000461
2. Source slide page 2 (extracted text): ELECTRONIC PRINCIPLES AND DEVICES Topic 11 - Clampers using Ideal diode 25-09-2026 2 EV0000462 · EV0000463 · EV0000464
3. Source slide page 3 (extracted text): ELECTRONIC PRINCIPLES AND DEVICES Clampers • A clamper is a circuit with a diode, resistor, and capacitor that shifts a waveform to a new DC level without altering its shape. • Also known as DC restorer. • Clamping networks have a capacitor connected directly from input Fig.1: Clamper • Resistive element connected in parallel with the output signal. • The diode is also in parallel with the output signal but may or may not have a series dc supply as an added element. 25-09-2026 3 EV0000465 · EV0000466 · EV0000467

4. Source slide page 4 (extracted text): ELECTRONIC PRINCIPLES AND DEVICES Steps used for analysis: Step 1: Examine the response of the portion of the input signal that will forward bias the diode. • For Fig. 2, the diode is forward biased during the positive portion of the signal. • The short-circuit equivalent for the diode results in Vo = 0 V for t = 0 to T/2 . Step 2: During the on period of the diode, assume that the capacitor will charge up instantaneously to a voltage Fig.2 Fig.3 level determined by the surrounding network. Step 3: Assume the capacitor retains its voltage while the diode is "off." Step 4: Track the position and polarity of v o to ensure correct levels. • When the input turns negative, the network appears as in Fig. 3, with the diode as an open circuit and voltage stored across the capacitor. Step 5: Check that the total swing of the output matches that of the input. (see Fig. 4) Fig. 4 25-09-2026 4 EV0000468 · EV0000469 · EV0000470
5. Source slide page 5 (extracted text): ELECTRONIC PRINCIPLES AND DEVICES Numerical Example1: Determine Vo for the network of shown below, for the input indicated in Fig. 1.1 Solution: Note: f = 1000 Hz => time period, T= 1 ms Step 1: For t1→ t2 of the vi since the diode is in its short- circuit state. The network appears as shown in Fig. 1.2 . Step 2: The output is across R (also directly across the 5-V Fig. 1.1 battery). So, v o = 5 V. Applying KVL around the input loop -20 V + VC - 5 V = 0 and VC = 25 V => the capacitor charges to 25 V. Step 3: For the period t2→t3 the network will appear as shown in Fig. 1.3 . Fig. 1.2 : Determining Vo Fig. 1.3: Determining v o Applying KVL around the outside loop of the network and VC with the with the diode +10 V + 25 V - vo = 0 diode in the “on” state. in the “off” state. vo = 35 V 25-09-2026 5 EV0000471 · EV0000472 · EV0000473
6. Source slide page 6 (extracted text): ELECTRONIC PRINCIPLES AND DEVICES Numerical Step 4: The time constant of the network of Fig. 1.3 is determined by the product RC and has the magnitude ζ = RC = (100 kΩ)(0.1 μF) = 0.01 s = 10 ms The total discharge time is: 5 ζ = 5(10 ms) = 50 ms Step 5: The resulting output appears in Fig. 1.4 with the input Fig. 1.4: Input and Output Waveform signal. Note: The output swing of 30 V matches the input swing. 25-09-2026 6 EV0000474 · EV0000475 · EV0000476
7. Source slide page 7 (extracted text): ELECTRONIC PRINCIPLES AND DEVICES Numerical Example 2: Repeat example 1, using a silicon diode with Vk = 0.7 V. Solution: For the short-circuit state, the network appears as in Fig. 2.1 , and v o is determined by KVL in the output section as: +5 V - 0.7 V - vo = 0 and vo = 4.3 V Fig.2.1 For the input section, KVL results in -20 V + VC + 0.7 V - 5 V = 0 and VC = 24.3 V For the period t2 →t3 the network will now appear as in Fig. 2.2 , with the only change being the voltage across the capacitor. Applying KVL yields +10 V + 24.3 V - vo = 0 and vo = 34.3 V Fig.2.2 25-09-2026 7 EV0000477 · EV0000478 · EV0000479
8. Source slide page 8 (extracted text): ELECTRONIC PRINCIPLES AND DEVICES Numerical The resulting output appears in Fig. 2.3 , verifying the statement that the input and output swings are the same. Fig. 2.3: Output Waveform 25-09-2026 8 EV0000480 · EV0000481 · EV0000482
9. Source slide page 9 (extracted text): ELECTRONIC PRINCIPLES AND DEVICES Clamping Networks Input Waveform Clamping Circuit Output Waveform 25-09-2026 9 EV0000483 · EV0000484 · EV0000485
10. Source slide page 10 (extracted text): THANK YOU Department of Electronics and Communication 25-09-2026 10 EV0000486 · EV0000487 · EV0000488

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:01.891] Good morning and welcome to Electronics Principle and Devices. [00:00:07.410] I'm Dr. Ananda, [00:00:09.871] Associate Professor in the Department of Electronics and Communication. [00:00:13.320] Today, let us see on clipper. [00:00:15.930] Today, we'll see clamper. Last class, [00:00:21.930] we have seen the clipper circuits, the series, and parallel configurations. [00:00:27.631] So what we have seen in the last class was, the clipper is a circuit where the part [00:00:31.890] of the output has been clipped off without disturbing the other part of the [00:00:35.130] waveform. Now, let us go ahead with the clamper circuit. [00:00:39.661] A clamper is a circuit with the diode resistors and capacitor that shift the [00:00:44.580] waveform to a new DC level without altering its shape. [00:00:48.239] It's also known as a DC restorer. The clamping circuits have capacitor connected [00:00:52.680] directly from input, as shown in the figure here. [00:00:57.000] Resistor element connected in parallel with the output signal. As shown. [00:01:01.710] The diode is also in parallel [00:01:05.700] with the output signal, but may or may not have a series DC supply as added [00:01:11.700] element. So may or may not have a series DC element. [00:01:16.950] So now let us go ahead with analyzing the steps for clamper. [00:01:22.380] So step [00:01:24.571] one. Examine the response of portion of an input signal that will forward by the [00:01:29.220] diode. As usual, like how we did for a clipper to the positive of cycle where the [00:01:34.530] diode is on. During the negative cycle, where the diode is off, we need to find [00:01:37.380] out. So now, the figure two. For figure two, we see that the applied voltage [00:01:43.320] here positive, and the diode is on, and the capacitor also is charging from plus to [00:01:48.420] minus. So the diode is forward biased here, and we represent, since we are using an [00:01:53.160] ideal diode, here VD is equal to zero for an ideal diode. [00:02:08.190] VD is equal to zero. [00:02:12.660] The short circuit for a diode results in Vout is equal to zero for T is equal to [00:02:19.020] zero to T/2. So since this line is shorted, from here to here it is shorted. [00:02:24.780] So definitely the Vout is zero. So here VD is zero, and hence voltage across the [00:02:29.760] resistance also is zero. So in the waveform now, for positive cycle, we represent [00:02:35.520] here [00:02:36.990] nothing. That is zero line. The zero line is there. [00:02:40.470] So we represent the zero line here. [00:02:42.750] Step two, during the on period of the diode, assume that the capacitor will charge [00:02:46.650] up to instantaneously to a voltage level determined by the surrounding network. [00:02:50.700] So now this is charged a voltage of V. That's why we represent here V. [00:02:54.810] Step three, assume the capacitor retains its voltage while the diode is off. [00:02:58.620] Okay, when the diode is off, it should retain a voltage. [00:03:01.770] That means to say with the negative cycle of the input, the [00:03:06.030] negative cycle diode is off, and we are represent as open circuit here. [00:03:10.171] So in this open circuit, the capacitor is retained. [00:03:13.171] We need to remember, capacitor is retaining the charge over there. [00:03:16.620] Whereas during the positive cycle, capacitor is charged to voltage. It is charged. [00:03:21.900] So the capacitor has a property of charging and discharging and also holding. [00:03:25.410] So these three properties plays an important role in the way the capacitor will [00:03:29.370] work. So it will retain the [00:03:32.370] voltage. So assume the capacitor retain. [00:03:35.550] So step three, track the position and polarity of the Vout to ensure the correct [00:03:39.300] level. So when the input turns negative, the network appears as shown in the figure [00:03:43.080] three with the diode as an open circuit and a voltage stored across the capacitor. [00:03:47.070] Then the step five, check the voltage swing of the output matches that of an input. [00:03:51.990] So we see that it is minus 2. How we get this minus 2? Let us see. [00:03:58.020] So for that now, we need to apply the KVL here. [00:04:01.440] So when we apply the KVL, the

applied voltage is V here. [00:04:05.820] So we give V, minus V across, then minus V across the capacitor and [00:04:12.270] minus Vout. Apply the KVL, this loop actually. [00:04:16.560] So we need to apply the KVL, this loop. Okay? So what is Vout is equal to? [00:04:22.531] Vout is equal to minus 2V. [00:04:29.040] Here represented as 2V. This is because of the capacitor. [00:04:32.490] So the previous case in the clipper, we looked at, we didn't have a capacitor. [00:04:36.870] So here capacitor is an added advantage where it stores the previous value, retains [00:04:40.410] the voltage. And hence, during the negative of cycle, the diode is off. [00:04:45.091] Since it's retaining the voltage, when you apply the KVL, it add it to the applied [00:04:48.480] voltage. And hence, Vout is equal to minus 2 volts. [00:04:53.880] Then let us solve a numerical for a different set of inputs being given over there. [00:04:59.520] I have a series of the input here, not

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Transcript block 2 (source interval 000300-000599 seconds): [00:05:02.880] one cycle. We have a two cycles here being given here and is represented as zero to [00:05:09.480] T1, T2, T3, and T4. [00:05:13.710] So we need to apply this signal to a clamper circuit, which has a [00:05:19.620] bias voltage also here, five volts being given with R is equal to 100 kilo ohm. [00:05:24.150] Let us start. [00:05:26.040] And there is also a given supply voltage here. [00:05:45.810] Sorry. Let us go back to the previous circuit. [00:05:51.690] Sorry. Yeah, we'll go to this. [00:05:57.240] So now note that F is equal to 1,000 hertz, so time period is equal to one [00:06:02.460] milliseconds. So to plot the output waveform, we need to note the [00:06:08.370] given T period here. Therefore, from T1 to [00:06:15.091] T2, there's a negative off cycle. Whereas from zero to T1, input voltage Vi make [00:06:20.701] the diode off, where the diode is off open circuit and Vout is equal to Vi. [00:06:27.331] So that is been taken care. But now from T1 to T2 of the wave, since diode is in [00:06:33.300] short circuit state, the network appears as shown in the figure 1.2. [00:06:37.350] So this is what T1 to T2 diode is in short circuit here. [00:06:40.800] So the output voltage is across R. So Vout is equal to five volts now, because here [00:06:46.561] VD is equal to zero. So VD is equal to zero. This will add to the 5 volts. [00:06:51.540] So 0 + 5, so V output is equal to 5 volts now. [00:06:57.480] This is for the period T1 to T2. [00:07:00.540] Similarly, apply the KVL, then find out. So this is been shown. [00:07:03.630] Similarly now, apply the KVL across this. [00:07:06.091] If I apply the KVL across this, minus 20, the input voltage, plus [00:07:12.120] VC, minus 5 volts, 0. So what is the voltage across capacitor? Twenty-five volts. [00:07:17.670] So capacitor charges to 25 volts. [00:07:20.970] Coming to step three. For the period from T2 to T3, the [00:07:26.311] next half cycle. The network will appear as shown in the figure 1.3 now. [00:07:31.710] That means that the diode is open circuit, and this part is, even though it's [00:07:35.820] present, it is absent. So now we need to apply the KVL to this loop. Okay? [00:07:41.160] So if I apply the KVL across this loop, plus 10 plus 25 minus Vout is equal to 0, [00:07:47.040] and hence Vout is equal to 35 volts. [00:07:50.701] So Vout is 35 volts now. So step four. [00:07:56.010] Now Vout. So what all we represented till now? [00:07:58.410] So T1 to T2 is 5 volts we represented, this is 35 volts we represented, and this [00:08:03.360] again, 5 volt T3 to T4, we need to find that. So let us see that. [00:08:09.300] But before that, we need to plot with a time duration. [00:08:12.390] So we calculate the time duration of 50 millisecond. [00:08:15.300] Then the resulting output appears in 1.4, the [00:08:19.530] input signal here. So we have calculated the output for all the steps from T1 to [00:08:24.841] T4. This is how it look like [00:08:28.951] the output for a clipper circuit. [00:08:32.880] Now, let us look at the example number two. [00:08:34.800] Repeat the example one using silicon diode. [00:08:37.201] So in the place of an ideal

diode, we have VK is equal to 0.7. [00:08:40.651] So replace the ideal diode by a 0.7 here. So let us start. [00:08:45.240] So for a short circuit, so for T1 to T2 first. [00:08:49.110] So Vout is equal to instead of 5 volts, we get 4.3. [00:08:51.931] We need to check that, because along with the 5 volts, we have 0.7 negative there. [00:08:56.580] So for the input section, KVL results in VC is equal to 24. [00:09:00.510] Then similarly, the charge stored, apply the KVL across the... [00:09:03.960] So instead of 25 volts, we get 24.3 as compared to the previous. [00:09:08.580] So for a period from T2 to T3, the network will now appear as shown in the figure [00:09:12.630] 2.2, with only the change being the voltage across the capacitor. [00:09:17.401] Correct. So apply the KVL. So it results in 34.3 output. [00:09:22.200] Because with the capacitor the short voltage is 24.3. [00:09:25.681] The previous case it was 25, and hence the voltage was 35 total. [00:09:30.000] Now the total voltage will be 34.3. 0.7 is reduced. [00:09:35.760] So that's how we represent the waveform here. [00:09:37.890] So now, so the voltage reduced value of the first is 4.3, and it is 34.3, [00:09:43.800] and again 4.3 is for the cycle from T3 to T4. [00:09:48.030] So the resulting voltage appears in figure 2.3, verifying the statement that input [00:09:51.570] and output things are the same. [00:09:54.660] Then there are various clamping networks are there, and we have to recognize.

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Transcript block 3 (source interval 000600-000899 seconds): [00:10:00.090] So for input of the signal input waveform and a clamping circuit, we get a negative [00:10:06.120] part here. The positive is clipped off here. [00:10:08.311] Similarly, for the diode, if you invert it, we get to the opposite manner. [00:10:12.901] Similarly, the diode is connected in this with the DC bias. [00:10:16.440] The negative part is been added here. The positive is there. [00:10:19.020] The negative is added over there. [00:10:22.410] Thank you.

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Full transcript: 1642 normalized words, 137 timestamped segments, 624.384 seconds duration. Low-confidence segments flagged: 0.

EV0000090 · EV0000091 · EV0000092 · EV0000093

13 Source slide deck: Experiential learning Unit 1

This entry preserves the complete slide deck titled Experiential learning Unit 1 as searchable extracted page text and page-faithful rendered images. Its association with a particular video is not confirmed; it remains a separate topic under Electronic Principles and Devices, Unit_01, to avoid an unsupported match. EV0000489 · EV0000490

Graphs and diagrams

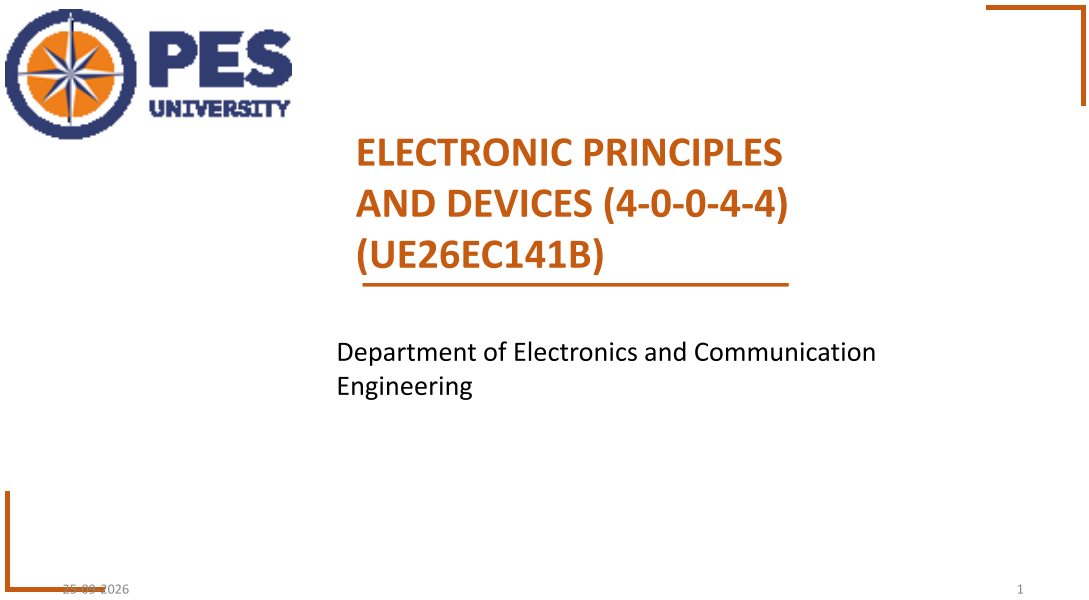

Figure G000130. Experiential learning Unit 1 · source slide page 1.

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How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

PES UNIVERSITY

ELECTRONIC PRINCIPLES AND DEVICES
(UE26EC141B)

Unit-1 –Experiential Learning-

Physically show all components in class: Passive and active components with Bread-board usage.
Signal-generator and measuring devices like Multimeter and Digital-Oscilloscope.
Give a task of measuring Resistance, Voltage, current etc

Department of Electronics and Communication.

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Figure G000131. Experiential learning Unit 1 · source slide page 2.

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ELECTRONIC PRINCIPLES AND DEVICES
Unit-1 –Experiential Learning-

PES UNIVERSITY

Components/ Equipment's required:
(i) (a) List of Passive Components

1. Resistors : 220Ω, 680Ω, 560Ω, 1MΩ

2. Capacitor : 1µF, 22µF

3. Potentiometer: 1KΩ, 10KΩ

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Figure G000132. Experiential learning Unit 1 · source slide page 3.

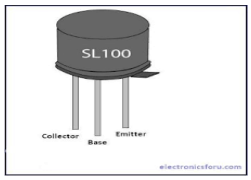
EV0000496

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

• (i) b List of Active components

(1) Diodes: BY127, IN4000

(2) BJT : SL100, CL100



(3) MOSFET: BS170, 2N700.

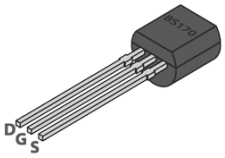


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Figure G000133. Experiential learning Unit 1 · source slide page 4.

EV0000499

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

• (i) c List of Measuring instruments & Functionality

(1) Multimeter : To measure Voltage, Current, Resistance etc

(2) Cathode ray oscilloscope(CRO): To measure voltage(p-p) and current and also to identify the type of waveform and compare it.

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Figure G000134. Experiential learning Unit 1 · source slide page 5.

EV0000502

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

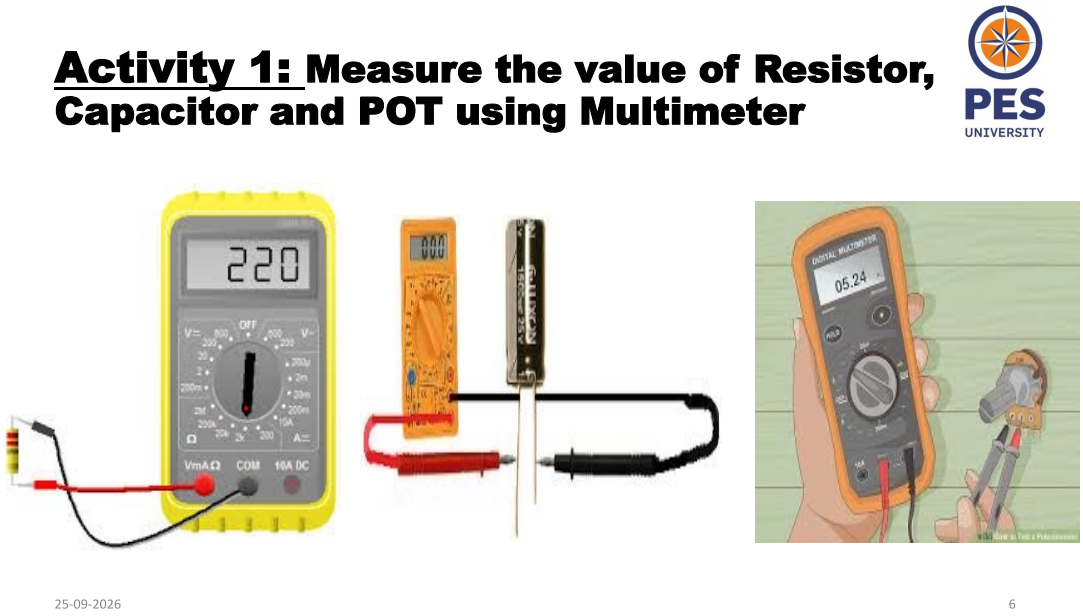

Figure G000135. Experiential learning Unit 1 · source slide page 6.

EV0000505

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

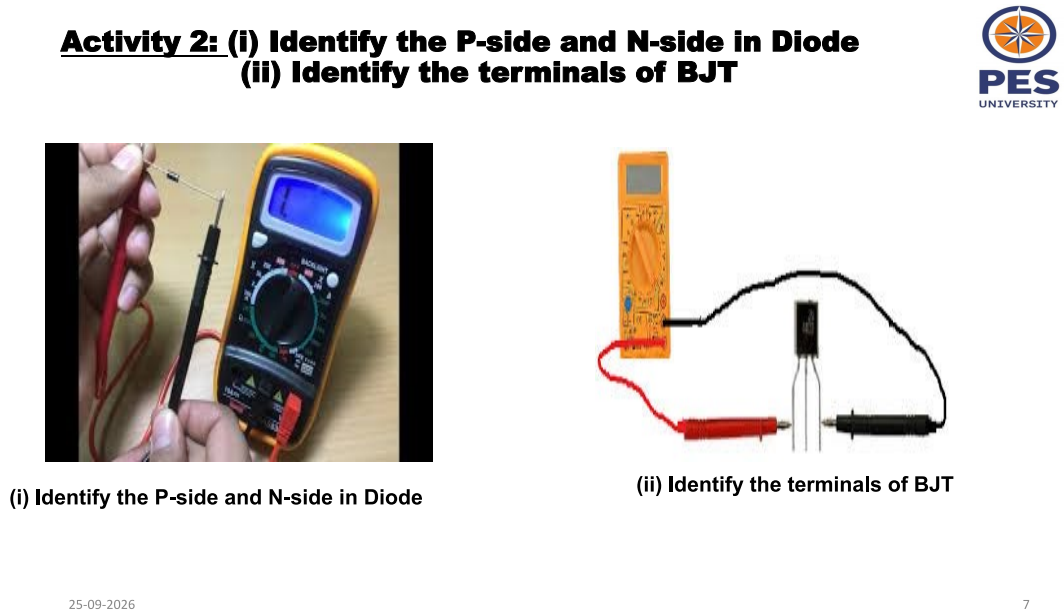

Figure G000136. Experiential learning Unit 1 · source slide page 7.

EV0000508

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Activity 3: How to set the input the input value like voltage & frequency in signal generator using CRO

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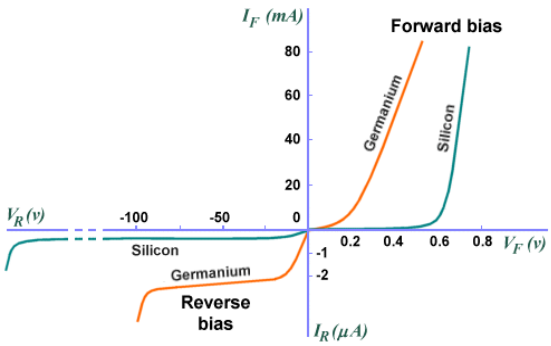
Figure G000137. Experiential learning Unit 1 · source slide page 8.

EV0000511

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Activity 4: Measure the voltage & current by forward biasing Diode/BJT using Multimeter



25-09-2026 9

Figure G000138. Experiential learning Unit 1 · source slide page 9.

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How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

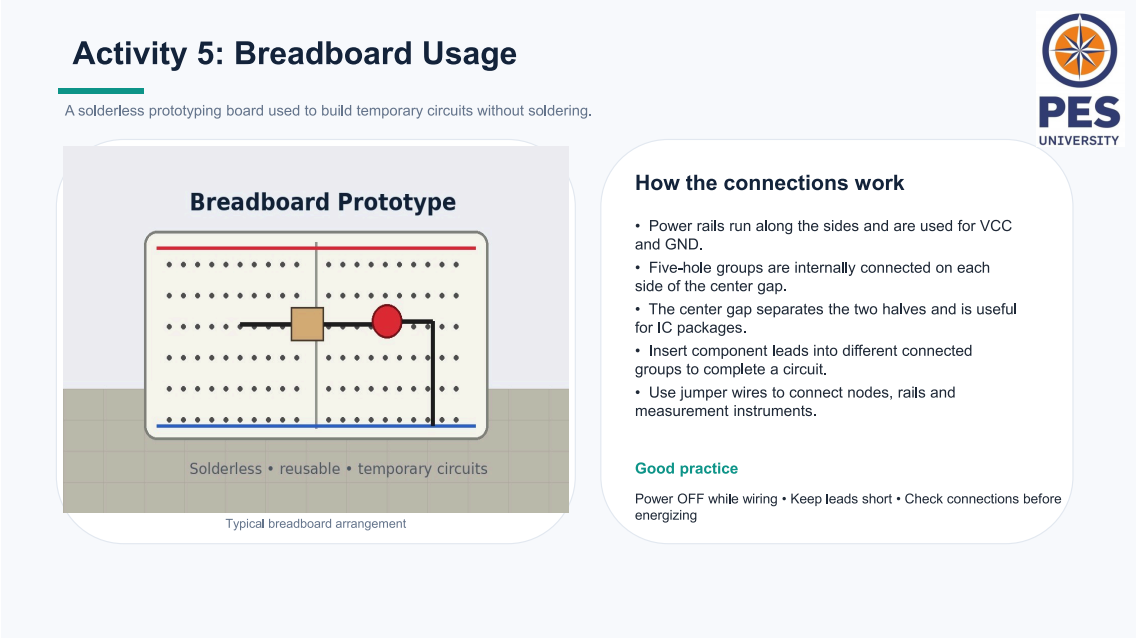

Figure G000139. Experiential learning Unit 1 · source slide page 10.

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How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Figure G000140. Experiential learning Unit 1 · source slide page 11.

EV0000520

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Important points and why

1. Source slide page 1 (extracted text): ELECTRONIC PRINCIPLES AND DEVICES (4-0-0-4-4) (UE26EC141B) Department of Electronics and Communication Engineering 25-09-2026 1 EV0000489 · EV0000490 · EV0000491
2. Source slide page 2 (extracted text): ELECTRONIC PRINCIPLES AND DEVICES (UE26EC141B) Unit-1 –Experiential Learning- Physically show all components in class: Passive and active components with Bread-board usage. Signal-generator and measuring devices like Multimeter and Digital-Oscilloscope. Give a task of measuring Resistance, Voltage, current etc Department of Electronics and Communication. 25-09-2026 2 EV0000492 · EV0000493 · EV0000494
3. Source slide page 3 (extracted text): ELECTRONIC PRINCIPLES AND DEVICES Unit-1 –Experiential Learning- Components/ Equipment's required: (i) (a) List of Passive Components 1. Resistors : 220Ω, 680Ω, 560Ω, 1MΩ 2. Capacitor : 1µF, 22µF 3. Potentiometer: 1KΩ, 10KΩ 25-09-2026 3 EV0000495 · EV0000496 · EV0000497
4. Source slide page 4 (extracted text): • (i) b List of Active components (1) Diodes: BY127, IN4000 (2) BJT : SL100, CL100 (3) MOSFET: BS170, 2N700. 25-09-2026 4 EV0000498 · EV0000499 · EV0000500
5. Source slide page 5 (extracted text): • (i) c List of Measuring instruments & Functionality (1) Multimeter : To measure Voltage, Current, Resistance etc (2) Cathode ray oscilloscope(CRO): To measure voltage(p-p) and current and also to identify the type of waveform and compare it. 25-09-2026 5 EV0000501 · EV0000502 · EV0000503
6. Source slide page 6 (extracted text): Activity 1: Measure the value of Resistor, Capacitor and POT using Multimeter 25-09-2026 6 EV0000504 · EV0000505 · EV0000506
7. Source slide page 7 (extracted text): Activity 2: (i) Identify the P-side and N-side in Diode (ii) Identify the terminals of BJT (ii) Identify the terminals of BJT (i) Identify the P-side and N-side in Diode 25-09-2026 7 EV0000507 · EV0000508 · EV0000509
8. Source slide page 8 (extracted text): Activity 3:How to set the input the input value like voltage & frequency in signal generator using CRO 25-09-2026 8 EV0000510 · EV0000511 · EV0000512
9. Source slide page 9 (extracted text): Activity 4: Measure the voltage & current by forward biasing Diode/BJT using Multimeter 25-09-2026 9 EV0000513 · EV0000514 · EV0000515
10. Source slide page 10 (extracted text): Activity 5: Breadboard Usage A solderless prototyping board used to build temporary circuits without soldering. How the connections work • Power rails run along the sides and are used for VCC and GND. • Five-hole groups are internally connected on each side of the center gap. • The center gap separates the two halves and is useful for IC packages. • Insert component leads into different connected groups to complete a circuit. • Use jumper wires to connect nodes, rails and measurement instruments. Good practice Power OFF while wiring • Keep leads short • Check connections before energizing Typical breadboard arrangement EV0000516 · EV0000517 · EV0000518
11. Source slide page 11 (extracted text): THANK YOU Department of Electronics and Communication 25-09-2026 11 EV0000519 · EV0000520 · EV0000521

SHORT-NOTE RECAP

Source deck preserved: Experiential learning Unit 1. Its extracted page text is listed as searchable points and all rendered source pages are attached as figures. Match to a specific video is candidate-based; this index does not claim a manual content review.

EV0000489

14 Source slide deck: Experiential learning Unit 1

This entry preserves the complete slide deck titled Experiential learning Unit 1 as searchable extracted page text and page-faithful rendered images. Its association with a particular video is not confirmed; it remains a separate topic under Electronic Principles and Devices, Unit_01, to avoid an unsupported match. EV0000522 · EV0000523

Graphs and diagrams

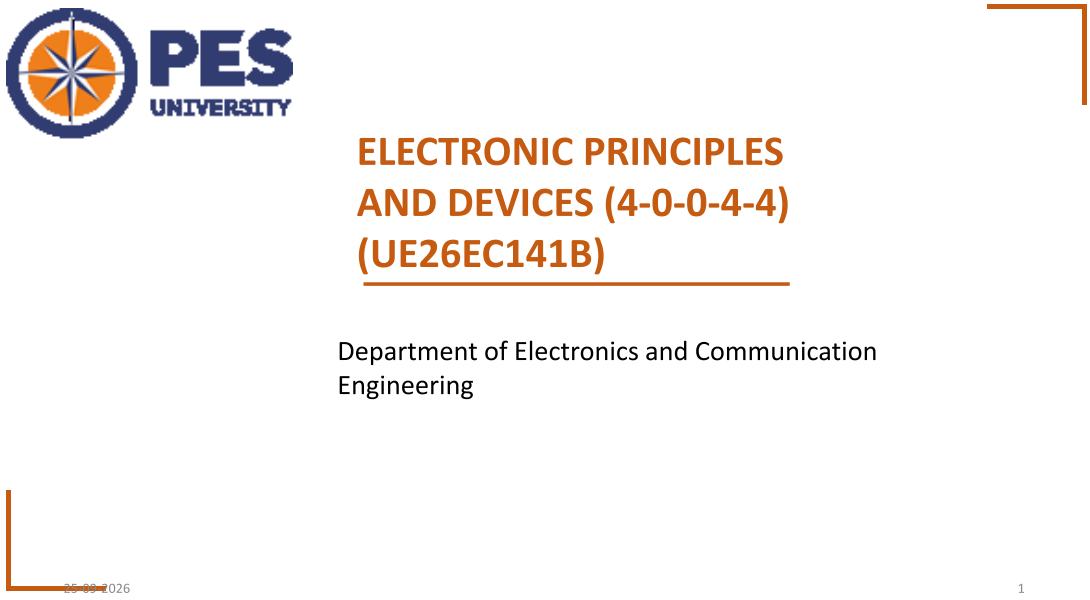

Figure G000141. Experiential learning Unit 1 · source slide page 1.

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How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

PES UNIVERSITY

ELECTRONIC PRINCIPLES AND DEVICES
(UE26EC141B)

Unit-1 –Experiential Learning-

Physically show all components in class: Passive and active components with Bread-board usage.
Signal-generator and measuring devices like Multimeter and Digital-Oscilloscope.
Give a task of measuring Resistance, Voltage, current etc

Department of Electronics and Communication.

25-09-2026 2

Figure G000142. Experiential learning Unit 1 · source slide page 2.

EV0000526

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ELECTRONIC PRINCIPLES AND DEVICES
Unit-1 –Experiential Learning-

PES UNIVERSITY

Components/ Equipment's required:
(i) (a) List of Passive Components

1. Resistors : 220Ω, 680Ω, 560Ω, 1MΩ

2. Capacitor : 1µF, 22µF

3. Potentiometer: 1KΩ, 10KΩ

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Figure G000143. Experiential learning Unit 1 · source slide page 3.

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How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

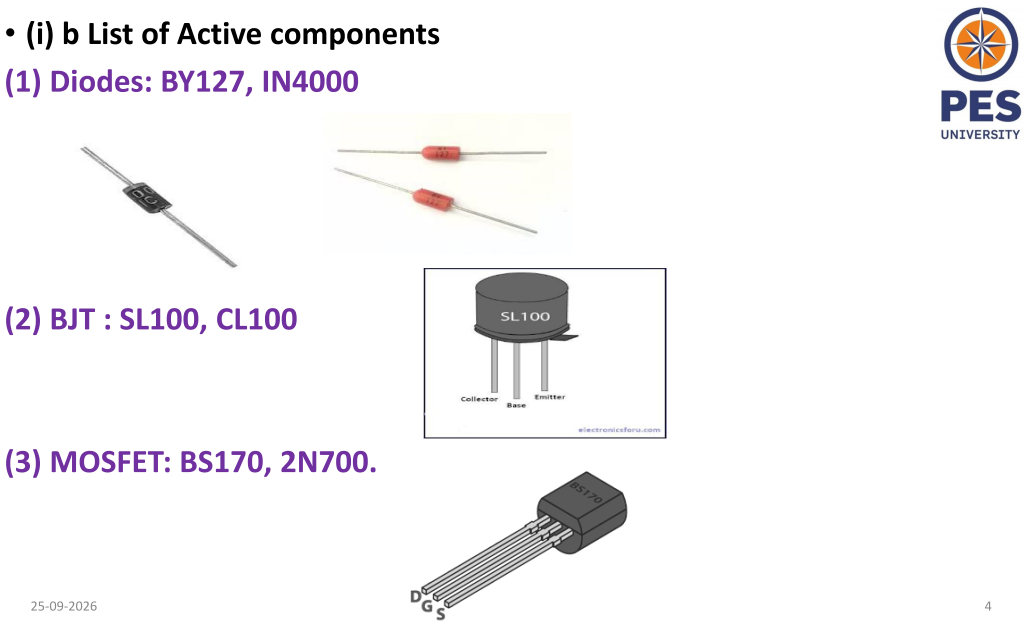


Figure G000144. Experiential learning Unit 1 · source slide page 4.

EV0000532

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

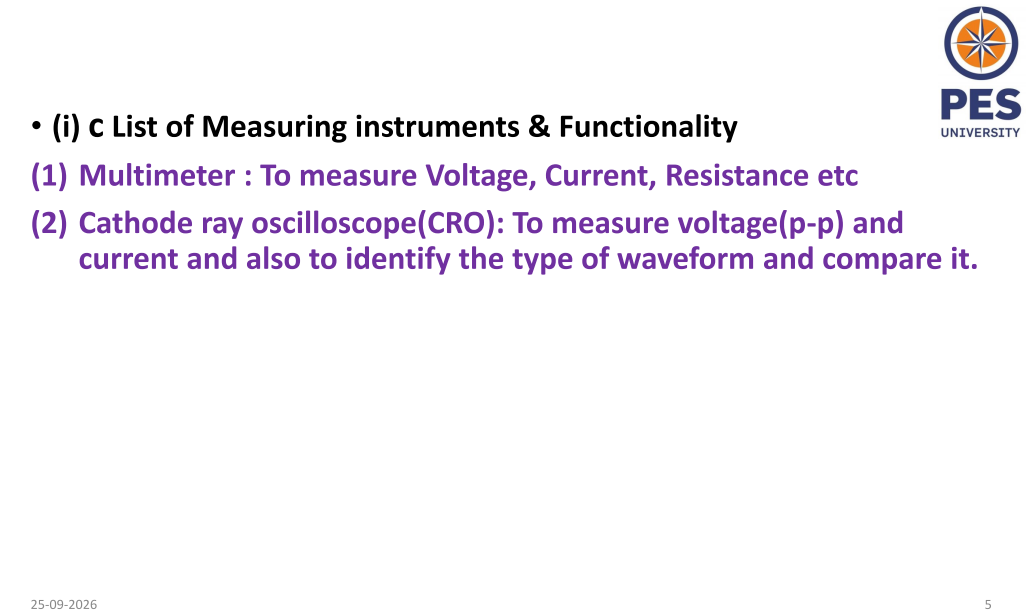


Figure G000145. Experiential learning Unit 1 · source slide page 5.

EV0000535

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

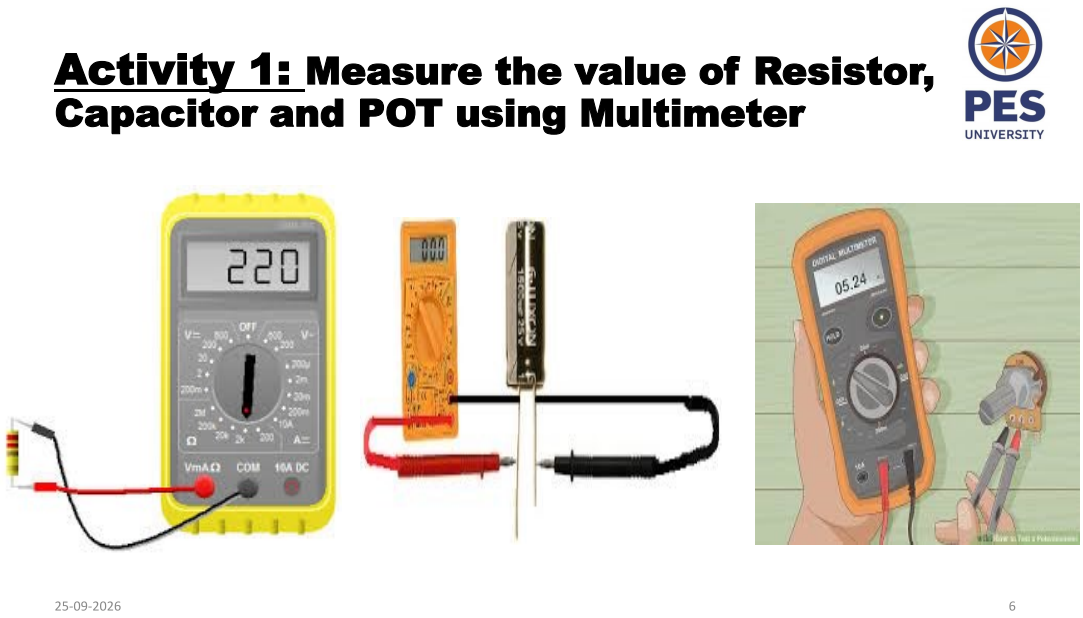


Figure G000146. Experiential learning Unit 1 · source slide page 6.

EV0000538

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

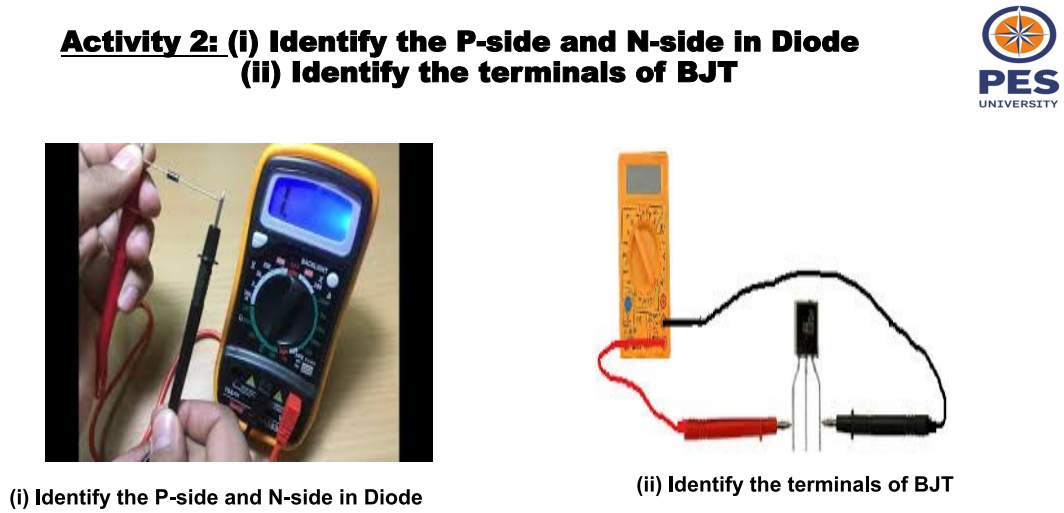


Figure G000147. Experiential learning Unit 1 · source slide page 7.

EV0000541

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Activity 3: How to set the input the input value like voltage & frequency in signal generator using CRO

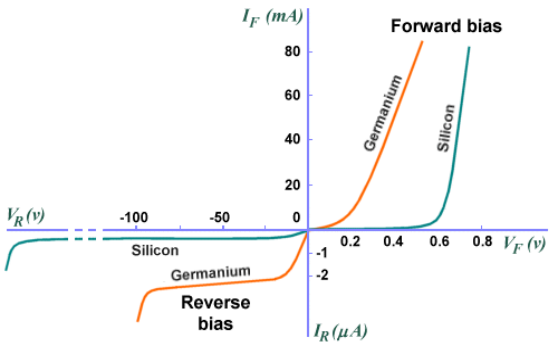
25-09-2026 8

Figure G000148. Experiential learning Unit 1 · source slide page 8.

EV0000544

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Activity 4: Measure the voltage & current by forward biasing Diode/BJT using Multimeter



25-09-2026 9

Figure G000149. Experiential learning Unit 1 · source slide page 9.

EV0000547

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

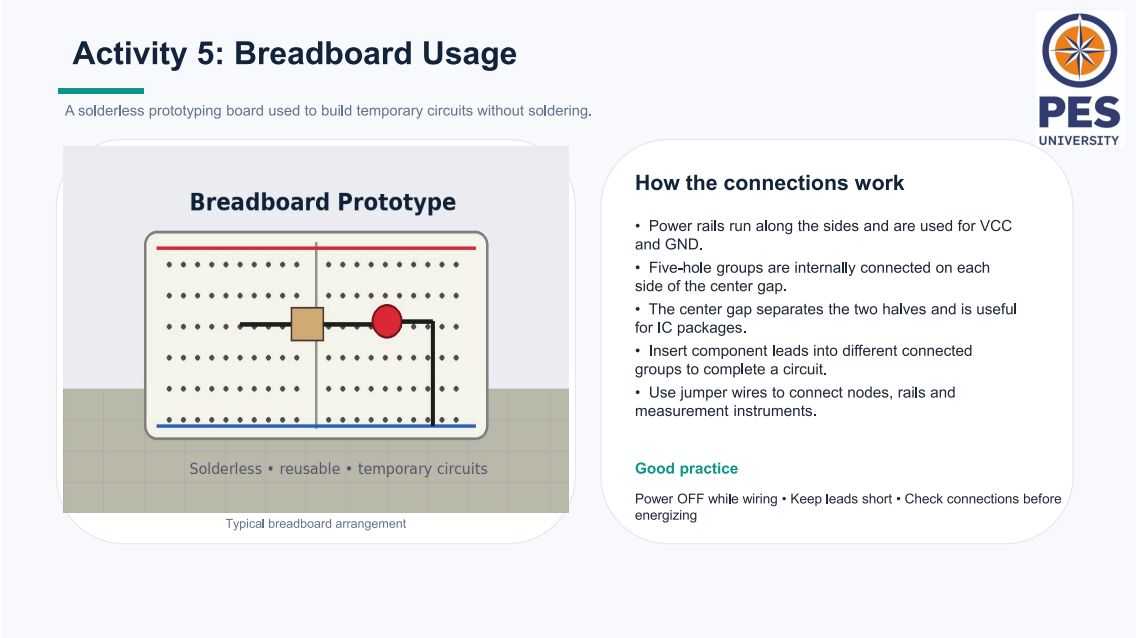


Figure G000150. Experiential learning Unit 1 · source slide page 10.

EV0000550

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Figure G000151. Experiential learning Unit 1 · source slide page 11.

EV0000553

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Important points and why

1. Source slide page 1 (extracted text): ELECTRONIC PRINCIPLES AND DEVICES (4-0-0-4-4) (UE26EC141B) Department of Electronics and Communication Engineering 25-09-2026 1 EV0000522 · EV0000523 · EV0000524
2. Source slide page 2 (extracted text): ELECTRONIC PRINCIPLES AND DEVICES (UE26EC141B) Unit-1 –Experiential Learning- Physically show all components in class: Passive and active components with Bread-board usage. Signal-generator and measuring devices like Multimeter and Digital-Oscilloscope. Give a task of measuring Resistance, Voltage, current etc Department of Electronics and Communication. 25-09-2026 2 EV0000525 · EV0000526 · EV0000527
3. Source slide page 3 (extracted text): ELECTRONIC PRINCIPLES AND DEVICES Unit-1 –Experiential Learning- Components/ Equipment's required: (i) (a) List of Passive Components 1. Resistors : 220Ω, 680Ω, 560Ω, 1MΩ 2. Capacitor : 1µF, 22µF 3. Potentiometer: 1KΩ, 10KΩ 25-09-2026 3 EV0000528 · EV0000529 · EV0000530
4. Source slide page 4 (extracted text): • (i) b List of Active components (1) Diodes: BY127, IN4000 (2) BJT : SL100, CL100 (3) MOSFET: BS170, 2N700. 25-09-2026 4 EV0000531 · EV0000532 · EV0000533
5. Source slide page 5 (extracted text): • (i) c List of Measuring instruments & Functionality (1) Multimeter : To measure Voltage, Current, Resistance etc (2) Cathode ray oscilloscope(CRO): To measure voltage(p-p) and current and also to identify the type of waveform and compare it. 25-09-2026 5 EV0000534 · EV0000535 · EV0000536
6. Source slide page 6 (extracted text): Activity 1: Measure the value of Resistor, Capacitor and POT using Multimeter 25-09-2026 6 EV0000537 · EV0000538 · EV0000539
7. Source slide page 7 (extracted text): Activity 2: (i) Identify the P-side and N-side in Diode (ii) Identify the terminals of BJT (ii) Identify the terminals of BJT (i) Identify the P-side and N-side in Diode 25-09-2026 7 EV0000540 · EV0000541 · EV0000542
8. Source slide page 8 (extracted text): Activity 3:How to set the input the input value like voltage & frequency in signal generator using CRO 25-09-2026 8 EV0000543 · EV0000544 · EV0000545
9. Source slide page 9 (extracted text): Activity 4: Measure the voltage & current by forward biasing Diode/BJT using Multimeter 25-09-2026 9 EV0000546 · EV0000547 · EV0000548
10. Source slide page 10 (extracted text): Activity 5: Breadboard Usage A solderless prototyping board used to build temporary circuits without soldering. How the connections work • Power rails run along the sides and are used for VCC and GND. • Five-hole groups are internally connected on each side of the center gap. • The center gap separates the two halves and is useful for IC packages. • Insert component leads into different connected groups to complete a circuit. • Use jumper wires to connect nodes, rails and measurement instruments. Good practice Power OFF while wiring • Keep leads short • Check connections before energizing Typical breadboard arrangement EV0000549 · EV0000550 · EV0000551
11. Source slide page 11 (extracted text): THANK YOU Department of Electronics and Communication 25-09-2026 11 EV0000552 · EV0000553 · EV0000554

SHORT-NOTE RECAP

Source deck preserved: Experiential learning Unit 1. Its extracted page text is listed as searchable points and all rendered source pages are attached as figures. Match to a specific video is candidate-based; this index does not claim a manual content review.

EV0000522

15 Historical question-source pages · unit mapping candidates

This section preserves historical paper/question-bank pages whose extracted terms matched Electronic Principles and Devices, Unit_01 (Unit-1 Semiconductor Diode) above the dataset's candidate threshold. The association has not been manually confirmed, source text/OCR may be incomplete, and the pages are not rewritten or presented as solved questions. EV0000555 · EV0000556

Important points and why

1. Archive page 1 from 2025 [1]__10viKVTE9W.pdf; unit match candidate score 0.6667; source text/OCR may be partial. 3/25/25, 10:58 AM ESA - Jan 2025 - UE24EC141A (set- 1) about:blank 1/5 (8.0 Marks) (9.0 Marks) PES University, Bengaluru (Established under Karnataka Act 16 of 2013) END SEMESTER ASSESSMENT (ESA) - Jan-Feb UE24EC141A - Electronic Principles And Devices Total Marks : 100.0 1.a. (i) Determine the dynamic resistance of a diode at a temperature of 40 C and a forward bias voltage of 0.6V, if the reverse saturation current is 5pA at 300K. n = 1 (4M) (ii) Determine I and V in the circuit shown below (4M) 1.b. What are the different approximations used to represent the diode? For each of them draw the equivalent circuit and VI Characteristics o D 0 5.77 mA 0.46107 m ohms EV0000555
2. Archive page 1 from 2025 [2]__1LrJQqKVte.pdf; unit match candidate score 0.5833; source text/OCR may be partial. 3/25/25, 10:58 AM ESA - Jan 2025 - UE24EC141A (set- 2) about:blank 1/7 (6.0 Marks) (4.0 Marks) PES University, Bengaluru (Established under Karnataka Act 16 of 2013) END SEMESTER ASSESSMENT (ESA) - Jan-Feb UE24EC141A - Electronic Principles And Devices Total Marks : 100.0 1.a. Describe the Temperature Effects on V-I Characteristics of a Diode with Forward and Reverse biased and suitable diagrams? (Diagram-2 M + Forward bised - 2 M + Reverse biased -2 M) 1.b. Determine the voltages and current through 20Ω and 10Ω resistor and also the total Current (I) for the configuration (2M + 1M +1M) 19.3 1.93 19.3 0.965 2.895 EV0000556

SHORT-NOTE RECAP

2 historical archive pages are included as question-source references. Inspect the original paper before relying on OCR or unit placement; no answers are newly authored here.

EV0000555

A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
Official unit catalogue	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/units.json	Official course/unit title and index.
Introduction to semiconductors, intrinsic and extrinsic semiconductors transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/AV_Summaries/001_Introduction_to_semiconductors_intrinsic_and_extrinsic_semiconductors__183ac314557f22c2.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Introduction to semiconductors, intrinsic and extrinsic semiconductors transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/AV_Summaries/001_Introduction_to_semiconductors_intrinsic_and_extrinsic_semiconductors__183ac314557f22c2.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Introduction to semiconductors, intrinsic and extrinsic semiconductors transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/AV_Summaries/001_Introduction_to_semiconductors_intrinsic_and_extrinsic_semiconductors__183ac314557f22c2.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9968; only the JSON segmentation supplies the displayed timestamps.
Introduction to semiconductors, intrinsic and extrinsic semiconductors transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/AV_Summaries/001_Introduction_to_semiconductors_intrinsic_and_extrinsic_semiconductors__183ac314557f22c2.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Semiconductor diode under no bias, forward bias and reverse bias conditions transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/AV_Summaries/002_Semiconductor_diode_under_no_bias_forward_bias_and_reverse_bias_conditions__1070a788810cfac7.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Semiconductor diode under no bias, forward bias and reverse bias conditions transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/AV_Summaries/002_Semiconductor_diode_under_no_bias_forward_bias_and_reverse_bias_conditions__1070a788810cfac7.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Semiconductor diode under no bias, forward bias and reverse bias conditions transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/AV_Summaries/002_Semiconductor_diode_under_no_bias_forward_bias_and_reverse_bias_conditions__1070a788810cfac7.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9939; only the JSON segmentation supplies the displayed timestamps.
Semiconductor diode under no bias, forward bias and reverse bias conditions transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/AV_Summaries/002_Semiconductor_diode_under_no_bias_forward_bias_and_reverse_bias_conditions__1070a788810cfac7.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Diode characteristics, Shockley's equation transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/AV_Summaries/003_Diode_characteristics_Shockley_s_equation__09b9cf730275f786.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Diode characteristics, Shockley's equation transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/AV_Summaries/003_Diode_characteristics_Shockley_s_equation__09b9cf730275f786.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Diode characteristics, Shockley's equation transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/AV_Summaries/003_Diode_characteristics_Shockley_s_equation__09b9cf730275f786.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9985; only the JSON segmentation supplies the displayed timestamps.
Diode characteristics, Shockley's equation transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/AV_Summaries/003_Diode_characteristics_Shockley_s_equation__09b9cf730275f786.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Zener and Avalanche breakdown, Temperature effects transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/AV_Summaries/004_Zener_and_Avalanche_breakdown_Temperature_effects__83cce28872db58b1.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Zener and Avalanche breakdown, Temperature effects transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/AV_Summaries/004_Zener_and_Avalanche_breakdown_Temperature_effects__83cce28872db58b1.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Zener and Avalanche breakdown, Temperature effects transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/AV_Summaries/004_Zener_and_Avalanche_breakdown_Temperature_effects__83cce28872db58b1.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9980; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Zener and Avalanche breakdown, Temperature effects transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/AV_Summaries/004_Zener_and_Avalanche_breakdown_Temperature_effects__83cce28872db58b1.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Ideal versus Practical diode, Diode resistances transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/AV_Summaries/005_Ideal_versus_Practical_diode_Diode_resistances__3b08a07358669379.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Ideal versus Practical diode, Diode resistances transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/AV_Summaries/005_Ideal_versus_Practical_diode_Diode_resistances__3b08a07358669379.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Ideal versus Practical diode, Diode resistances transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/AV_Summaries/005_Ideal_versus_Practical_diode_Diode_resistances__3b08a07358669379.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9976; only the JSON segmentation supplies the displayed timestamps.
Ideal versus Practical diode, Diode resistances transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/AV_Summaries/005_Ideal_versus_Practical_diode_Diode_resistances__3b08a07358669379.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Diode equivalent circuits Series diode configurations transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/AV_Summaries/006_Diode_equivalent_circuits_Series_diode_configurations__91f64c479cb44c60.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Diode equivalent circuits Series diode configurations transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/AV_Summaries/006_Diode_equivalent_circuits_Series_diode_configurations__91f64c479cb44c60.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Diode equivalent circuits Series diode configurations transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/AV_Summaries/006_Diode_equivalent_circuits_Series_diode_configurations__91f64c479cb44c60.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9966; only the JSON segmentation supplies the displayed timestamps.
Diode equivalent circuits Series diode configurations transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/AV_Summaries/006_Diode_equivalent_circuits_Series_diode_configurations__91f64c479cb44c60.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Parallel and Series- Parallel configurations transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/AV_Summaries/007_Parallel_and_Series-_Parallel_configurations__f6ab675a2f96218e.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Parallel and Series- Parallel configurations transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/AV_Summaries/007_Parallel_and_Series-_Parallel_configurations__f6ab675a2f96218e.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Parallel and Series- Parallel configurations transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/AV_Summaries/007_Parallel_and_Series-_Parallel_configurations__f6ab675a2f96218e.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9981; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Parallel and Series- Parallel configurations transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/AV_Summaries/007_Parallel_and_Series-_Parallel_configurations__f6ab675a2f96218e.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Logical Operations using diode transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/AV_Summaries/008_Logical_Operations_using_diode__a5ae846222c4060c.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Logical Operations using diode transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/AV_Summaries/008_Logical_Operations_using_diode__a5ae846222c4060c.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Logical Operations using diode transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/AV_Summaries/008_Logical_Operations_using_diode__a5ae846222c4060c.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9977; only the JSON segmentation supplies the displayed timestamps.
Logical Operations using diode transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/AV_Summaries/008_Logical_Operations_using_diode__a5ae846222c4060c.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Diode Clippers : Series using Ideal diode transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/AV_Summaries/009_Diode_Clippers_Series_using_Ideal_diode__bd9b40ddcbd2d089.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Diode Clippers : Series using Ideal diode transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/AV_Summaries/009_Diode_Clippers_Series_using_Ideal_diode__bd9b40ddcbd2d089.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Diode Clippers : Series using Ideal diode transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/AV_Summaries/009_Diode_Clippers_Series_using_Ideal_diode__bd9b40ddcbd2d089.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9987; only the JSON segmentation supplies the displayed timestamps.
Diode Clippers : Series using Ideal diode transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/AV_Summaries/009_Diode_Clippers_Series_using_Ideal_diode__bd9b40ddcbd2d089.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Diode Clippers : Parallel using Ideal diode transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/AV_Summaries/010_Diode_Clippers_Parallel_using_Ideal_diode__b4558739e39a139a.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Diode Clippers : Parallel using Ideal diode transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/AV_Summaries/010_Diode_Clippers_Parallel_using_Ideal_diode__b4558739e39a139a.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Diode Clippers : Parallel using Ideal diode transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/AV_Summaries/010_Diode_Clippers_Parallel_using_Ideal_diode__b4558739e39a139a.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9977; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Diode Clippers : Parallel using Ideal diode transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/AV_Summaries/010_Diode_Clippers_Parallel_using_Ideal_diode__b4558739e39a139a.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Clampers using Ideal diode transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/AV_Summaries/011_Clampers_using_Ideal_diode__a50c9b8c3f829f04.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Clampers using Ideal diode transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/AV_Summaries/011_Clampers_using_Ideal_diode__a50c9b8c3f829f04.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Clampers using Ideal diode transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/AV_Summaries/011_Clampers_using_Ideal_diode__a50c9b8c3f829f04.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9988; only the JSON segmentation supplies the displayed timestamps.
Clampers using Ideal diode transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/AV_Summaries/011_Clampers_using_Ideal_diode__a50c9b8c3f829f04.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Slide page text: Introduction to semiconductors intrinsic and extrinsic semiconductors	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/001_Introduction_to_semiconductors_intrinsic_and_extrinsic_semiconductors__349c88a00d515771.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.

Source	Path	Use
Alternate extracted slide page text: Introduction to semiconductors intrinsic and extrinsic semiconductors	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/001_Introduction_to_semiconductors_intrinsic_and_extrinsic_semiconductors__349c88a00d515771.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Introduction to semiconductors intrinsic and extrinsic semiconductors (converted PDF; original hash 10836acb38251f1a6456c6ee7c31fb424695e79f4c7ec0bf52052c6c790e33e0)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/001_Introduction_to_semiconductors_intrinsic_and_extrinsic_semiconductors__349c88a00d515771.pdf	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Semiconductor diode under no bias forward bias and reverse bias conditions	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/002_Semiconductor_diode_under_no_bias_forward_bias_and_reverse_bias_conditions__63662d27d393ceed.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Semiconductor diode under no bias forward bias and reverse bias conditions	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/002_Semiconductor_diode_under_no_bias_forward_bias_and_reverse_bias_conditions__63662d27d393ceed.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Semiconductor diode under no bias forward bias and reverse bias conditions (converted PDF; original hash 7316e46912b815d4fabf620c52e93e433e55f0cd560ff2059481868325749cf2)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/002_Semiconductor_diode_under_no_bias_forward_bias_and_reverse_bias_conditions__63662d27d393ceed.pdf	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Diode characteristics Shockley s equation	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/003_Diode_characteristics_Shockley_s_equation__87b47de24c62e1d8.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.

Source	Path	Use
Alternate extracted slide page text: Diode characteristics Shockley s equation	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/003_Diode_characteristics_Shockley_s_equation__87b47de24c62e1d8.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Diode characteristics Shockley s equation (converted PDF; original hash 8175b4e63ef1b61c0f6e35bdc122fae79376c17503d529a988b2efaf83987ea2)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/003_Diode_characteristics_Shockley_s_equation__87b47de24c62e1d8.pdf	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Zener and Avalanche breakdown Temperature effects	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/004_Zener_and_Avalanche_breakdown_Temperature_effects__e2302d9472a3245c.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Zener and Avalanche breakdown Temperature effects	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/004_Zener_and_Avalanche_breakdown_Temperature_effects__e2302d9472a3245c.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Zener and Avalanche breakdown Temperature effects (converted PDF; original hash cdec8f804bb21361720719598181 4ec3927fd064950ca7549ea330aa7 57ea27d)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/004_Zener_and_Avalanche_breakdown_Temperature_effects__e2302d9472a3245c.pdf	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Ideal versus Practical diode Diode resistances	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/005_Ideal_versus_Practical_diode_Diode_resistances__14c863657e4b9e42.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.

Source	Path	Use
Alternate extracted slide page text: Ideal versus Practical diode Diode resistances	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/005_Ideal_versus_Practical_diode_Diode_resistances__14c863657e4b9e42.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Ideal versus Practical diode Diode resistances (converted PDF; original hash fd04b9a2689d48164c1b684fadfa9dddd18ded5098eac76a684a6ac33d1c120d)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/005_Ideal_versus_Practical_diode_Diode_resistances__14c863657e4b9e42.pdf	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Diode equivalent circuits Series diode configurations	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/006_Diode_equivalent_circuits_Series_diode_configurations__5453dc2f4e4957f0.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Diode equivalent circuits Series diode configurations	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/006_Diode_equivalent_circuits_Series_diode_configurations__5453dc2f4e4957f0.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Diode equivalent circuits Series diode configurations (converted PDF; original hash 51f1afe81317c1c49f809f003fd54686c0951fcd952ea07a58d6eb4d0b9c5385)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/006_Diode_equivalent_circuits_Series_diode_configurations__5453dc2f4e4957f0.pdf	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Parallel and Series Parallel configurations	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/007_Parallel_and_Series-_Parallel_configurations__aa18f7794f270520.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.

Source	Path	Use
Alternate extracted slide page text: Parallel and Series Parallel configurations	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/007_Parallel_and_Series-_Parallel_configurations__aa18f7794f270520.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Parallel and Series Parallel configurations (converted PDF; original hash 27d54c28dbffe0fd12cd5a4ffa3f28944f3957d04c83a2412a72e88ca1ef3835)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/007_Parallel_and_Series-_Parallel_configurations__aa18f7794f270520.pdf	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Logical Operations using diode	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/008_Logical_Operations_using_diode__69541ca2fb5e21d2.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Logical Operations using diode	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/008_Logical_Operations_using_diode__69541ca2fb5e21d2.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Logical Operations using diode (converted PDF; original hash 4201923b33b3c6c2e2b1586fd2dae0eef75e584057a8c7aadd60c41f1669e4da)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/008_Logical_Operations_using_diode__69541ca2fb5e21d2.pdf	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Diode Clippers Series using Ideal diode	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/009_Diode_Clippers_Series_using_Ideal_diode__7fc19e7302be29a7.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.

Source	Path	Use
Alternate extracted slide page text: Diode Clippers Series using Ideal diode	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/009_Diode_Clippers_Series_using_Ideal_diode__7fc19e7302be29a7.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Diode Clippers Series using Ideal diode (converted PDF; original hash 634c9aee0eedd6dd35f4d1eb18596f004701f9fea9f4f840f40aa146edbe3697)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/009_Diode_Clippers_Series_using_Ideal_diode__7fc19e7302be29a7.pdf	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Diode Clippers Parallel using Ideal diode	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/010_Diode_Clippers_Parallel_using_Ideal_diode__20dfefa028bcfd77.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Diode Clippers Parallel using Ideal diode	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/010_Diode_Clippers_Parallel_using_Ideal_diode__20dfefa028bcfd77.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Diode Clippers Parallel using Ideal diode (converted PDF; original hash c03093630fb1f356cc8af8dae8671a4d77ba2c903c269e9d6be09f45f6b9fbe8)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/010_Diode_Clippers_Parallel_using_Ideal_diode__20dfefa028bcfd77.pdf	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Clampers using Ideal diode	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/011_Clampers_using_Ideal_diode__f7d3c8caf08bbb8f.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.

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Alternate extracted slide page text: Clampers using Ideal diode	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/011_Clampers_using_Ideal_diode__f7d3c8caf08bbb8f.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
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Slide page text: Experiential learning Unit 1	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/013_Experiential_learning_Unit-1__3854d81f81e55ac8.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Experiential learning Unit 1	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/013_Experiential_learning_Unit-1__3854d81f81e55ac8.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Experiential learning Unit 1 (converted PDF; original hash 59479989b0f50a3c5fa2b0df49cc737f7c6d0cf73fd5cbe45bb06dbebf4fc5d9)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/013_Experiential_learning_Unit-1__3854d81f81e55ac8.pdf	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Experiential learning Unit 1	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/014_Experiential_learning_Unit-1__0d18720b1b592cfc.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.

Source	Path	Use
Alternate extracted slide page text: Experiential learning Unit 1	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/014_Experiential_learning_Unit-1__0d18720b1b592cfc.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
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Extracted historical pages: 2025 [1]__10viKVTE9W.pdf	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/First_Year_8enterprise/Files/EPD/Papers/ESA Papers/2025 [1]__10viKVTE9W.pdf	Candidate-matched page text excerpt; original PDF bytes, hash and path remain in datasets/question-archive.json.
Extracted historical pages: 2025 [2]__1LrJQqKVte.pdf	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/First_Year_8enterprise/Files/EPD/Papers/ESA Papers/2025 [2]__1LrJQqKVte.pdf	Candidate-matched page text excerpt; original PDF bytes, hash and path remain in datasets/question-archive.json.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.
2. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
3. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.

Coverage

Every inventoried source unit has an explicit coverage decision. Covered means reviewed and mapped to content; exclusions and gaps remain visible.

Evidence	Source unit	Status	Decision
EV0000001	SRC00001/catalogue	covered	Unit title and index copied from the official units.json catalogue; human content review is pending.

Evidence	Source unit	Status	Decision
EV00000002	SRC00002/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0063. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000003	SRC00003/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0063. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000004	SRC00004/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0063. Sidecar variants compare at similarity 0.9968; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000005	SRC00005/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0063. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000006	SRC00002/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0064 to s0105. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000007	SRC00003/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0064 to s0105. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000008	SRC00004/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0064 to s0105. Sidecar variants compare at similarity 0.9968; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV00000009	SRC00005/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0064 to s0105. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000010	SRC00006/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0066. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000011	SRC00007/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0066. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000012	SRC00008/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0066. Sidecar variants compare at similarity 0.9939; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000013	SRC00009/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0066. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000014	SRC00006/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0067 to s0080. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000015	SRC00007/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0067 to s0080. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000016	SRC00008/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0067 to s0080. Sidecar variants compare at similarity 0.9939; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000017	SRC00009/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0067 to s0080. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000018	SRC00010/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0073. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000019	SRC00011/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0073. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000020	SRC00012/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0073. Sidecar variants compare at similarity 0.9985; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000021	SRC00013/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0073. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000022	SRC00010/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0074 to s0140. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV00000023	SRC00011/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0074 to s0140. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000024	SRC00012/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0074 to s0140. Sidecar variants compare at similarity 0.9985; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000025	SRC00013/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0074 to s0140. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000026	SRC00010/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0141 to s0145. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000027	SRC00011/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0141 to s0145. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000028	SRC00012/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0141 to s0145. Sidecar variants compare at similarity 0.9985; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000029	SRC00013/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0141 to s0145. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV00000030	SRC00014/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0081. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000031	SRC00015/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0081. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000032	SRC00016/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0081. Sidecar variants compare at similarity 0.9980; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000033	SRC00017/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0081. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000034	SRC00014/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0082 to s0164. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000035	SRC00015/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0082 to s0164. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000036	SRC00016/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0082 to s0164. Sidecar variants compare at similarity 0.9980; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000037	SRC00017/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0082 to s0164. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000038	SRC00018/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0065. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000039	SRC00019/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0065. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000040	SRC00020/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0065. Sidecar variants compare at similarity 0.9976; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000041	SRC00021/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0065. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000042	SRC00018/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0066 to s0111. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000043	SRC00019/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0066 to s0111. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV00000044	SRC00020/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0066 to s0111. Sidecar variants compare at similarity 0.9976; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000045	SRC00021/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0066 to s0111. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000046	SRC00022/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0064. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000047	SRC00023/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0064. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000048	SRC00024/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0064. Sidecar variants compare at similarity 0.9966; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000049	SRC00025/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0064. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000050	SRC00022/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0065 to s0075. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV00000051	SRC00023/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0065 to s0075. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000052	SRC00024/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0065 to s0075. Sidecar variants compare at similarity 0.9966; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000053	SRC00025/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0065 to s0075. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000054	SRC00026/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0063. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000055	SRC00027/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0063. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000056	SRC00028/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0063. Sidecar variants compare at similarity 0.9981; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000057	SRC00029/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0063. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000058	SRC00026/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0064 to s0114. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000059	SRC00027/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0064 to s0114. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000060	SRC00028/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0064 to s0114. Sidecar variants compare at similarity 0.9981; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000061	SRC00029/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0064 to s0114. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000062	SRC00030/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0065. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000063	SRC00031/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0065. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000064	SRC00032/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0065. Sidecar variants compare at similarity 0.9977; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV00000065	SRC00033/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0065. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000066	SRC00030/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0066 to s0075. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000067	SRC00031/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0066 to s0075. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000068	SRC00032/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0066 to s0075. Sidecar variants compare at similarity 0.9977; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000069	SRC00033/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0066 to s0075. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000070	SRC00034/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0067. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000071	SRC00035/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0067. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000072	SRC00036/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0067. Sidecar variants compare at similarity 0.9987; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000073	SRC00037/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0067. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000074	SRC00034/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0068 to s0141. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000075	SRC00035/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0068 to s0141. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000076	SRC00036/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0068 to s0141. Sidecar variants compare at similarity 0.9987; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000077	SRC00037/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0068 to s0141. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000078	SRC00034/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0142 to s0200. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV00000079	SRC00035/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0142 to s0200. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000080	SRC00036/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0142 to s0200. Sidecar variants compare at similarity 0.9987; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000081	SRC00037/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0142 to s0200. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000082	SRC00038/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0069. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000083	SRC00039/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0069. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000084	SRC00040/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0069. Sidecar variants compare at similarity 0.9977; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000085	SRC00041/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0069. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000086	SRC00038/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0070 to s0109. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000087	SRC00039/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0070 to s0109. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000088	SRC00040/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0070 to s0109. Sidecar variants compare at similarity 0.9977; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000089	SRC00041/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0070 to s0109. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000090	SRC00042/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0067. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000091	SRC00043/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0067. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000092	SRC00044/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0067. Sidecar variants compare at similarity 0.9988; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000093	SRC00045/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0067. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000094	SRC00042/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0068 to s0130. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000095	SRC00043/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0068 to s0130. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000096	SRC00044/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0068 to s0130. Sidecar variants compare at similarity 0.9988; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000097	SRC00045/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0068 to s0130. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000098	SRC00042/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0131 to s0137. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000099	SRC00043/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0131 to s0137. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV00001 00	SRC00044/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0131 to s0137. Sidecar variants compare at similarity 0.9988; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001 01	SRC00045/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0131 to s0137. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001 02	SRC00046/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/001_Introduction_to_semiconductors_intrinsic_and_extrinsic_semiconductors__349c88a00d515771.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00001 03	SRC00048/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/001_Introduction_to_semiconductors_intrinsic_and_extrinsic_semiconductors__349c88a00d515771.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00001 04	SRC00047/p1	covered	Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human comparison is pending.
EV00001 05	SRC00046/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/001_Introduction_to_semiconductors_intrinsic_and_extrinsic_semiconductors__349c88a00d515771.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000106	SRC00048/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/001_Introduction_to_semiconductors_intrinsic_and_extrinsic_semiconductors__349c88a00d515771.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000107	SRC00047/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.
EV0000108	SRC00046/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/001_Introduction_to_semiconductors_intrinsic_and_extrinsic_semiconductors__349c88a00d515771.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000109	SRC00048/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/001_Introduction_to_semiconductors_intrinsic_and_extrinsic_semiconductors__349c88a00d515771.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000110	SRC00047/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.
EV0000111	SRC00046/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/001_Introduction_to_semiconductors_intrinsic_and_extrinsic_semiconductors__349c88a00d515771.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000112	SRC00048/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/001_Introduction_to_semiconductors_intrinsic_and_extrinsic_semiconductors__349c88a00d515771.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000113	SRC00047/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.
EV0000114	SRC00046/p5	covered	Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/001_Introduction_to_semiconductors_intrinsic_and_extrinsic_semiconductors__349c88a00d515771.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000115	SRC00048/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/001_Introduction_to_semiconductors_intrinsic_and_extrinsic_semiconductors__349c88a00d515771.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000116	SRC00047/p5	covered	Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending.
EV0000117	SRC00046/p6	covered	Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/001_Introduction_to_semiconductors_intrinsic_and_extrinsic_semiconductors__349c88a00d515771.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000118	SRC00048/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/001_Introduction_to_semiconductors_intrinsic_and_extrinsic_semiconductors__349c88a00d515771.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000119	SRC00047/p6	covered	Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending.
EV0000120	SRC00046/p7	covered	Extracted slide page 7 from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/001_Introduction_to_semiconductors_intrinsic_and_extrinsic_semiconductors__349c88a00d515771.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000121	SRC00048/p7	covered	Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/001_Introduction_to_semiconductors_intrinsic_and_extrinsic_semiconductors__349c88a00d515771.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000122	SRC00047/p7	covered	Alternate extracted slide text page 7; normalized text matches the .slides.md page. Human comparison is pending.
EV0000123	SRC00046/p8	covered	Extracted slide page 8 from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/001_Introduction_to_semiconductors_intrinsic_and_extrinsic_semiconductors__349c88a00d515771.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000124	SRC00048/p8	covered	Original slide page 8 rendered from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/001_Introduction_to_semiconductors_intrinsic_and_extrinsic_semiconductors__349c88a00d515771.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000125	SRC00047/p8	covered	Alternate extracted slide text page 8; normalized text matches the .slides.md page. Human comparison is pending.
EV0000126	SRC00049/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/002_Semiconductor_diode_under_no_bias_forward_bias_and_reverse_bias_conditions__63662d27d393ceed.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000127	SRC00051/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/002_Semiconductor_diode_under_no_bias_forward_bias_and_reverse_bias_conditions__63662d27d393ceed.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000128	SRC00050/p1	covered	Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human comparison is pending.
EV0000129	SRC00049/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/002_Semiconductor_diode_under_no_bias_forward_bias_and_reverse_bias_conditions__63662d27d393ceed.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000130	SRC00051/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/002_Semiconductor_diode_under_no_bias_forward_bias_and_reverse_bias_conditions__63662d27d393ceed.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000131	SRC00050/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.
EV0000132	SRC00049/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/002_Semiconductor_diode_under_no_bias_forward_bias_and_reverse_bias_conditions__63662d27d393ceed.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000133	SRC00051/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/002_Semiconductor_diode_under_no_bias_forward_bias_and_reverse_bias_conditions__63662d27d393ceed.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000134	SRC00050/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.
EV0000135	SRC00049/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/002_Semiconductor_diode_under_no_bias_forward_bias_and_reverse_bias_conditions__63662d27d393ceed.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000136	SRC00051/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/002_Semiconductor_diode_under_no_bias_forward_bias_and_reverse_bias_conditions__63662d27d393ceed.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000137	SRC00050/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.
EV0000138	SRC00049/p5	covered	Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/002_Semiconductor_diode_under_no_bias_forward_bias_and_reverse_bias_conditions__63662d27d393ceed.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000139	SRC00051/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/002_Semiconductor_diode_under_no_bias_forward_bias_and_reverse_bias_conditions__63662d27d393ceed.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000140	SRC00050/p5	covered	Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending.
EV0000141	SRC00049/p6	covered	Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/002_Semiconductor_diode_under_no_bias_forward_bias_and_reverse_bias_conditions__63662d27d393ceed.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000142	SRC00051/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/002_Semiconductor_diode_under_no_bias_forward_bias_and_reverse_bias_conditions__63662d27d393ceed.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000143	SRC00050/p6	covered	Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending.
EV0000144	SRC00049/p7	covered	Extracted slide page 7 from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/002_Semiconductor_diode_under_no_bias_forward_bias_and_reverse_bias_conditions__63662d27d393ceed.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000145	SRC00051/p7	covered	Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/002_Semiconductor_diode_under_no_bias_forward_bias_and_reverse_bias_conditions__63662d27d393ceed.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000146	SRC00050/p7	covered	Alternate extracted slide text page 7; normalized text matches the .slides.md page. Human comparison is pending.
EV0000147	SRC00049/p8	covered	Extracted slide page 8 from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/002_Semiconductor_diode_under_no_bias_forward_bias_and_reverse_bias_conditions__63662d27d393ceed.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV00001148	SRC00051/p8	covered	Original slide page 8 rendered from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/002_Semiconductor_diode_under_no_bias_forward_bias_and_reverse_bias_conditions__63662d27d393ceed.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00001149	SRC00050/p8	covered	Alternate extracted slide text page 8; normalized text matches the .slides.md page. Human comparison is pending.
EV00001150	SRC00049/p9	covered	Extracted slide page 9 from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/002_Semiconductor_diode_under_no_bias_forward_bias_and_reverse_bias_conditions__63662d27d393ceed.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00001151	SRC00051/p9	covered	Original slide page 9 rendered from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/002_Semiconductor_diode_under_no_bias_forward_bias_and_reverse_bias_conditions__63662d27d393ceed.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00001152	SRC00050/p9	covered	Alternate extracted slide text page 9; normalized text matches the .slides.md page. Human comparison is pending.
EV00001153	SRC00052/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/003_Diode_characteristics_Shockley_s_equation__87b47de24c62e1d8.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000154	SRC00054/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/003_Diode_characteristics_Shockley_s_equation__87b47de24c62e1d8.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000155	SRC00053/p1	covered	Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human comparison is pending.
EV0000156	SRC00052/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/003_Diode_characteristics_Shockley_s_equation__87b47de24c62e1d8.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000157	SRC00054/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/003_Diode_characteristics_Shockley_s_equation__87b47de24c62e1d8.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000158	SRC00053/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.
EV0000159	SRC00052/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/003_Diode_characteristics_Shockley_s_equation__87b47de24c62e1d8.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000160	SRC00054/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/003_Diode_characteristics_Shockley_s_equation__87b47de24c62e1d8.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000161	SRC00053/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.
EV0000162	SRC00052/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/003_Diode_characteristics_Shockley_s_equation__87b47de24c62e1d8.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000163	SRC00054/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/003_Diode_characteristics_Shockley_s_equation__87b47de24c62e1d8.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000164	SRC00053/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.
EV0000165	SRC00052/p5	covered	Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/003_Diode_characteristics_Shockley_s_equation__87b47de24c62e1d8.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000166	SRC00054/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/003_Diode_characteristics_Shockley_s_equation__87b47de24c62e1d8.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000167	SRC00053/p5	covered	Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending.
EV0000168	SRC00052/p6	covered	Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/003_Diode_characteristics_Shockley_s_equation__87b47de24c62e1d8.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000169	SRC00054/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/003_Diode_characteristics_Shockley_s_equation__87b47de24c62e1d8.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000170	SRC00053/p6	covered	Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending.
EV0000171	SRC00052/p7	covered	Extracted slide page 7 from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/003_Diode_characteristics_Shockley_s_equation__87b47de24c62e1d8.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000172	SRC00054/p7	covered	Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/003_Diode_characteristics_Shockley_s_equation__87b47de24c62e1d8.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000173	SRC00053/p7	covered	Alternate extracted slide text page 7; normalized text matches the .slides.md page. Human comparison is pending.
EV0000174	SRC00052/p8	covered	Extracted slide page 8 from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/003_Diode_characteristics_Shockley_s_equation__87b47de24c62e1d8.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000175	SRC00054/p8	covered	Original slide page 8 rendered from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/003_Diode_characteristics_Shockley_s_equation__87b47de24c62e1d8.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000176	SRC00053/p8	covered	Alternate extracted slide text page 8; normalized text matches the .slides.md page. Human comparison is pending.
EV0000177	SRC00052/p9	covered	Extracted slide page 9 from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/003_Diode_characteristics_Shockley_s_equation__87b47de24c62e1d8.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000178	SRC00054/p9	covered	Original slide page 9 rendered from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/003_Diode_characteristics_Shockley_s_equation__87b47de24c62e1d8.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000179	SRC00053/p9	covered	Alternate extracted slide text page 9; normalized text matches the .slides.md page. Human comparison is pending.
EV0000180	SRC00052/p10	covered	Extracted slide page 10 from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/003_Diode_characteristics_Shockley_s_equation__87b47de24c62e1d8.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000181	SRC00054/p10	covered	Original slide page 10 rendered from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/003_Diode_characteristics_Shockley_s_equation__87b47de24c62e1d8.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000182	SRC00053/p10	covered	Alternate extracted slide text page 10; normalized text matches the .slides.md page. Human comparison is pending.
EV0000183	SRC00052/p11	covered	Extracted slide page 11 from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/003_Diode_characteristics_Shockley_s_equation__87b47de24c62e1d8.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000184	SRC00054/p11	covered	Original slide page 11 rendered from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/003_Diode_characteristics_Shockley_s_equation__87b47de24c62e1d8.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000185	SRC00053/p11	covered	Alternate extracted slide text page 11; normalized text matches the .slides.md page. Human comparison is pending.
EV0000186	SRC00052/p12	covered	Extracted slide page 12 from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/003_Diode_characteristics_Shockley_s_equation__87b47de24c62e1d8.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000187	SRC00054/p12	covered	Original slide page 12 rendered from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/003_Diode_characteristics_Shockley_s_equation__87b47de24c62e1d8.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000188	SRC00053/p12	covered	Alternate extracted slide text page 12; normalized text matches the .slides.md page. Human comparison is pending.
EV0000189	SRC00055/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/004_Zener_and_Avalanche_breakdown_Temperature_effects__e2302d9472a3245c.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV00001190	SRC00057/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/004_Zener_and_Avalanche_breakdown_Temperature_effects__e2302d9472a3245c.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00001191	SRC00056/p1	covered	Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human comparison is pending.
EV00001192	SRC00055/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/004_Zener_and_Avalanche_breakdown_Temperature_effects__e2302d9472a3245c.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00001193	SRC00057/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/004_Zener_and_Avalanche_breakdown_Temperature_effects__e2302d9472a3245c.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00001194	SRC00056/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.
EV00001195	SRC00055/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/004_Zener_and_Avalanche_breakdown_Temperature_effects__e2302d9472a3245c.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV00001 96	SRC00057/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/004_Zener_and_Avalanche_breakdown_Temperature_effects__e2302d9472a3245c.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00001 97	SRC00056/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.
EV00001 98	SRC00055/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/004_Zener_and_Avalanche_breakdown_Temperature_effects__e2302d9472a3245c.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00001 99	SRC00057/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/004_Zener_and_Avalanche_breakdown_Temperature_effects__e2302d9472a3245c.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00002 00	SRC00056/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.
EV00002 01	SRC00055/p5	covered	Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/004_Zener_and_Avalanche_breakdown_Temperature_effects__e2302d9472a3245c.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000202	SRC00057/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/004_Zener_and_Avalanche_breakdown_Temperature_effects__e2302d9472a3245c.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000203	SRC00056/p5	covered	Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending.
EV0000204	SRC00055/p6	covered	Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/004_Zener_and_Avalanche_breakdown_Temperature_effects__e2302d9472a3245c.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000205	SRC00057/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/004_Zener_and_Avalanche_breakdown_Temperature_effects__e2302d9472a3245c.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000206	SRC00056/p6	covered	Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending.
EV0000207	SRC00055/p7	covered	Extracted slide page 7 from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/004_Zener_and_Avalanche_breakdown_Temperature_effects__e2302d9472a3245c.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000208	SRC00057/p7	covered	Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/004_Zener_and_Avalanche_breakdown_Temperature_effects__e2302d9472a3245c.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000209	SRC00056/p7	covered	Alternate extracted slide text page 7; normalized text matches the .slides.md page. Human comparison is pending.
EV0000210	SRC00055/p8	covered	Extracted slide page 8 from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/004_Zener_and_Avalanche_breakdown_Temperature_effects__e2302d9472a3245c.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000211	SRC00057/p8	covered	Original slide page 8 rendered from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/004_Zener_and_Avalanche_breakdown_Temperature_effects__e2302d9472a3245c.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000212	SRC00056/p8	covered	Alternate extracted slide text page 8; normalized text matches the .slides.md page. Human comparison is pending.
EV0000213	SRC00055/p9	covered	Extracted slide page 9 from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/004_Zener_and_Avalanche_breakdown_Temperature_effects__e2302d9472a3245c.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000214	SRC00057/p9	covered	Original slide page 9 rendered from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/004_Zener_and_Avalanche_breakdown_Temperature_effects__e2302d9472a3245c.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000215	SRC00056/p9	covered	Alternate extracted slide text page 9; normalized text matches the .slides.md page. Human comparison is pending.
EV0000216	SRC00055/p10	covered	Extracted slide page 10 from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/004_Zener_and_Avalanche_breakdown_Temperature_effects__e2302d9472a3245c.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000217	SRC00057/p10	covered	Original slide page 10 rendered from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/004_Zener_and_Avalanche_breakdown_Temperature_effects__e2302d9472a3245c.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000218	SRC00056/p10	covered	Alternate extracted slide text page 10; normalized text matches the .slides.md page. Human comparison is pending.
EV0000219	SRC00055/p11	covered	Extracted slide page 11 from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/004_Zener_and_Avalanche_breakdown_Temperature_effects__e2302d9472a3245c.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000220	SRC00057/p11	covered	Original slide page 11 rendered from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/004_Zener_and_Avalanche_breakdown_Temperature_effects__e2302d9472a3245c.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000221	SRC00056/p11	covered	Alternate extracted slide text page 11; normalized text matches the .slides.md page. Human comparison is pending.
EV0000222	SRC00055/p12	covered	Extracted slide page 12 from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/004_Zener_and_Avalanche_breakdown_Temperature_effects__e2302d9472a3245c.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000223	SRC00057/p12	covered	Original slide page 12 rendered from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/004_Zener_and_Avalanche_breakdown_Temperature_effects__e2302d9472a3245c.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000224	SRC00056/p12	covered	Alternate extracted slide text page 12; normalized text matches the .slides.md page. Human comparison is pending.
EV0000225	SRC00055/p13	covered	Extracted slide page 13 from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/004_Zener_and_Avalanche_breakdown_Temperature_effects__e2302d9472a3245c.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000226	SRC00057/p13	covered	Original slide page 13 rendered from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/004_Zener_and_Avalanche_breakdown_Temperature_effects__e2302d9472a3245c.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000227	SRC00056/p13	covered	Alternate extracted slide text page 13; normalized text matches the .slides.md page. Human comparison is pending.
EV0000228	SRC00055/p14	covered	Extracted slide page 14 from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/004_Zener_and_Avalanche_breakdown_Temperature_effects__e2302d9472a3245c.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000229	SRC00057/p14	covered	Original slide page 14 rendered from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/004_Zener_and_Avalanche_breakdown_Temperature_effects__e2302d9472a3245c.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000230	SRC00056/p14	covered	Alternate extracted slide text page 14; normalized text matches the .slides.md page. Human comparison is pending.
EV0000231	SRC00055/p15	covered	Extracted slide page 15 from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/004_Zener_and_Avalanche_breakdown_Temperature_effects__e2302d9472a3245c.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000232	SRC00057/p15	covered	Original slide page 15 rendered from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/004_Zener_and_Avalanche_breakdown_Temperature_effects__e2302d9472a3245c.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000233	SRC00056/p15	covered	Alternate extracted slide text page 15; normalized text matches the .slides.md page. Human comparison is pending.
EV0000234	SRC00055/p16	covered	Extracted slide page 16 from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/004_Zener_and_Avalanche_breakdown_Temperature_effects__e2302d9472a3245c.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000235	SRC00057/p16	covered	Original slide page 16 rendered from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/004_Zener_and_Avalanche_breakdown_Temperature_effects__e2302d9472a3245c.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000236	SRC00056/p16	covered	Alternate extracted slide text page 16; normalized text matches the .slides.md page. Human comparison is pending.
EV0000237	SRC00058/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/005_Ideal_versus_Practical_diode_Diode_resistances__14c863657e4b9e42.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000238	SRC00060/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/005_Ideal_versus_Practical_diode_Diode_resistances__14c863657e4b9e42.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000239	SRC00059/p1	covered	Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human comparison is pending.
EV0000240	SRC00058/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/005_Ideal_versus_Practical_diode_Diode_resistances__14c863657e4b9e42.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000241	SRC00060/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/005_Ideal_versus_Practical_diode_Diode_resistances__14c863657e4b9e42.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000242	SRC00059/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.
EV0000243	SRC00058/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/005_Ideal_versus_Practical_diode_Diode_resistances__14c863657e4b9e42.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000244	SRC00060/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/005_Ideal_versus_Practical_diode_Diode_resistances__14c863657e4b9e42.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000245	SRC00059/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.
EV0000246	SRC00058/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/005_Ideal_versus_Practical_diode_Diode_resistances__14c863657e4b9e42.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000247	SRC00060/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/005_Ideal_versus_Practical_diode_Diode_resistances__14c863657e4b9e42.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000248	SRC00059/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.
EV0000249	SRC00058/p5	covered	Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/005_Ideal_versus_Practical_diode_Diode_resistances__14c863657e4b9e42.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV00002250	SRC00060/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/005_Ideal_versus_Practical_diode_Diode_resistances__14c863657e4b9e42.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00002251	SRC00059/p5	covered	Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending.
EV00002252	SRC00058/p6	covered	Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/005_Ideal_versus_Practical_diode_Diode_resistances__14c863657e4b9e42.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00002253	SRC00060/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/005_Ideal_versus_Practical_diode_Diode_resistances__14c863657e4b9e42.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00002254	SRC00059/p6	covered	Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending.
EV00002255	SRC00058/p7	covered	Extracted slide page 7 from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/005_Ideal_versus_Practical_diode_Diode_resistances__14c863657e4b9e42.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV00002256	SRC00060/p7	covered	Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/005_Ideal_versus_Practical_diode_Diode_resistances__14c863657e4b9e42.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00002257	SRC00059/p7	covered	Alternate extracted slide text page 7; normalized text matches the .slides.md page. Human comparison is pending.
EV00002258	SRC00058/p8	covered	Extracted slide page 8 from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/005_Ideal_versus_Practical_diode_Diode_resistances__14c863657e4b9e42.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00002259	SRC00060/p8	covered	Original slide page 8 rendered from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/005_Ideal_versus_Practical_diode_Diode_resistances__14c863657e4b9e42.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00002260	SRC00059/p8	covered	Alternate extracted slide text page 8; normalized text matches the .slides.md page. Human comparison is pending.
EV00002261	SRC00058/p9	covered	Extracted slide page 9 from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/005_Ideal_versus_Practical_diode_Diode_resistances__14c863657e4b9e42.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000262	SRC00060/p9	covered	Original slide page 9 rendered from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/005_Ideal_versus_Practical_diode_Diode_resistances__14c863657e4b9e42.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000263	SRC00059/p9	covered	Alternate extracted slide text page 9; normalized text matches the .slides.md page. Human comparison is pending.
EV0000264	SRC00058/p10	covered	Extracted slide page 10 from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/005_Ideal_versus_Practical_diode_Diode_resistances__14c863657e4b9e42.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000265	SRC00060/p10	covered	Original slide page 10 rendered from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/005_Ideal_versus_Practical_diode_Diode_resistances__14c863657e4b9e42.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000266	SRC00059/p10	covered	Alternate extracted slide text page 10; normalized text matches the .slides.md page. Human comparison is pending.
EV0000267	SRC00058/p11	covered	Extracted slide page 11 from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/005_Ideal_versus_Practical_diode_Diode_resistances__14c863657e4b9e42.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000268	SRC00060/p11	covered	Original slide page 11 rendered from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/005_Ideal_versus_Practical_diode_Diode_resistances__14c863657e4b9e42.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000269	SRC00059/p11	covered	Alternate extracted slide text page 11; normalized text matches the .slides.md page. Human comparison is pending.
EV0000270	SRC00058/p12	covered	Extracted slide page 12 from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/005_Ideal_versus_Practical_diode_Diode_resistances__14c863657e4b9e42.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000271	SRC00060/p12	covered	Original slide page 12 rendered from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/005_Ideal_versus_Practical_diode_Diode_resistances__14c863657e4b9e42.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000272	SRC00059/p12	covered	Alternate extracted slide text page 12; normalized text matches the .slides.md page. Human comparison is pending.
EV0000273	SRC00058/p13	covered	Extracted slide page 13 from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/005_Ideal_versus_Practical_diode_Diode_resistances__14c863657e4b9e42.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000274	SRC00060/p13	covered	Original slide page 13 rendered from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/005_Ideal_versus_Practical_diode_Diode_resistances__14c863657e4b9e42.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000275	SRC00059/p13	covered	Alternate extracted slide text page 13; normalized text matches the .slides.md page. Human comparison is pending.
EV0000276	SRC00058/p14	covered	Extracted slide page 14 from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/005_Ideal_versus_Practical_diode_Diode_resistances__14c863657e4b9e42.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000277	SRC00060/p14	covered	Original slide page 14 rendered from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/005_Ideal_versus_Practical_diode_Diode_resistances__14c863657e4b9e42.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000278	SRC00059/p14	covered	Alternate extracted slide text page 14; normalized text matches the .slides.md page. Human comparison is pending.
EV0000279	SRC00058/p15	covered	Extracted slide page 15 from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/005_Ideal_versus_Practical_diode_Diode_resistances__14c863657e4b9e42.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV00002280	SRC00060/p15	covered	Original slide page 15 rendered from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/005_Ideal_versus_Practical_diode_Diode_resistances__14c863657e4b9e42.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00002281	SRC00059/p15	covered	Alternate extracted slide text page 15; normalized text matches the .slides.md page. Human comparison is pending.
EV00002282	SRC00058/p16	covered	Extracted slide page 16 from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/005_Ideal_versus_Practical_diode_Diode_resistances__14c863657e4b9e42.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00002283	SRC00060/p16	covered	Original slide page 16 rendered from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/005_Ideal_versus_Practical_diode_Diode_resistances__14c863657e4b9e42.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00002284	SRC00059/p16	covered	Alternate extracted slide text page 16; normalized text matches the .slides.md page. Human comparison is pending.
EV00002285	SRC00058/p17	covered	Extracted slide page 17 from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/005_Ideal_versus_Practical_diode_Diode_resistances__14c863657e4b9e42.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV00002 86	SRC00060/p17	covered	Original slide page 17 rendered from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/005_Ideal_versus_Practical_diode_Diode_resistances__14c863657e4b9e42.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00002 87	SRC00059/p17	covered	Alternate extracted slide text page 17; normalized text matches the .slides.md page. Human comparison is pending.
EV00002 88	SRC00058/p18	covered	Extracted slide page 18 from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/005_Ideal_versus_Practical_diode_Diode_resistances__14c863657e4b9e42.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00002 89	SRC00060/p18	covered	Original slide page 18 rendered from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/005_Ideal_versus_Practical_diode_Diode_resistances__14c863657e4b9e42.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00002 90	SRC00059/p18	covered	Alternate extracted slide text page 18; normalized text matches the .slides.md page. Human comparison is pending.
EV00002 91	SRC00061/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/006_Diode_equivalent_circuits_Series_diode_configurations__5453dc2f4e4957f0.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000292	SRC00063/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/006_Diode_equivalent_circuits_Series_diode_configurations__5453dc2f4e4957f0.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000293	SRC00062/p1	covered	Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human comparison is pending.
EV0000294	SRC00061/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/006_Diode_equivalent_circuits_Series_diode_configurations__5453dc2f4e4957f0.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000295	SRC00063/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/006_Diode_equivalent_circuits_Series_diode_configurations__5453dc2f4e4957f0.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000296	SRC00062/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.
EV0000297	SRC00061/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/006_Diode_equivalent_circuits_Series_diode_configurations__5453dc2f4e4957f0.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000298	SRC00063/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/006_Diode_equivalent_circuits_Series_diode_configurations__5453dc2f4e4957f0.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000299	SRC00062/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.
EV0000300	SRC00061/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/006_Diode_equivalent_circuits_Series_diode_configurations__5453dc2f4e4957f0.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000301	SRC00063/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/006_Diode_equivalent_circuits_Series_diode_configurations__5453dc2f4e4957f0.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000302	SRC00062/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.
EV0000303	SRC00061/p5	covered	Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/006_Diode_equivalent_circuits_Series_diode_configurations__5453dc2f4e4957f0.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000304	SRC00063/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/006_Diode_equivalent_circuits_Series_diode_configurations__5453dc2f4e4957f0.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000305	SRC00062/p5	covered	Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending.
EV0000306	SRC00061/p6	covered	Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/006_Diode_equivalent_circuits_Series_diode_configurations__5453dc2f4e4957f0.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000307	SRC00063/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/006_Diode_equivalent_circuits_Series_diode_configurations__5453dc2f4e4957f0.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000308	SRC00062/p6	covered	Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending.
EV0000309	SRC00061/p7	covered	Extracted slide page 7 from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/006_Diode_equivalent_circuits_Series_diode_configurations__5453dc2f4e4957f0.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000310	SRC00063/p7	covered	Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/006_Diode_equivalent_circuits_Series_diode_configurations__5453dc2f4e4957f0.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000311	SRC00062/p7	covered	Alternate extracted slide text page 7; normalized text matches the .slides.md page. Human comparison is pending.
EV0000312	SRC00061/p8	covered	Extracted slide page 8 from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/006_Diode_equivalent_circuits_Series_diode_configurations__5453dc2f4e4957f0.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000313	SRC00063/p8	covered	Original slide page 8 rendered from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/006_Diode_equivalent_circuits_Series_diode_configurations__5453dc2f4e4957f0.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000314	SRC00062/p8	covered	Alternate extracted slide text page 8; normalized text matches the .slides.md page. Human comparison is pending.
EV0000315	SRC00061/p9	covered	Extracted slide page 9 from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/006_Diode_equivalent_circuits_Series_diode_configurations__5453dc2f4e4957f0.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000316	SRC00063/p9	covered	Original slide page 9 rendered from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/006_Diode_equivalent_circuits_Series_diode_configurations__5453dc2f4e4957f0.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000317	SRC00062/p9	covered	Alternate extracted slide text page 9; normalized text matches the .slides.md page. Human comparison is pending.
EV0000318	SRC00061/p10	covered	Extracted slide page 10 from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/006_Diode_equivalent_circuits_Series_diode_configurations__5453dc2f4e4957f0.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000319	SRC00063/p10	covered	Original slide page 10 rendered from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/006_Diode_equivalent_circuits_Series_diode_configurations__5453dc2f4e4957f0.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000320	SRC00062/p10	covered	Alternate extracted slide text page 10; normalized text matches the .slides.md page. Human comparison is pending.
EV0000321	SRC00061/p11	covered	Extracted slide page 11 from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/006_Diode_equivalent_circuits_Series_diode_configurations__5453dc2f4e4957f0.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000322	SRC00063/p11	covered	Original slide page 11 rendered from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/006_Diode_equivalent_circuits_Series_diode_configurations__5453dc2f4e4957f0.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000323	SRC00062/p11	covered	Alternate extracted slide text page 11; normalized text matches the .slides.md page. Human comparison is pending.
EV0000324	SRC00061/p12	covered	Extracted slide page 12 from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/006_Diode_equivalent_circuits_Series_diode_configurations__5453dc2f4e4957f0.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000325	SRC00063/p12	covered	Original slide page 12 rendered from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/006_Diode_equivalent_circuits_Series_diode_configurations__5453dc2f4e4957f0.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000326	SRC00062/p12	covered	Alternate extracted slide text page 12; normalized text matches the .slides.md page. Human comparison is pending.
EV0000327	SRC00061/p13	covered	Extracted slide page 13 from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/006_Diode_equivalent_circuits_Series_diode_configurations__5453dc2f4e4957f0.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000328	SRC00063/p13	covered	Original slide page 13 rendered from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/006_Diode_equivalent_circuits_Series_diode_configurations__5453dc2f4e4957f0.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000329	SRC00062/p13	covered	Alternate extracted slide text page 13; normalized text matches the .slides.md page. Human comparison is pending.
EV0000330	SRC00064/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/007_Parallel_and_Series-_Parallel_configurations__aa18f7794f270520.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000331	SRC00066/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/007_Parallel_and_Series-_Parallel_configurations__aa18f7794f270520.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000332	SRC00065/p1	covered	Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human comparison is pending.
EV0000333	SRC00064/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/007_Parallel_and_Series-_Parallel_configurations__aa18f7794f270520.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000334	SRC00066/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/007_Parallel_and_Series-_Parallel_configurations__aa18f7794f270520.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000335	SRC00065/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.
EV0000336	SRC00064/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/007_Parallel_and_Series-_Parallel_configurations__aa18f7794f270520.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000337	SRC00066/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/007_Parallel_and_Series-_Parallel_configurations__aa18f7794f270520.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000338	SRC00065/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.
EV0000339	SRC00064/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/007_Parallel_and_Series-_Parallel_configurations__aa18f7794f270520.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000340	SRC00066/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/007_Parallel_and_Series-_Parallel_configurations__aa18f7794f270520.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000341	SRC00065/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.
EV0000342	SRC00064/p5	covered	Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/007_Parallel_and_Series-_Parallel_configurations__aa18f7794f270520.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000343	SRC00066/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/007_Parallel_and_Series-_Parallel_configurations__aa18f7794f270520.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000344	SRC00065/p5	covered	Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending.
EV0000345	SRC00064/p6	covered	Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/007_Parallel_and_Series-_Parallel_configurations__aa18f7794f270520.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000346	SRC00066/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/007_Parallel_and_Series-_Parallel_configurations__aa18f7794f270520.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000347	SRC00065/p6	covered	Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending.
EV0000348	SRC00064/p7	covered	Extracted slide page 7 from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/007_Parallel_and_Series-_Parallel_configurations__aa18f7794f270520.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000349	SRC00066/p7	covered	Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/007_Parallel_and_Series-_Parallel_configurations__aa18f7794f270520.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000350	SRC00065/p7	covered	Alternate extracted slide text page 7; normalized text matches the .slides.md page. Human comparison is pending.
EV0000351	SRC00064/p8	covered	Extracted slide page 8 from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/007_Parallel_and_Series-_Parallel_configurations__aa18f7794f270520.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000352	SRC00066/p8	covered	Original slide page 8 rendered from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/007_Parallel_and_Series-_Parallel_configurations__aa18f7794f270520.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000353	SRC00065/p8	covered	Alternate extracted slide text page 8; normalized text matches the .slides.md page. Human comparison is pending.
EV0000354	SRC00064/p9	covered	Extracted slide page 9 from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/007_Parallel_and_Series-_Parallel_configurations__aa18f7794f270520.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000355	SRC00066/p9	covered	Original slide page 9 rendered from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/007_Parallel_and_Series-_Parallel_configurations__aa18f7794f270520.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000356	SRC00065/p9	covered	Alternate extracted slide text page 9; normalized text matches the .slides.md page. Human comparison is pending.
EV0000357	SRC00064/p10	covered	Extracted slide page 10 from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/007_Parallel_and_Series-_Parallel_configurations__aa18f7794f270520.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000358	SRC00066/p10	covered	Original slide page 10 rendered from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/007_Parallel_and_Series-_Parallel_configurations__aa18f7794f270520.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000359	SRC00065/p10	covered	Alternate extracted slide text page 10; normalized text matches the .slides.md page. Human comparison is pending.
EV0000360	SRC00064/p11	covered	Extracted slide page 11 from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/007_Parallel_and_Series-_Parallel_configurations__aa18f7794f270520.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000361	SRC00066/p11	covered	Original slide page 11 rendered from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/007_Parallel_and_Series-_Parallel_configurations__aa18f7794f270520.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000362	SRC00065/p11	covered	Alternate extracted slide text page 11; normalized text matches the .slides.md page. Human comparison is pending.
EV0000363	SRC00064/p12	covered	Extracted slide page 12 from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/007_Parallel_and_Series-_Parallel_configurations__aa18f7794f270520.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000364	SRC00066/p12	covered	Original slide page 12 rendered from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/007_Parallel_and_Series-_Parallel_configurations__aa18f7794f270520.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000365	SRC00065/p12	covered	Alternate extracted slide text page 12; normalized text matches the .slides.md page. Human comparison is pending.
EV0000366	SRC00067/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/008_Logical_Operations_using_diode__69541ca2fb5e21d2.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000367	SRC00069/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/008_Logical_Operations_using_diode__69541ca2fb5e21d2.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000368	SRC00068/p1	covered	Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human comparison is pending.
EV0000369	SRC00067/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/008_Logical_Operations_using_diode__69541ca2fb5e21d2.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV00003 70	SRC00069/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/008_Logical_Operations_using_diode__69541ca2fb5e21d2.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00003 71	SRC00068/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.
EV00003 72	SRC00067/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/008_Logical_Operations_using_diode__69541ca2fb5e21d2.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00003 73	SRC00069/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/008_Logical_Operations_using_diode__69541ca2fb5e21d2.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00003 74	SRC00068/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.
EV00003 75	SRC00067/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/008_Logical_Operations_using_diode__69541ca2fb5e21d2.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV00003 76	SRC00069/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/008_Logical_Operations_using_diode__69541ca2fb5e21d2.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00003 77	SRC00068/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.
EV00003 78	SRC00067/p5	covered	Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/008_Logical_Operations_using_diode__69541ca2fb5e21d2.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00003 79	SRC00069/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/008_Logical_Operations_using_diode__69541ca2fb5e21d2.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00003 80	SRC00068/p5	covered	Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending.
EV00003 81	SRC00067/p6	covered	Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/008_Logical_Operations_using_diode__69541ca2fb5e21d2.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000382	SRC00069/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/008_Logical_Operations_using_diode__69541ca2fb5e21d2.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000383	SRC00068/p6	covered	Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending.
EV0000384	SRC00067/p7	covered	Extracted slide page 7 from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/008_Logical_Operations_using_diode__69541ca2fb5e21d2.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000385	SRC00069/p7	covered	Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/008_Logical_Operations_using_diode__69541ca2fb5e21d2.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000386	SRC00068/p7	covered	Alternate extracted slide text page 7; normalized text matches the .slides.md page. Human comparison is pending.
EV0000387	SRC00067/p8	covered	Extracted slide page 8 from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/008_Logical_Operations_using_diode__69541ca2fb5e21d2.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000388	SRC00069/p8	covered	Original slide page 8 rendered from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/008_Logical_Operations_using_diode__69541ca2fb5e21d2.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000389	SRC00068/p8	covered	Alternate extracted slide text page 8; normalized text matches the .slides.md page. Human comparison is pending.
EV0000390	SRC00067/p9	covered	Extracted slide page 9 from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/008_Logical_Operations_using_diode__69541ca2fb5e21d2.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000391	SRC00069/p9	covered	Original slide page 9 rendered from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/008_Logical_Operations_using_diode__69541ca2fb5e21d2.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000392	SRC00068/p9	covered	Alternate extracted slide text page 9; normalized text matches the .slides.md page. Human comparison is pending.
EV0000393	SRC00070/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/009_Diode_Clippers_Series_using_Ideal_diode__7fc19e7302be29a7.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000394	SRC00072/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/009_Diode_Clippers_Series_using_Ideal_diode__7fc19e7302be29a7.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000395	SRC00071/p1	covered	Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human comparison is pending.
EV0000396	SRC00070/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/009_Diode_Clippers_Series_using_Ideal_diode__7fc19e7302be29a7.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000397	SRC00072/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/009_Diode_Clippers_Series_using_Ideal_diode__7fc19e7302be29a7.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000398	SRC00071/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.
EV0000399	SRC00070/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/009_Diode_Clippers_Series_using_Ideal_diode__7fc19e7302be29a7.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000400	SRC00072/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/009_Diode_Clippers_Series_using_Ideal_diode__7fc19e7302be29a7.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000401	SRC00071/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.
EV0000402	SRC00070/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/009_Diode_Clippers_Series_using_Ideal_diode__7fc19e7302be29a7.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000403	SRC00072/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/009_Diode_Clippers_Series_using_Ideal_diode__7fc19e7302be29a7.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000404	SRC00071/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.
EV0000405	SRC00070/p5	covered	Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/009_Diode_Clippers_Series_using_Ideal_diode__7fc19e7302be29a7.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000406	SRC00072/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/009_Diode_Clippers_Series_using_Ideal_diode__7fc19e7302be29a7.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000407	SRC00071/p5	covered	Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending.
EV0000408	SRC00070/p6	covered	Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/009_Diode_Clippers_Series_using_Ideal_diode__7fc19e7302be29a7.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000409	SRC00072/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/009_Diode_Clippers_Series_using_Ideal_diode__7fc19e7302be29a7.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000410	SRC00071/p6	covered	Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending.
EV0000411	SRC00070/p7	covered	Extracted slide page 7 from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/009_Diode_Clippers_Series_using_Ideal_diode__7fc19e7302be29a7.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000412	SRC00072/p7	covered	Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/009_Diode_Clippers_Series_using_Ideal_diode__7fc19e7302be29a7.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000413	SRC00071/p7	covered	Alternate extracted slide text page 7; normalized text matches the .slides.md page. Human comparison is pending.
EV0000414	SRC00070/p8	covered	Extracted slide page 8 from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/009_Diode_Clippers_Series_using_Ideal_diode__7fc19e7302be29a7.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000415	SRC00072/p8	covered	Original slide page 8 rendered from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/009_Diode_Clippers_Series_using_Ideal_diode__7fc19e7302be29a7.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000416	SRC00071/p8	covered	Alternate extracted slide text page 8; normalized text matches the .slides.md page. Human comparison is pending.
EV0000417	SRC00070/p9	covered	Extracted slide page 9 from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/009_Diode_Clippers_Series_using_Ideal_diode__7fc19e7302be29a7.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000418	SRC00072/p9	covered	Original slide page 9 rendered from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/009_Diode_Clippers_Series_using_Ideal_diode__7fc19e7302be29a7.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000419	SRC00071/p9	covered	Alternate extracted slide text page 9; normalized text matches the .slides.md page. Human comparison is pending.
EV0000420	SRC00070/p10	covered	Extracted slide page 10 from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/009_Diode_Clippers_Series_using_Ideal_diode__7fc19e7302be29a7.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000421	SRC00072/p10	covered	Original slide page 10 rendered from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/009_Diode_Clippers_Series_using_Ideal_diode__7fc19e7302be29a7.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000422	SRC00071/p10	covered	Alternate extracted slide text page 10; normalized text matches the .slides.md page. Human comparison is pending.
EV0000423	SRC00070/p11	covered	Extracted slide page 11 from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/009_Diode_Clippers_Series_using_Ideal_diode__7fc19e7302be29a7.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000424	SRC00072/p11	covered	Original slide page 11 rendered from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/009_Diode_Clippers_Series_using_Ideal_diode__7fc19e7302be29a7.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000425	SRC00071/p11	covered	Alternate extracted slide text page 11; normalized text matches the .slides.md page. Human comparison is pending.
EV0000426	SRC00070/p12	covered	Extracted slide page 12 from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/009_Diode_Clippers_Series_using_Ideal_diode__7fc19e7302be29a7.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000427	SRC00072/p12	covered	Original slide page 12 rendered from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/009_Diode_Clippers_Series_using_Ideal_diode__7fc19e7302be29a7.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000428	SRC00071/p12	covered	Alternate extracted slide text page 12; normalized text matches the .slides.md page. Human comparison is pending.
EV0000429	SRC00070/p13	covered	Extracted slide page 13 from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/009_Diode_Clippers_Series_using_Ideal_diode__7fc19e7302be29a7.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000430	SRC00072/p13	covered	Original slide page 13 rendered from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/009_Diode_Clippers_Series_using_Ideal_diode__7fc19e7302be29a7.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000431	SRC00071/p13	covered	Alternate extracted slide text page 13; normalized text matches the .slides.md page. Human comparison is pending.
EV0000432	SRC00070/p14	covered	Extracted slide page 14 from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/009_Diode_Clippers_Series_using_Ideal_diode__7fc19e7302be29a7.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000433	SRC00072/p14	covered	Original slide page 14 rendered from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/009_Diode_Clippers_Series_using_Ideal_diode__7fc19e7302be29a7.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000434	SRC00071/p14	covered	Alternate extracted slide text page 14; normalized text matches the .slides.md page. Human comparison is pending.
EV0000435	SRC00073/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/010_Diode_Clippers_Parallel_using_Ideal_diode__20dfefa028bcfd77.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000436	SRC00075/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/010_Diode_Clippers_Parallel_using_Ideal_diode__20dfefa028bcfd77.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000437	SRC00074/p1	covered	Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human comparison is pending.
EV0000438	SRC00073/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/010_Diode_Clippers_Parallel_using_Ideal_diode__20dfefa028bcfd77.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000439	SRC00075/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/010_Diode_Clippers_Parallel_using_Ideal_diode__20dfefa028bcfd77.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000440	SRC00074/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.
EV0000441	SRC00073/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/010_Diode_Clippers_Parallel_using_Ideal_diode__20dfefa028bcfd77.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000442	SRC00075/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/010_Diode_Clippers_Parallel_using_Ideal_diode__20dfefa028bcfd77.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000443	SRC00074/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.
EV0000444	SRC00073/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/010_Diode_Clippers_Parallel_using_Ideal_diode__20dfefa028bcfd77.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000445	SRC00075/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/010_Diode_Clippers_Parallel_using_Ideal_diode__20dfefa028bcfd77.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000446	SRC00074/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.
EV0000447	SRC00073/p5	covered	Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/010_Diode_Clippers_Parallel_using_Ideal_diode__20dfefa028bcfd77.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000448	SRC00075/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/010_Diode_Clippers_Parallel_using_Ideal_diode__20dfefa028bcfd77.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000449	SRC00074/p5	covered	Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending.
EV0000450	SRC00073/p6	covered	Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/010_Diode_Clippers_Parallel_using_Ideal_diode__20dfefa028bcfd77.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000451	SRC00075/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/010_Diode_Clippers_Parallel_using_Ideal_diode__20dfefa028bcfd77.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000452	SRC00074/p6	covered	Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending.
EV0000453	SRC00073/p7	covered	Extracted slide page 7 from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/010_Diode_Clippers_Parallel_using_Ideal_diode__20dfefa028bcfd77.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000454	SRC00075/p7	covered	Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/010_Diode_Clippers_Parallel_using_Ideal_diode__20dfefa028bcfd77.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000455	SRC00074/p7	covered	Alternate extracted slide text page 7; normalized text matches the .slides.md page. Human comparison is pending.
EV0000456	SRC00073/p8	covered	Extracted slide page 8 from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/010_Diode_Clippers_Parallel_using_Ideal_diode__20dfefa028bcfd77.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000457	SRC00075/p8	covered	Original slide page 8 rendered from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/010_Diode_Clippers_Parallel_using_Ideal_diode__20dfefa028bcfd77.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000458	SRC00074/p8	covered	Alternate extracted slide text page 8; normalized text matches the .slides.md page. Human comparison is pending.
EV0000459	SRC00076/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/011_Clampers_using_Ideal_diode__f7d3c8caf08bbb8f.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000460	SRC00078/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/011_Clampers_using_Ideal_diode__f7d3c8caf08bbb8f.pdf; visual page included as vector figure. Human layout/content review is pending.

Evidence	Source unit	Status	Decision
EV0000461	SRC00077/p1	covered	Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human comparison is pending.
EV0000462	SRC00076/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/011_Clampers_using_Ideal_diode__f7d3c8caf08bbb8f.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000463	SRC00078/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/011_Clampers_using_Ideal_diode__f7d3c8caf08bbb8f.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000464	SRC00077/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.
EV0000465	SRC00076/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/011_Clampers_using_Ideal_diode__f7d3c8caf08bbb8f.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000466	SRC00078/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/011_Clampers_using_Ideal_diode__f7d3c8caf08bbb8f.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000467	SRC00077/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000468	SRC00076/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/011_Clampers_using_Ideal_diode__f7d3c8caf08bbb8f.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000469	SRC00078/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/011_Clampers_using_Ideal_diode__f7d3c8caf08bbb8f.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000470	SRC00077/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.
EV0000471	SRC00076/p5	covered	Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/011_Clampers_using_Ideal_diode__f7d3c8caf08bbb8f.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000472	SRC00078/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/011_Clampers_using_Ideal_diode__f7d3c8caf08bbb8f.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000473	SRC00077/p5	covered	Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending.
EV0000474	SRC00076/p6	covered	Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/011_Clampers_using_Ideal_diode__f7d3c8caf08bbb8f.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000475	SRC00078/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/011_Clampers_using_Ideal_diode__f7d3c8caf08bbb8f.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000476	SRC00077/p6	covered	Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending.
EV0000477	SRC00076/p7	covered	Extracted slide page 7 from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/011_Clampers_using_Ideal_diode__f7d3c8caf08bbb8f.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000478	SRC00078/p7	covered	Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/011_Clampers_using_Ideal_diode__f7d3c8caf08bbb8f.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000479	SRC00077/p7	covered	Alternate extracted slide text page 7; normalized text matches the .slides.md page. Human comparison is pending.
EV0000480	SRC00076/p8	covered	Extracted slide page 8 from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/011_Clampers_using_Ideal_diode__f7d3c8caf08bbb8f.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000481	SRC00078/p8	covered	Original slide page 8 rendered from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/011_Clampers_using_Ideal_diode__f7d3c8caf08bbb8f.pdf; visual page included as vector figure. Human layout/content review is pending.

Evidence	Source unit	Status	Decision
EV0000482	SRC00077/p8	covered	Alternate extracted slide text page 8; normalized text matches the .slides.md page. Human comparison is pending.
EV0000483	SRC00076/p9	covered	Extracted slide page 9 from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/011_Clampers_using_Ideal_diode__f7d3c8caf08bbb8f.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000484	SRC00078/p9	covered	Original slide page 9 rendered from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/011_Clampers_using_Ideal_diode__f7d3c8caf08bbb8f.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000485	SRC00077/p9	covered	Alternate extracted slide text page 9; normalized text matches the .slides.md page. Human comparison is pending.
EV0000486	SRC00076/p10	covered	Extracted slide page 10 from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/011_Clampers_using_Ideal_diode__f7d3c8caf08bbb8f.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000487	SRC00078/p10	covered	Original slide page 10 rendered from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/011_Clampers_using_Ideal_diode__f7d3c8caf08bbb8f.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000488	SRC00077/p10	covered	Alternate extracted slide text page 10; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000489	SRC00079/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/013_Experiential_learning_Unit-1__3854d81f81e55ac8.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000490	SRC00081/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/013_Experiential_learning_Unit-1__3854d81f81e55ac8.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000491	SRC00080/p1	covered	Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human comparison is pending.
EV0000492	SRC00079/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/013_Experiential_learning_Unit-1__3854d81f81e55ac8.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000493	SRC00081/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/013_Experiential_learning_Unit-1__3854d81f81e55ac8.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000494	SRC00080/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV00004 95	SRC00079/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/013_Experiential_learning_Unit-1__3854d81f81e55ac8.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00004 96	SRC00081/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/013_Experiential_learning_Unit-1__3854d81f81e55ac8.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00004 97	SRC00080/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.
EV00004 98	SRC00079/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/013_Experiential_learning_Unit-1__3854d81f81e55ac8.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00004 99	SRC00081/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/013_Experiential_learning_Unit-1__3854d81f81e55ac8.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00005 00	SRC00080/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV00005 01	SRC00079/p5	covered	Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/013_Experiential_learning_Unit-1__3854d81f81e55ac8.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00005 02	SRC00081/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/013_Experiential_learning_Unit-1__3854d81f81e55ac8.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00005 03	SRC00080/p5	covered	Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending.
EV00005 04	SRC00079/p6	covered	Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/013_Experiential_learning_Unit-1__3854d81f81e55ac8.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00005 05	SRC00081/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/013_Experiential_learning_Unit-1__3854d81f81e55ac8.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00005 06	SRC00080/p6	covered	Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000507	SRC00079/p7	covered	Extracted slide page 7 from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/013_Experiential_learning_Unit-1__3854d81f81e55ac8.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000508	SRC00081/p7	covered	Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/013_Experiential_learning_Unit-1__3854d81f81e55ac8.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000509	SRC00080/p7	covered	Alternate extracted slide text page 7; normalized text matches the .slides.md page. Human comparison is pending.
EV0000510	SRC00079/p8	covered	Extracted slide page 8 from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/013_Experiential_learning_Unit-1__3854d81f81e55ac8.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000511	SRC00081/p8	covered	Original slide page 8 rendered from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/013_Experiential_learning_Unit-1__3854d81f81e55ac8.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000512	SRC00080/p8	covered	Alternate extracted slide text page 8; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000513	SRC00079/p9	covered	Extracted slide page 9 from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/013_Experiential_learning_Unit-1__3854d81f81e55ac8.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000514	SRC00081/p9	covered	Original slide page 9 rendered from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/013_Experiential_learning_Unit-1__3854d81f81e55ac8.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000515	SRC00080/p9	covered	Alternate extracted slide text page 9; normalized text matches the .slides.md page. Human comparison is pending.
EV0000516	SRC00079/p10	covered	Extracted slide page 10 from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/013_Experiential_learning_Unit-1__3854d81f81e55ac8.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000517	SRC00081/p10	covered	Original slide page 10 rendered from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/013_Experiential_learning_Unit-1__3854d81f81e55ac8.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000518	SRC00080/p10	covered	Alternate extracted slide text page 10; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000519	SRC00079/p11	covered	Extracted slide page 11 from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/013_Experiential_learning_Unit-1__3854d81f81e55ac8.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000520	SRC00081/p11	covered	Original slide page 11 rendered from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/013_Experiential_learning_Unit-1__3854d81f81e55ac8.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000521	SRC00080/p11	covered	Alternate extracted slide text page 11; normalized text matches the .slides.md page. Human comparison is pending.
EV0000522	SRC00082/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/014_Experiential_learning_Unit-1__0d18720b1b592cfc.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000523	SRC00084/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/014_Experiential_learning_Unit-1__0d18720b1b592cfc.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000524	SRC00083/p1	covered	Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000525	SRC00082/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/014_Experiential_learning_Unit-1__0d18720b1b592cfc.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000526	SRC00084/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/014_Experiential_learning_Unit-1__0d18720b1b592cfc.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000527	SRC00083/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.
EV0000528	SRC00082/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/014_Experiential_learning_Unit-1__0d18720b1b592cfc.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000529	SRC00084/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/014_Experiential_learning_Unit-1__0d18720b1b592cfc.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000530	SRC00083/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000531	SRC00082/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/014_Experiential_learning_Unit-1__0d18720b1b592cfc.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000532	SRC00084/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/014_Experiential_learning_Unit-1__0d18720b1b592cfc.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000533	SRC00083/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.
EV0000534	SRC00082/p5	covered	Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/014_Experiential_learning_Unit-1__0d18720b1b592cfc.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000535	SRC00084/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/014_Experiential_learning_Unit-1__0d18720b1b592cfc.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000536	SRC00083/p5	covered	Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000537	SRC00082/p6	covered	Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/014_Experiential_learning_Unit-1__0d18720b1b592cfc.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000538	SRC00084/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/014_Experiential_learning_Unit-1__0d18720b1b592cfc.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000539	SRC00083/p6	covered	Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending.
EV0000540	SRC00082/p7	covered	Extracted slide page 7 from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/014_Experiential_learning_Unit-1__0d18720b1b592cfc.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000541	SRC00084/p7	covered	Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/014_Experiential_learning_Unit-1__0d18720b1b592cfc.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000542	SRC00083/p7	covered	Alternate extracted slide text page 7; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000543	SRC00082/p8	covered	Extracted slide page 8 from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/014_Experiential_learning_Unit-1__0d18720b1b592cfc.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000544	SRC00084/p8	covered	Original slide page 8 rendered from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/014_Experiential_learning_Unit-1__0d18720b1b592cfc.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000545	SRC00083/p8	covered	Alternate extracted slide text page 8; normalized text matches the .slides.md page. Human comparison is pending.
EV0000546	SRC00082/p9	covered	Extracted slide page 9 from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/014_Experiential_learning_Unit-1__0d18720b1b592cfc.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000547	SRC00084/p9	covered	Original slide page 9 rendered from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/014_Experiential_learning_Unit-1__0d18720b1b592cfc.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000548	SRC00083/p9	covered	Alternate extracted slide text page 9; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV00005549	SRC00082/p10	covered	Extracted slide page 10 from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/014_Experiential_learning_Unit-1__0d18720b1b592cfc.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00005550	SRC00084/p10	covered	Original slide page 10 rendered from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/014_Experiential_learning_Unit-1__0d18720b1b592cfc.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00005551	SRC00083/p10	covered	Alternate extracted slide text page 10; normalized text matches the .slides.md page. Human comparison is pending.
EV00005552	SRC00082/p11	covered	Extracted slide page 11 from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/014_Experiential_learning_Unit-1__0d18720b1b592cfc.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00005553	SRC00084/p11	covered	Original slide page 11 rendered from pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/014_Experiential_learning_Unit-1__0d18720b1b592cfc.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00005554	SRC00083/p11	covered	Alternate extracted slide text page 11; normalized text matches the .slides.md page. Human comparison is pending.
EV00005555	SRC00085/q1	covered	Original archive PDF page 1; candidate unit association is text-based and unconfirmed.
EV00005556	SRC00086/q1	covered	Original archive PDF page 1; candidate unit association is text-based and unconfirmed.

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